



Workshop Manual

Crafter 2017 ➤

TGE 2017 ➤

e-Crafter 2019 ➤

e-TGE 2019 ➤

General body repairs, exterior

Edition 12.2018



List of Workshop Manual Repair Groups

Repair Group

- 00 - Technical data
- 50 - Body - front
- 51 - Body - centre
- 55 - Bonnet, rear lid
- 57 - Front doors, door components, central locking
- 58 - Rear doors, door components
- 63 - Bumpers
- 64 - Glazing
- 66 - Exterior equipment

Technical information should always be available to the foremen and mechanics, because their careful and constant adherence to the instructions is essential to ensure vehicle road-worthiness and safety. In addition, the normal basic safety precautions for working on motor vehicles must, as a matter of course, be observed.



Contents

00 - Technical data	1
1 Safety information	1
1.1 Safety precautions when working in the vicinity of high-voltage components	1
1.2 Safety precautions when working on high-voltage system	1
1.3 Safety measures when working on vehicles with a start/stop system	2
2 Hazard classification for the high-voltage system	3
50 - Body - front	10
1 Lock carrier	10
1.1 Assembly overview - lock carrier	10
1.2 Removing and installing lock carrier	13
1.3 Moving to and back from service position	16
1.4 Removing and installing air duct	23
1.5 Removing and installing connecting part for impact bar/subframe	27
1.6 Removing and installing lock carrier struts	27
2 Wings	30
2.1 Assembly overview - wing	30
2.2 Removing and installing wing	31
3 Bulkhead	36
3.1 Assembly overview - plenum chamber cover	36
3.2 Removing and installing plenum chamber cover	39
4 Carrier frame for high-voltage components	44
4.1 Assembly overview - carrier frame for high-voltage components	44
51 - Body - centre	45
1 Carrier frame for high-voltage battery 1 AX2	45
1.1 Assembly overview - carrier frame for high-voltage battery 1 AX2	45
1.2 Removing and installing high-voltage battery 1 AX2	46
2 Deformation elements	47
2.1 Assembly overview - deformation elements	47
2.2 Remove and install deformation elements	47
55 - Bonnet, rear lid	49
1 Bonnet (hood)	49
1.1 Assembly overview - bonnet	49
1.2 Assembly overview - Bowden cable	52
1.3 Removing and installing bonnet	53
1.4 Adjusting bonnet	55
1.5 Removing and installing bonnet lock	57
1.6 Removing and installing bonnet release lever	58
1.7 Removing and installing striker pin	59
1.8 Removing and installing hinges	60
1.9 Removing and installing insulation	61
1.10 Removing and installing Bowden cable	63
1.11 Removing and installing support rod	64
2 Tank flap unit	65
2.1 Assembly overview - tank flap unit	65
2.2 Removing and installing fuel tank flap	66
57 - Front doors, door components, central locking	69
1 Door	69
1.1 Assembly overview - door	69



1.2	Assembly overview - door seals	70
1.3	Removing and installing inner door seal	73
1.4	Removing and installing door	75
1.5	Adjusting door	77
1.6	Adjusting striker pin	78
1.7	Removing and installing door hinge	79
1.8	Removing and installing door arrester	80
2	Door components	82
2.1	Assembly overview - door components	82
2.2	Assembly overview - window regulator	84
2.3	Assembly overview - door handle and door lock	85
2.4	Assembly overview - window channels and window slot (aperture) seals	86
2.5	Removing and installing window regulator motor	86
2.6	Removing and installing window regulator	87
2.7	Removing and installing inner door cover	88
2.8	Removing and installing striker pin	89
2.9	Removing and installing door handle	90
2.10	Removing and installing door handle bracket	91
2.11	Removing and installing door lock	92
2.12	Removing and installing window channel	94
2.13	Removing and installing lock cylinder	95
2.14	Removing and installing cap on door handle	97
2.15	Removing and installing inner window slot seal	97
2.16	Removing and installing outer window slot seal	98
3	Central locking	99
3.1	Overview of fitting locations - central locking	99
3.2	Removing and installing driver door control unit J386 and front passenger door control unit J387	100
3.3	Removing and installing ignition key battery	100
3.4	Removing and installing cap on ignition key	101
3.5	Removing and installing bonnet contact switch F387	103
58	Rear doors, door components	104
1	Door	104
1.1	Assembly overview - door	104
1.2	Assembly overview - door seals	105
1.3	Removing and installing door	105
1.4	Adjusting door	106
1.5	Adjusting striker pin	108
1.6	Removing and installing door hinges	110
1.7	Removing and installing door arrester	111
1.8	Removing and installing inner door seal	113
2	Door components	114
2.1	Assembly overview - door handle and door lock	114
2.2	Removing and installing door components	120
2.3	Removing and installing striker pin	128
2.4	Removing and installing door handle	130
2.5	Removing and installing door handle bracket	131
2.6	Removing and installing door lock	134
2.7	Removing and installing cap on door handle	141
2.8	Removing and installing lock cylinder	143
3	Sliding door	145
3.1	Assembly overview - sliding door	145
3.2	Assembly overview - door seals	148
3.3	Removing and installing sliding door	149



3.4	Adjusting sliding door	151
3.5	Removing and installing add-on parts	156
3.6	Removing and installing inner door seal	160
4	Wing door	162
4.1	Assembly overview - wing door	162
4.2	Assembly overview - door seals	165
4.3	Removing and installing wing door	165
4.4	Adjusting wing door	166
4.5	Removing and installing hinges	167
4.6	Removing and installing door attachments, wing door	169
4.7	Removing and installing outer door seal	174
4.8	Removing and installing inner door seal	175
63	Bumpers	177
1	Bumper, front	177
1.1	Assembly overview - bumper cover	177
1.2	Removing and installing bumper cover	179
1.3	Removing and installing impact bar	182
1.4	Removing and installing add-on parts	184
2	Bumper, rear	188
2.1	Assembly overview - bumper cover	188
2.2	Assembly overview - impact bar	190
2.3	Removing and installing bumper cover	192
2.4	Removing and installing impact bar	196
2.5	Removing and installing add-on parts	197
2.6	Installing holder for parking aid sender	199
64	Glazing	201
1	Repair notes	201
1.1	Minimum curing periods for bonded windows	201
1.2	Window repair	201
1.3	Installation instructions for bonded windows	202
1.4	Preparing old undamaged windows for fitting	204
1.5	Preparing new windows for installing	204
1.6	Preparing body flange for fitting	205
1.7	Cleaning off excess adhesive	206
2	Windscreen	207
2.1	Assembly overview - windscreen	207
2.2	Removing and installing windscreen	207
3	Rear windscreen	213
3.1	Assembly overview - rear window	213
3.2	Removing and installing rear window	216
4	Side window	228
4.1	Assembly overview - front side window	228
4.2	Assembly overview - rear side windows	230
4.3	Removing and installing front side window	232
4.4	Removing and installing rear side window	237
5	Door windows	241
5.1	Assembly overview - rear door window	241
5.2	Removing and installing front door window	241
5.3	Removing and installing fixed front door window	242
5.4	Removing and installing rear door window	243
66	Exterior equipment	247
1	Radiator grille and front trim	247



1.1	Assembly overview - radiator grille	247
1.2	Removing and installing radiator grille	248
2	Mouldings, trims, extensions	249
2.1	Assembly overview - covers	249
2.2	Assembly overview - wheel arch covers	251
2.3	Assembly overview - heat shield	252
2.4	Removing and installing cover	253
2.5	Removing and installing trim strip	264
2.6	Removing and installing B-pillar trim	265
2.7	Removing and installing wheel arch covers	266
2.8	Removing and installing underbody heat shield	268
3	Noise insulation	270
3.1	Assembly overview - noise insulation	270
3.2	Removing and installing noise insulation	271
4	Underbody cladding	274
4.1	Overview of fitting locations - underbody cladding	274
4.2	Assembly overview - underbody cladding	277
4.3	Removing and installing underbody cladding	279
4.4	Removing and installing underbody cladding, e-Crafter or e-TGE	287
5	Roof rails for roof load carrier	292
5.1	Assembly overview - roof rails for roof load carrier	292
6	Exterior mirror	293
6.1	Assembly overview - exterior mirror	293
6.2	Removing and installing exterior mirror	295
6.3	Removing and installing mirror glass	295
6.4	Removing and installing mirror cover	296
6.5	Removing and installing mirror trim	297
7	Wheel housing liner	299
7.1	Assembly overview - front wheel housing liner	299
7.2	Removing and installing front wheel housing liner	299
8	Lettering and emblems	302
8.1	Assembly overview - lettering and emblems	302
8.2	Dimensions - lettering and emblems on rear	308
8.3	Dimensions - lettering and emblems on sides	311
8.4	Renewing lettering and badges at rear	311
8.5	Renewing lettering and badges	314
8.6	Removing and installing lettering and emblems for radiator grille	315
9	Towing bracket	318
9.1	Assembly overview - towing bracket	318
9.2	Removing and installing towing bracket	319
10	Spare wheel bracket	324
10.1	Assembly overview - spare wheel bracket	324
10.2	Removing and installing spare wheel bracket	325
10.3	Removing and installing tether	327
10.4	Removing and installing spare wheel winch bracket	328
11	Mudflap	330
11.1	Assembly overview - mudflap	330
11.2	Removing and installing mudflaps	330
12	Load floor body	333
12.1	Contact corrosion	333
12.2	Assembly overview - dropside bed	333
12.3	Assembly overview - dropside body	337
12.4	Assembly overview - rear cross member	338
12.5	Assembly overview - carrier	339



12.6	Assembly overview - skid plate	340
12.7	Assembly overview - wheel housing	341
12.8	Removing and installing dropside bed	341
12.9	Removing and installing tailboard, front	342
12.10	Removing and installing tailgate	343
12.11	Removing and installing console	343
12.12	Removing and installing carrier	344
12.13	Removing and installing skid plate	344
12.14	Removing and installing strut	345
12.15	Removing and installing wheel housing	346





00 – Technical data

1 Safety information

(VRL012371; Edition 12.2018)

⇒ ["1.1 Safety precautions when working in the vicinity of high-voltage components", page 1](#)

⇒ ["1.2 Safety precautions when working on high-voltage system", page 1](#)

⇒ ["1.3 Safety measures when working on vehicles with a start/stop system", page 2](#)

1.1 Safety precautions when working in the vicinity of high-voltage components

Danger to life from high voltage

The high-voltage system is under high voltage. Damage to high-voltage components can result in severe or fatal injury from electric shock.

- Perform visual check of high-voltage components and high-voltage cables.
- Never use cutting or forming tools, or any other sharp-edged tools.
- Never use heat sources such as welding, brazing, soldering, hot air or thermal bonding equipment.

1.2 Safety precautions when working on high-voltage system

Danger to life from high voltage

The high-voltage system is under high voltage. Risk of severe or fatal injury due to electric shock.

- People having electronic medical devices for life support or health maintenance in or on their bodies are prohibited from working on the high-voltage system. Such medical devices include internal analgesic pumps, implanted defibrillators, pacemakers, insulin pumps and hearing aids.
- The high-voltage system must be de-energised by a suitably qualified technician.

Risk of injury from motor starting unexpectedly

On electric and hybrid vehicles, it can easily be missed that the vehicle is in "ready" mode. There is a risk of parts of the body becoming trapped or drawn in.

- Switch off ignition.
- Always store the ignition key outside the vehicle.



Risk of damage to high-voltage cables

Improper handling of high-voltage cables or high-voltage connectors may result in damage to their insulation.

- Never support body weight on high-voltage cables or high-voltage connectors.
- Never support any tools on high-voltage cables or high-voltage connectors.
- Never kink or severely bend high-voltage cables.
- Always observe the coding when connecting high-voltage connectors.

Risk of injury from activated stationary air conditioning

On electric and hybrid vehicles with active stationary air conditioning, the stationary air conditioning could switch on unintentionally. Risk of limbs becoming trapped or drawn in by the radiator fan starting automatically.

- Deactivate the stationary air conditioning.

1.3 Safety measures when working on vehicles with a start/stop system

Risk of injury from motor starting unexpectedly

If the vehicle's start/stop system is activated, the engine can start unexpectedly. A message in the dash panel insert indicates whether the Start/Stop system is activated.

- Deactivate Start/Stop system by switching off the ignition.



2 Hazard classification for the high-voltage system

DANGER

The vehicle's high-voltage system and the high-voltage battery are dangerous and can cause burns or other injuries and even lead to a fatal electric shock.

- Any work on the high-voltage system, or on systems which could be indirectly affected by it, must only be carried out by properly trained and qualified expert personnel.
- In the event of queries or uncertainties regarding the terms "high-voltage technician" or "high-voltage expert", or those concerning the high-voltage system, the responsible importer must be contacted prior to the start of any work.
- Any repair work must be performed in accordance with applicable laws and regulations, the state-of-the-art technology, any relevant accident prevention regulations (in Germany, including but not limited to the BGI/GUV-I 8686 – Training for work on vehicles with high voltage systems), as well as this workshop manual.

Before work on the high-voltage system is started, a high-voltage technician must de-energise the high-voltage system ➔ Electric drive motor (270, LS2); Rep. gr. 93 ; De-energising high-voltage system .

The types of work for which the high-voltage system has to be de-energised are indicated in the list entitled "Work on the high-voltage system" ➔ [page 3](#) .

Work for which the high-voltage system has to be de-energised:

- ◆ Only the HVT ➔ [page 4](#) is authorised to de-energise the high-voltage system in a certified manner.
- ◆ All work on an electric drive vehicle must be carried out only by those who are at least qualified as electrically instructed persons (EIP) ➔ [page 4](#) .
- ◆ Regardless of the work to be performed, visually inspect high-voltage components in the work area.
- ◆ High-voltage cables must not be extensively bent or kinked.
- ◆ In the event of noticeable problems or if in any doubt, the high-voltage technician (HVT) or the high-voltage expert (HVE) ➔ [page 4](#) must be consulted.
- ◆ Any work involving metal-removing, deforming and sharp-edged tools or heat sources such as welding, soldering, hot air, thermal bonding and infrared drying in the vicinity of high-voltage components and cables is prohibited. In this case, the high-voltage system must be de-energised and the respective component removed or sufficiently protected.
- ◆ All listed work refers to the removal and installation or the renewal of the individual components.
- ◆ For safety reasons, no further work should be performed in or on the vehicle during the charging process.
- ◆ For regular maintenance work, the high-voltage system does not have to be de-energised.



Explanation of qualifications

Working on high-voltage system

Qualification VW	Qualification MAN	Qualification DGUV-I 200-005	Scenario
EIP (electrically instructed person)	Person made aware of high-voltage issues in connection with MAN Truck & Bus AG	Chapter 2	Induction training for non-electrical work (< 60 VDC)
HVT (high-voltage technician)	Qualified vehicle electrician of the MAN Truck & Bus AG (for vehicles not being intrinsically safe in terms of high voltage)	Chapter 3.1 b	Work on inherently safe HV production vehicles ♦ Vehicles are de-energised exclusively with certification. ♦ Complete contact protection in place.
HVE level 1 (high-voltage expert)	Qualified vehicle electrician with authorisation to work in energised vehicles of MAN Truck & Bus AG (for vehicles not being intrinsically safe in terms of high voltage)	Chapter 3.2 b	Work on not inherently safe HV production vehicles ♦ Vehicles are also de-energised without certification or there is no complete contact protection in place, e.g. in the event of an accident.
HVE level 2 (high-voltage expert)		Chapter 3.3	Work on live accumulators ♦ Working on parts supplied with voltage, inevitably with no contact protection, for fault finding, component replacement, etc.

Observe the minimum qualification for the following work:	Minimum qualification:
De-energise high-voltage system	HVT ➔ page 4

Work on power unit

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➔ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Three-phase current drive - VX54-	X			EIP
Electric drive motor - V141-	X			EIP
Drive motor temperature sender/rotor position sender -G712- / -G713-	X			EIP



Work on power and control electronics for electric drive

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Power and control electronics for electric drive - JX1- with ♦ Electric drive control unit - J841- ♦ Intermediate circuit capacitor 1 - C25- ♦ Voltage converter - A19- ♦ DC/AC converter for drive motor - A37-	X			EIP
Air conditioner compressor fuse - S355-	X			EIP

Working on high-voltage system

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Potential equalisation lines (earth) with connection to HV components	X			HVT
Measure insulation resistance		X		HVT
TW – maintenance connector for high-voltage system (service connector, service disconnection)		X		HVT
PX – high-voltage cables (orange) in vehicle	X			HVT

Work on heating and air conditioning system

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Electrical air conditioner compressor - V470-		X		EIP
High-voltage heater (PTC) - Z115-		X		EIP
Work on heating and air conditioning system in vehicle interior			X	EIP
Refrigerant lines in the vehicle periphery (work which is not directly on the AC compressor and can be carried out without opening the refrigerant circuit, e.g. loosening and securing refrigerant lines)			X	EIP



When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➔ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Air conditioning performance test (for checking the pressure levels in the refrigerant circuit by means of air conditioning service equipment)			X	EIP
Refrigerant lines directly on electrical air conditioner compressor - V470-		X		EIP
Extract, evacuate or fill refrigerant			X	EIP

Work on charging connection

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➔ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
High-voltage battery charging socket 1 - UX4- with ◆ Temperature sender for charging socket 1 - G853- ◆ LED module for charging socket 1 - L263- ◆ Actuator for locking		X		EIP
Immediate charge button - E766-			X	EIP
Manual release mechanism for charging connector		X		EIP
Charge high-voltage battery in workshop area			X	EIP

Work on charging unit

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➔ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Charging unit 1 for high-voltage battery - AX4- with control unit for high-voltage battery charging unit - J1050-		X		EIP



Work on high-voltage battery

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Remove and install underbody cladding			X	EIP
Remove and install high-voltage battery 1 - AX2-		X		HVT
Disconnect high-voltage network connection		X		HVT
Disconnect low-voltage connection of high-voltage battery			X	EIP
Battery regulation control unit - J840- (service flap)		X		HVT
Module monitor control unit for batteries - J497- (battery removed)				HVE
Switching unit for high-voltage battery - SX6- (battery removed)				HVE
Open high-voltage battery 1 - AX2- (battery removed)				HVE
Bond high-voltage battery 1 - AX2- (battery removed)				HVE
Battery module 1 - J991- and other battery modules (battery removed)				HVE

Work on accident vehicles

When working on the following components:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Preliminary evaluation of degree of danger				HVT
Body work (with straightening jig)	X			EIP
Paint vehicle – observe instructions in paintwork manual			X	EIP

Work in vicinity of high-voltage components

When working on the following components or for following work:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➤ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Body work (assembly work as well as glass and dent repairs)			X	EIP
Gearbox with electric drive motor	X			EIP
Steering rack		X		EIP
Subframe, front		X		EIP



When working on the following components or for following work:	Before work is started, high-voltage system de-energised by HVT?			Minimum qualification: ➔ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Front brakes			X	EIP
Rear brakes			X	EIP
Brake servo			X	EIP
Brake pressure accumulator for energy recovery of brake system	X			EIP
Rear axle and running gear			X	EIP
Underbody cladding			X	EIP
When welding, cover high-voltage components and visually inspect afterwards	X			EIP
Work involving metal-removing, deforming and sharp-edged tools or heat sources such as welding, soldering, hot air, thermal bonding and infrared drying in the vicinity of high-voltage components and cables	X			EIP
Work for which the motor is detached (e.g. motor mounting)		X		EIP
Control units and electric components of 12-V system (except airbag)			X	EIP
Battery - A-			X	EIP
Front left headlight - MX1-			X	EIP
Front right headlight - MX2-			X	EIP
Change bulb in headlight			X	EIP

General work

When working on the following components:	De-energise high-voltage system before work is started?			Minimum qualification: ➔ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Drain and fill fluids (coolant and oils)			X	EIP
Coolant circuit and coolant expansion tank			X	EIP
Change tyres			X	EIP
Miscellaneous work on 12-V system (excluding airbag)			X	EIP
Airbag		X		EIP
Work on earth points of 12-V system (without potential equalisation lines)			X	EIP
Tail lights			X	EIP
Renew/repair windows			X	EIP
Repairs in vehicle interior			X	EIP
Repairs in roof area			X	EIP
Repairs on rear lid			X	EIP
Repairs on bumpers			X	EIP



When working on the following components:	De-energise high-voltage system before work is started?			Minimum qualification: ➤ page 4
	yes (diagnostic and manual power disconnection)	yes (only diagnostic power disconnection)	No	
Move lock carrier to service position			X	EIP
Removing and installing lock carrier			X	EIP
Repair work on lock carrier			X	EIP
Disconnect high-voltage battery 1 - AX2- from retaining frame		X		HVT
Remove and install deformation elements			X	EIP



50 – Body - front

1 Lock carrier

⇒ [“1.1 Assembly overview - lock carrier”, page 10](#)

⇒ [“1.2 Removing and installing lock carrier”, page 13](#)

⇒ [“1.3 Moving to and back from service position”, page 16](#)

⇒ [“1.4 Removing and installing air duct”, page 23](#)

⇒ [“1.5 Removing and installing connecting part for impact bar/subframe”, page 27](#)

⇒ [“1.6 Removing and installing lock carrier struts”, page 27](#)

1.1 Assembly overview - lock carrier

⇒ [“1.1.1 Assembly overview - lock carrier”, page 10](#)

⇒ [“1.1.2 Assembly overview - lock carrier, e-Crafter or e-TGE”, page 12](#)

1.1.1 Assembly overview - lock carrier

1 - Lock carrier

- ☐ Removing and installing
⇒ [page 13](#)
- ☐ Moving to and back from
service position
⇒ [page 16](#)

2 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 70 Nm +180°

3 - Connecting parts

- ☐ For frame
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 27](#)

4 - Bolts

- ☐ Qty. 3 on each side
- ☐ 8 Nm

5 - Bolts

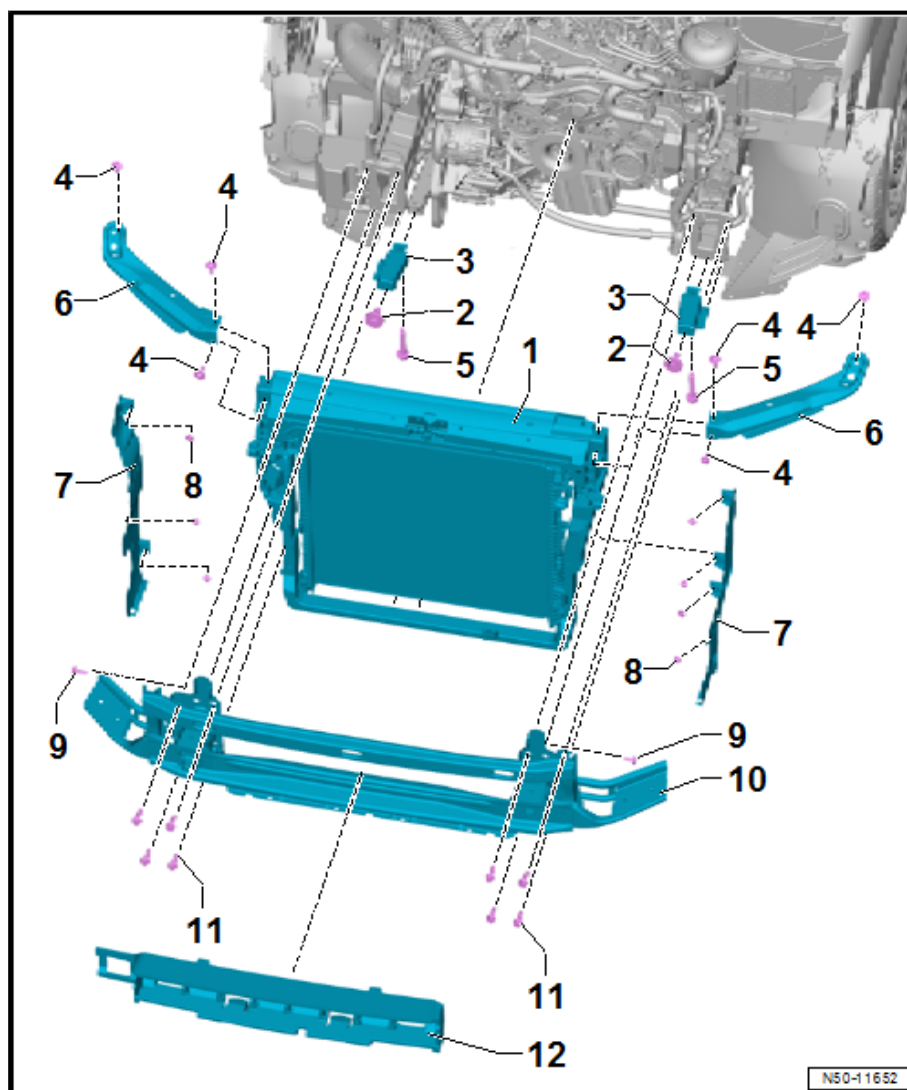
- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 70 Nm +180°

6 - Struts

- ☐ Lock carrier
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 27](#)

7 - Air ducts

- ☐ Qty. 2



Volkswagen Technical Site: <https://vwts.ru>



- ☐ Removing and installing
⇒ [page 23](#)

8 - Bolts

- ☐ Qty. 4 left
- ☐ Qty. 3 right
- ☐ 4 Nm

9 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

10 - Impact bar

- ☐ Front
- ☐ Removing and installing ⇒ [page 182](#)

11 - Bolts

- ☐ Qty. 8
- ☐ 60 Nm

12 - Foam filler piece

- ☐ Removing and installing ⇒ [page 184](#)



1.1.2 Assembly overview - lock carrier, e-Crafter or e-TGE

1 - Bolts

- ☐ Qty. 6
- ☐ 8 Nm

2 - Struts

- ☐ Lock carrier
- ☐ Qty. 2
- ☐ Removing and installing
[⇒ page 27](#)

3 - Connecting parts

- ☐ For frame
- ☐ Qty. 2
- ☐ Removing and installing
[⇒ page 27](#)

4 - Lock carrier

- ☐ Removing and installing
[⇒ page 13](#)
- ☐ Moving to and back from service position
[⇒ page 16](#)

5 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 70 Nm +180°

6 - Air duct

- ☐ Left
- ☐ Removing and installing
[⇒ page 23](#)

7 - Bolts

- ☐ Qty. 4
- ☐ 8 Nm

8 - Bolts

- ☐ Qty. 8
- ☐ 60 Nm

9 - Foam filler piece

- ☐ Removing and installing [⇒ page 184](#)

10 - Bolts

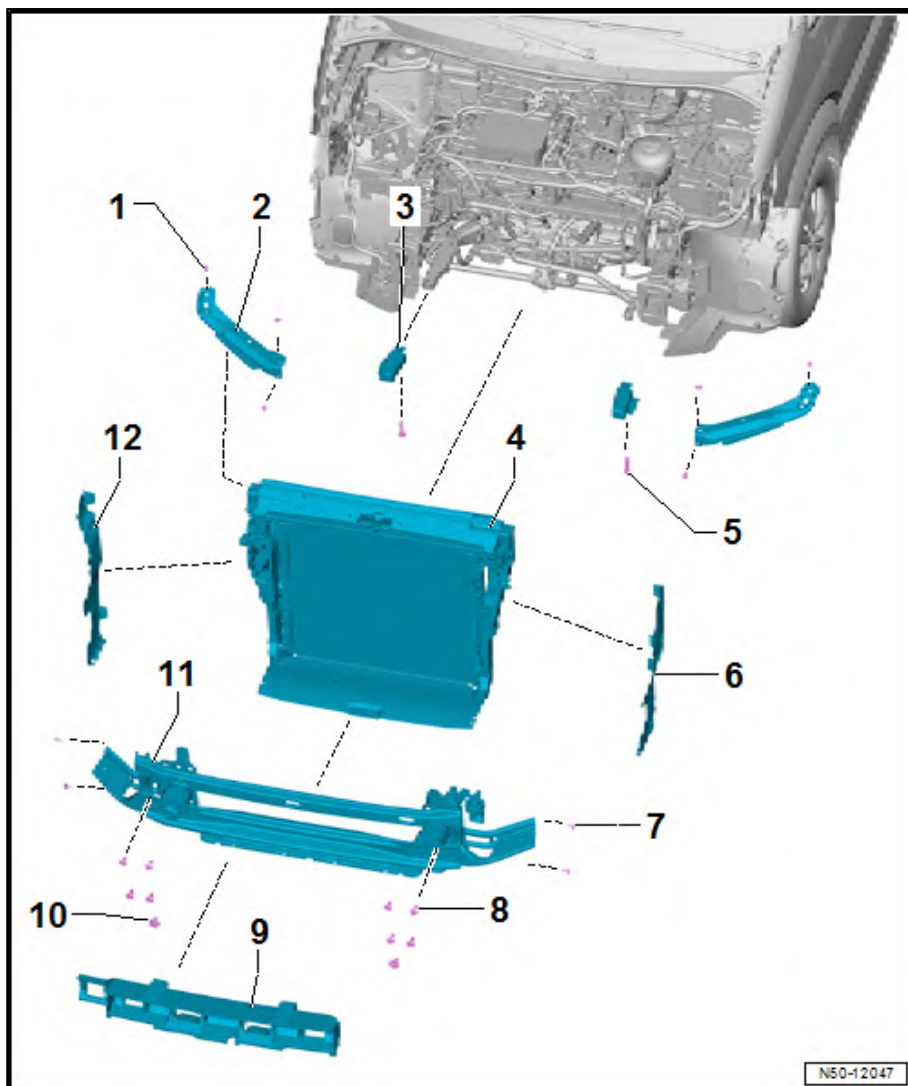
- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 70 Nm +180°

11 - Bumper carrier

- ☐ Front
- ☐ Removing and installing [⇒ page 182](#)

12 - Air duct

- ☐ Right
- ☐ Removing and installing [⇒ page 23](#)





1.2 Removing and installing lock carrier

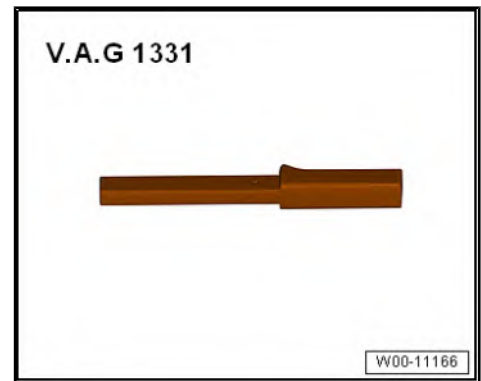
⇒ ["1.2.1 Removing and installing lock carrier", page 13](#)

⇒ ["1.2.2 Removing and installing lock carrier, e-Crafter or e-TGE", page 14](#)

1.2.1 Removing and installing lock carrier

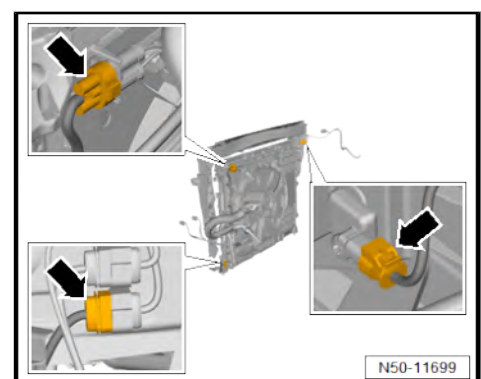
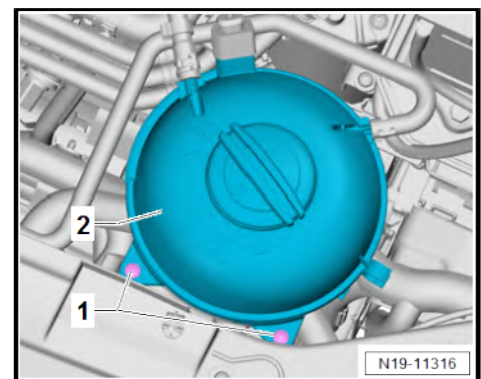
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

- Move to service position ⇒ [page 16](#) .
- Drain coolant ⇒ Rep. gr. 19 ; Cooling system/coolant; Draining and filling coolant .
- Unscrew bolts -1-.
- Lay coolant expansion tank -2- to one side.
- Loosen clamps of coolant hoses at top right of radiator.
- Pull off coolant hoses at top right of radiator.
- Disconnect refrigerant lines from condenser ⇒ Heating, air conditioning; Rep. gr. 87 ; Refrigerant circuit; Disconnecting refrigerant lines from and connecting to condenser .
- Separate electrical connectors -arrows-.





- Unscrew bolts -3- on left and right of adapter - T10467- -4-.
- Unscrew guide pin - T 10093- -2- from left and right longitudinal members with the aid of a second mechanic.
- Remove lock carrier -1- from front end.

Installing

Install in reverse order of removal, observing the following:

Vehicles with adaptive cruise control

- Calibrate adaptive cruise control ⇒ Running gear, axles, steering; Rep. gr. 44 ; Adaptive cruise control .

Continued for all vehicles



Note

- ♦ *Always fit lock carrier on the bolting points at the foremost position in the elongated hole. Only then can the lock carrier move in a crash situation.*
- ♦ *Lock carrier must be installed stress-free.*
- Adjust headlights ⇒ Electrical system; Rep. gr. 94 ; Headlights; Adjusting headlights .

Specified torques

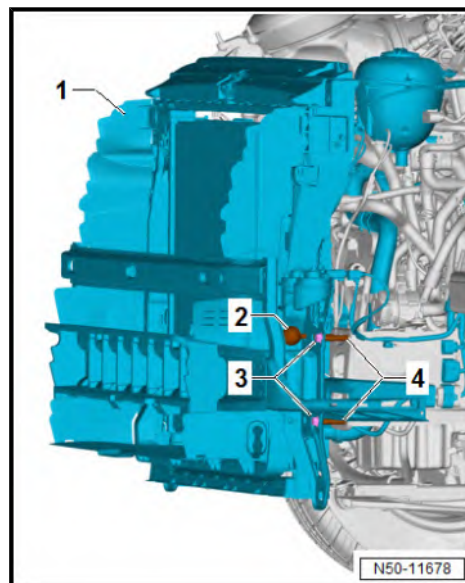
- ♦ ⇒ [“1.1 Assembly overview - lock carrier”, page 10](#)

1.2.2 Removing and installing lock carrier, e-Crafter or e-TGE

Special tools and workshop equipment required

- ♦ Hose clamps up to 25 mm - 3094-

- ♦ Torque wrench - V.A.G 1331-





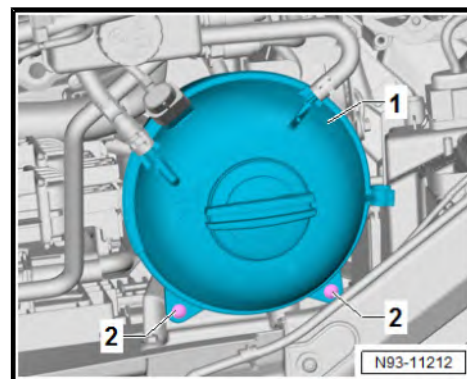
- ◆ Drip tray for workshop hoist - VAS 6208-



- ◆ Vehicle diagnostic tester

Removing

- Move to service position ➔ [page 19](#) .
- Unscrew bolts -2-.
- Lay coolant expansion tank -1- to one side.



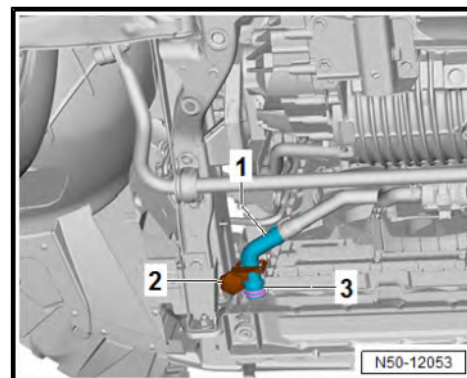
- Clamp off coolant hoses -1- using hose clamps up to 25 mm - 3094- -2-.
- Place drip tray for workshop hoist - VAS 6208- underneath.

CAUTION

On a warm motor, the cooling system is under high pressure. Hot steam/hot coolant can escape - risk of scalding.

Skin and other parts of the body may be scalded.

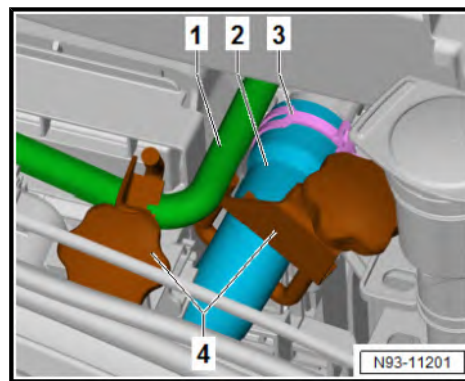
- Wear protective gloves.
- Wear safety goggles.
- Reduce excess pressure by covering cap of coolant expansion tank with cloths and opening it carefully.



- Loosen clamp -3-.
- Pull off coolant hose -1-.



- Clamp off coolant hoses -1- and -2- with hose clamps up to 25 mm - 3094- -4-.
- Loosen clamp of coolant hose -1-.
- Loosen clamp -3-.
- Pull off coolant hoses -1- and -2-.
- Disconnect electrical connectors on the following components:
 - ◆ Bonnet contact switch - F387-
 - ◆ Ambient temperature sensor - G17- ,
 - ◆ Front airbag crash sensor for driver side - G283- ,
 - ◆ Front airbag crash sensor for front passenger side - G284- ,
 - ◆ Treble horn - H2- and bass horn - H7- ,
 - ◆ Radiator fan - V7- .
- Disconnect refrigerant lines from condenser ⇒ Heating, air conditioning; Rep. gr. 87 ; Refrigerant circuit; Disconnecting refrigerant lines from and connecting to condenser .
- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure lock carrier -1-.
- Unscrew bolts -2-.
- Unscrew guide pins - T10093- -3-. Remove lock carrier -1- together with second mechanic.



Installing

Install in reverse order of removal, observing the following:

Vehicles with adaptive cruise control

- Calibrate adaptive cruise control ⇒ Running gear, axles, steering; Rep. gr. 44 ; Adaptive cruise control .

Continued for all vehicles

- Check coolant level and top up as needed ⇒ Rep. gr. 93 ; Cooling system, vehicles with high voltage system; Draining and filling coolant .

Specified torques

- ◆ ⇒ ["1.1 Assembly overview - lock carrier", page 10](#)

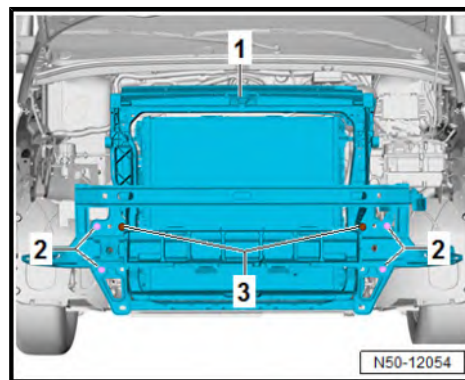
1.3 Moving to and back from service position

⇒ ["1.3.1 Moving to and back from service position", page 16](#)

⇒ ["1.3.2 Moving to and back from service position, e-Crafter or e-TGE", page 19](#)

1.3.1 Moving to and back from service position

Special tools and workshop equipment required

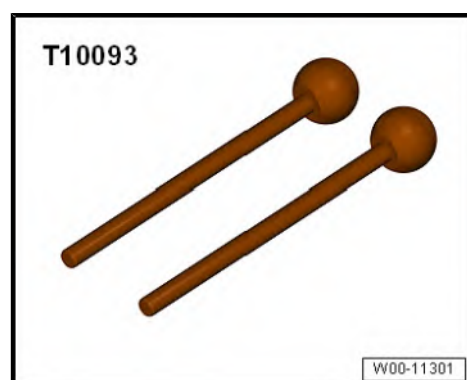




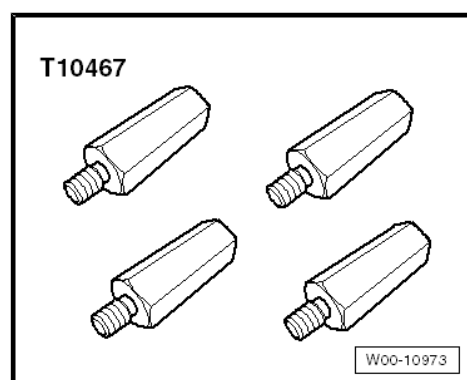
- ◆ Setting gauge - 3371-



- ◆ Guide pins - T10093-



- ◆ Adapter - T10467-



- ◆ Torque wrench - V.A.G 1331-



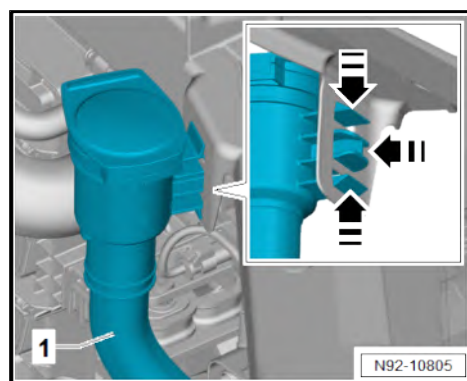


◆ Torque wrench - V.A.G 1332-

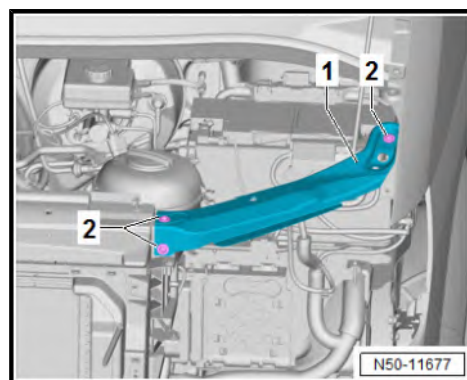


Moving to service position

- Remove front bumper cover ➤ [page 179](#) .
- Remove left and right headlights ➤ Electrical system; Rep. gr. 94 ; Headlights; Removing and installing headlights .
- Remove connecting hose for air filter from lock carrier ➤ Rep. gr. 23 ; Air filter; Removing and installing air filter housing .
- Brace the bonnet open.
- Unclip filler neck -1- of washer fluid reservoir on lock carrier in -direction of arrow-.

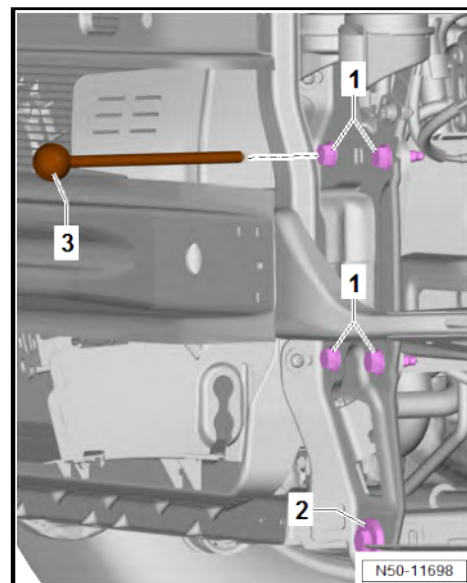


- Unclip Bowden cable coupling from support arm -1-, and disconnect Bowden cable ➤ [page 63](#) .
- Unscrew bolts -2- of support arm -1- on left and right.
- Remove support arm -1- on left and right from vehicle.





- Remove bolts -1- on left and right side of longitudinal members.
- Screw guide pin - T 10093- -3- at top and on inside into left and right longitudinal members.
- Unscrew bolts -2- on left and right at bottom on connecting piece.



- Lock carrier -1- can be pulled forwards about 10 cm on guide pin - T 10093- -2-.
- Screw in adapter - T10467- -4- on left and right between lock carrier and longitudinal member to secure lock carrier.
- Use bolts -3- to bolt lock carrier to adapter - T10467- on left and right.

Moving from service position

Install in reverse order of removal, observing the following:

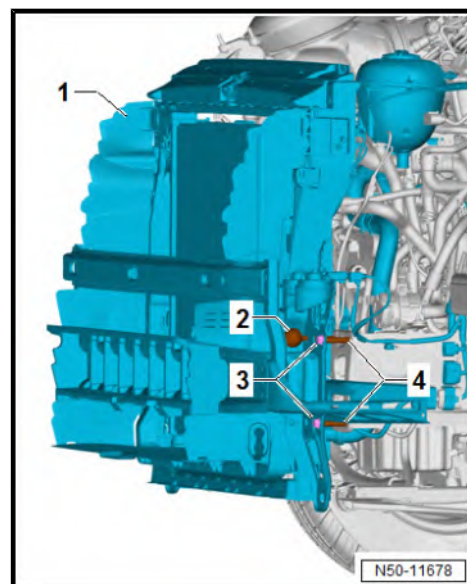
- Use setting gauge - 3371- to check gaps and height difference; adjust as necessary ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .

Vehicles with adaptive cruise control

- Calibrate adaptive cruise control ⇒ Running gear, axles, steering; Rep. gr. 44 ; Adaptive cruise control .

Specified torques

- ◆ ⇒ [“1.1 Assembly overview - lock carrier”, page 10](#)



1.3.2 Moving to and back from service position, e-Crafter or e-TGE

Special tools and workshop equipment required

- ◆ Setting gauge - 3371-

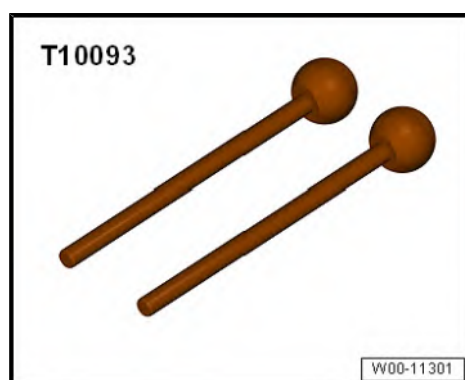




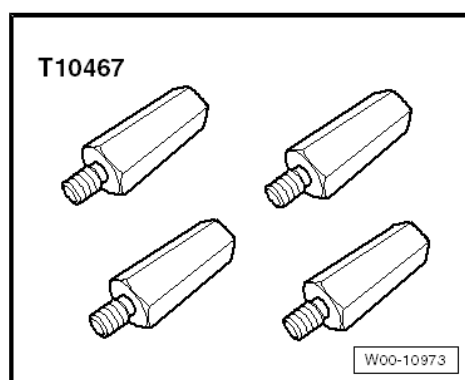
◆ Removal wedge - 3409-



◆ Guide pins - T10093-



◆ Adapter - T10467-



◆ Torque wrench - V.A.G 1331-





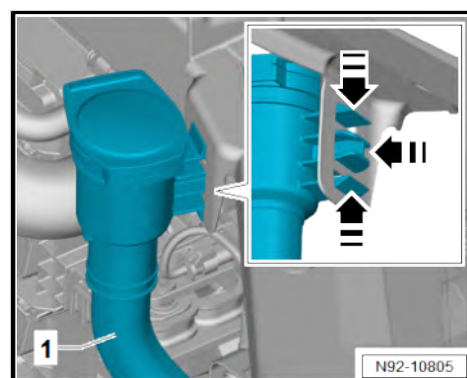
◆ Torque wrench - V.A.G 1332-



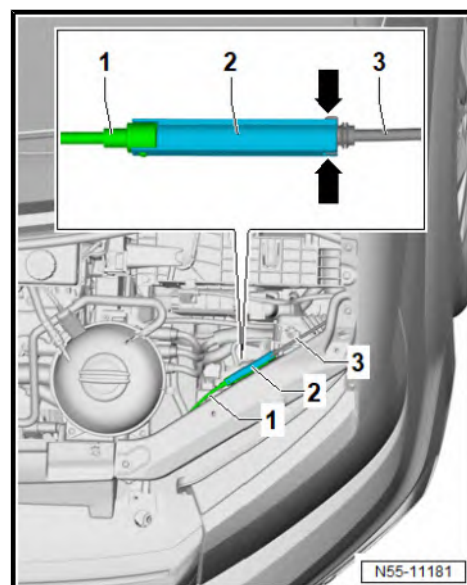
◆ Vehicle diagnostic tester

Moving to service position

- Remove front bumper cover ➔ [page 179](#) .
- Remove left and right headlights ➔ Electrical system; Rep. gr. 94 ; Headlights; Removing and installing headlights .
- Brace the bonnet open.
- Remove support rod ➔ [page 64](#) .
- Release locking lugs on filler pipe for washer fluid reservoir -1- in -directions of arrow-.
- Lay filler pipe for washer fluid reservoir -1- aside.

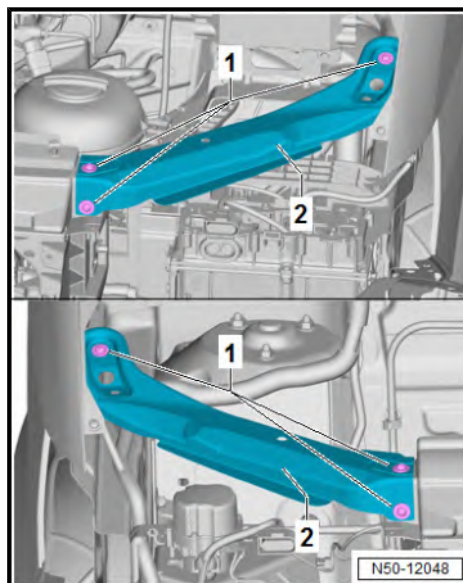


- Unclip Bowden cable coupling -1- at lock carrier strut.
- Use removal wedge - 3409- to push apart cap -2- in area marked with -arrows- until it has been released from the two projections.
- Detach Bowden cable -3- from Bowden cable coupling -1-.
- Disconnect refrigerant lines from condenser ➔ Heating, air conditioning; Rep. gr. 87 ; Refrigerant circuit; Disconnecting refrigerant lines from and connecting to condenser .

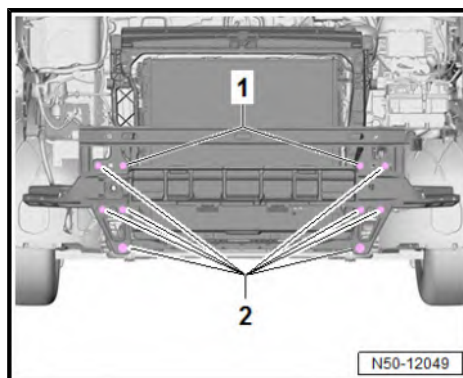




- Unscrew bolts -1-.
- Remove lock carrier struts -2-.



- Unscrew bolts -1-.
- Screw guide pins - T10093- into resulting free holes.
- Unscrew bolts -2-.
- Pull lock carrier on guide pins - T10093- by approx. 10 cm in direction of travel. Hold it in this position.
- The following work step must be performed by a second mechanic.



- Screw adapter - T10467- -1- holes between lock carrier -2- and longitudinal member, as shown in illustration.
- Use bolts -3- to bolt lock carrier -2- to adapter - T10467- -1-.

Moving from service position

Move back from service position in the reverse order observing the following:

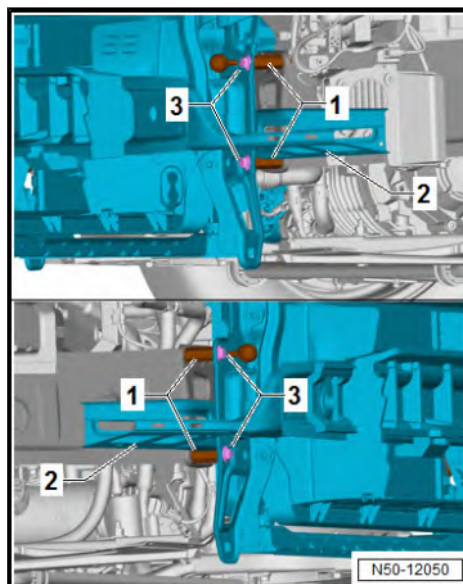
- Use setting gauge - 3371- to check gaps and height difference; adjust as necessary ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .

Vehicles with adaptive cruise control

- Calibrate adaptive cruise control ⇒ Running gear, axles, steering; Rep. gr. 44 ; Adaptive cruise control .

Specified torques

- ♦ ⇒ ["1.1 Assembly overview - lock carrier", page 10](#)





1.4 Removing and installing air duct

⇒ ["1.4.1 Removing and installing air duct, left", page 23](#)

⇒ ["1.4.2 Removing and installing air duct, left, e-Crafter or e-TGE", page 23](#)

⇒ ["1.4.3 Removing and installing air duct, right", page 24](#)

⇒ ["1.4.4 Removing and installing air duct, right, e-Crafter or e-TGE", page 25](#)

⇒ ["1.4.5 Removing and installing air duct, bottom", page 25](#)

⇒ ["1.4.6 Removing and installing air duct, bottom, e-Crafter or e-TGE", page 26](#)

1.4.1 Removing and installing air duct, left

Special tools and workshop equipment required

- ◆ Removal wedge - 3409-

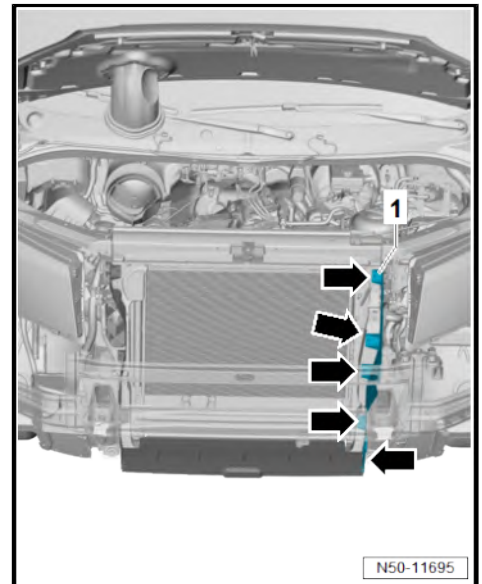


Removing

- Remove bumper cover ⇒ [page 179](#) .
- Unclip air duct -1- in area marked with -arrows- using removal wedge - 3409- .
- Swing out air duct -1-.

Installing

Install in reverse order.



1.4.2 Removing and installing air duct, left, e-Crafter or e-TGE

Special tools and workshop equipment required



◆ Removal wedge - 3409-

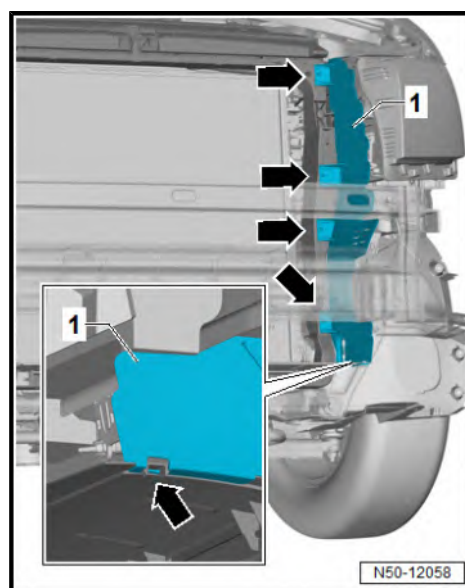


Removing

- Remove bumper cover ⇒ [page 179](#) .
- Unclip air duct -1- in area marked with -arrows- using removal wedge - 3409- .
- Swing out air duct -1-.

Installing

Install in reverse order.



1.4.3 Removing and installing air duct, right

Special tools and workshop equipment required

◆ Removal wedge - 3409-



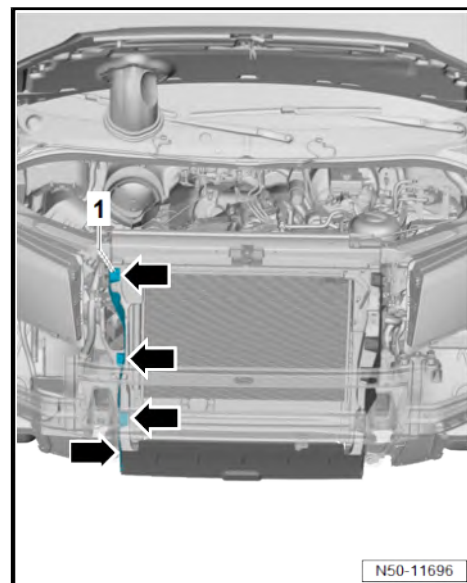


Removing

- Remove bumper cover ➔ [page 179](#) .
- Unclip air duct -1- in area marked with -arrows- using removal wedge - 3409- .
- Swing out air duct -1-.

Installing

Install in reverse order.



1.4.4 Removing and installing air duct, right, e-Crafter or e-TGE

Special tools and workshop equipment required

- ◆ Removal wedge - 3409-

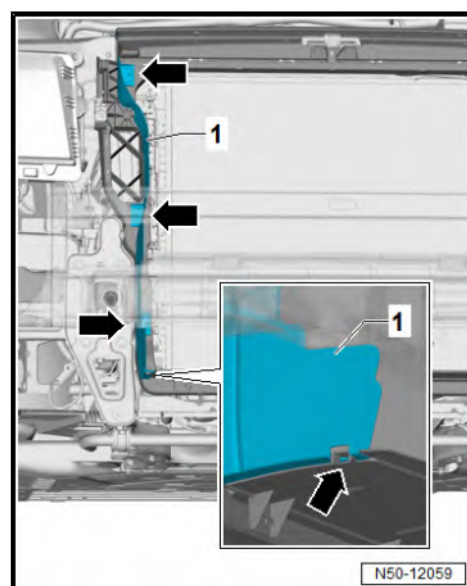


Removing

- Remove bumper cover ➔ [page 179](#) .
- Unclip air duct -1- in area marked with -arrows- using removal wedge - 3409- .
- Swing out air duct -1-.

Installing

Install in reverse order.



1.4.5 Removing and installing air duct, bottom

Special tools and workshop equipment required



◆ Removal wedge - 3409-

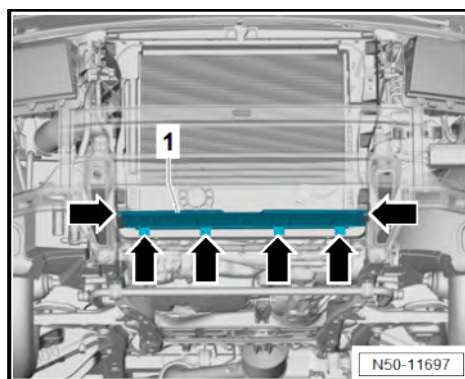


Removing

- Remove bumper cover ➤ [page 179](#) .
- Unclip air duct -1- in area marked with -arrows- using removal wedge - 3409- .
- Remove air duct -1-.

Installing

Install in reverse order.



1.4.6 Removing and installing air duct, bottom, e-Crafter or e-TGE

Special tools and workshop equipment required

◆ Removal wedge - 3409-

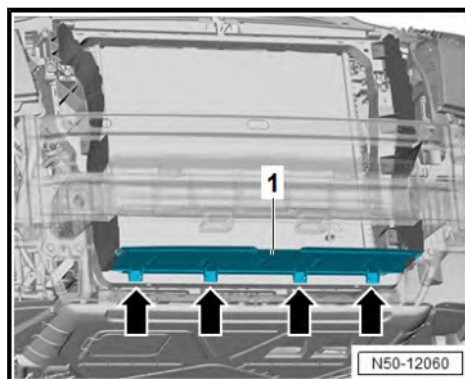


Removing

- Remove bumper cover ➤ [page 179](#) .
- Unclip air duct -1- in area marked with -arrows- using removal wedge - 3409- .
- Remove air duct -1-.

Installing

Install in reverse order.





1.5 Removing and installing connecting part for impact bar/subframe

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-



Removing

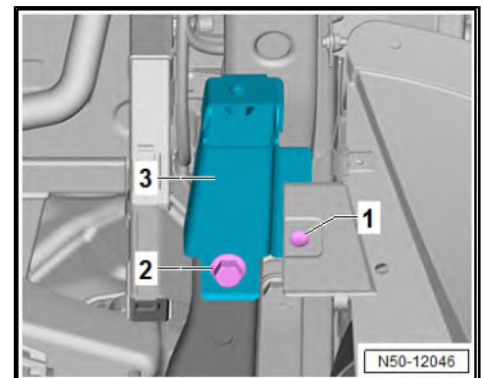
- Move to service position ⇒ [page 16](#) .
- Unscrew bolts -1- and -2-.
- Remove connecting part -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“1.1 Assembly overview - lock carrier”, page 10](#)



1.6 Removing and installing lock carrier struts

⇒ [“1.6.1 Removing and installing lock carrier struts, left side of vehicle”, page 27](#)

⇒ [“1.6.2 Removing and installing lock carrier struts, right side of vehicle”, page 28](#)

1.6.1 Removing and installing lock carrier struts, left side of vehicle

Special tools and workshop equipment required

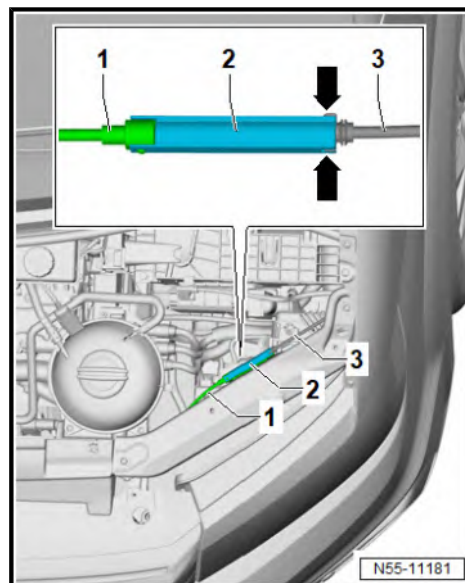
- ◆ Torque wrench - V.A.G 1331-





Removing

- Remove support rod ➔ [page 64](#) .
- Unclip Bowden cable coupling -1- at lock carrier strut.
- Use removal wedge - 3409- to push apart cap -2- in area marked with -arrows- until it has been released from the two projections.
- Detach Bowden cable -3- from Bowden cable coupling -1-.



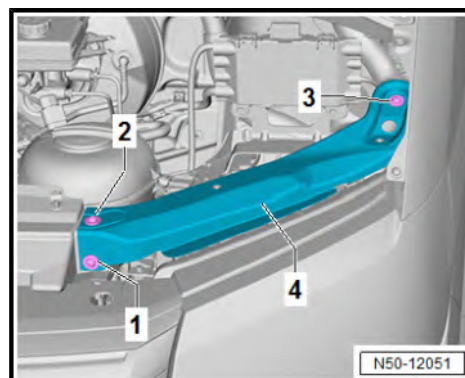
- Unscrew bolts -1 ... 3-.
- Remove lock carrier strut -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ➔ ["1.1 Assembly overview - lock carrier", page 10](#)



1.6.2 Removing and installing lock carrier struts, right side of vehicle

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-





Removing

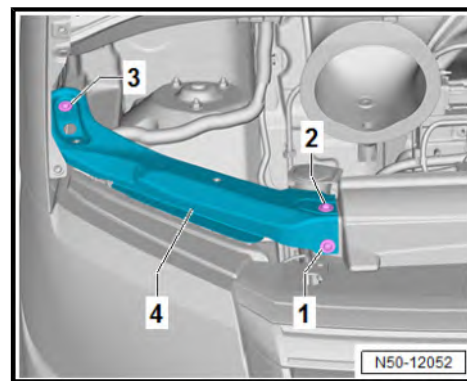
- Unscrew bolts -1 ... 3-.
- Remove lock carrier strut -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["1.1 Assembly overview - lock carrier", page 10](#)





2 Wings

⇒ "2.1 Assembly overview - wing", page 30

⇒ "2.2 Removing and installing wing", page 31

2.1 Assembly overview - wing

1 - Wings

- ❑ Removing and installing
⇒ [page 31](#)

2 - Bolts

- ❑ Qty. 2
- ❑ 6 Nm

3 - Bolts

- ❑ Qty. 2
- ❑ 6 Nm

4 - Support

- ❑ Wings
- ❑ Removing and installing
⇒ [page 31](#)

5 - Bracket

- ❑ Wing, centre
- ❑ Removing and installing
⇒ [page 31](#)

6 - Bolts

- ❑ Qty. 3
- ❑ 6 Nm

7 - Bolt

- ❑ 6 Nm

8 - Bolts

- ❑ Qty. 2
- ❑ 6 Nm

9 - Web part

- ❑ For wing

10 - Wings

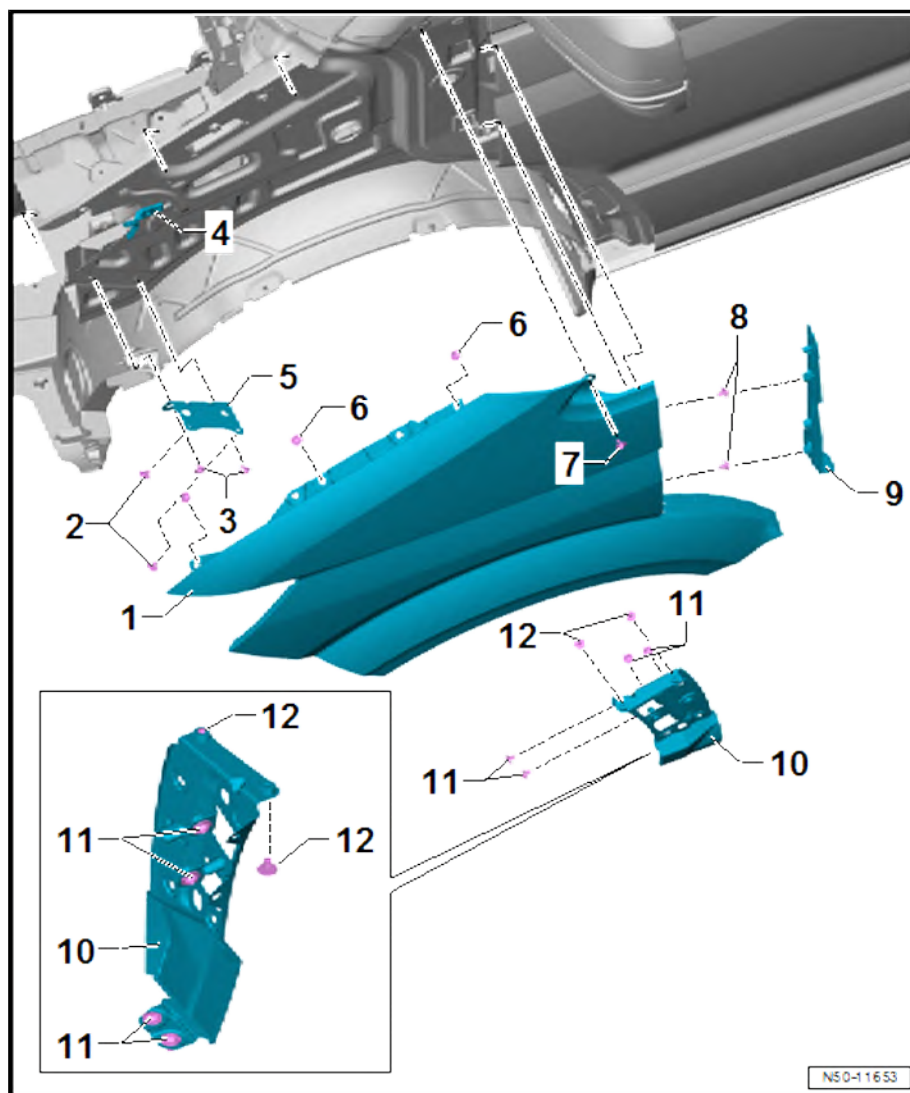
- ❑ Front, bottom
- ❑ Removing and installing ⇒ [page 31](#)

11 - Bolts

- ❑ Qty. 4
- ❑ 6 Nm

12 - Bolts

- ❑ Qty. 2
- ❑ 6 Nm





2.2 Removing and installing wing

⇒ [“2.2.1 Removing and installing wing”, page 31](#)

⇒ [“2.2.2 Removing and installing wing bracket, middle”, page 33](#)

⇒ [“2.2.3 Removing and installing wing at bottom”, page 34](#)

⇒ [“2.2.4 Removing and installing wing support”, page 35](#)

2.2.1 Removing and installing wing

Special tools and workshop equipment required

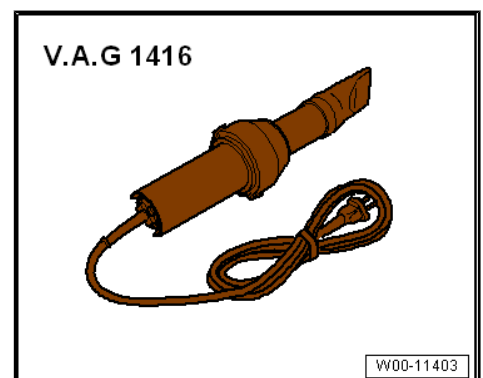
◆ Setting gauge - 3371-



◆ Removal wedge - 3409-



◆ Hot-air blower - V.A.G 1416-



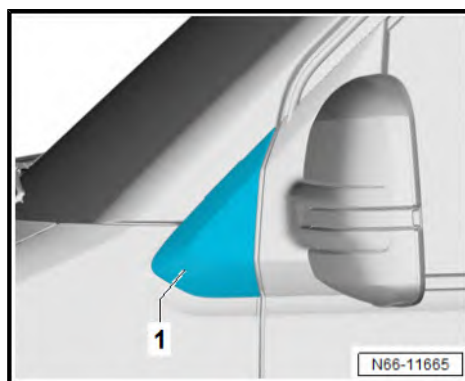


◆ Torque wrench - V.A.G 1783-

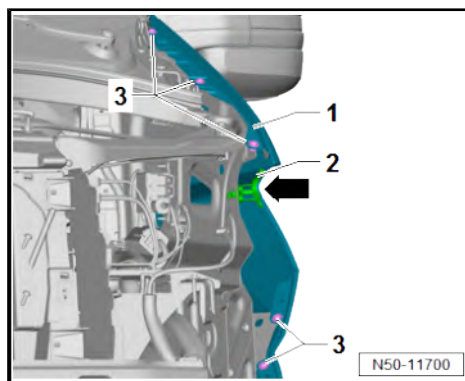


Removing

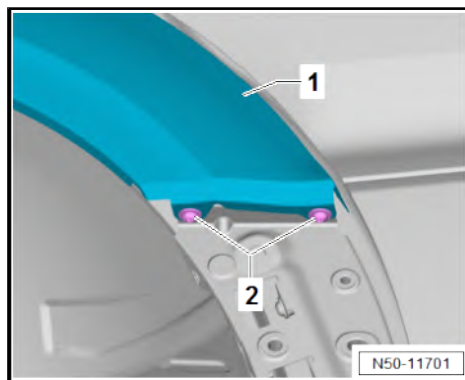
- Remove front bumper cover ➤ [page 179](#) .
- Remove wheel housing liner ➤ [page 299](#) .
- Remove trim strip of wing ➤ [page 249](#) .
- Use removal wedge - 3409- to unclip cap -1-.
- The following work steps must be performed by a second mechanic.



- Remove bolts -3- in wing -1-
- Remove bonnet on left hinge ➤ [page 53](#) .
- Detach insulation -arrow- between bracket -2- and wing -1-.



- Unscrew bolts -2- at bottom on wing -1-
- Open door.

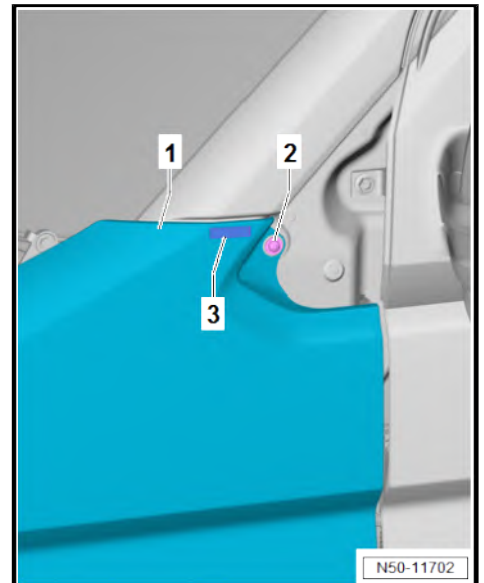




- Remove bolts -2- in wing -1-



- Unscrew bolt -2-
- Heat insulation -3- with hot air blower - V.A.G 1416- .
- Remove wing -1-.



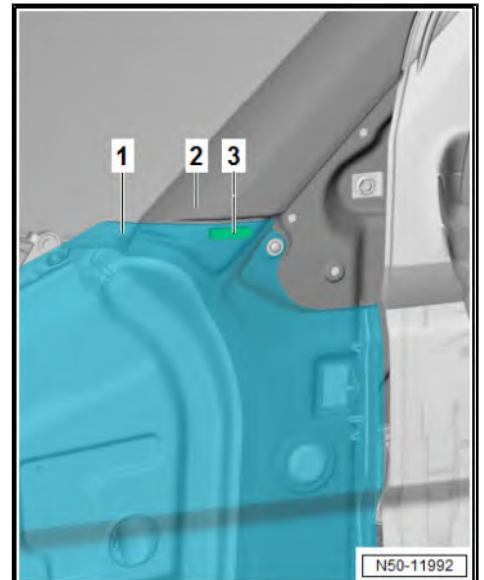
Installing

Install in reverse order of removal, observing the following:

- Renew insulation -3- between wing -1- and A-pillar -2- according to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
- If necessary, carry out corrosion protection measures.
- Use setting gauge - 3371- to check gaps and height difference of wing; adjust as necessary ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - wing”, page 30](#)



2.2.2 Removing and installing wing bracket, middle

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

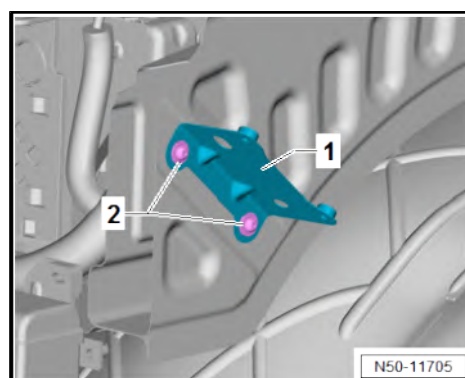
- Remove wing ⇒ [page 31](#) .
- Unscrew bolts -2-.
- Remove bracket -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - wing”, page 30](#)



2.2.3 Removing and installing wing at bottom

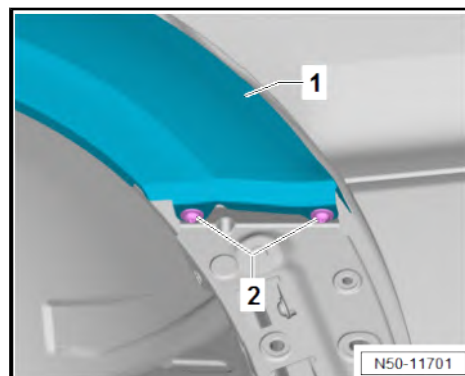
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove trim strip of wing ⇒ [page 249](#) .
- Remove bolts -2- in wing -1-





- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Remove wing at bottom -1-.



Note

If present, the zinc film between the bottom wing -1- and side member is damaged during removal.

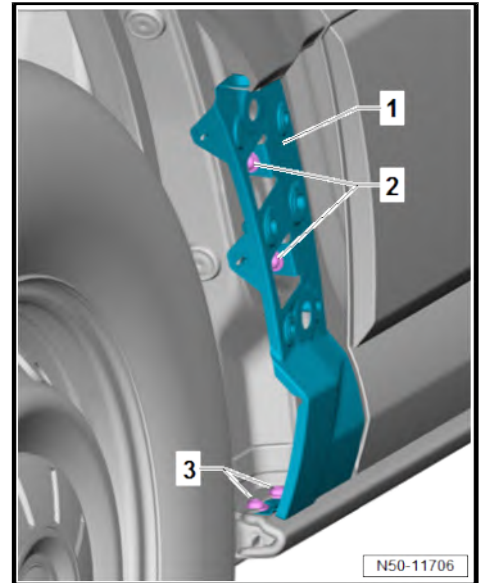
Installing

Install in reverse order of removal, observing the following:

- Make sure that self-adhesive zinc film - D 381 ZN1 A2- is applied between bottom wing -1- and side member without fail.

Specified torques

- ◆ ⇒ ["2.1 Assembly overview - wing", page 30](#)



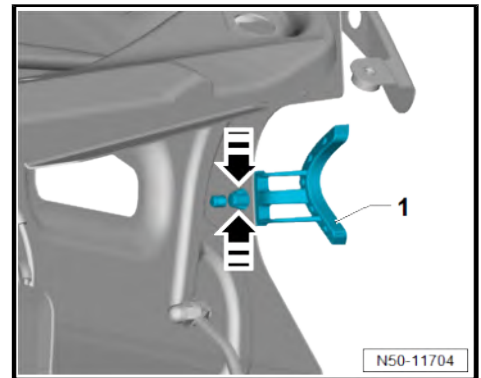
2.2.4 Removing and installing wing support

Removing

- Remove wing ⇒ [page 31](#) .
- Release fasteners in -direction of arrow-.
- Remove wing support -1-.

Installing

Install in reverse order.





3 Bulkhead

⇒ [“3.1 Assembly overview - plenum chamber cover”, page 36](#)

⇒ [“3.2 Removing and installing plenum chamber cover”, page 39](#)

3.1 Assembly overview - plenum chamber cover

⇒ [“3.1.1 Assembly overview - plenum chamber cover”, page 36](#)

⇒ [“3.1.2 Assembly overview - plenum chamber cover, as of calendar week 13/2017”, page 37](#)

⇒ [“3.1.3 Assembly overview - plenum chamber cover, e-Crafter or e-TGE”, page 38](#)

3.1.1 Assembly overview - plenum chamber cover

1 - Plenum chamber cover

- ☐ Right
- ☐ Removing and installing
⇒ [page 39](#)

2 - Plenum chamber cover

- ☐ Left
- ☐ Removing and installing
⇒ [page 39](#)

3 - Bolt

- ☐ 8 Nm

4 - Support

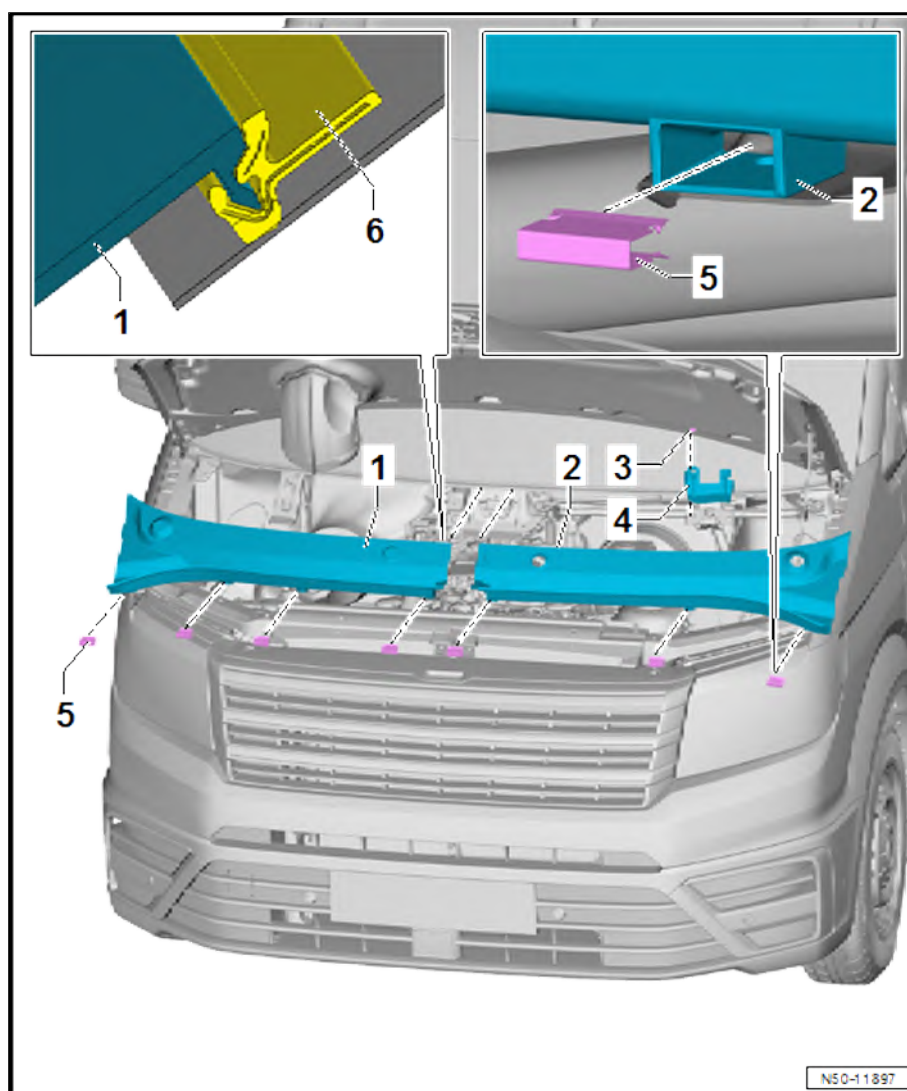
- ☐ Plenum chamber cover
- ☐ Removing and installing
⇒ [page 39](#)

5 - Clips

- ☐ Qty. 6

6 - Windscreen seal

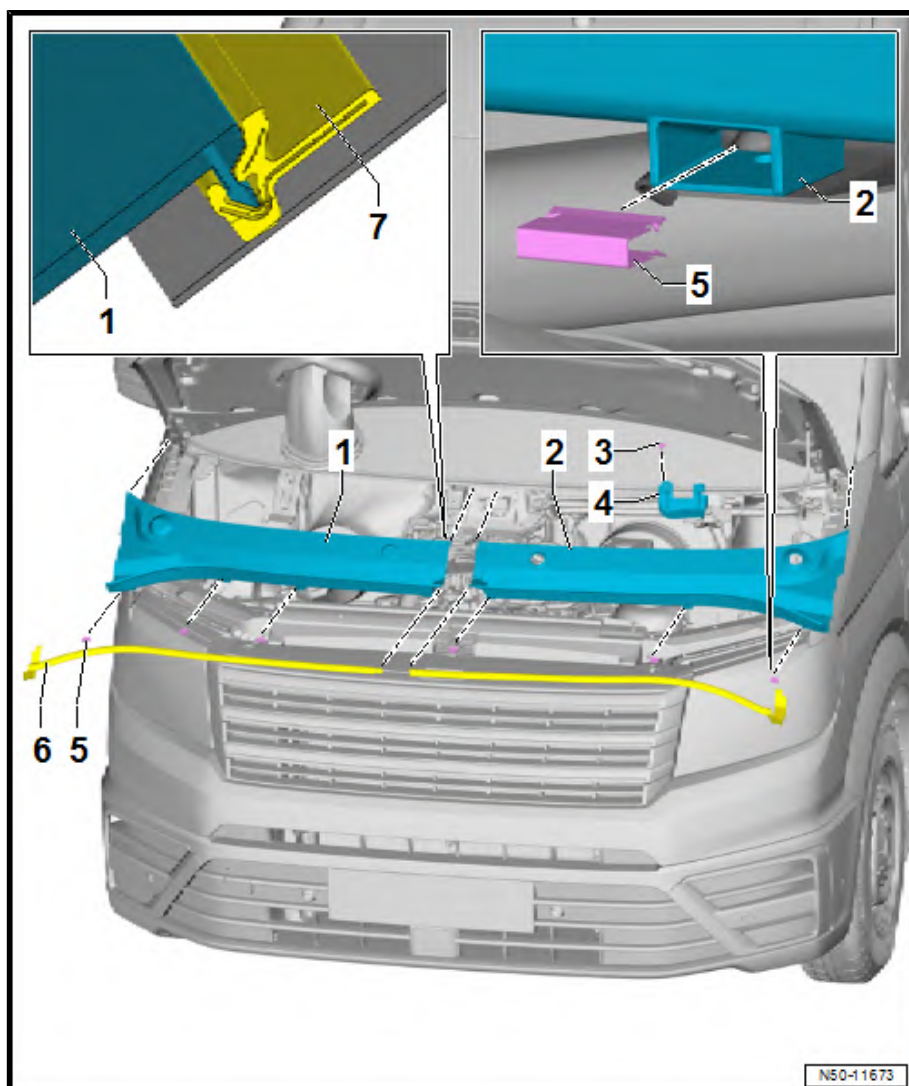
- ☐ Part of windscreen





3.1.2 Assembly overview - plenum chamber cover, as of calendar week 13/2017

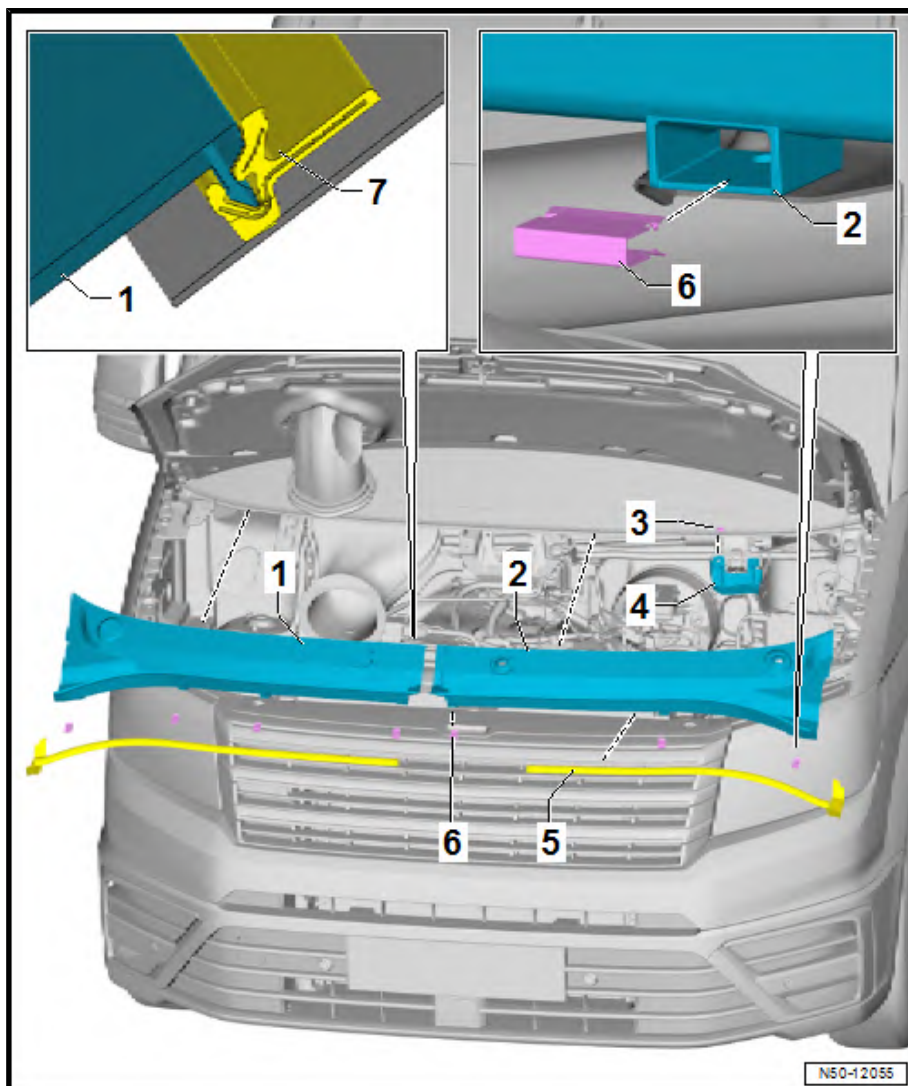
- 1 - Plenum chamber, right
 - ❑ Removing and installing
⇒ [page 40](#)
- 2 - Plenum chamber cover, left
 - ❑ Removing and installing
⇒ [page 40](#)
- 3 - Bolt
 - ❑ 8 Nm
- 4 - Support
 - ❑ Plenum chamber cover
 - ❑ Removing and installing
⇒ [page 39](#)
- 5 - Clips
 - ❑ Qty. 6
- 6 - Seal
- 7 - Windscreen seal
 - ❑ On windscreen





3.1.3 Assembly overview - plenum chamber cover, e-Crafter or e-TGE

- 1 - Plenum chamber, right
 - ❑ Removing and installing
⇒ [page 40](#)
- 2 - Plenum chamber cover, left
 - ❑ Removing and installing
⇒ [page 40](#)
- 3 - Bolt
 - ❑ 8 Nm
- 4 - Support
 - ❑ Plenum chamber cover
 - ❑ Removing and installing
⇒ [page 39](#)
- 5 - Seal
- 6 - Clips
 - ❑ Qty. 7
- 7 - Windscreen seal
 - ❑ On windscreen





3.2 Removing and installing plenum chamber cover

⇒ [“3.2.1 Removing and installing plenum chamber cover”, page 39](#)

⇒ [“3.2.2 Removing and installing plenum chamber cover, as of calendar week 13/2017”, page 40](#)

⇒ [“3.2.3 Removing and installing plenum chamber cover, e-Crafter or e-TGE”, page 41](#)

⇒ [“3.2.4 Sealing plenum chamber cover”, page 41](#)

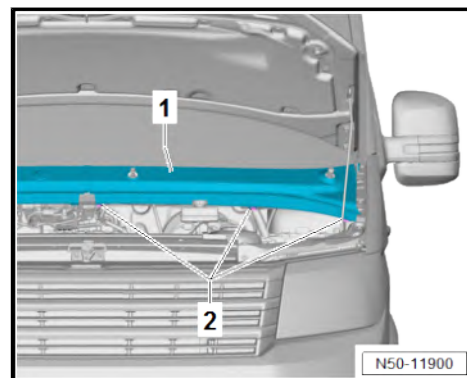
⇒ [“3.2.5 Removing and installing plenum chamber cover support”, page 42](#)

3.2.1 Removing and installing plenum chamber cover

Removing

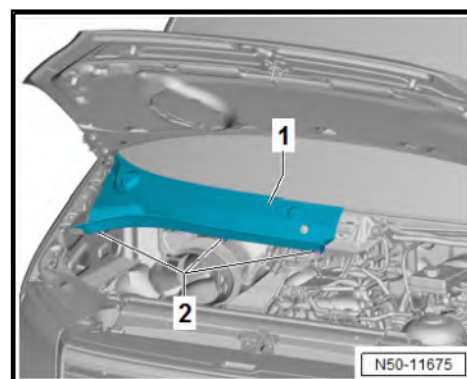
Plenum chamber cover, left

- Remove wiper arms ⇒ Electrical system; Rep. gr. 92 ; Windscreen wiper system; Removing and installing wiper arms .
- Unclip clips -2-.
- Starting from edge of windscreen, pull plenum chamber cover -1- upwards and out of windscreen seal on bottom edge of windscreen.
- Remove plenum chamber cover -1-.



Plenum chamber, right

- Removing left plenum chamber cover ⇒ [page 39](#) .
- Unclip clips -2-.
- Starting from edge of windscreen, pull plenum chamber cover -1- upwards and out of windscreen seal on bottom edge of windscreen.
- Remove plenum chamber cover -1-.



Installing

Install in reverse order of removal, observing the following:

Procedure for new windscreen

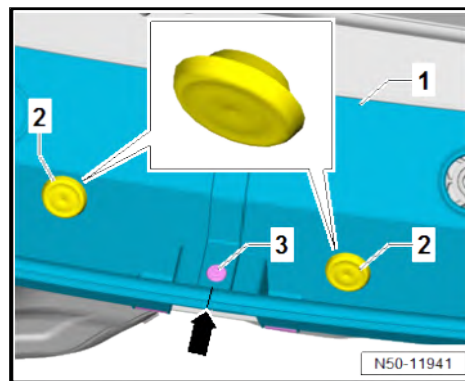
- Remove insert from windscreen seal of new windscreen.

Continued for all vehicles

- Seal plenum chamber cover ⇒ [page 41](#) .
- Spray windscreen seal with soapy water.



- Make sure installation position of plenum chamber cover -1- is correct. Recess for wiper arm shaft must be aligned centrally.
- Seal spreader rivet -3- with butyl adhesive sealing cord .
- Install spreader rivet -3- in plenum chamber cover -1-.
- Replace caps with new plugs -2-.
- Make sure installation position -arrow- of seal for plenum chamber cover is correct.
- Seal plenum chamber cover seal on left and right with projecting butyl adhesive sealing cord -arrow-.
- Wiper arms ⇒ Electrical system; Rep. gr. 92 ; Windscreen wiper system; Removing and installing wiper arms .

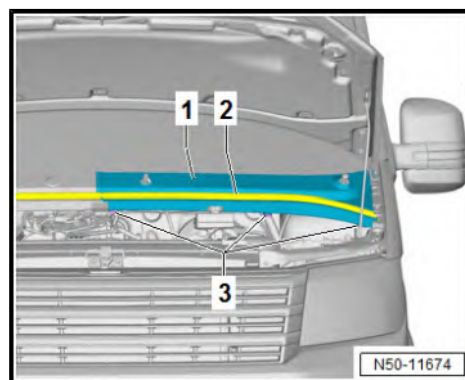


3.2.2 Removing and installing plenum chamber cover, as of calendar week 13/2017

Removing

Plenum chamber cover, left

- Remove wiper arms ⇒ Electrical system; Rep. gr. 92 ; Windscreen wiper system; Removing and installing wiper arms .
- Pull seal -2- off plenum chamber cover -1-.
- Unclip clips -3-.
- Starting from edge of windscreen, pull plenum chamber cover -1- upwards and out of windscreen seal on bottom edge of windscreen.
- Remove plenum chamber cover -1-.



Plenum chamber, right

- Removing left plenum chamber cover ⇒ [page 40](#) .
- Unclip clips -2-.
- Starting from edge of windscreen, pull plenum chamber cover -1- upwards and out of windscreen seal on bottom edge of windscreen.
- Remove plenum chamber cover -1-.

Installing

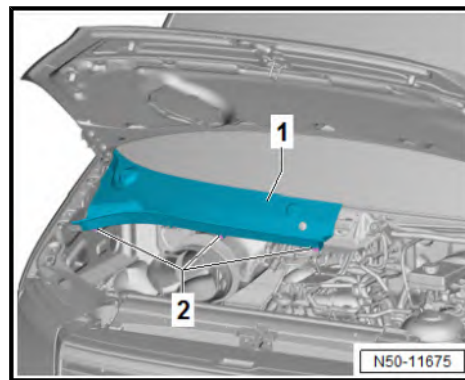
Install in reverse order of removal, observing the following:

Procedure for new windscreen

- Remove insert from windscreen seal of new windscreen.

Continued for all vehicles

- Spray windscreen seal with soapy water.



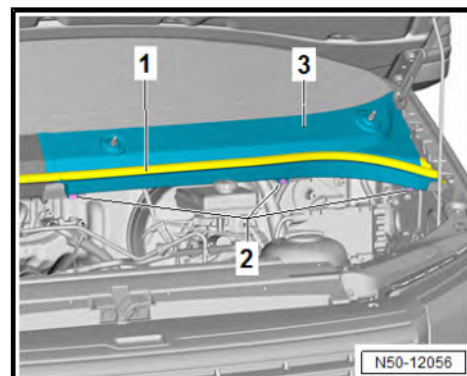


3.2.3 Removing and installing plenum chamber cover, e-Crafter or e-TGE

Removing

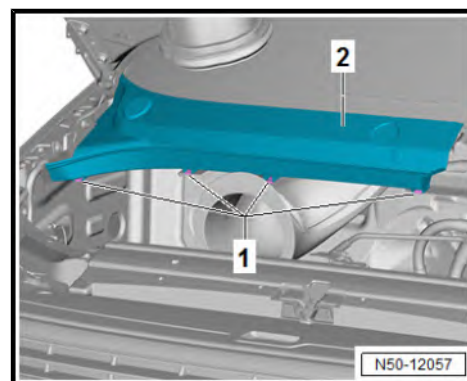
Plenum chamber cover, left

- Remove wiper arms ⇒ Electrical system; Rep. gr. 92 ; Windscreen wiper system; Removing and installing wiper arms .
- Pull off seal -1-.
- Pull off clips -2-.
- Starting from edge of windscreen, pull plenum chamber cover -3- upwards and out of windscreen seal on bottom edge of windscreen.



Plenum chamber, right

- Removing left plenum chamber cover ⇒ [page 41](#) .
- Pull off clips -1-.
- Starting from edge of windscreen, pull plenum chamber cover -2- upwards and out of windscreen seal on bottom edge of windscreen.



Installing

Install in reverse order of removal, observing the following:

Procedure for new windscreen

- Remove insert from windscreen seal of new windscreen.

Continued for all vehicles

- Spray windscreen seal with soapy water.

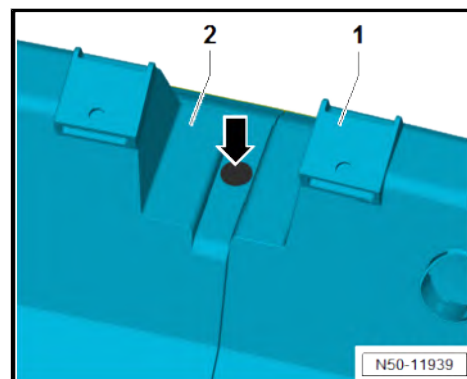
3.2.4 Sealing plenum chamber cover

Special tools and workshop equipment required

- ◆ Hand drill

Procedure

- Place left -2- and right -1- plenum chamber covers upside down onto suitable surface.
- Push together left -2- and right -1- plenum chamber covers as shown in illustration.
- When drilling, secure plenum chamber covers to prevent them from turning.
- Wear safety goggles when drilling.
- Using drill , make a 6 mm hole in connecting tabs -arrow-, and deburr hole.
- Push apart left -2- and right -1- plenum chamber covers.



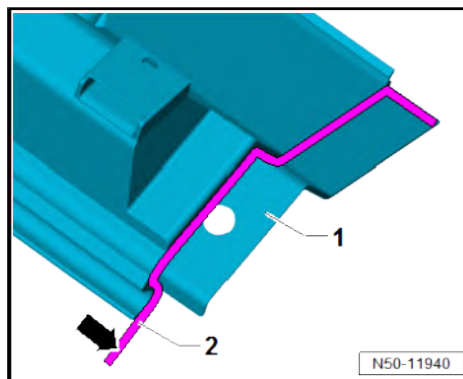


- Attach butyl adhesive sealing cord -2- to underside of left plenum chamber cover -1- as shown in illustration.



Note

Let butyl adhesive sealing cord -2- project approx. 2 cm -arrow-.



- Apply profiled foam -2- (approx. 13 cm in length) on underside -A- of left plenum chamber cover -1- as shown in illustration.

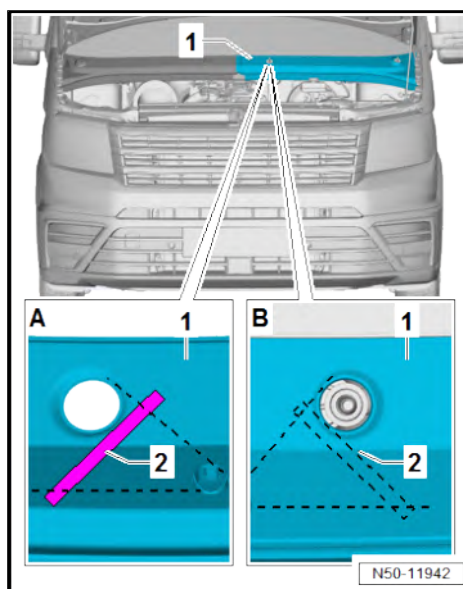
-A- underside of left plenum chamber cover -1-

-B- upper side of left plenum chamber cover -1-



Note

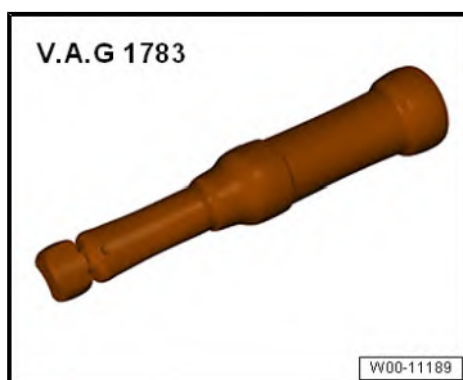
- ♦ Profiled foam -2- is used to deflect intruding rainwater.
 - ♦ Make sure to apply the profiled foam -2- in exactly the same direction as illustrated.
- Install plenum chamber covers ⇒ [page 39](#) .



3.2.5 Removing and installing plenum chamber cover support

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-





Removing

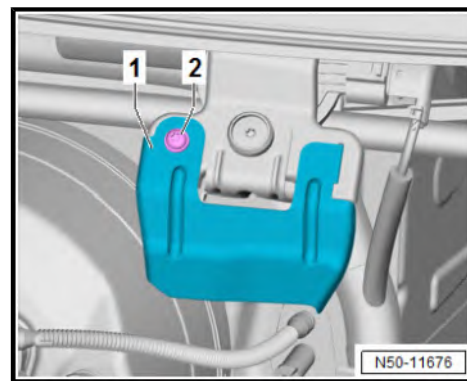
- Removing left plenum chamber cover ➔ [page 39](#) .
- Unscrew bolt -2-. Remove plenum chamber cover -1- when doing this.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“3.1 Assembly overview - plenum chamber cover”](#), [page 36](#)





4 Carrier frame for high-voltage components

⇒ ["4.1 Assembly overview - carrier frame for high-voltage components", page 44](#)

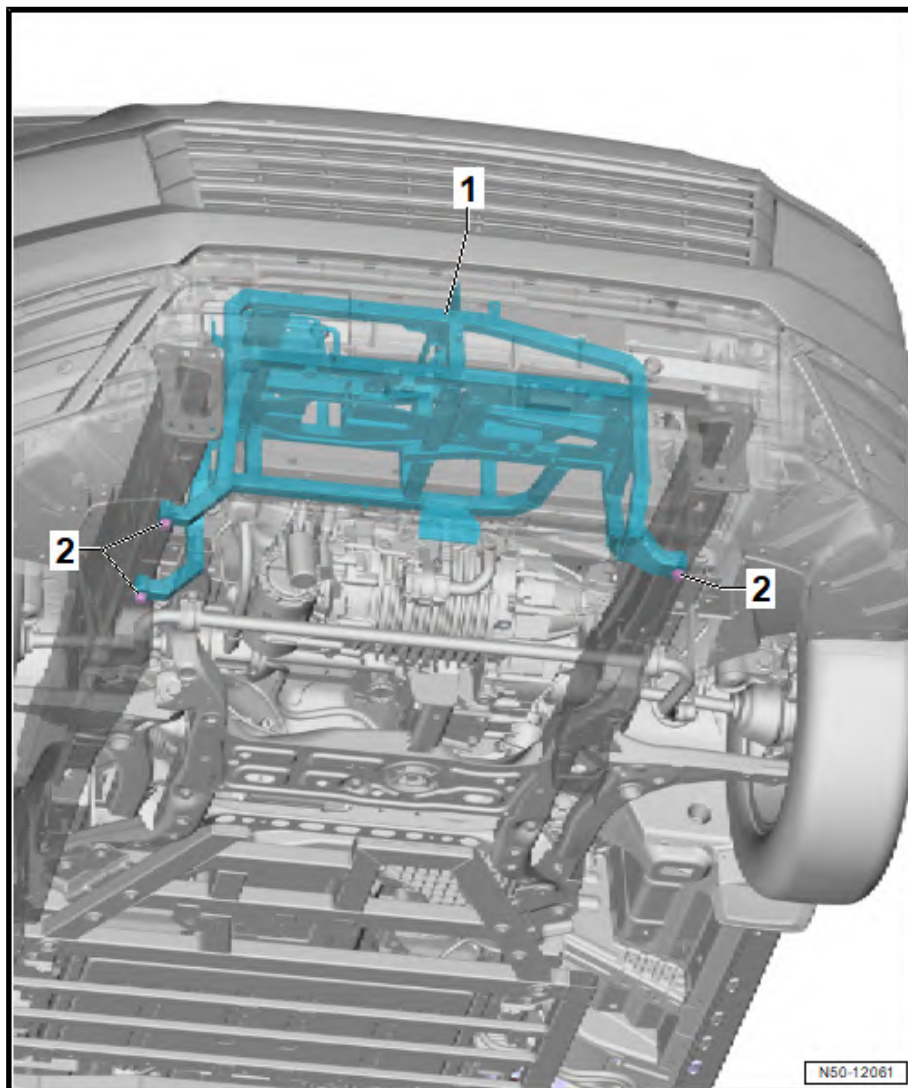
4.1 Assembly overview - carrier frame for high-voltage components

1 - Carrier frame

- ☐ High-voltage components

2 - Bolts

- ☐ Renew after removal
- ☐ Qty. 3
- ☐ 50 Nm +180°



51 – Body - centre

1 Carrier frame for high-voltage battery 1 - AX2-

⇒ [“1.1 Assembly overview - carrier frame for high-voltage battery 1 AX2”, page 45](#)

⇒ [“1.2 Removing and installing high-voltage battery 1 AX2”, page 46](#)

1.1 Assembly overview - carrier frame for high-voltage battery 1 - AX2-

1 - Nuts

- ☐ Renew after removal
- ☐ Qty. 2

2 - Subframe

- ☐ Assembly overview ⇒ Running gear, axles, steering; Rep. gr. 40 ; Subframe; Assembly overview - subframe
- ☐ Removing and installing ⇒ Running gear, axles, steering; Rep. gr. 40 ; Subframe; Removing and installing subframe with steering rack
- ☐ Repairing ⇒ Running gear, axles, steering; Rep. gr. 40 ; Subframe; Repairing subframe

3 - Bracket

- ☐ Qty. 2

4 - Bolts

- ☐ Qty. 2
- ☐ Specified torque ⇒ Running gear, axles, steering; Rep. gr. 40 ; Subframe; Assembly overview - subframe

5 - High-voltage battery 1 - AX2-

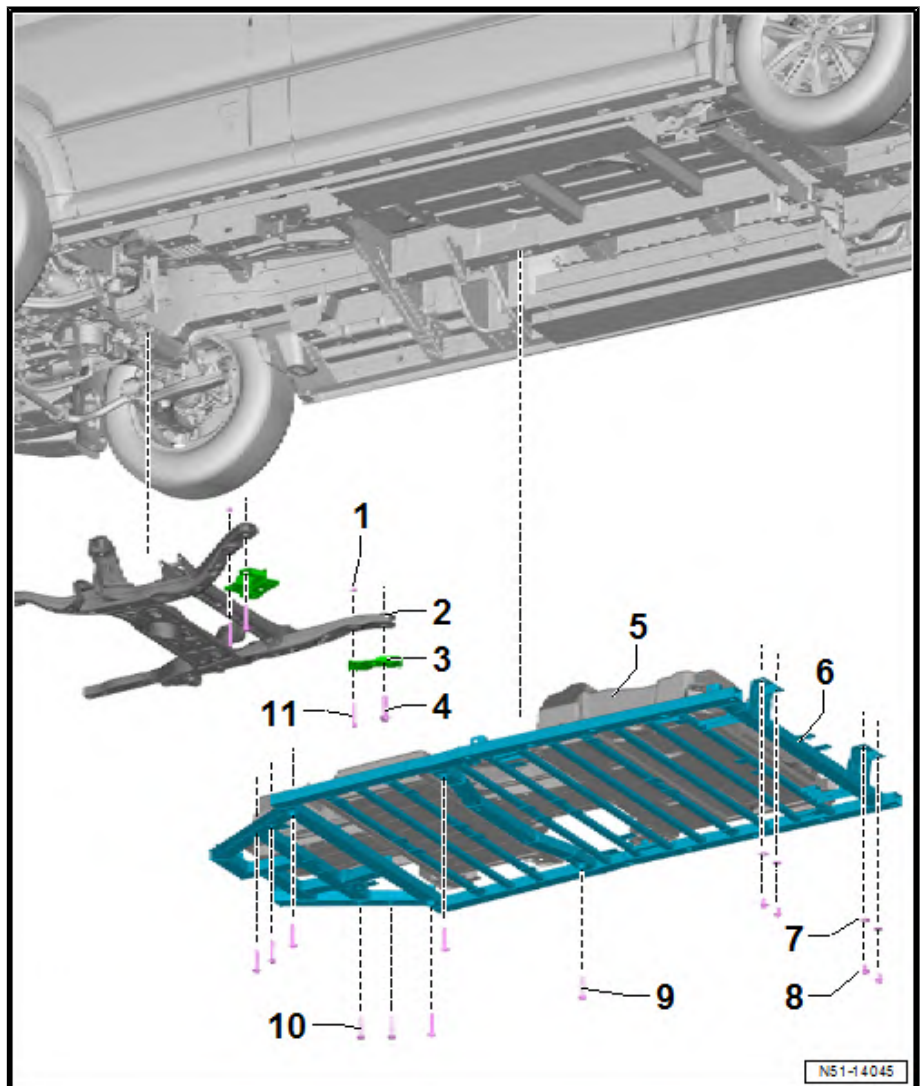
- ☐ Assembly overview ⇒ Rep. gr. 93 ; High-voltage battery unit; Assembly overview - high-voltage battery
- ☐ Removing and installing ⇒ Rep. gr. 93 ; High-voltage battery unit; Removing and installing high-voltage battery 1 -AX2-

6 - Carrier frame

- ☐ For high-voltage battery 1 - AX2-
- ☐ Removing and installing ⇒ [page 46](#)

7 - Washers

- ☐ Renew after removal
- ☐ Qty. 4





8 - Bolts

- ☐ Renew after removal
- ☐ Qty. 4
- ☐ 70 Nm +45°

9 - Bolts

- ☐ Renew after removal
- ☐ Qty. 4
- ☐ 70 Nm +90°

10 - Bolts

- ☐ Renew after removal
- ☐ Qty. 4
- ☐ 50 Nm +180°

11 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 70 Nm +180°

1.2 Removing and installing high-voltage battery 1 - AX2-

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-



Removing

- Remove high-voltage battery 1 - AX2- ⇒ Rep. gr. 93 ; High-voltage battery unit; Removing and installing high-voltage battery 1 -AX2- .
- Detach carrier frame from high-voltage battery 1 - AX2- ⇒ Rep. gr. 93 ; High-voltage battery unit; Detaching carrier frame from high-voltage battery 1 -AX2- .

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["1.1 Assembly overview - carrier frame for high-voltage battery 1 AX2", page 45](#)
- ◆ High-voltage battery unit; Assembly overview - high-voltage battery ⇒ Rep. gr. 93 ; High-voltage battery unit; Assembly overview - high-voltage battery

2 Deformation elements

⇒ [“2.1 Assembly overview - deformation elements”, page 47](#)

⇒ [“2.2 Remove and install deformation elements”, page 47](#)

2.1 Assembly overview - deformation elements

1 - Bolts

- ☐ Qty. 7
- ☐ 40 Nm

2 - Deformation element

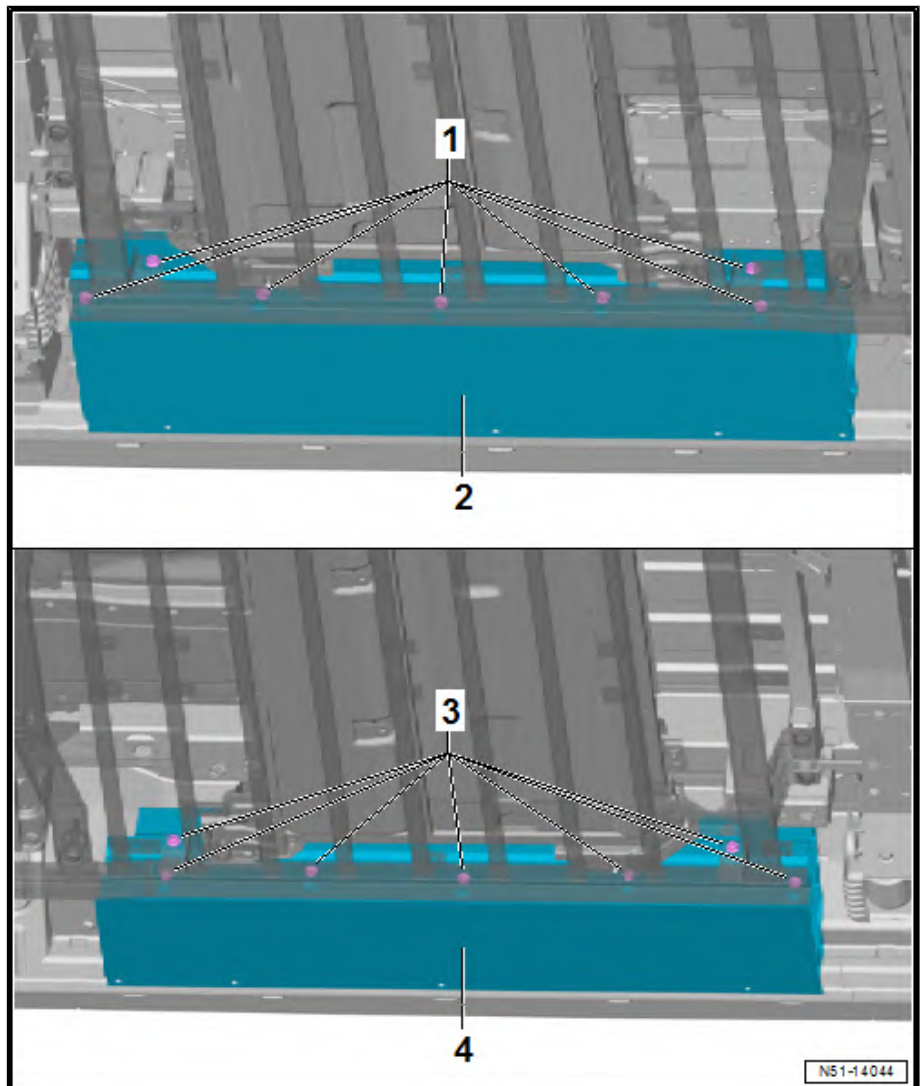
- ☐ Left
- ☐ Removing and installing
⇒ [page 47](#)

3 - Bolts

- ☐ Qty. 7
- ☐ 40 Nm

4 - Deformation element

- ☐ Right
- ☐ Removing and installing
⇒ [page 47](#)



2.2 Remove and install deformation elements

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331-



Removing

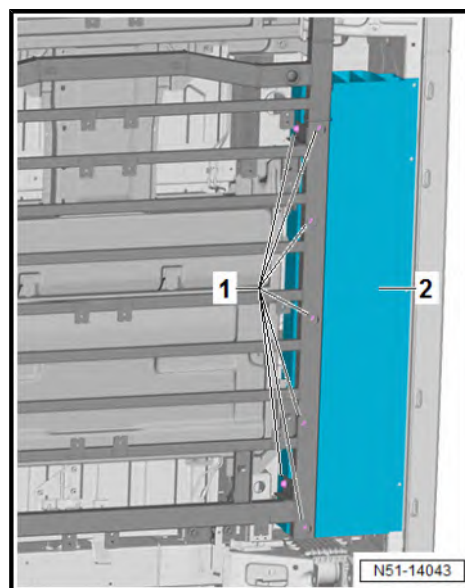
- Remove front and rear underbody cladding on the relevant side ⇒ [page 287](#) .
- The following work steps must be performed by a second mechanic.
- Unscrew bolts -1-. Have a second mechanic secure deformation element -2-.
- Swing out deformation element -2- towards centre of vehicle together with a second mechanic.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - deformation elements”, page 47](#)





55 – Bonnet, rear lid

1 Bonnet (hood)

- ⇒ [“1.1 Assembly overview - bonnet”, page 49](#)
- ⇒ [“1.2 Assembly overview - Bowden cable”, page 52](#)
- ⇒ [“1.3 Removing and installing bonnet”, page 53](#)
- ⇒ [“1.4 Adjusting bonnet”, page 55](#)
- ⇒ [“1.5 Removing and installing bonnet lock”, page 57](#)
- ⇒ [“1.6 Removing and installing bonnet release lever”, page 58](#)
- ⇒ [“1.7 Removing and installing striker pin”, page 59](#)
- ⇒ [“1.8 Removing and installing hinges”, page 60](#)
- ⇒ [“1.9 Removing and installing insulation”, page 61](#)
- ⇒ [“1.10 Removing and installing Bowden cable”, page 63](#)
- ⇒ [“1.11 Removing and installing support rod”, page 64](#)

1.1 Assembly overview - bonnet

- ⇒ [“1.1.1 Assembly overview - bonnet”, page 49](#)
- ⇒ [“1.1.2 Assembly overview - bonnet, e-Crafter or e-TGE”, page 51](#)

1.1.1 Assembly overview - bonnet



1 - Bonnet (hood)

- ☐ Removing and installing ⇒ [page 53](#)
- ☐ Adjusting ⇒ [page 55](#)

2 - Stop buffer

- ☐ Qty. 4

3 - Seals

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 62](#)

4 - Striker

- ☐ Removing and installing ⇒ [page 59](#)
- ☐ Adjusting ⇒ [page 55](#)

5 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 10 Nm

6 - Dampers

- ☐ Removing and installing ⇒ [page 61](#)

7 - Bolts

- ☐ Qty. 4
- ☐ 20 Nm

8 - Nuts

- ☐ Qty. 4
- ☐ 20 Nm

9 - Hinges

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 60](#)

10 - Support rod

- ☐ Removing and installing ⇒ [page 64](#)

11 - Wiring harness

- ☐ For bonnet contact switch - F266-

12 - Bolts

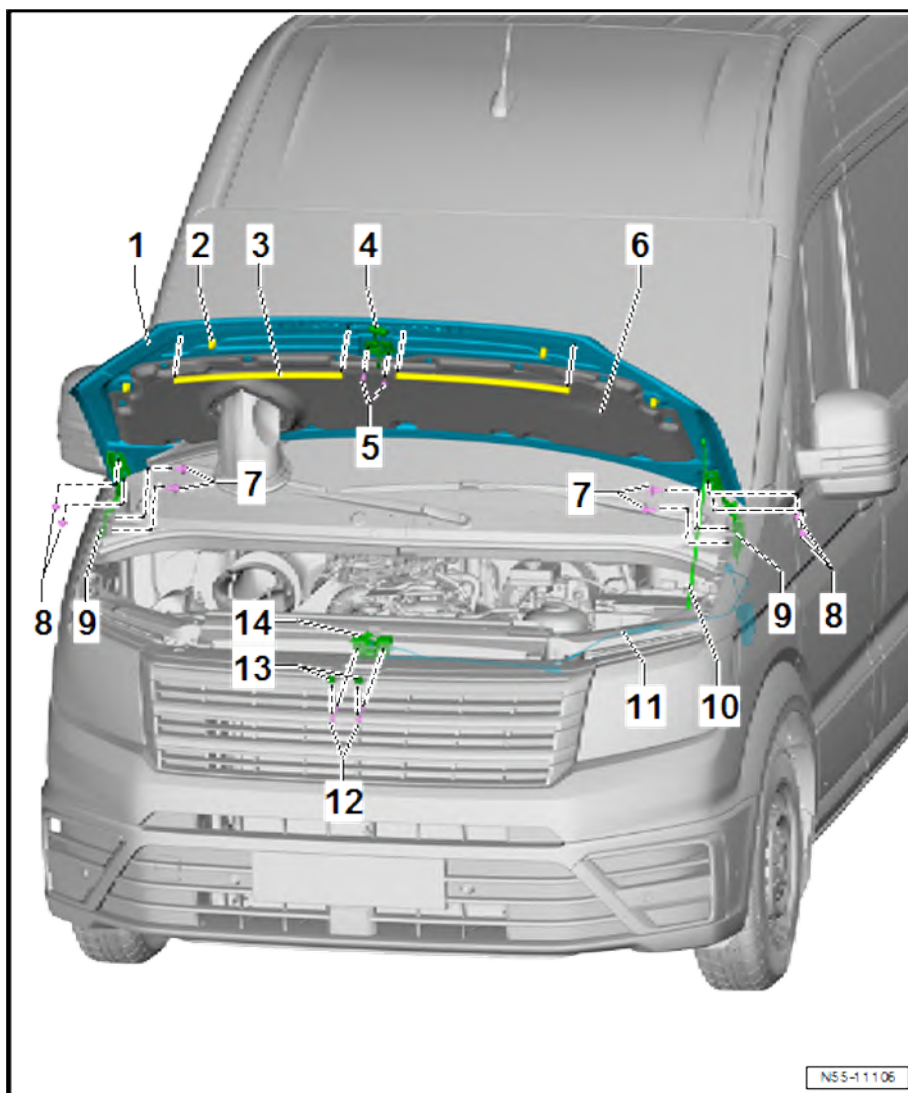
- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 12 Nm

13 - Covers

- ☐ Qty. 2

14 - Bonnet lock

- ☐ Removing and installing ⇒ [page 57](#)
- ☐ Adjusting ⇒ [page 55](#)





1.1.2 Assembly overview - bonnet, e-Crafter or e-TGE

1 - Bonnet (hood)

- ☐ Removing and installing
⇒ [page 53](#)
- ☐ Adjusting ⇒ [page 55](#)

2 - Stop buffer

- ☐ Qty. 4

3 - Striker

- ☐ Removing and installing
⇒ [page 59](#)

4 - Seals

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 62](#)

5 - Insulation

- ☐ Removing and installing
⇒ [page 61](#)

6 - Hinges

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 60](#)

7 - Bolts

- ☐ Qty. 4
- ☐ 20 Nm

8 - Support rod

- ☐ Removing and installing
⇒ [page 64](#)

9 - Nuts

- ☐ Qty. 4
- ☐ 20 Nm

10 - Operating lever

- ☐ Removing and installing ⇒ [page 58](#)

11 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 12 Nm

12 - Covers

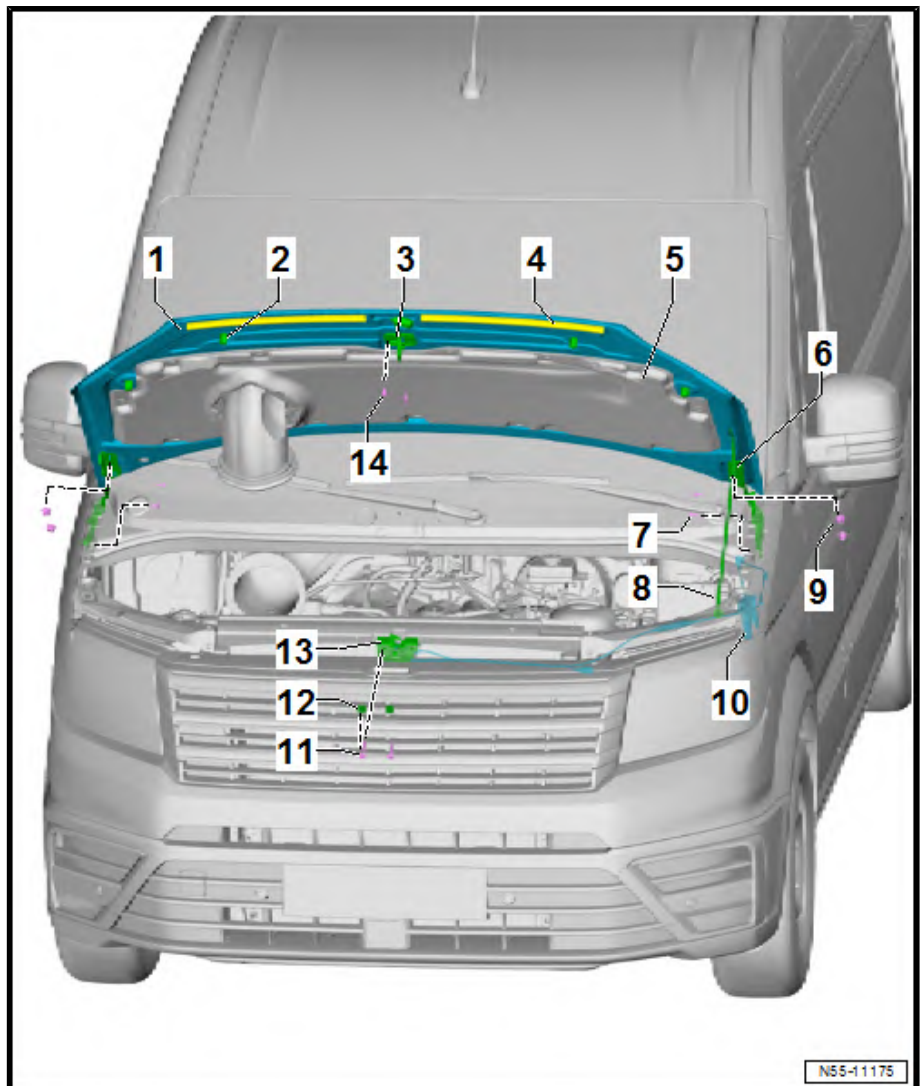
- ☐ Qty. 2

13 - Bonnet lock

- ☐ Removing and installing ⇒ [page 57](#)

14 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 10 Nm





1.2 Assembly overview - Bowden cable

⇒ [“1.2.1 Assembly overview - Bowden cable”, page 52](#)

⇒ [“1.2.2 Assembly overview - Bowden cable, e-Crafter or e-TGE”, page 53](#)

1.2.1 Assembly overview - Bowden cable

1 - Operating lever

- ❑ Removing and installing
⇒ [page 58](#)

2 - Bolt

- ❑ 2 Nm

3 - Mounting bracket

4 - Expansion nut

5 - Bowden cable

- ❑ From mounting bracket to Bowden cable coupling
- ❑ Removing and installing
⇒ [page 63](#)

6 - Seal

- ❑ For Bowden cable

7 - Bowden cable coupling

8 - Bowden cable

- ❑ From Bowden cable coupling to bonnet lock
- ❑ Removing and installing
⇒ [page 64](#)

9 - Bolts

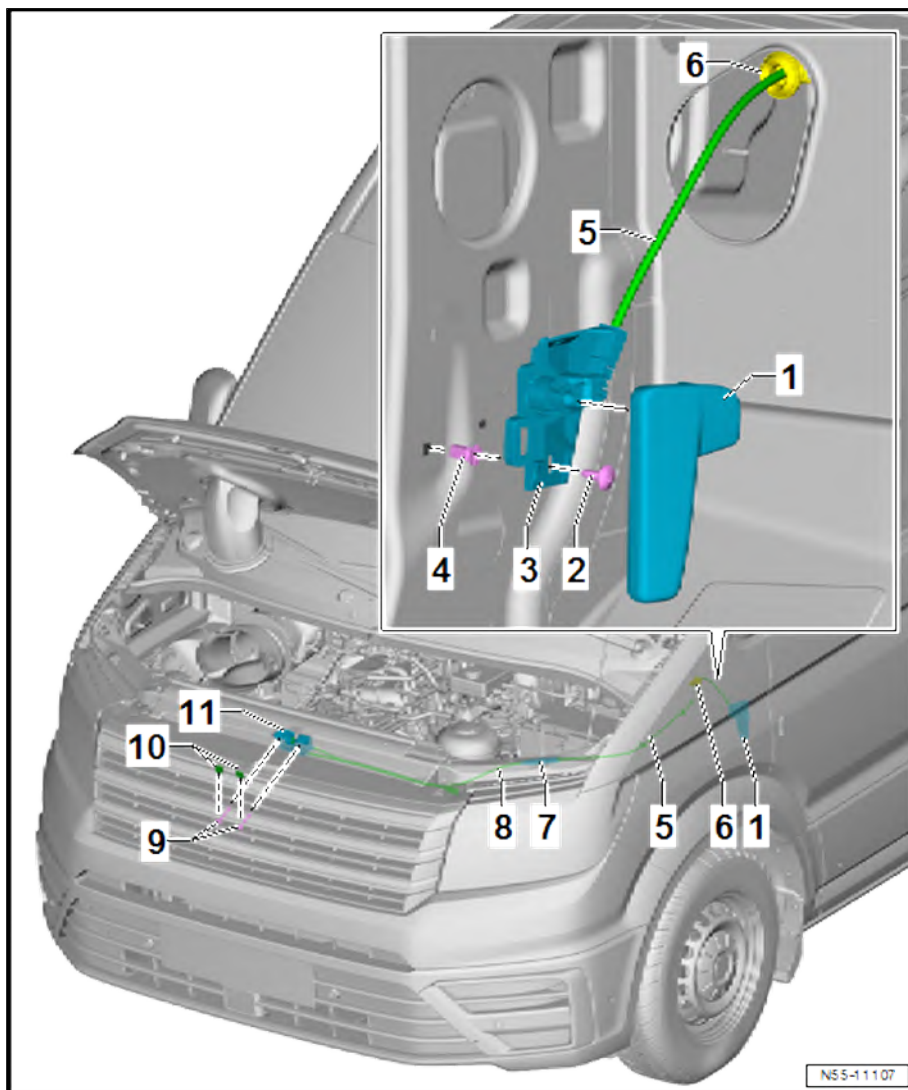
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 12 Nm

10 - Covers

- ❑ Qty. 2

11 - Bonnet lock

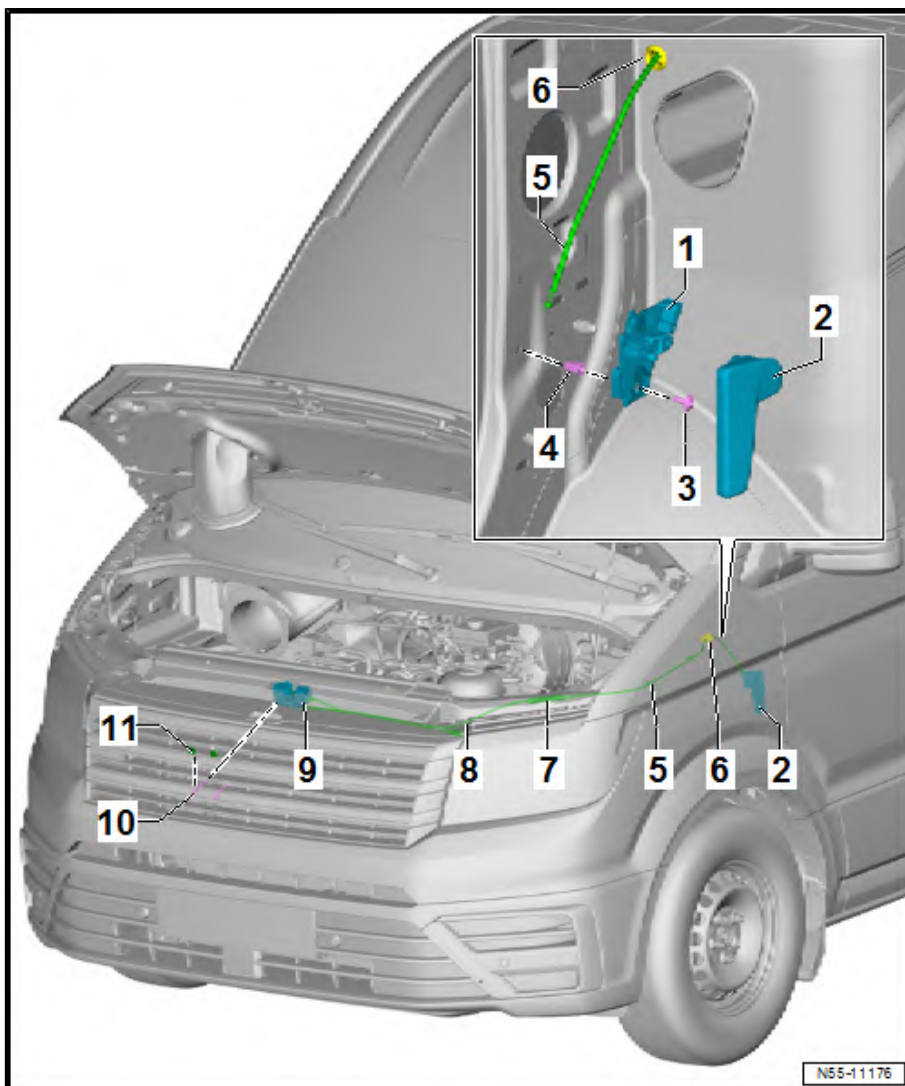
- ❑ Removing and installing
⇒ [page 57](#)





1.2.2 Assembly overview - Bowden cable, e-Crafter or e-TGE

- 1 - Mounting bracket
- 2 - Operating lever
 - ☐ Removing and installing
⇒ [page 58](#)
- 3 - Bolt
 - ☐ 2 Nm
- 4 - Expansion nut
- 5 - Bowden cable
 - ☐ From mounting bracket to Bowden cable coupling
 - ☐ Removing and installing
⇒ [page 63](#)
- 6 - Seal
 - ☐ For Bowden cable
- 7 - Bowden cable coupling
- 8 - Bowden cable
 - ☐ From Bowden cable coupling to bonnet lock
 - ☐ Removing and installing
⇒ [page 64](#)
- 9 - Bonnet lock
 - ☐ Removing and installing
⇒ [page 57](#)
- 10 - Bolts
 - ☐ Renew after removal
 - ☐ Qty. 2
 - ☐ 12 Nm
- 11 - Covers
 - ☐ Qty. 2



1.3 Removing and installing bonnet

⇒ ["1.3.1 Removing and installing bonnet", page 53](#)

⇒ ["1.3.2 Removing and installing bonnet, e-Crafter or e-TGE", page 54](#)

1.3.1 Removing and installing bonnet

Removing

- Remove fresh air intake duct from bonnet ⇒ Heating, air conditioning; Rep. gr. 87 ; Air duct; Removing and installing fresh air intake duct .



Vehicles with heated spray jets

- Disconnect electrical connector -3-.

Continued for all vehicles

- Release catch -2- in -direction of arrow B-. When doing this, pull hose connection -4- in -direction of arrow A- off spray jet -1-.
- If fitted, unclip wiring harness and hose for washer fluid from bonnet.
- If fitted, guide wiring harness together with hose for washer fluid out of bonnet.
- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure bonnet -3-.
- Unscrew nuts -2- on the left and right-hand side.
- With the aid of a second mechanic, remove bonnet -3- from hinges -1- on left and right side.

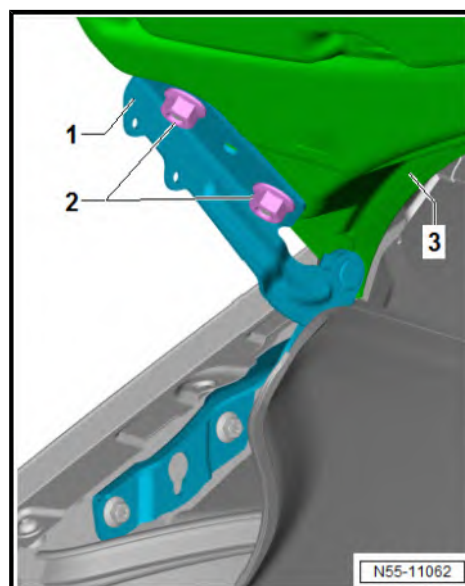
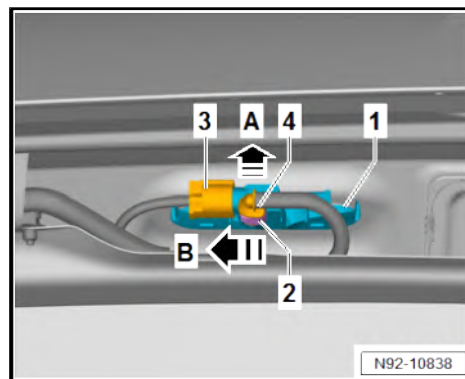
Installing

Install in reverse order of removal, observing the following:

- Adjusting bonnet ⇒ [page 55](#) .
- Adjust spray jets ⇒ Electrical system; Rep. gr. 92 ; Wind-screen washer system; Adjusting spray jets .

Specified torques

- ♦ ⇒ [“1.1 Assembly overview - bonnet”, page 49](#)



1.3.2 Removing and installing bonnet, e-Crafter or e-TGE

Removing

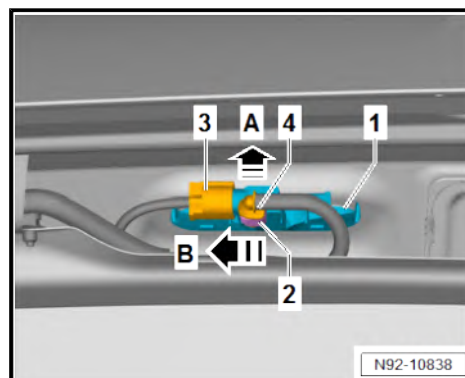
- Remove fresh air intake duct from bonnet ⇒ Heating, air conditioning; Rep. gr. 87 ; Air duct; Removing and installing fresh air intake duct .

Vehicles with heated spray jets

- Disconnect electrical connector -3-.
- Unclip wiring harness and hose for washer fluid from bonnet.
- Guide wiring harness together with hose for washer fluid out of bonnet.

Continued for all vehicles

- Release catch -2- in -direction of arrow B-. When doing this, pull hose connection -4- in -direction of arrow A- off spray jet -1-.
- Unclip hose for washer fluid from bonnet. Guide it out of bonnet when doing this.
- The following work steps must be performed by a second mechanic.





- Have a second mechanic secure bonnet -1-.
- Unscrew nuts -2-.
- With the aid of a second mechanic, remove bonnet -1- from hinges -3-.

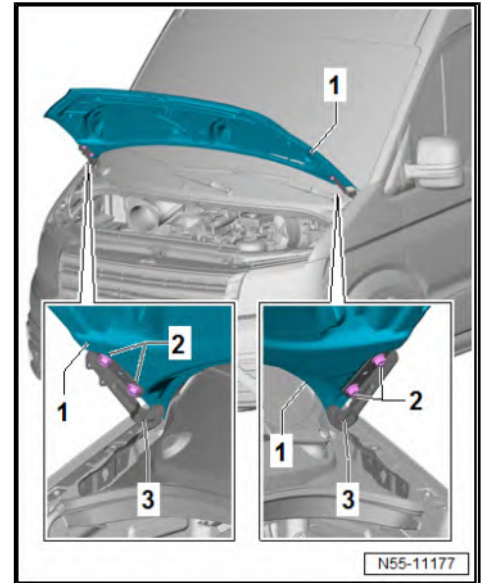
Installing

Install in reverse order of removal, observing the following:

- Adjusting bonnet ⇒ [page 55](#) .
- Adjust spray jets ⇒ Electrical system; Rep. gr. 92 ; Wind-screen washer system; Adjusting spray jets .

Specified torques

- ◆ ⇒ ["1.1 Assembly overview - bonnet", page 49](#)



1.4 Adjusting bonnet

Special tools and workshop equipment required

- ◆ Setting gauge - 3371-



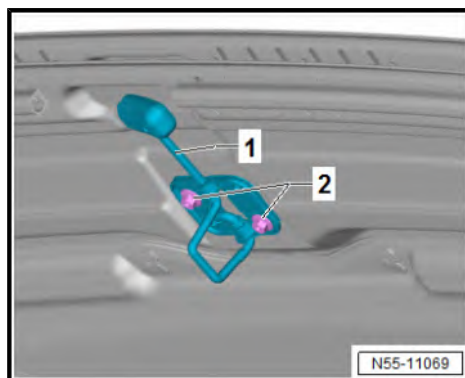
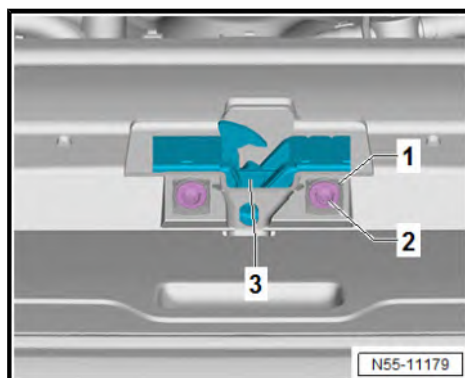
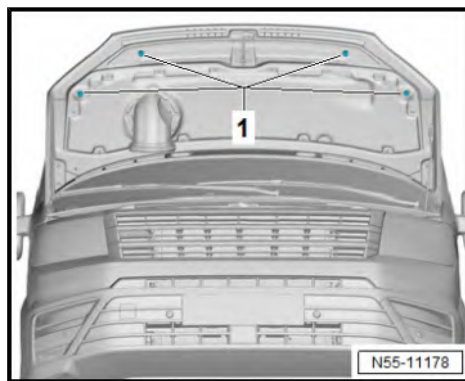
- ◆ Torque wrench - V.A.G 1331-





Procedure

- Vehicle must be standing on all wheels.
- Screw in stop buffer -1- clockwise to stop.
- Remove covers -1-.
- Unscrew bolts -2-.
- Screw in new bolts -2- to stop by hand.
- Align bonnet lock -3- centrally in horizontal and vertical direction. When doing this, tighten bolts -2- to specified torque ⇒ [page 49](#) .
- Loosen bolts -2-.
- Close bonnet.
- The striker pin -1- must enter the bonnet lock in the centre.
- Open bonnet.
- Hold striker pin -1- in this position. When doing this, tighten bolts -2- to specified torque ⇒ [page 49](#) .
- Use setting gauge - 3371- to check gaps and height difference of bonnet/wing ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .





Gaps and height difference outside the tolerance range

- Adjust gaps of bonnet in longitudinal direction and between wings by loosening nuts -2- on hinge -1-.
- Adjust height difference of bonnet by loosening bolts -3- on hinge -1-.
- Repeat work steps on left side of vehicle.
- Check height difference of bonnet/wing ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .
- Adjust height difference by screwing in or out stop buffers, if necessary.
- The stop buffers must make contact with the lock carrier struts.

Continued for all vehicles

- Close bonnet.
- Unscrew front stop buffers without tension until they make contact with lock carrier.
- Open bonnet.
- After successful adjustment, check specified torques again.
- If necessary, carry out corrosion protection measures.

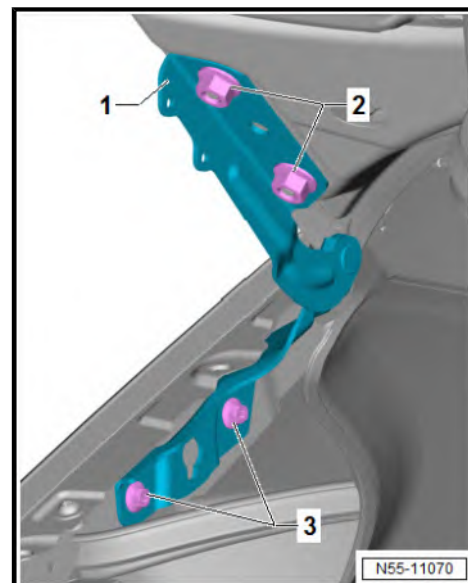
Specified torques

- ◆ ⇒ ["1.1 Assembly overview - bonnet", page 49](#)

1.5 Removing and installing bonnet lock

Special tools and workshop equipment required

- ◆ Setting gauge - 3371-



- ◆ Removal wedge - 3409-



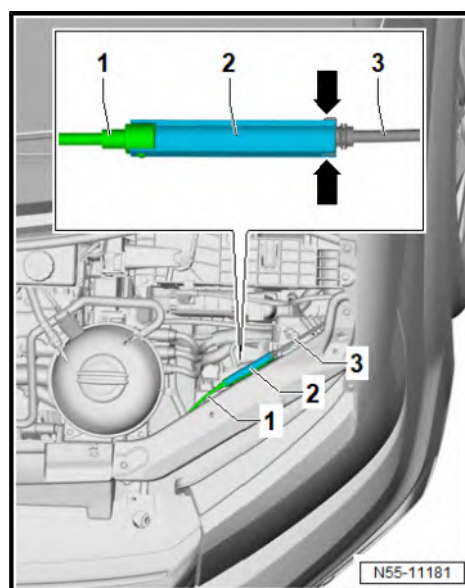


◆ Torque wrench - V.A.G 1331-

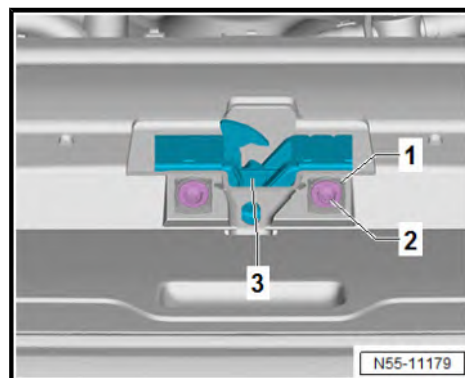


Removing

- Unclip Bowden cable coupling -1- at lock carrier strut.
- Use removal wedge - 3409- to push apart cap -2- in area marked with -arrows- until it has been released from the two projections.
- Detach Bowden cable -3- from Bowden cable coupling -1-.



- Lever off covers -1- using removal wedge - 3409- .
- Unscrew bolts -2-.
- Disconnect electrical connector for bonnet contact switch - F387- .
- Unclip wiring harness from lock carrier.
- Remove bonnet lock -3- until Bowden cable on bonnet lock -3- can be detached.
- Detach Bowden cable on bonnet lock -3-. Remove bonnet lock -3- when doing this.



Installing

Install in reverse order of removal, observing the following:

- Use setting gauge - 3371- to check gaps and height difference of bonnet ➔ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .
- If necessary, adjust bonnet ➔ [page 55](#) .

Specified torques

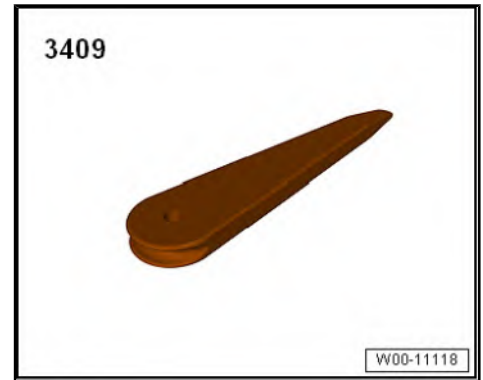
- ◆ ➔ [“1.1 Assembly overview - bonnet”, page 49](#)

1.6 Removing and installing bonnet release lever

Special tools and workshop equipment required



◆ Removal wedge - 3409-



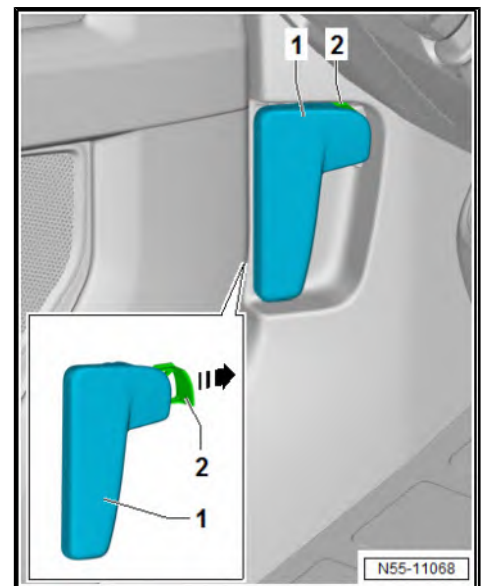
Removing

- Pull release lever -1-. When doing this, use removal wedge - 3409- to lever out retaining clip -2- in -direction of arrow-.
- Pull release lever -1- off mounting bracket.

Installing

Install in reverse order of removal, observing the following:

- Insert retaining clip -2- into release lever -1-.
- Push release lever -1- onto mounting bracket.
- Perform functional check.



1.7 Removing and installing striker pin

Special tools and workshop equipment required

◆ Setting gauge - 3371-





- ◆ Torque wrench - V.A.G 1410-



Removing

- Unscrew bolts -2-. When doing this, remove striker pin -1-.

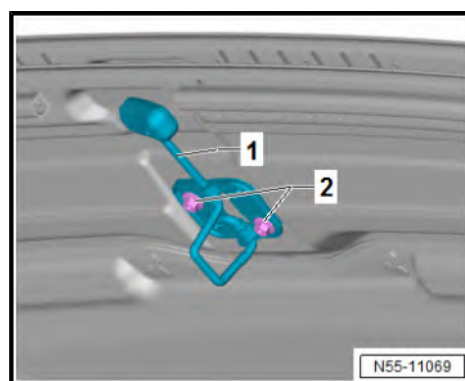
Installing

Install in reverse order of removal, observing the following:

- Use setting gauge - 3371- to check gaps and height difference of bonnet ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .
- If necessary, adjust bonnet ⇒ [page 55](#) .

Specified torques

- ◆ ⇒ [“1.1 Assembly overview - bonnet”, page 49](#)



1.8 Removing and installing hinges

Special tools and workshop equipment required

- ◆ Setting gauge - 3371-



- ◆ Torque wrench - V.A.G 1331-





Removing

- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure bonnet.
- Unscrew nuts -2-.
- Unscrew bolts -3-.
- Pull off hinge -1-.

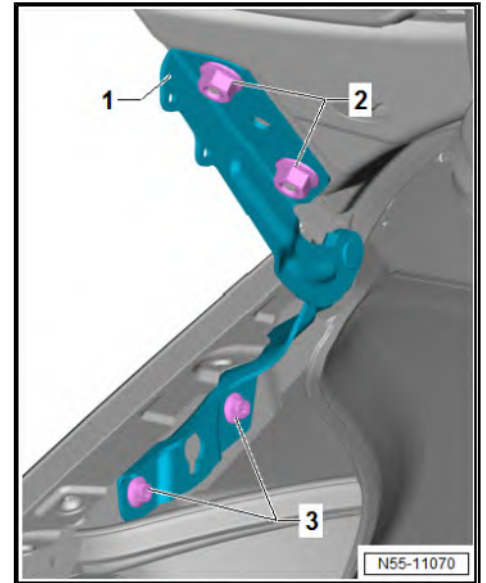
Installing

Install in reverse order of removal, observing the following:

- Use setting gauge - 3371- to check gaps and height difference of bonnet ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .
- If necessary, adjust bonnet ⇒ [page 55](#) .

Specified torques

- ◆ ⇒ [“1.1 Assembly overview - bonnet”, page 49](#)



1.9 Removing and installing insulation

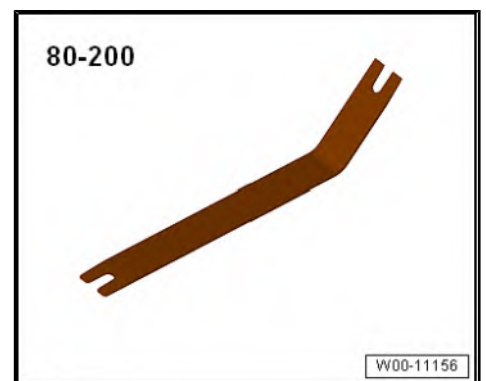
⇒ [“1.9.1 Removing and installing insulation”, page 61](#)

⇒ [“1.9.2 Removing and installing bonnet seal”, page 62](#)

1.9.1 Removing and installing insulation

Special tools and workshop equipment required

- ◆ Release lever - 80-200-



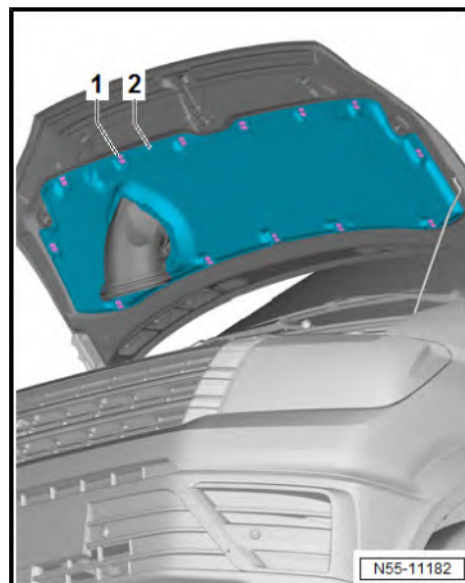


Removing

- Lever out retaining clips -1- using release lever - 80-200- .
- Remove insulation -2-.

Installing

Install in reverse order.



1.9.2 Removing and installing bonnet seal

Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



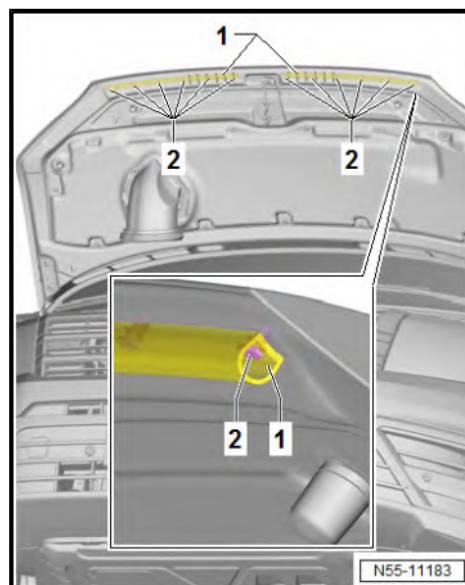
Removing

- Guide removal wedge - 3409- behind seal -1-. When doing this, unclip clips -2-.

Installing

Install in reverse order of removal, observing the following:

- Make sure that clips -2- are seated correctly in seal -1-. Correct them as necessary.





1.10 Removing and installing Bowden cable

⇒ [“1.10.1 Removing and installing Bowden cable, part section from mounting bracket”, page 63](#)

⇒ [“1.10.2 Removing and installing Bowden cable, part section to bonnet lock”, page 64](#)

1.10.1 Removing and installing Bowden cable, part section from mounting bracket

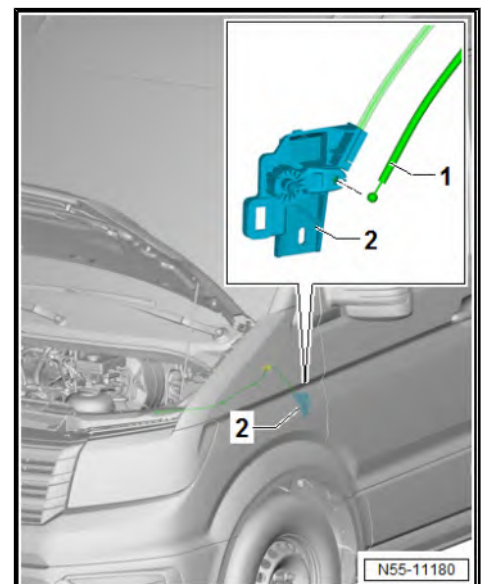
Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



Removing

- Remove lower A-pillar trim ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing A-pillar trim .
- Detach Bowden cable -1- on mounting bracket -2-.



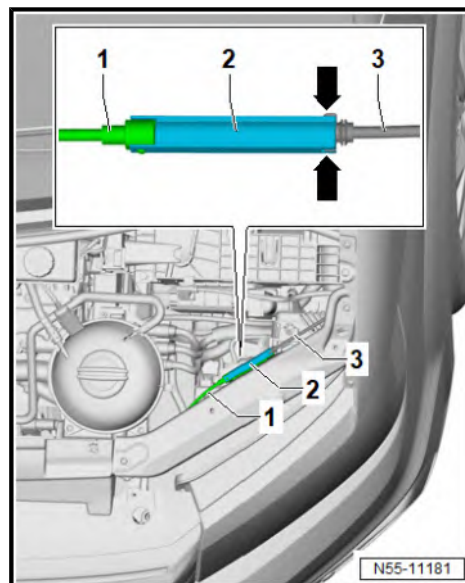


- Unclip Bowden cable coupling -1- at lock carrier strut.
- Use removal wedge - 3409- to push apart cap -2- in area marked with -arrows- until it has been released from the two projections.
- Detach Bowden cable -3- from Bowden cable coupling -1-.
- Detach Bowden cable -3- from retainers in area of wing.
- Pull grommet out of bulkhead.
- Remove Bowden cable together with grommet.

Installing

Install in reverse order of removal, observing the following:

- Ensure proper seating of grommet.
- Perform functional check.



1.10.2 Removing and installing Bowden cable, part section to bonnet lock

Removing

- Remove bonnet lock ➔ [page 57](#) .
- Unclip Bowden cable from lock carrier. Remove Bowden cable when doing this.

Installing

Install in reverse order of removal, observing the following:

- Perform functional check.

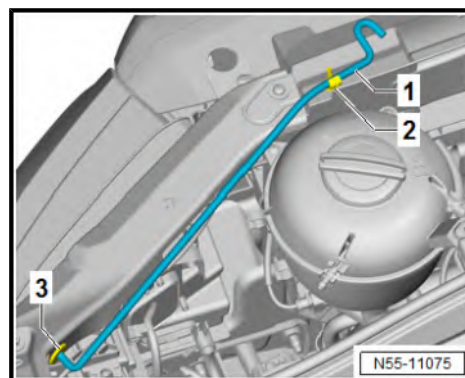
1.11 Removing and installing support rod

Removing

- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure bonnet.
- Unclip support rod -1- from clip -2-.
- Guide support rod -1- out of rubber mounting -3-.

Installing

Install in reverse order.





2 Tank flap unit

⇒ [“2.1 Assembly overview - tank flap unit”, page 65](#)

⇒ [“2.2 Removing and installing fuel tank flap”, page 66](#)

2.1 Assembly overview - tank flap unit

⇒ [“2.1.1 Assembly overview - tank flap unit”, page 65](#)

⇒ [“2.1.2 Assembly overview - tank flap unit, e-Crafter or e-TGE”, page 66](#)

2.1.1 Assembly overview - tank flap unit

1 - Tank flap

- ☐ Removing and installing
⇒ [page 66](#)

2 - Buffer cap

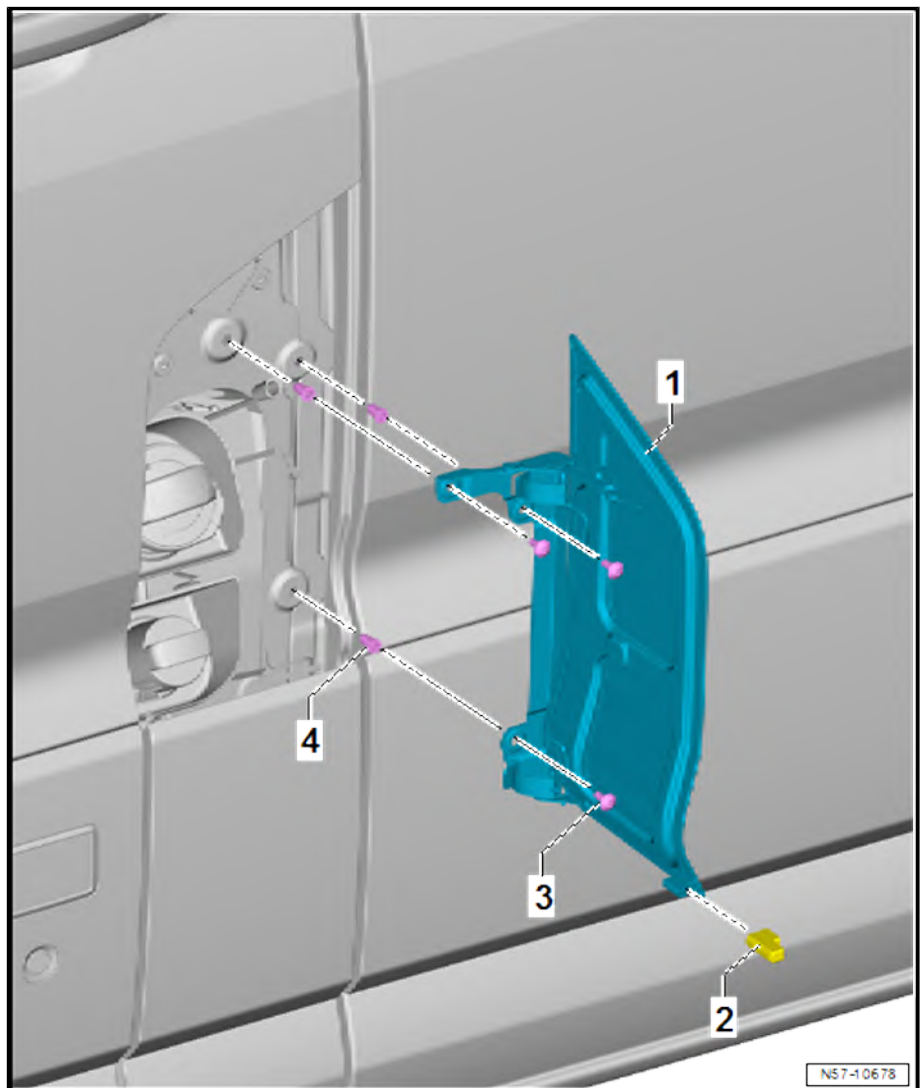
- ☐ Tank flap

3 - Bolts

- ☐ Qty. 3
- ☐ 4.5 Nm

4 - Pop rivet nut

- ☐ Qty. 3





2.1.2 Assembly overview - tank flap unit, e-Crafter or e-TGE

1 - Tank flap

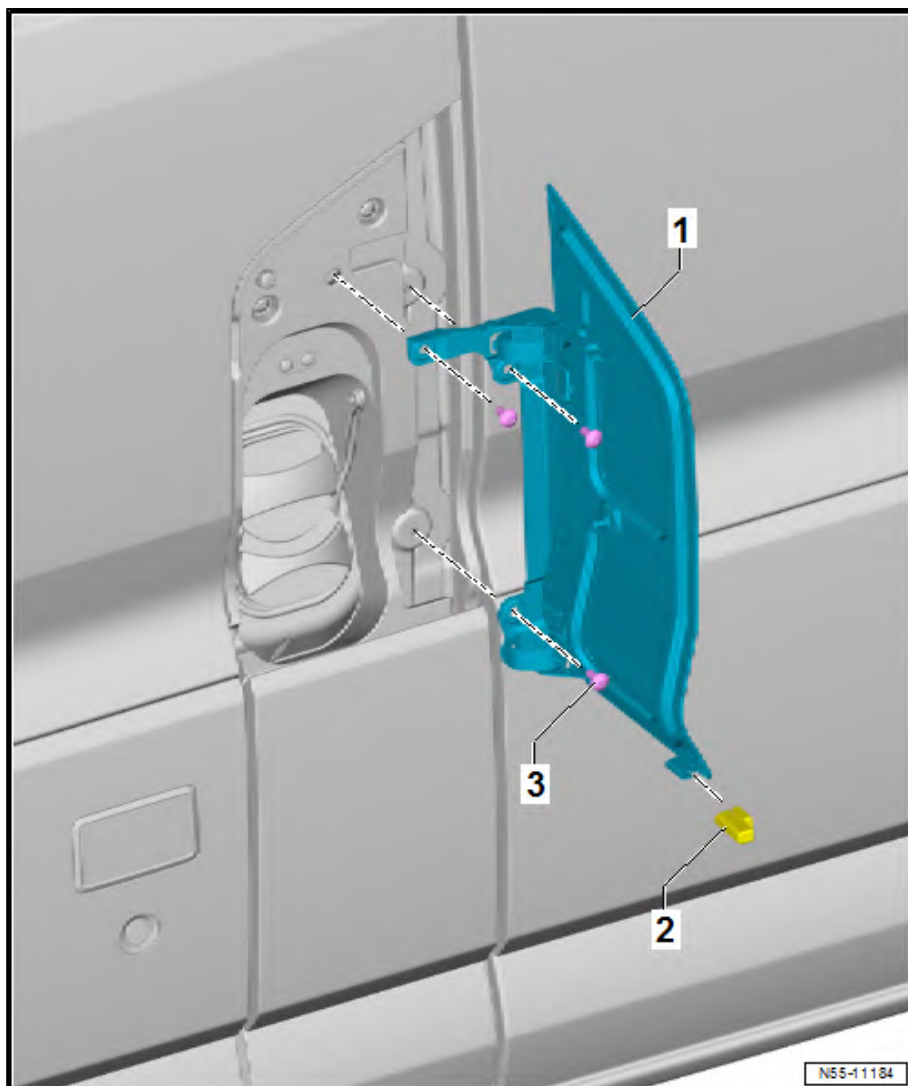
- ❑ Removing and installing
⇒ [page 66](#)

2 - Buffer cap

- ❑ Tank flap

3 - Bolts

- ❑ Qty. 3
- ❑ 4.5 Nm



2.2 Removing and installing fuel tank flap

⇒ [“2.2.1 Removing and installing fuel tank flap”, page 66](#)

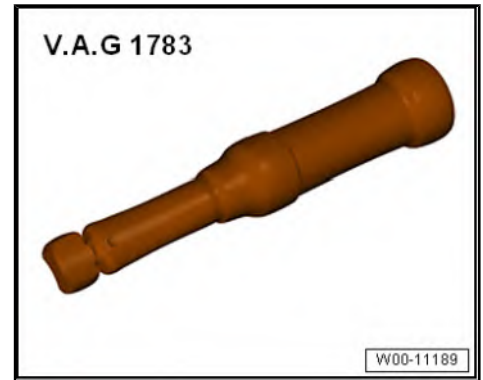
⇒ [“2.2.2 Removing and installing tank flap, e-Crafter or e-TGE”, page 67](#)

2.2.1 Removing and installing fuel tank flap

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

- Open tank flap -1-.
- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Detach filler neck -4-
- Unscrew tank cap.
- Remove tank flap -1- together with tank cap.

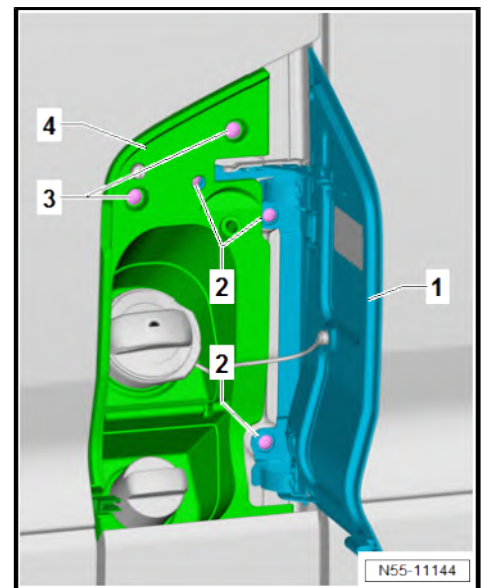
Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["2.1 Assembly overview - tank flap unit", page 65](#)

Component	Specified torque
Filler neck	4.5 Nm



2.2.2 Removing and installing tank flap, e-Crafter or e-TGE

Special tools and workshop equipment required

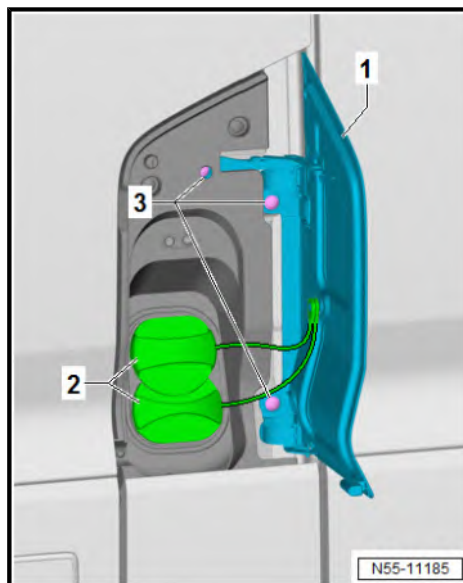
- ◆ Torque wrench - V.A.G 1783-





Removing

- Open tank flap -1-.
- Pull off plug -2-.
- Unscrew bolts -3-.



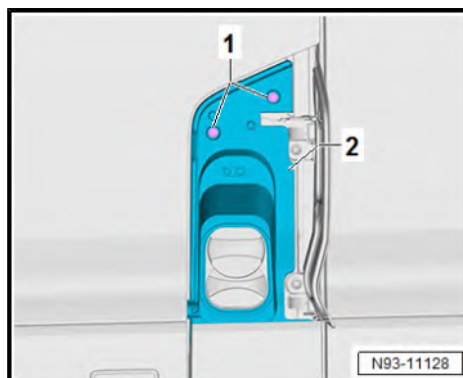
- Unscrew bolts -1-.
- Pull cover -2- towards outer side of vehicle. When doing this, remove tank flap together with cap.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - tank flap unit”, page 65](#)
- ◆ Charging socket; Assembly overview - charging socket ⇒ Rep. gr. 93 ; Charging socket; Assembly overview - charging socket



57 – Front doors, door components, central locking

1 Door

- ⇒ ["1.1 Assembly overview - door", page 69](#)
- ⇒ ["1.2 Assembly overview - door seals", page 70](#)
- ⇒ ["1.3 Removing and installing inner door seal", page 73](#)
- ⇒ ["1.4 Removing and installing door", page 75](#)
- ⇒ ["1.5 Adjusting door", page 77](#)
- ⇒ ["1.6 Adjusting striker pin", page 78](#)
- ⇒ ["1.7 Removing and installing door hinge", page 79](#)
- ⇒ ["1.8 Removing and installing door arrester", page 80](#)

1.1 Assembly overview - door

1 - Door

- ☐ Front
- ☐ Removing and installing
⇒ [page 75](#)
- ☐ Adjusting ⇒ [page 77](#)

2 - Door hinge

- ☐ Top
- ☐ Removing and installing
⇒ [page 79](#)

3 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 20 Nm +90°

4 - Bolts

- ☐ Qty. 2
- ☐ 25 Nm

5 - Caps

- ☐ Qty. 2

6 - Bolt

- ☐ Renew after removal
- ☐ 20 Nm +90°

7 - Bolts

- ☐ Qty. 2
- ☐ 10 Nm

8 - Door arrester

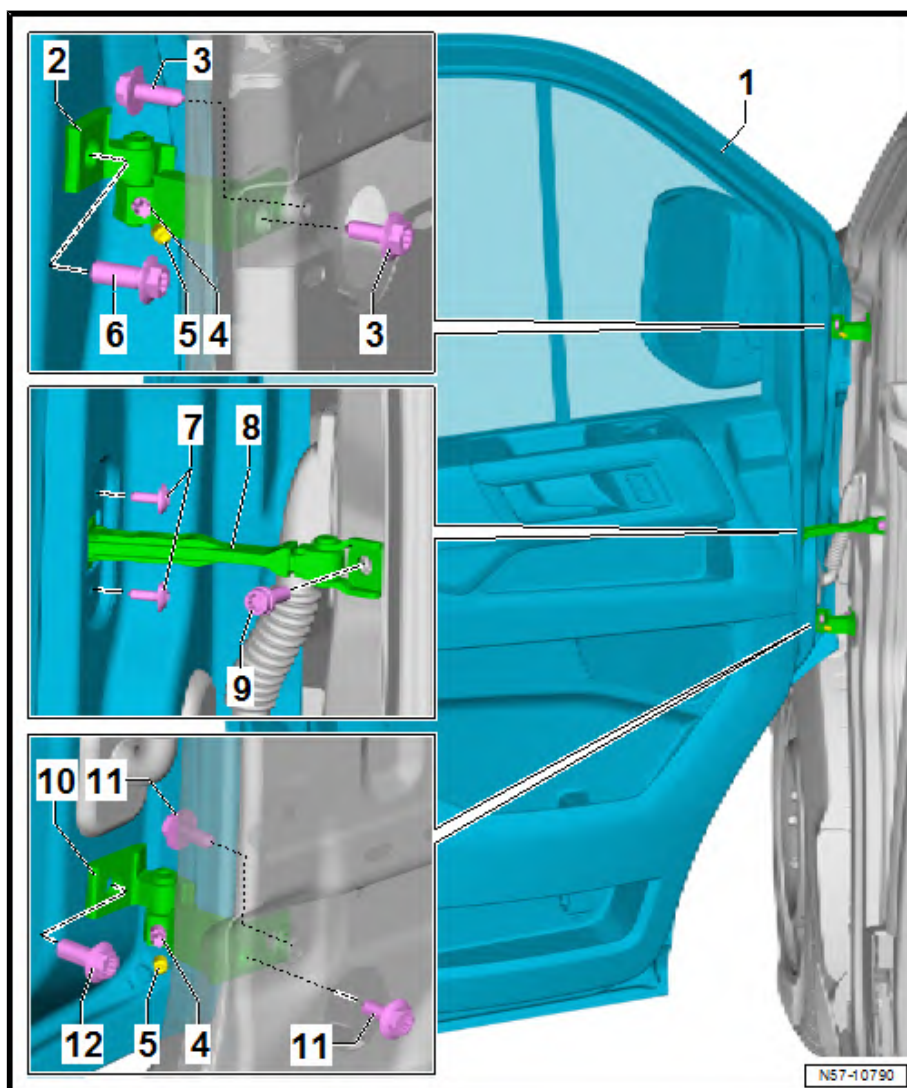
- ☐ Removing and installing
⇒ [page 80](#)

9 - Bolt

- ☐ 30 Nm

10 - Door hinge

- ☐ Bottom
- ☐ Removing and installing ⇒ [page 79](#)





11 - Bolt

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 20 Nm +90°

12 - Bolt

- ☐ Renew after removal
- ☐ 20 Nm +90°

1.2 Assembly overview - door seals

⇒ ["1.2.1 Assembly overview - door seals, left-hand drive vehicles", page 70](#)

⇒ ["1.2.2 Assembly overview - door seals, right-hand drive vehicles", page 71](#)

⇒ ["1.2.3 Assembly overview - door seals, e-Crafter or e-TGE", page 72](#)

1.2.1 Assembly overview - door seals, left-hand drive vehicles

1 - Door seals

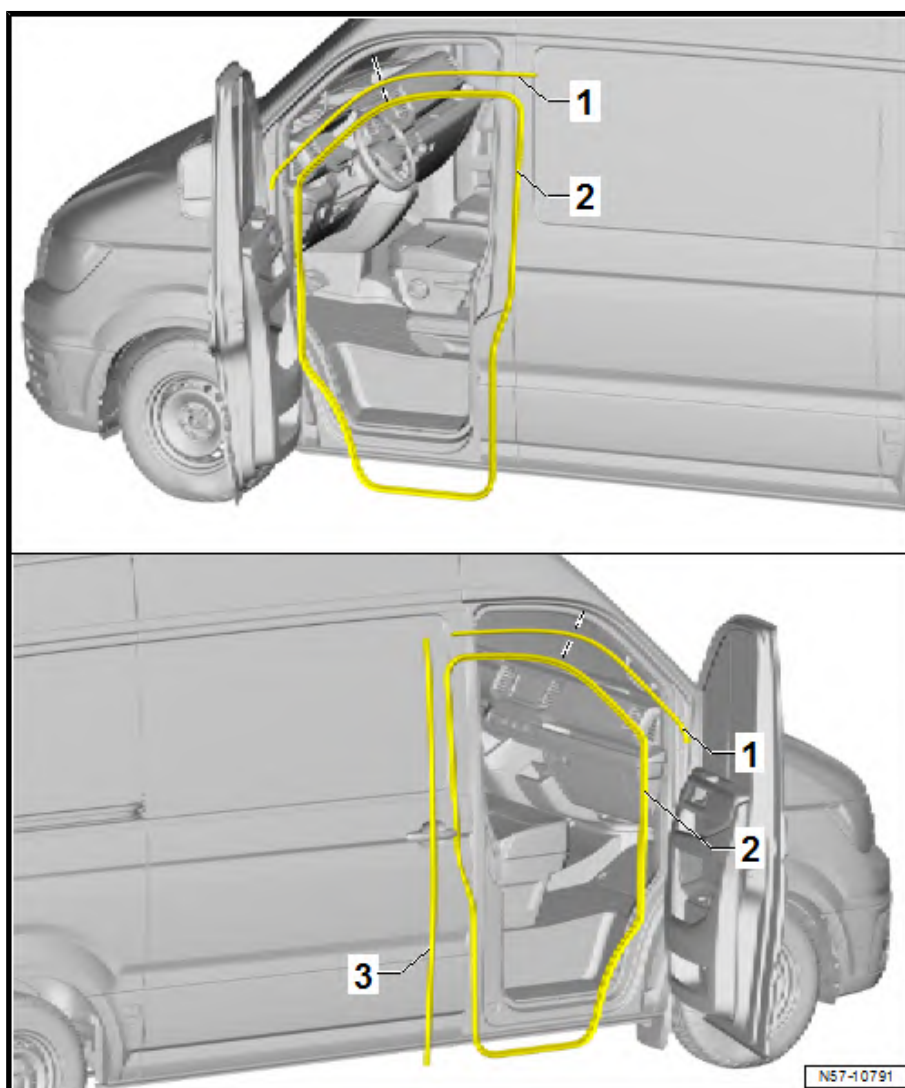
- ☐ Renew after removal
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 73](#)

2 - Pre-seal

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 73](#)

3 - Insulation

- ☐ Renew after removal
- ☐ Removing and installing
⇒ [page 73](#)





1.2.2 Assembly overview - door seals, right-hand drive vehicles

1 - Door seals

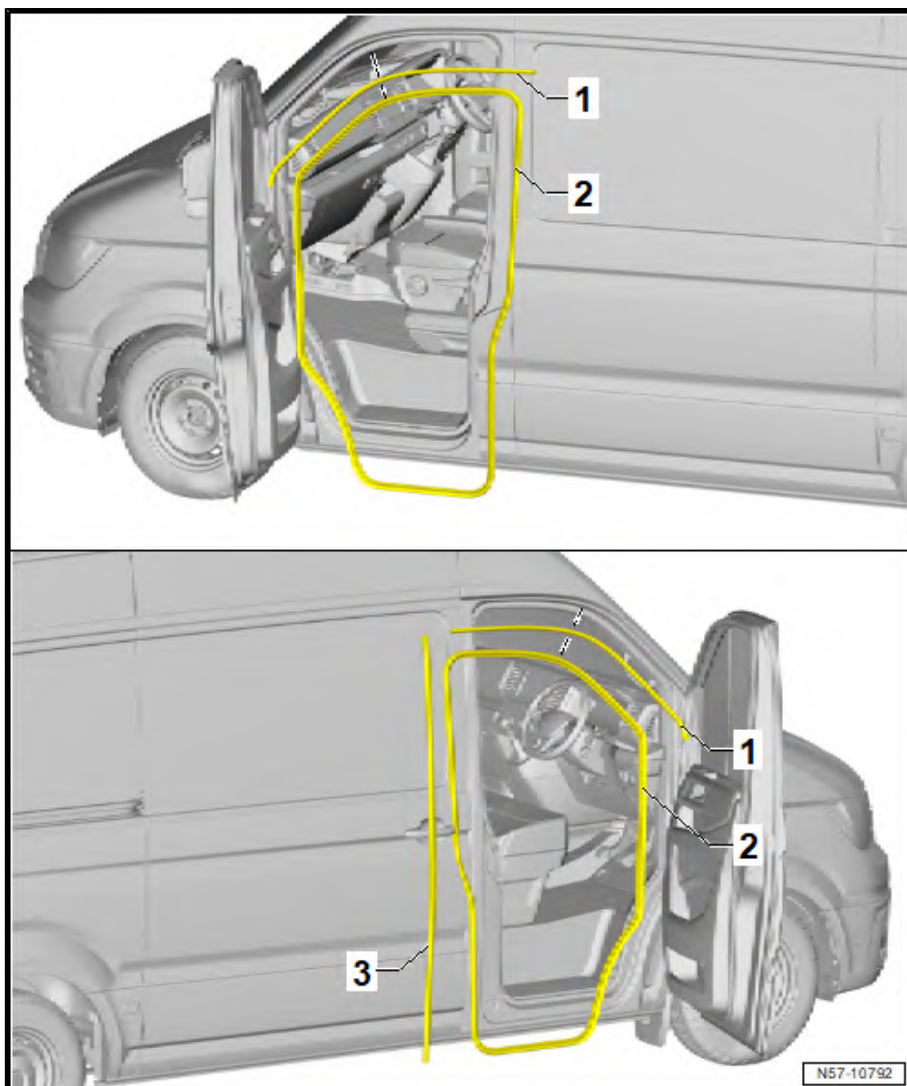
- ☐ Renew after removal
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 73](#)

2 - Pre-seal

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 73](#)

3 - Insulation

- ☐ Renew after removal
- ☐ Removing and installing
⇒ [page 73](#)





1.2.3 Assembly overview - door seals, e-Crafter or e-TGE

1 - Door seals

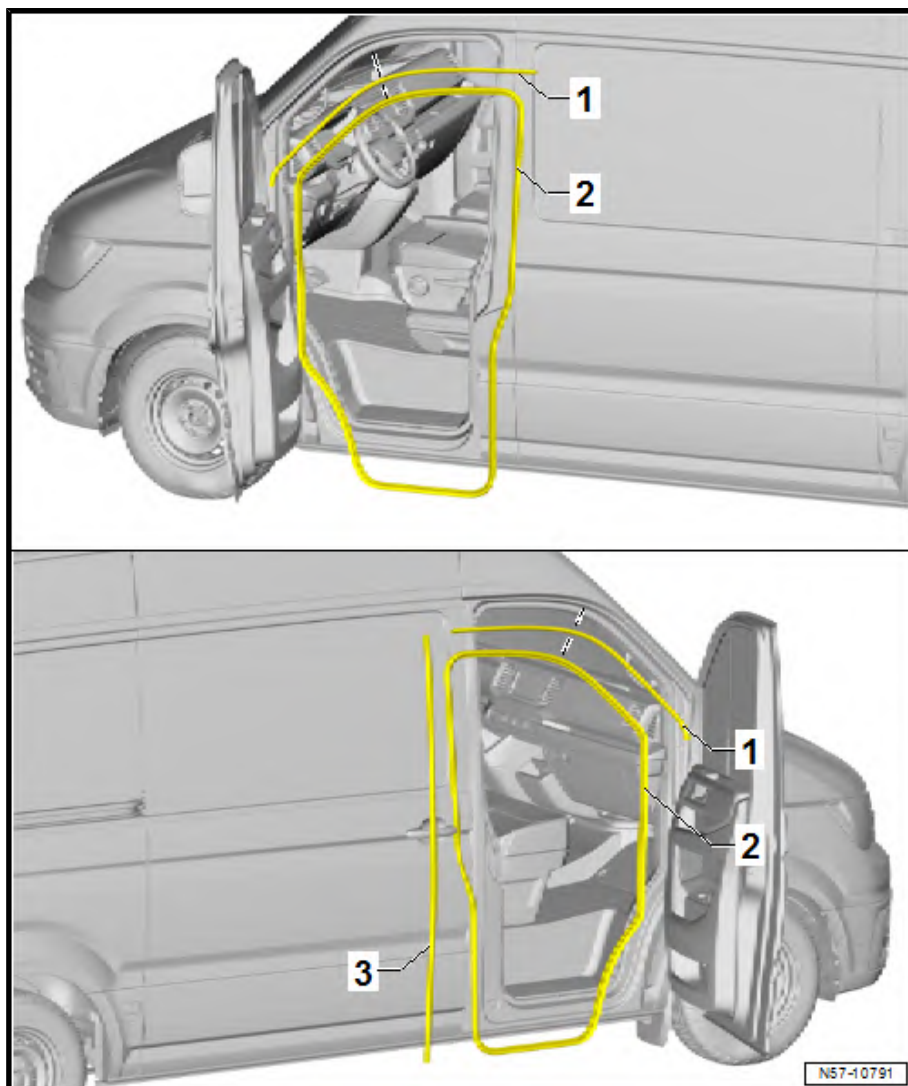
- ☐ Renew after removal
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 73](#)

2 - Pre-seal

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 73](#)

3 - Insulation

- ☐ Renew after removal
- ☐ Removing and installing
⇒ [page 73](#)





1.3 Removing and installing inner door seal

⇒ [“1.3.1 Removing and installing inner door seal”, page 73](#)

⇒ [“1.3.2 Removing and installing door seal”, page 73](#)

⇒ [“1.3.3 Removing and installing insulation”, page 74](#)

1.3.1 Removing and installing inner door seal

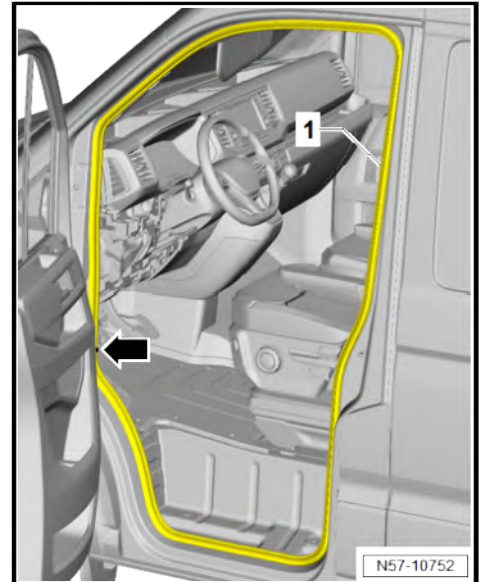
Removing

- Remove step on relevant side ⇒ General body repairs, interior; Rep. gr. 70 ; Interior trims; Removing and installing step .
- Pull off inner door seal -1-.

Installing

Install in reverse order of removal, observing the following:

- Fit new inner door seal -1- with vulcanisation point -arrow- at height of lower door hinge.
- Vulcanisation point -arrow- must not be kinked.
- Install inner door seal -1- all round.



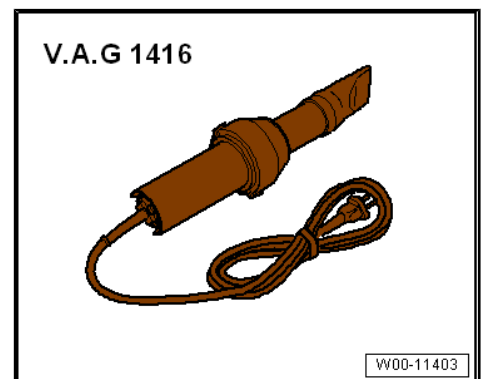
1.3.2 Removing and installing door seal

Special tools and workshop equipment required

- ◆ Roller for door insulation foil - 3356-



- ◆ Hot-air blower - V.A.G 1416-





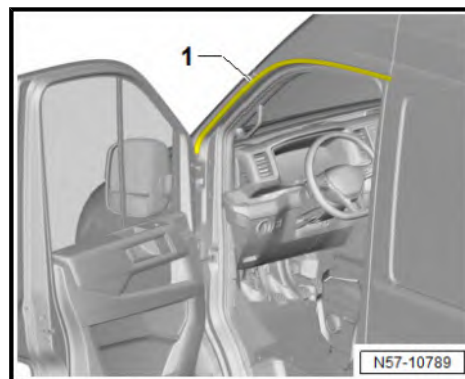
Removing

- Heat pre-seal -1- with hot air blower - V.A.G 1416- . When doing this, pull off pre-seal -1-.

Installing

Install in reverse order of removal, observing the following:

- Clean bonding surface on body using adhesive remover.
- Fit pre-seal -1- on B-pillar, and position it.
- Press on pre-seal -1- using roller - 3356- .



1.3.3 Removing and installing insulation

Special tools and workshop equipment required

- ◆ Roller for door insulation foil - 3356-



- ◆ Hot-air blower - V.A.G 1416-



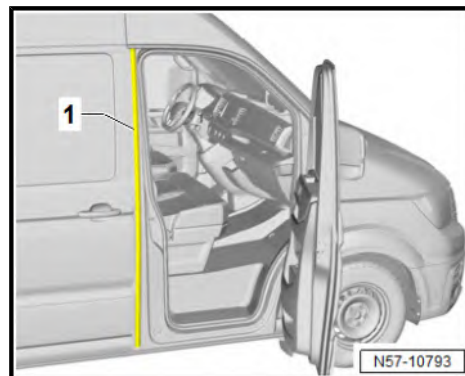
Removing

- Heat insulation -1- with hot air blower - V.A.G 1416- . When doing this, pull off insulation -1-.

Installing

Install in reverse order of removal, observing the following:

- Clean bonding surface on body using adhesive remover.
- Observe the following installation position:

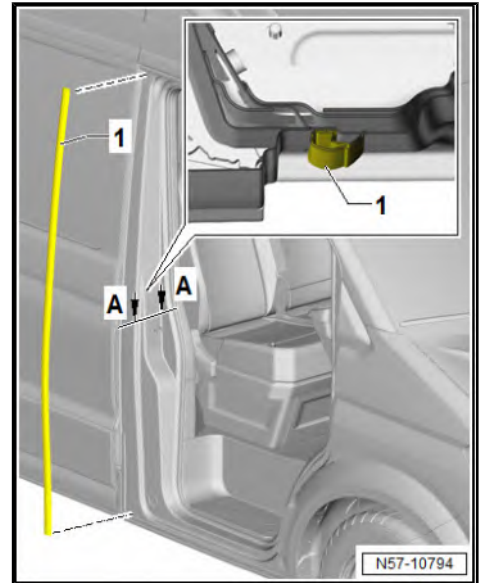




Installation position – insulation, vehicles without sliding door



The sectional view -A- illustrates the installation position of the insulation -1-.



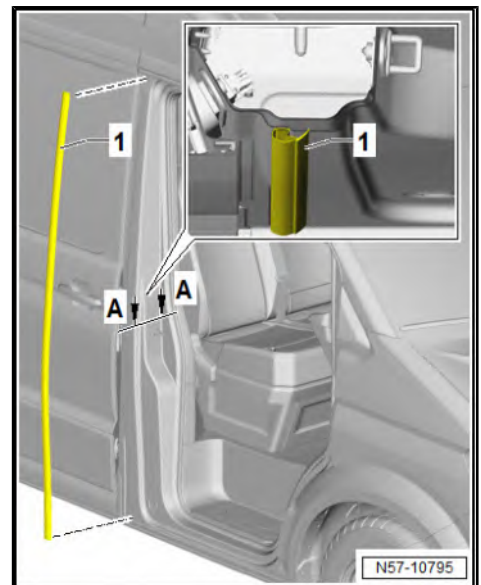
Installation position – insulation, vehicles with sliding door



The sectional view -A- illustrates the installation position of the insulation -1-.

Continued for all vehicles

- Fit insulation -1- on B-pillar, and position it.
- Press on insulation -1- using roller - 3356- .



1.4 Removing and installing door

Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



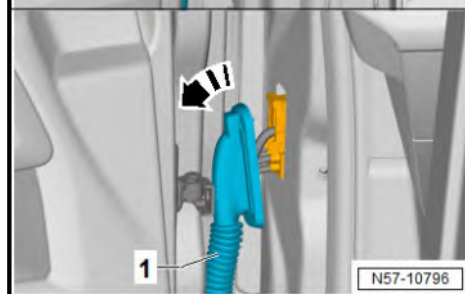
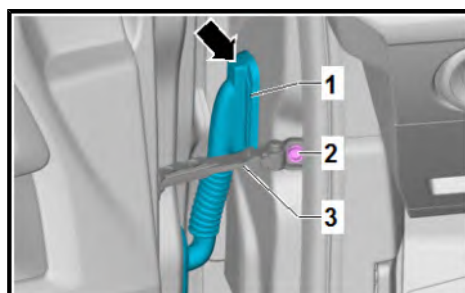


◆ Torque wrench - V.A.G 1331-

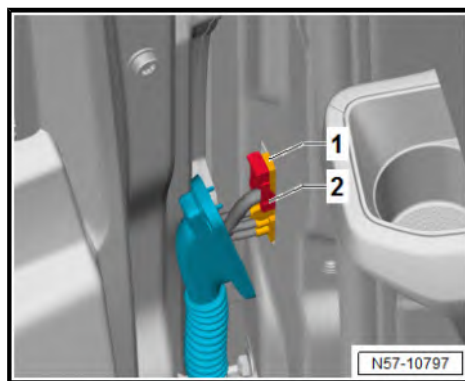


Removing

- Release boot -1- in area marked with -arrow-. When doing this, use removal wedge - 3409- to lever off boot at top of A-pillar.
- Unscrew bolt -2-
- Insert door arrester -3- into door.
- Swing boot -1- in -direction of arrow- out of A-pillar.



- Disconnect electrical connector -1- by swinging retaining clip -2- downwards.





- Lever off caps -1- using removal wedge - 3409- .
- Loosen bolts -2-.
- The following work step must be performed by a second mechanic.
- With the aid of a second mechanic, lift door -3- out of hinges. Remove door -3- when doing this.

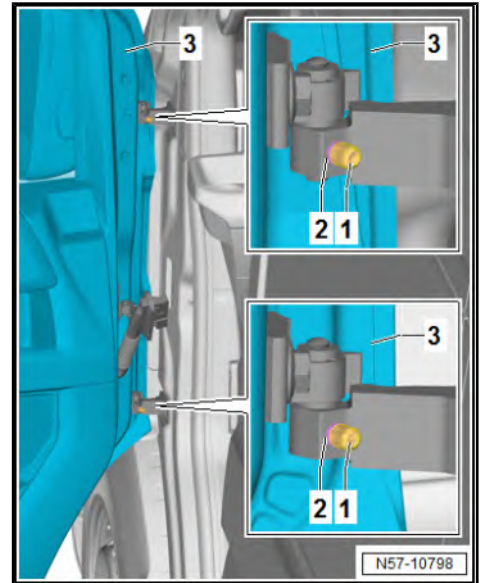
Installing

Install in reverse order of removal, observing the following:

- Use setting gauge - 3371- to check gaps and height difference of door ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .
- If necessary, adjust door ⇒ [page 77](#) .

Specified torques

- ◆ ⇒ ["1.1 Assembly overview - door", page 69](#)



1.5 Adjusting door

Special tools and workshop equipment required

- ◆ Setting gauge - 3371-



- ◆ Torque wrench - V.A.G 1331-





Procedure

- Vehicle must be standing on all wheels.

Adjustment requires the aid of a second mechanic.

- Loosen bolt -2-
- Use setting gauge - 3371- to check and adjust panel gaps of door -1- ⇒ Body Repairs; Rep. gr. 00 ; Technical data; Panel gaps/shutlines; Front body .

Upper door hinge on A-pillar

- Remove central tube for dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Central tube for dash panel; Removing and installing central tube for dash panel .

Lower door hinge on A-pillar

- Remove lower A-pillar trim ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing A-pillar trim .
- Loosen bolts -3- and -4- in upper or lower area.
- Use setting gauge - 3371- to check and adjust panel gaps of door -1- ⇒ Body Repairs; Rep. gr. 00 ; Technical data; Panel gaps/shutlines; Front body .
- Tighten bolts to specified torque.
- Following assembly or adjustment work, perform corrosion protection measures on hinges and bolts.

Specified torques

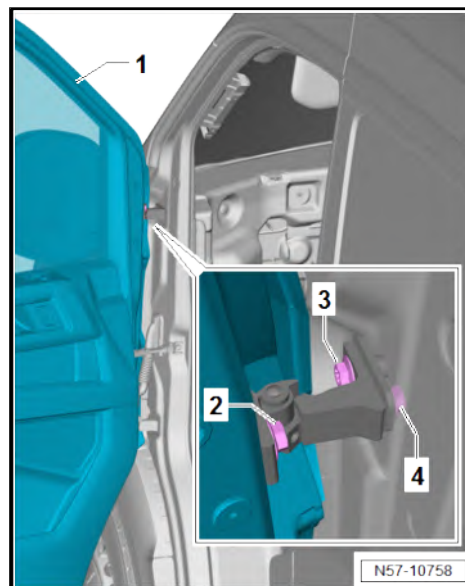
- ♦ ⇒ [“1.1 Assembly overview - door”, page 69](#)

1.6 Adjusting striker pin

Special tools and workshop equipment required

- ♦ Setting gauge - 3371-

- ♦ Torque wrench - V.A.G 1331-





Procedure

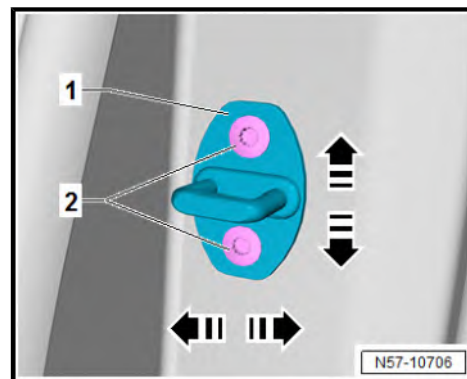
- Vehicle must be standing on all wheels.
- Loosen bolts -2-.
- Adjust striker pin -1- by moving it in the respective -direction of arrow-. Hold it in the correct position, and tighten it to specified torque ➔ [page 85](#) .
- Use setting gauge - 3371- to check gaps and height difference of door ➔ Body Repairs; Rep. gr. 00 ; Gaps; Body, front . Repeat adjustment procedure as necessary.

Gaps and height difference of door outside the tolerance range

- Adjust door ➔ [page 77](#) .

Specified torques

- ◆ ➔ [“2.3 Assembly overview - door handle and door lock”, page 85](#)



1.7 Removing and installing door hinge

➔ [“1.7.1 Removing and installing door hinge, at top or bottom on door”, page 79](#)

➔ [“1.7.2 Removing and installing door hinge, at top or bottom on A-pillar”, page 79](#)

1.7.1 Removing and installing door hinge, at top or bottom on door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

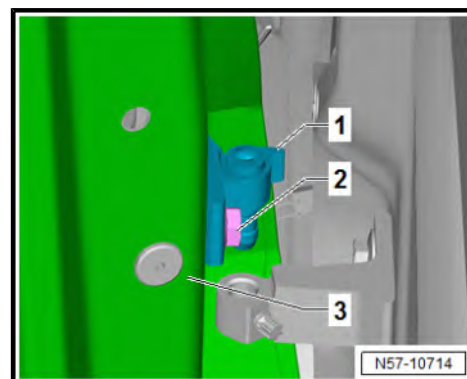
- Remove door ➔ [page 75](#) .
- Unscrew bolt -2-. When doing this, remove door hinge -1- from door -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“1.1 Assembly overview - door”, page 69](#)



1.7.2 Removing and installing door hinge, at top or bottom on A-pillar

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331-



Removing

- Remove door ⇒ [page 75](#) .

Door upper hinge

- Remove central tube for dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Central tube for dash panel; Removing and installing central tube for dash panel .

Door lower hinge

- Remove lower A-pillar trim ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing A-pillar trim .

Continued for all vehicles

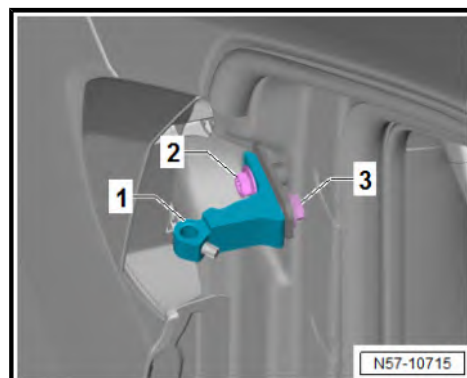
- Unscrew bolts -2- and -3-. When doing this, remove door hinge -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“1.1 Assembly overview - door”, page 69](#)



1.8 Removing and installing door arrester

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





Removing

- Remove front bass loudspeakers -R21- / -R23- ⇒ Rep. gr. 91 ; Sound system; Removing and installing front bass loudspeakers R21/R23 .
- Unscrew bolt -2-
- Unscrew bolts -3-. When doing this, remove door arrester -1- through loudspeaker opening.

Installing

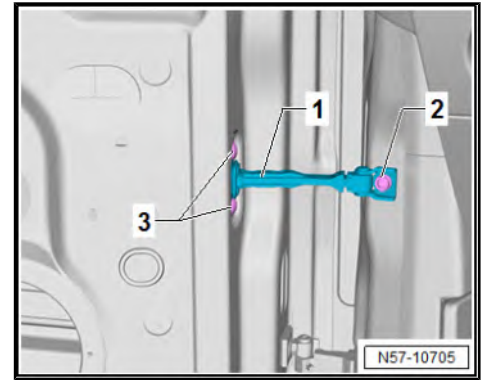
Install in reverse order of removal, observing the following:

Vehicles with door arrester cover

- Observe position and identification of door arrester cover:
 - ◆ Left door: "L top" readable and at top
 - ◆ Right door: "R top" readable and at top

Specified torques

- ◆ ⇒ ["1.1 Assembly overview - door", page 69](#)





2 Door components

- ⇒ ["2.1 Assembly overview - door components", page 82](#)
- ⇒ ["2.2 Assembly overview - window regulator", page 84](#)
- ⇒ ["2.3 Assembly overview - door handle and door lock", page 85](#)
- ⇒ ["2.4 Assembly overview - window channels and window slot \(aperture\) seals", page 86](#)
- ⇒ ["2.5 Removing and installing window regulator motor", page 86](#)
- ⇒ ["2.6 Removing and installing window regulator", page 87](#)
- ⇒ ["2.7 Removing and installing inner door cover", page 88](#)
- ⇒ ["2.8 Removing and installing striker pin", page 89](#)
- ⇒ ["2.9 Removing and installing door handle", page 90](#)
- ⇒ ["2.10 Removing and installing door handle bracket", page 91](#)
- ⇒ ["2.11 Removing and installing door lock", page 92](#)
- ⇒ ["2.12 Removing and installing window channel", page 94](#)
- ⇒ ["2.13 Removing and installing lock cylinder", page 95](#)
- ⇒ ["2.14 Removing and installing cap on door handle", page 97](#)
- ⇒ ["2.15 Removing and installing inner window slot seal", page 97](#)
- ⇒ ["2.16 Removing and installing outer window slot seal", page 98](#)

2.1 Assembly overview - door components



1 - Protective film

- ☐ Cannot be removed without damage
- ☐ Removing and installing
⇒ [page 88](#)

2 - Plugs

- ☐ Qty. 2
- ☐ 30 x 35 mm

3 - Plugs

- ☐ 15.5 x 42.5 mm

4 - Plugs

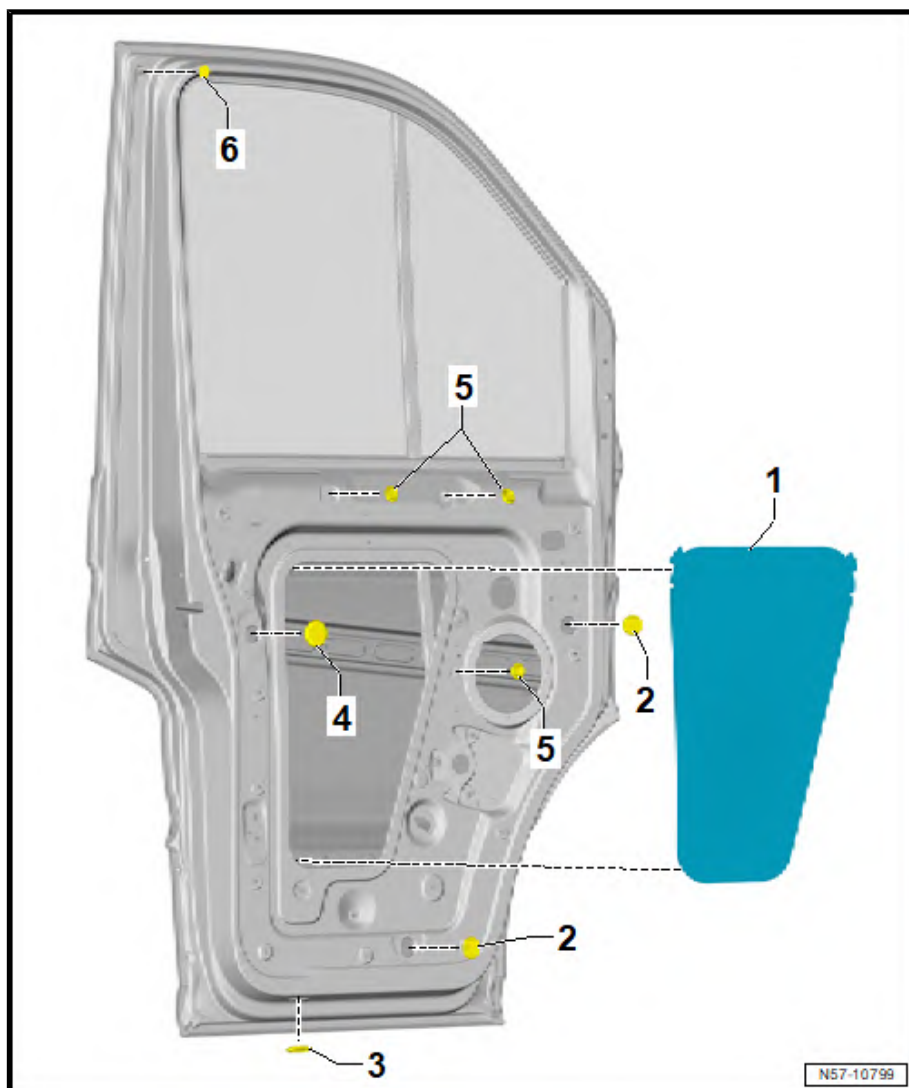
- ☐ Diameter: 30 mm

5 - Plugs

- ☐ Qty. 3
- ☐ Diameter: 23 mm

6 - Plugs

- ☐ Diameter: 12 mm





2.2 Assembly overview - window regulator

1 - Window regulator

- ❑ Removing and installing
⇒ [page 87](#)

2 - Window regulator motor - V14/V15-

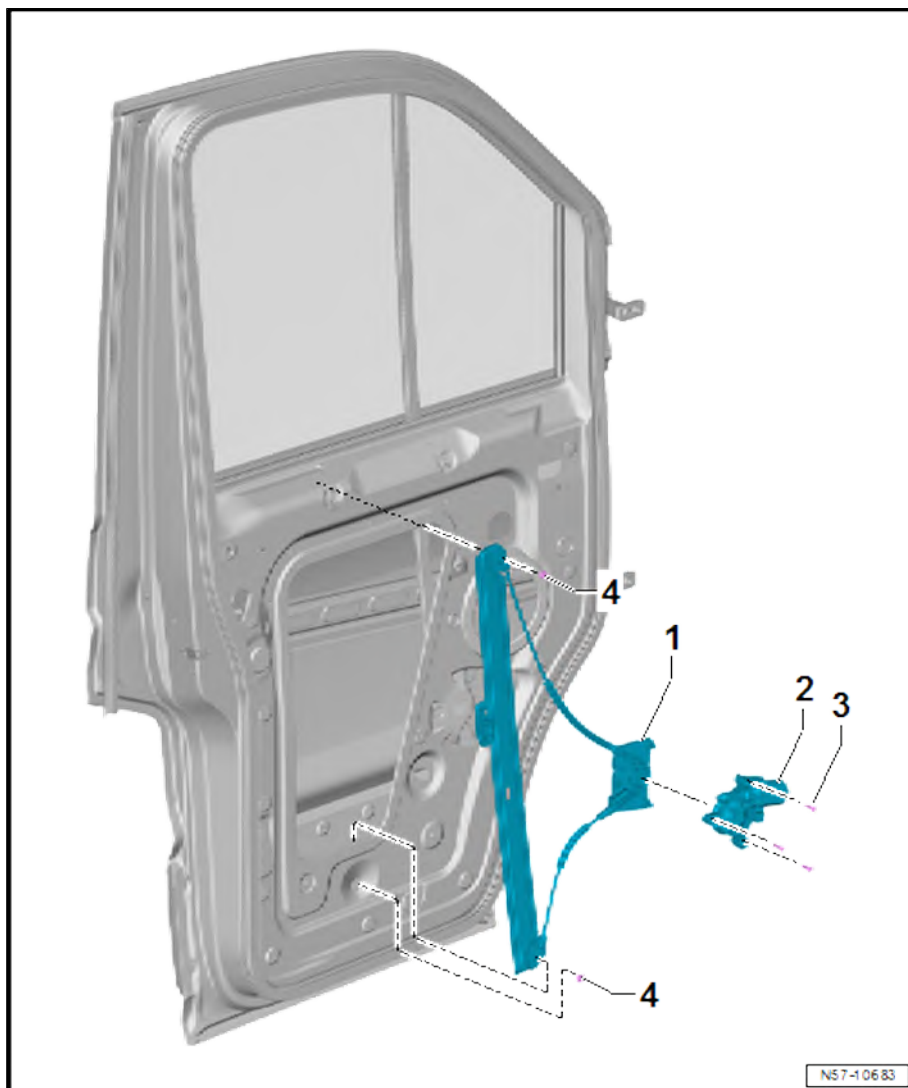
- ❑ Removing and installing
⇒ [page 86](#)

3 - Bolts

- ❑ Qty. 3
- ❑ 3 Nm

4 - Nuts

- ❑ Qty. 2
- ❑ 8 Nm





2.3 Assembly overview - door handle and door lock

1 - Cap

- ☐ For lock cylinder
- ☐ Removing and installing
⇒ [page 95](#)

2 - Door handle

- ☐ Removing and installing
⇒ [page 90](#)

3 - Bolt

- ☐ 2.5 Nm

4 - Carrier

- ☐ Front
- ☐ Removing and installing
⇒ [page 90](#)

5 - Mounting bracket

- ☐ Removing and installing
⇒ [page 91](#)

6 - Bowden cable

- ☐ From door lock to mounting bracket.
- ☐ Removing and installing
⇒ [page 92](#)

7 - Bowden cable

- ☐ From inner door handle to door lock
- ☐ Removing and installing
⇒ [page 92](#)

8 - Door lock

- ☐ Removing and installing
⇒ [page 92](#)

9 - Striker

- ☐ Removing and installing
⇒ [page 89](#)
- ☐ Adjusting ⇒ [page 78](#)

10 - Bolts

- ☐ Qty. 2
- ☐ 25 Nm

11 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

12 - Bolt

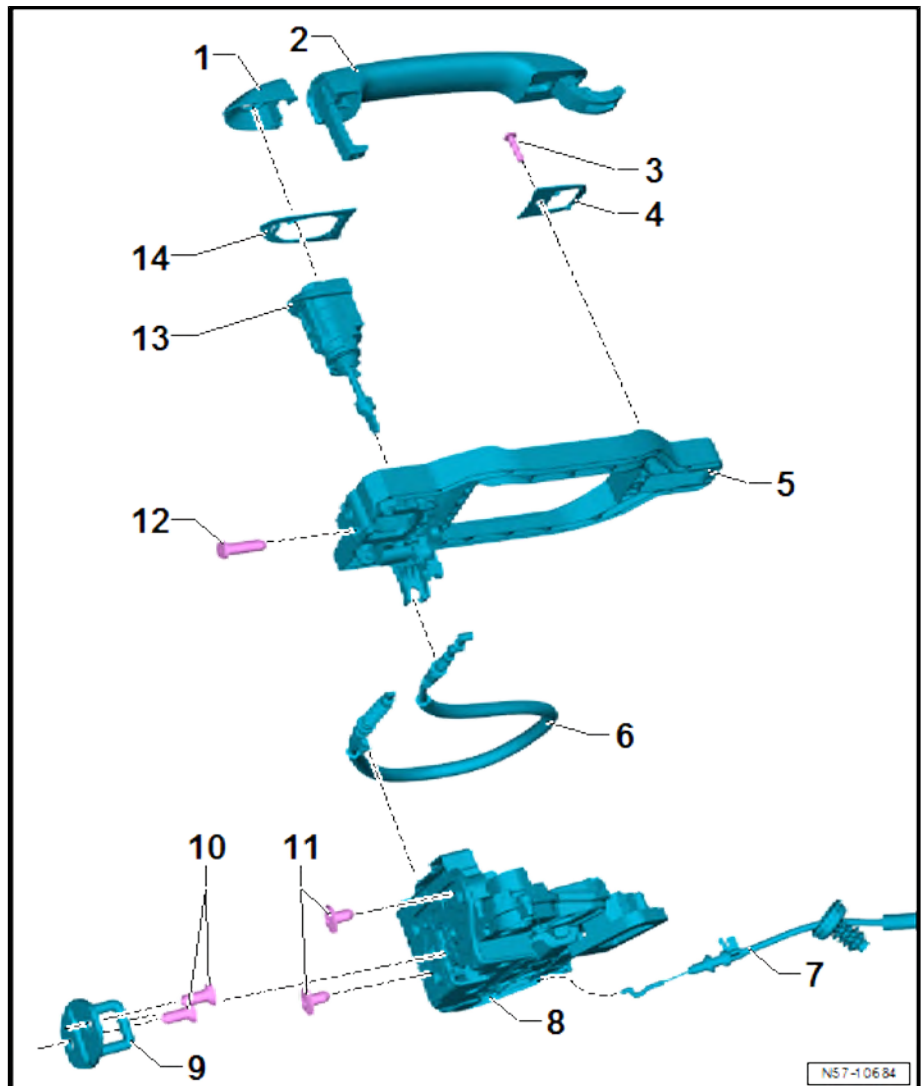
- ☐ 2.5 Nm

13 - Lock cylinder

- ☐ Removing and installing ⇒ [page 95](#)

14 - Carrier

- ☐ Rear
- ☐ Removing and installing ⇒ [page 90](#)





2.4 Assembly overview - window channels and window slot (aperture) seals

1 - Door

- ☐ Removing and installing
⇒ [page 75](#)
- ☐ Adjusting ⇒ [page 78](#)

2 - Window channel

- ☐ Removing and installing
⇒ [page 94](#)

3 - Guide rail

- ☐ Removing and installing
⇒ [page 242](#)

4 - Window slot seal

- ☐ On outside
- ☐ Renew after removal
- ☐ Removing and installing
⇒ [page 98](#)

5 - Window slot seal

- ☐ On inside
- ☐ Removing and installing
⇒ [page 97](#)

6 - Bolts

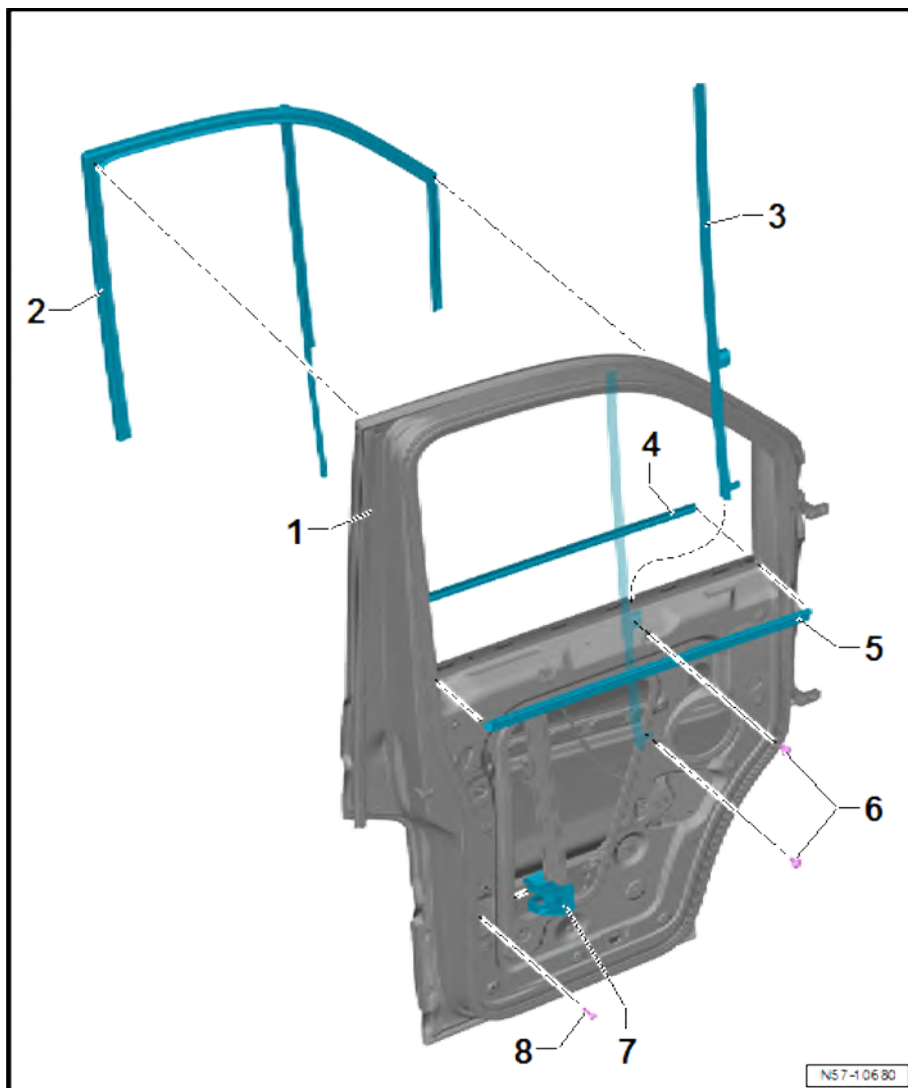
- ☐ Qty. 2
- ☐ 10 Nm

7 - Guide rail

- ☐ Door windows
- ☐ Removing and installing
⇒ [page 94](#)

8 - Bolt

- ☐ 3 Nm



2.5 Removing and installing window regulator motor

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

- Remove front door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .
- Disconnect electrical connector -2-.
- Unscrew bolts -3-.
- Remove window regulator motor -1-.

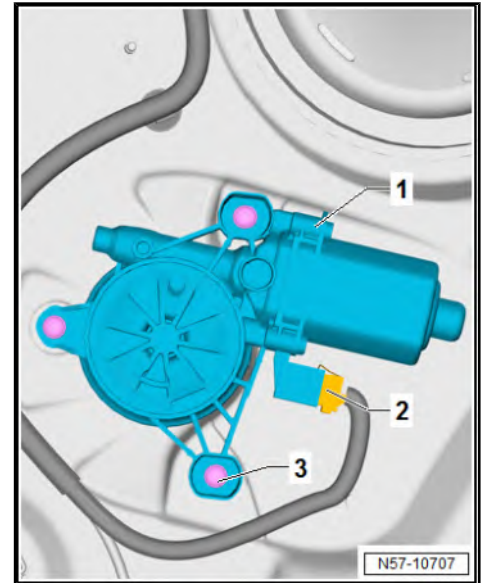
Installing

Install in reverse order of removal, observing the following:

- Fit window regulator motor, and correct the tooth overlap by turning the yoke on the window regulator if necessary.
- Observe tightening sequence of bolts:
 - ◆ 1. Bolt, top
 - ◆ 2. Bolt, centre
 - ◆ 3. Bolt, bottom.

Specified torques

- ◆ ⇒ [“2.2 Assembly overview - window regulator”, page 84](#)



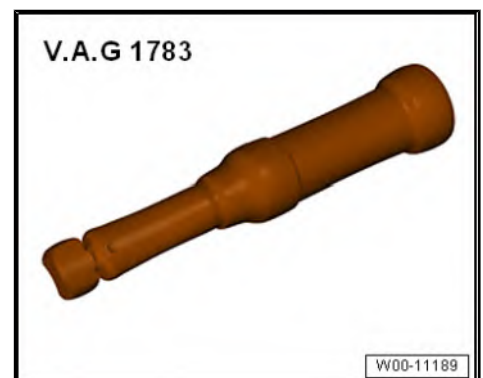
2.6 Removing and installing window regulator

Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



- ◆ Torque wrench - V.A.G 1783-





Removing

- Remove front door window ⇒ [page 241](#) .
- Remove window regulator motor ⇒ [page 86](#) .
- Use removal wedge - 3409- to release cable drum housing from door.
- Loosen nut -2-.
- Unscrew nut -3-.
- Guide window regulator -1- in lower area out of hole. When doing this, push window regulator -1- upwards inside of door.
- Swing out window regulator -1- through opening.

Installing

Install in reverse order of removal, observing the following:

- Make sure that the cable drum housing can be heard to engage.

Specified torques

- ♦ ⇒ ["2.2 Assembly overview - window regulator", page 84](#)

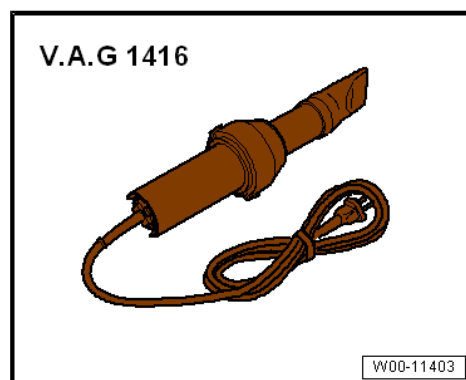
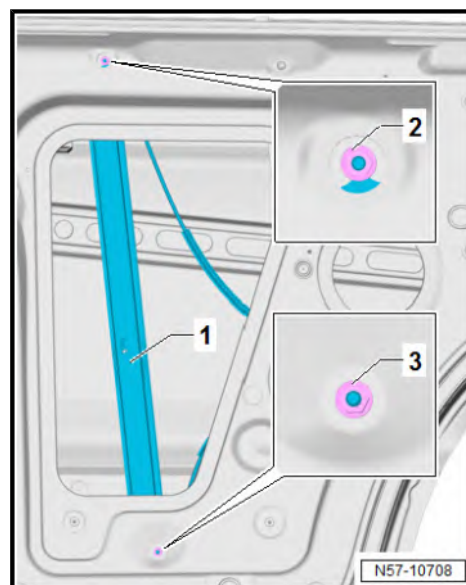
2.7 Removing and installing inner door cover

⇒ ["2.7.1 Removing and installing protective film", page 88](#)

2.7.1 Removing and installing protective film

Special tools and workshop equipment required

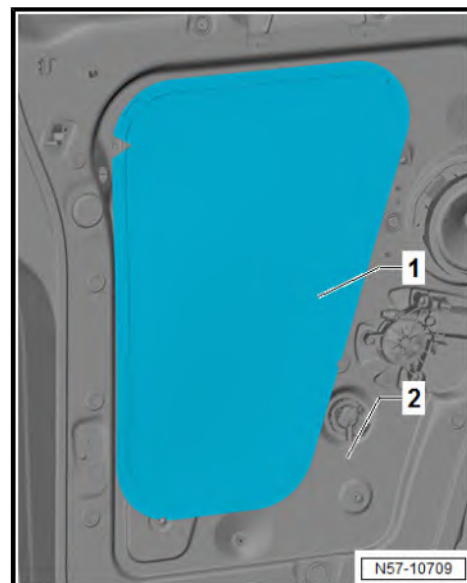
- ♦ Hot-air blower - V.A.G 1416-





Removing

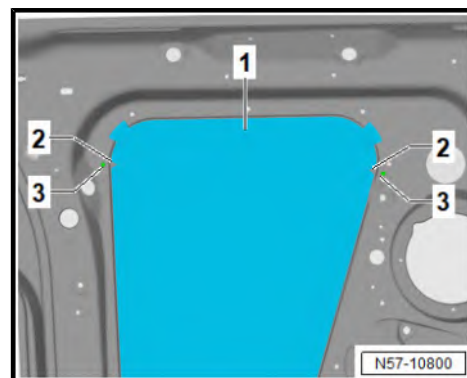
- Remove front door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .
- Heat bonding surface of inner door film -1- with hot air blower - V.A.G 1416- . When doing this, detach inner door film -1- from door -2-.



Installing

Install in reverse order of removal, observing the following:

- Clean bonding surface with adhesive remover.
- Fit protective film -1-.
- Markings -2- and -3- must align.
- The protective film -1- must be bonded in free of any tension and kinks.



2.8 Removing and installing striker pin

Special tools and workshop equipment required

- ◆ Setting gauge - 3371-





- ◆ Torque wrench - V.A.G 1331-



Removing

- Unscrew bolts -2-. When doing this, remove striker pin -1-.

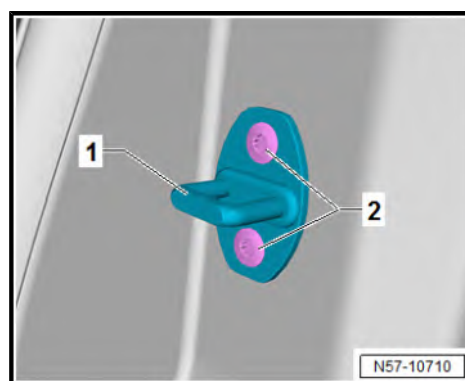
Installing

Install in reverse order of removal, observing the following:

- Use setting gauge - 3371- to check gaps and height difference of door ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .
- If necessary, adjust striker pin ⇒ [page 78](#) .

Specified torques

- ◆ ⇒ [“2.3 Assembly overview - door handle and door lock”, page 85](#)



2.9 Removing and installing door handle

⇒ [“2.9.1 Removing and installing door handle”, page 90](#)

⇒ [“2.9.2 Removing and installing door handle underlays”, page 90](#)

2.9.1 Removing and installing door handle

Removing

Driver side

- Remove lock cylinder ⇒ [page 95](#) .

Front passenger side

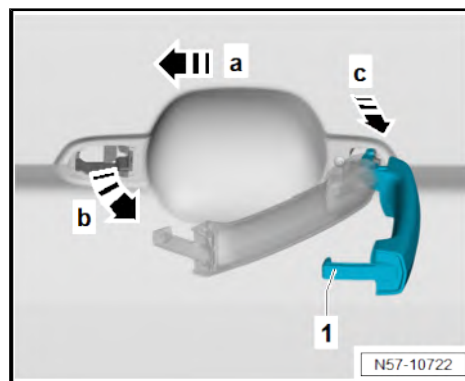
- Removing cap on door handle ⇒ [page 97](#) .

Continued for all vehicles

- Pull door handle -1- in -direction of arrow a- out of mounting in mounting bracket. When doing this, swing it in -direction of arrow b-.
- Remove door handle -1- in -direction of arrow c- from mounting bracket, keeping door handle at right angles to mounting bracket.

Installing

Install in reverse order.

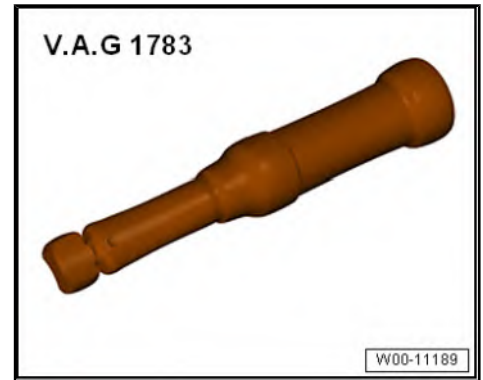


2.9.2 Removing and installing door handle underlays

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

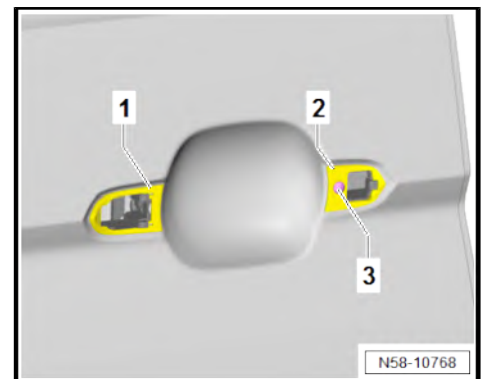
- Remove door handle ➔ [page 90](#) .
- Unclip rear underlay -1-.
- Unscrew bolt -3-
- Unclip front underlay -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“2.3 Assembly overview - door handle and door lock”, page 85](#)



2.10 Removing and installing door handle bracket

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

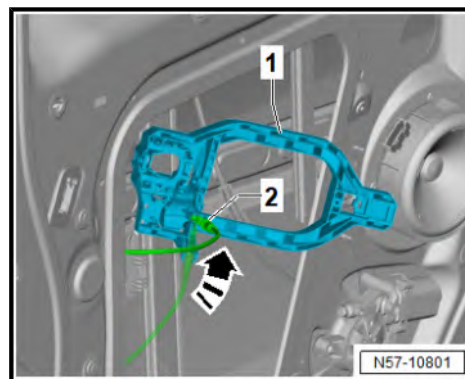
- Door window in upper position.
- Remove protective film ➔ [page 88](#) .
- Remove underlays for door handle ➔ [page 90](#) .
- If fitted, unclip Bowden cable from clip on outer door panel.



- Remove mounting bracket -1- from door.
- Pull Bowden cable -2- out of retainer. When doing this, swing it in -direction of arrow- by approx. 90°.
- Remove Bowden cable -2- from guide on mounting bracket -1-.

Installing

Install in reverse order.



2.11 Removing and installing door lock

⇒ [“2.11.1 Removing and installing door lock”, page 92](#)

⇒ [“2.11.2 Removing and installing Bowden cable for door lock/interior door handle”, page 93](#)

⇒ [“2.11.3 Removing and installing Bowden cable for door lock/mounting bracket”, page 93](#)

2.11.1 Removing and installing door lock

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-



Removing

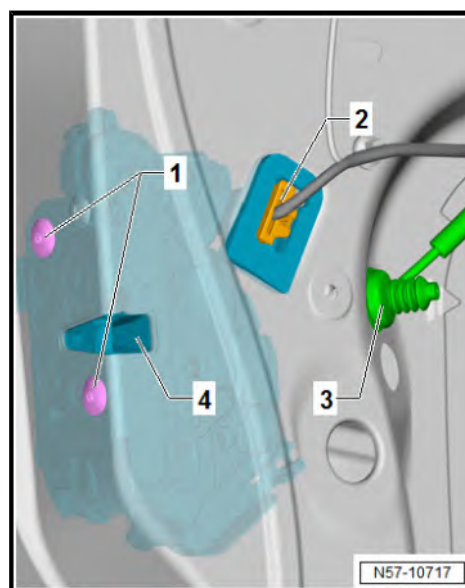
- Remove mounting bracket ⇒ [page 91](#) .
- Disconnect electrical connector -2-.
- Push grommet -3- together with Bowden cable into door.
- Unscrew bolts -1-. When doing this, remove door lock -4- together with Bowden cables.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [“2.3 Assembly overview - door handle and door lock”, page 85](#)





2.11.2 Removing and installing Bowden cable for door lock/interior door handle

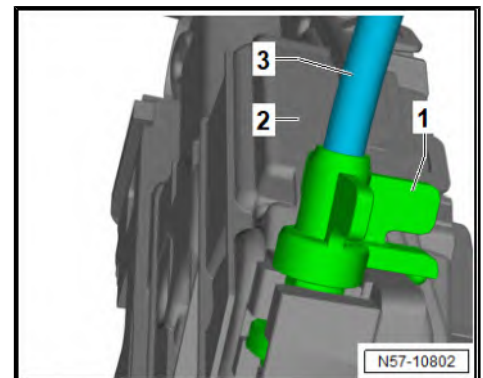
Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



Removing

- Remove door lock ⇒ [page 92](#) .
- Fit removal wedge - 3409- between Bowden cable retainer -1- and door lock -2-.
- Use removal wedge - 3409- to lift Bowden cable retainer -1-, and hold it in this position.
- Turn Bowden cable -3- by 90°. When doing this, pull Bowden cable -3- out of retainer on door lock -2-.
- Swing Bowden cable -3- until it can be guided out of eye in door lock -2-.



Installing

Install in reverse order.

2.11.3 Removing and installing Bowden cable for door lock/mounting bracket

Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



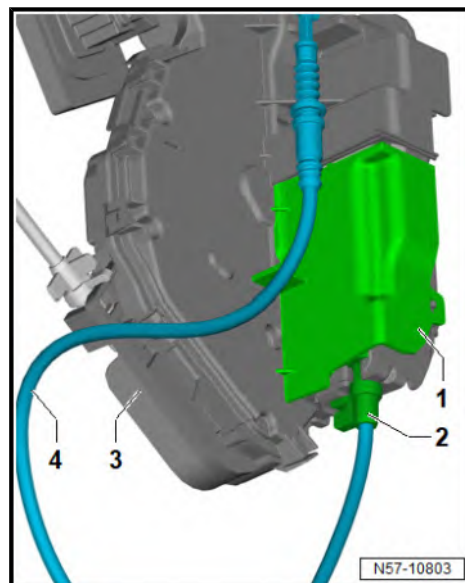


Removing

- Remove door lock ➔ [page 92](#) .
- Open flap -1-.
- Fit removal wedge - 3409- between Bowden cable retainer -2- and door lock -3-.
- Use removal wedge - 3409- to lift Bowden cable retainer -2-, and hold it in this position.
- Turn Bowden cable -4- clockwise by 90°. When doing this, pull Bowden cable -4- out of retainer on door lock -3-.
- Swing Bowden cable -4- until it can be guided out of eye in door lock -3-.

Installing

Install in reverse order.



2.12 Removing and installing window channel

➔ [“2.12.1 Removing and installing window channel”, page 94](#)

➔ [“2.12.2 Removing and installing guide rail”, page 94](#)

2.12.1 Removing and installing window channel

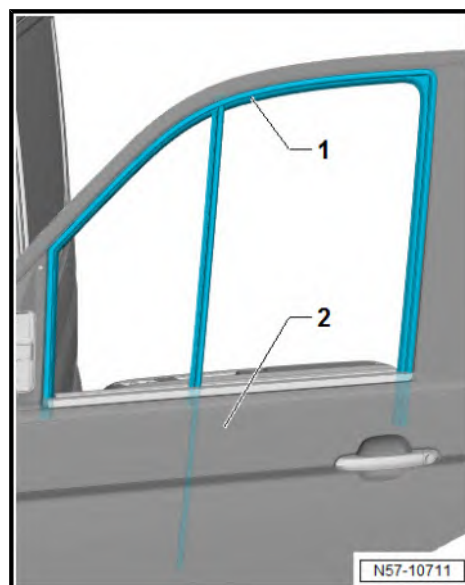
Removing

- Remove door window ➔ [page 241](#) .
- Remove fixed front door window ➔ [page 242](#) .
- Pull window channel -1- off door -2- all around.

Installing

Install in reverse order of removal, observing the following:

- Spray window guide -1- with soapy water.

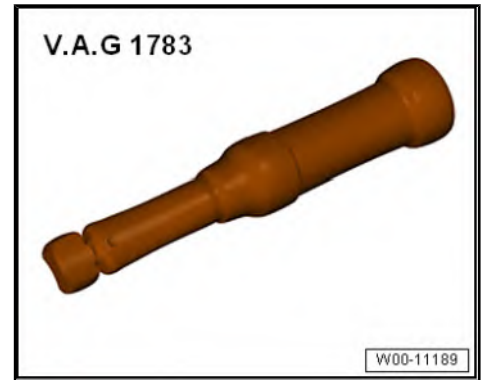


2.12.2 Removing and installing guide rail

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

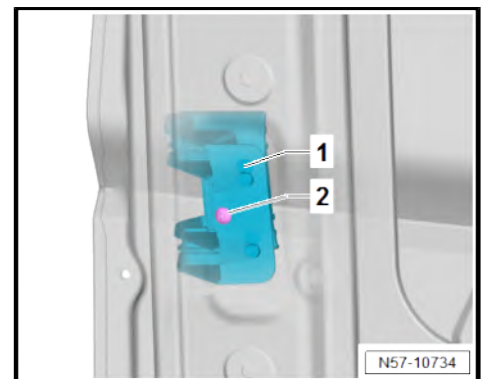
- Remove protective film ⇒ [page 88](#) .
- Unscrew bolt -2-
- Remove guide rail -1-

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.4 Assembly overview - window channels and window slot \(aperture\) seals”, page 86](#)



2.13 Removing and installing lock cylinder

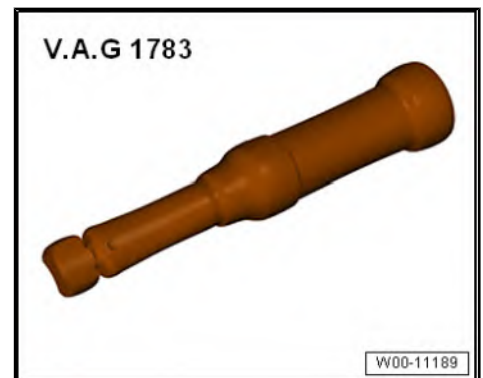
⇒ [“2.13.1 Removing and installing lock cylinder”, page 95](#)

⇒ [“2.13.2 Removing and installing lock cylinder cap”, page 96](#)

2.13.1 Removing and installing lock cylinder

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





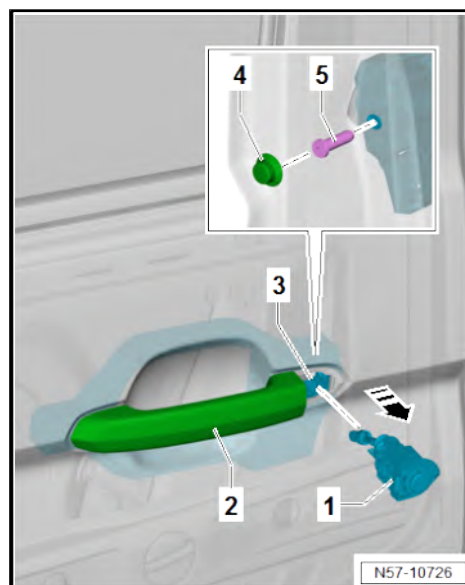
Removing

- Remove cap for lock cylinder ➔ [page 96](#) .
- Remove cap -4-.
- Pull door handle -2- in -direction of arrow-. When doing this, hold door handle -2- in this position.
- Unscrew bolt -5- by approximately 2 turns.
- Push bolt -5- in direction of travel. When doing this, remove lock cylinder -1-.

Installing

Install in reverse order of removal, observing the following:

- During installation, lock cylinder must be pressed against outside door panel.
- Tighten bolt -5- to specified torque.
- When tightening, push lock cylinder -1- towards outer door panel.



Specified torques

- ♦ ➔ [“2.3 Assembly overview - door handle and door lock”, page 85](#)

2.13.2 Removing and installing lock cylinder cap

Special tools and workshop equipment required

- ♦ Removal wedge - 3409-



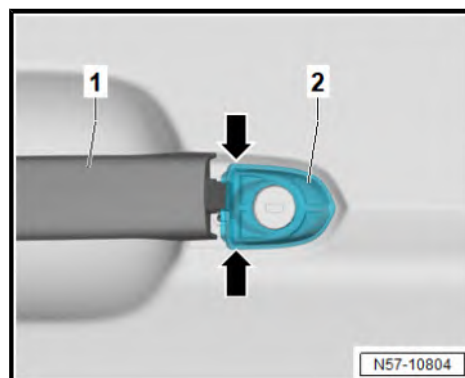
Removing

- Pull door handle -1-. When doing this, hold door handle -1- in this position.
- Use removal wedge - 3409- to lever off lock cylinder cap -arrows- in area marked with -2-.

Installing

Install in reverse order of removal, observing the following:

- Push on lock cylinder cap -2-. When doing this, engage cap in lock cylinder.

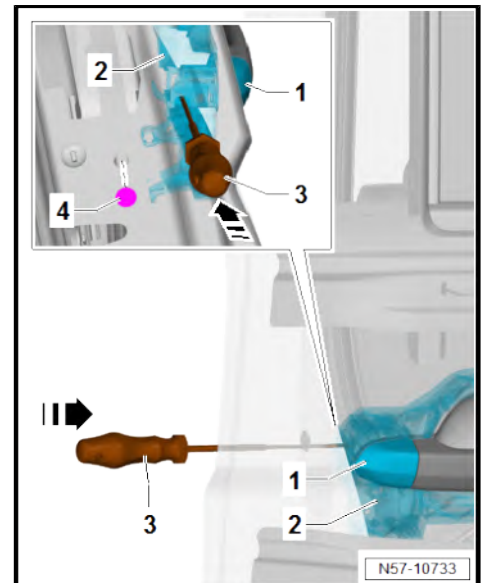




2.14 Removing and installing cap on door handle

Removing

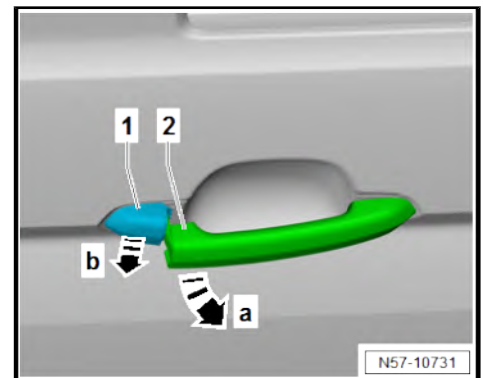
- If fitted, pull off film -4-.
- The following work steps must be performed by a second mechanic.
- Use a screwdriver -3- to release locking hook of mounting bracket -2- through exposed hole in -direction of arrow-.
- Have a second mechanic push cap -1- against outer door panel.



- Pull door handle -2- in direction of -arrow a-. When doing this, hold door handle -2- in this position.
- Remove cap -1- in -direction of arrow b-.

Installing

Install in reverse order.



2.15 Removing and installing inner window slot seal

Special tools and workshop equipment required

- ◆ Removal wedge - 3409-





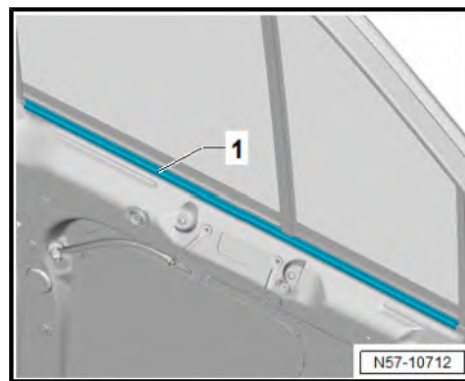
Removing

- Remove front door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .
- Use removal wedge - 3409- to lever inner window slot seal -1- off inner door panel.

Installing

Install in reverse order of removal, observing the following:

- Position inner window slot seal -1- on centre web. Push onto inner door panel when doing this.



2.16 Removing and installing outer window slot seal

Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



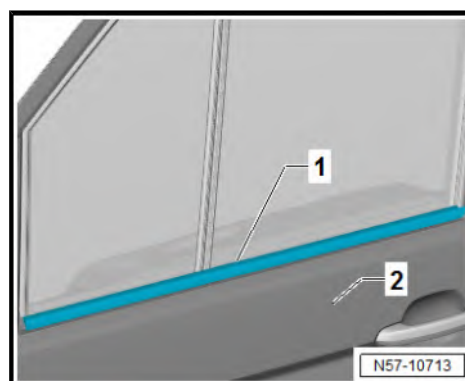
Removing

- Use removal wedge - 3409- to lever outer window slot seal -1- off outer door panel -2-.

Installing

Install in reverse order of removal, observing the following:

- Spray lubricant onto new outer window slot seal -1-.
- Position outer window slot seal -1- on centre web. Guide it under exterior mirror when doing this.
- Push outer window slot seal -1- onto outer door panel.



3 Central locking

⇒ [“3.1 Overview of fitting locations - central locking”, page 99](#)

⇒ [“3.2 Removing and installing driver door control unit J386 and front passenger door control unit J387”, page 100](#)

⇒ [“3.3 Removing and installing ignition key battery”, page 100](#)

⇒ [“3.4 Removing and installing cap on ignition key”, page 101](#)

⇒ [“3.5 Removing and installing bonnet contact switch F387”, page 103](#)

3.1 Overview of fitting locations - central locking

1 - Front passenger door control unit - J387-

- ❑ Removing and installing
⇒ [page 100](#)

2 - Window regulator switch in front passenger door - E107-

- ❑ Removing and installing
⇒ Electrical system;
Rep. gr. 96 ; Controls;
Removing and installing
window regulator switch
in front passenger door
-E107-

3 - Front passenger door lock unit - VX22-

- ❑ Removing and installing
⇒ [page 92](#)

4 - Rear right door lock unit - VX24-

- ❑ Removing and installing
⇒ [page 134](#)

5 - Rear wing door central locking motor - V96-

- ❑ Removing and installing
⇒ [page 134](#)

6 - Rear left door lock unit - VX23-

- ❑ Removing and installing
⇒ [page 134](#)

7 - Driver door lock unit - VX21-

- ❑ Removing and installing
⇒ [page 92](#)

8 - Front driver side window regulator button - E710-

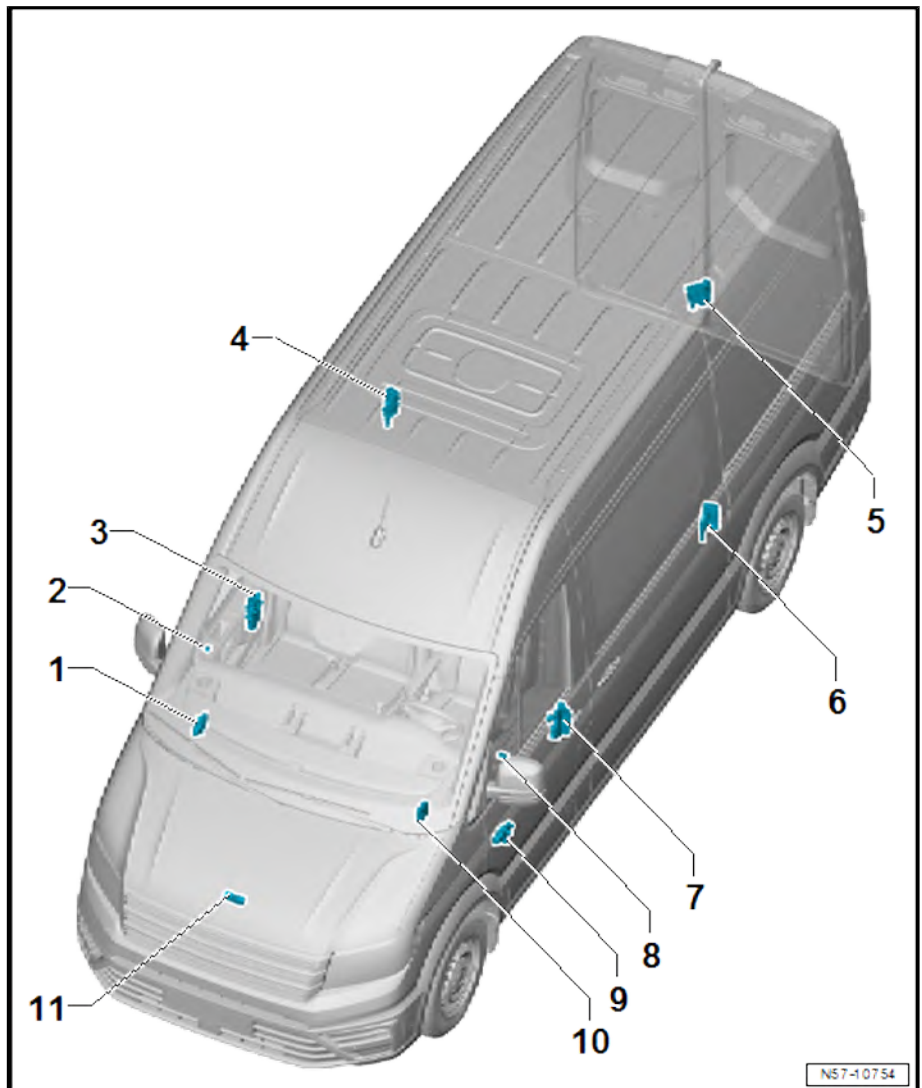
- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 96 ; Controls; Removing and installing central switch for window regulators in driver door -E189- .

9 - Driver door control unit - J386-

- ❑ Removing and installing ⇒ [page 100](#)

10 - Onboard supply control unit - J519-

- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 97 ; Control units; Removing and installing onboard supply control unit -J519-





11 - Bonnet contact switch - F387-

- Removing and installing ⇒ [page 103](#)

3.2 Removing and installing driver door control unit - J386- and front passenger door control unit - J387-

Removing

- Remove front door trim on relevant side ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .

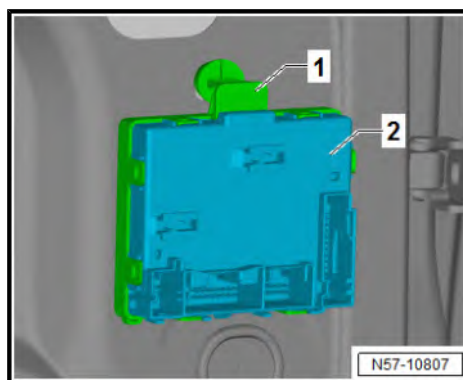
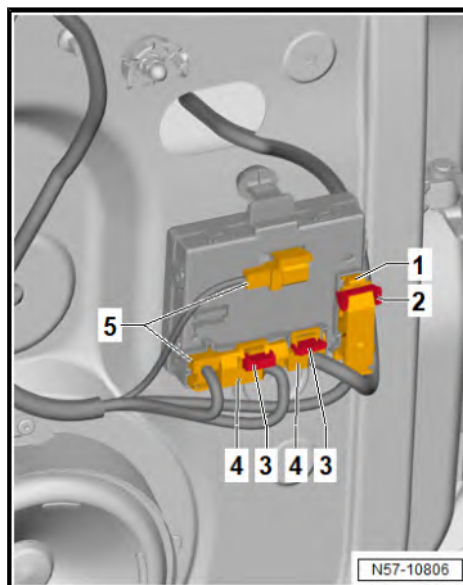


Note

- ◆ *A door control unit with the maximum number of electrical connectors is shown below.*
- ◆ *The number of electrical connectors may vary depending on the equipment level.*
- Disconnect electrical connector -1- by swinging retaining clip -2- upwards.
- Pull out connector locks -3-.
- Disconnect electrical connectors -4- and -5-.
- Push bracket -1- downwards. When doing this, release door control unit -2- at top from door.
- Detach door control unit -2- from door.

Installing

Install in reverse order.



3.3 Removing and installing ignition key battery

⇒ [“3.3.1 Removing and installing ignition key battery”, page 100](#)

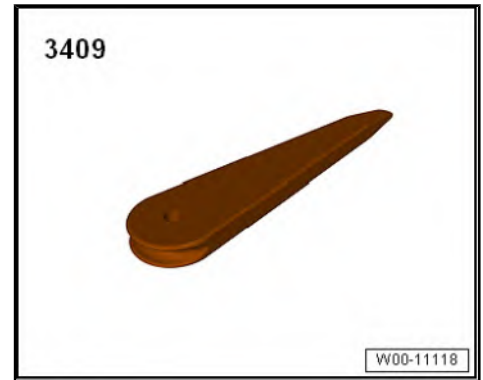
⇒ [“3.3.2 Synchronizing ignition key”, page 101](#)

3.3.1 Removing and installing ignition key battery

Special tools and workshop equipment required



◆ Removal wedge - 3409-



Removing

- Fold out key bit.
- Use removal wedge - 3409- to lever off battery compartment cover -1-.
- Fit a small screwdriver -2-, as shown in illustration.
- Lever battery -3- out of battery compartment.

Installing

Install in reverse order of removal, observing the following:

Procedure when renewing battery

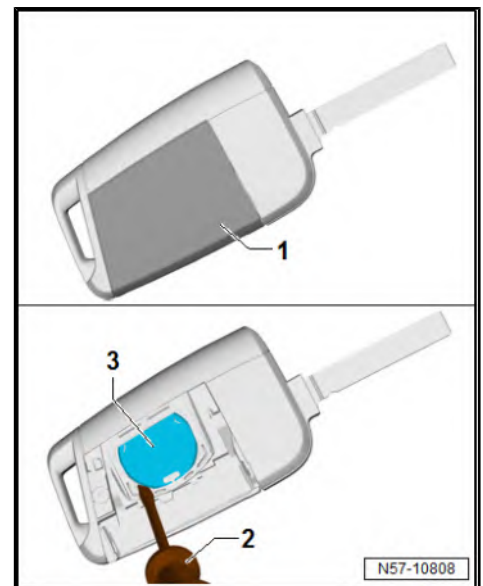
- Press any button on ignition key.

Continued for all vehicles

- Insert battery with visible positive terminal at an angle of approx. 45° into battery compartment. When doing this, push battery into battery compartment.
- Synchronise ignition key ➤ [page 101](#) .

Installing

Install in reverse order.



3.3.2 Synchronizing ignition key

Procedure

- Hold ignition key to lock cylinder. When doing this, press button for opening.
- Unlock vehicle via lock cylinder within a minute.

3.4 Removing and installing cap on ignition key

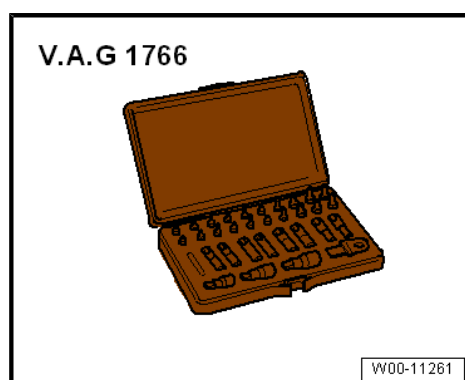
Special tools and workshop equipment required



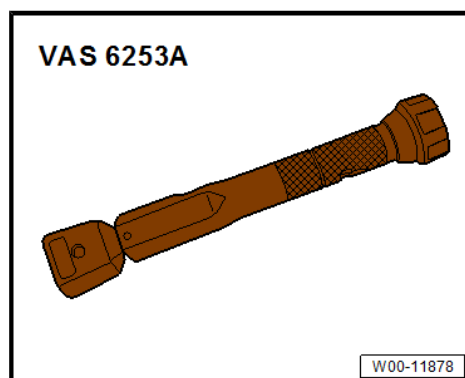
◆ Removal wedge - 3409-



◆ TORX bit -T7- from TORX bit set - V.A.G 1766-



◆ Torque wrench - VAS 6253A-



Removing

- Fold out key bit.
- Use removal wedge - 3409- to lever off battery compartment cover -1-.
- Use TORX bit -T7- from TORX bit set - V.A.G 1766- to unscrew bolt -2-.
- Push cap -3- in -direction of arrow- off ignition key.

Installing

Install in reverse order of removal, observing the following:

Component	Specified torque
Bolt -2-	0.4 Nm or 40 Ncm





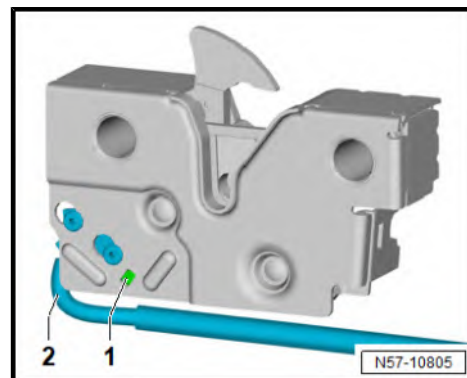
3.5 Removing and installing bonnet contact switch - F387-

Removing

- Remove bonnet lock ⇒ [page 57](#) .
- Release locking lug -1- using screwdriver. When doing this, push bonnet contact switch - F387- -2- out of elongated holes.

Installing

Install in reverse order.



58 – Rear doors, door components

1 Door

- ⇒ [“1.1 Assembly overview - door”, page 104](#)
- ⇒ [“1.2 Assembly overview - door seals”, page 105](#)
- ⇒ [“1.3 Removing and installing door”, page 105](#)
- ⇒ [“1.4 Adjusting door”, page 106](#)
- ⇒ [“1.5 Adjusting striker pin”, page 108](#)
- ⇒ [“1.6 Removing and installing door hinges”, page 110](#)
- ⇒ [“1.7 Removing and installing door arrester”, page 111](#)
- ⇒ [“1.8 Removing and installing inner door seal”, page 113](#)

1.1 Assembly overview - door

1 - Cab door

- ☐ Removing and installing
⇒ [page 105](#)
- ☐ Adjusting ⇒ [page 106](#)

2 - Bolts

- ☐ Renew after removal
- ☐ Qty. 4
- ☐ 20 Nm +90°

3 - Foam element

4 - Door hinges

- ☐ Upper and lower
- ☐ Removing and installing
⇒ [page 110](#)

5 - Stop

6 - Bolts

- ☐ Renew after removal
- ☐ Qty. 4
- ☐ 20 Nm +90°

7 - Door arrester

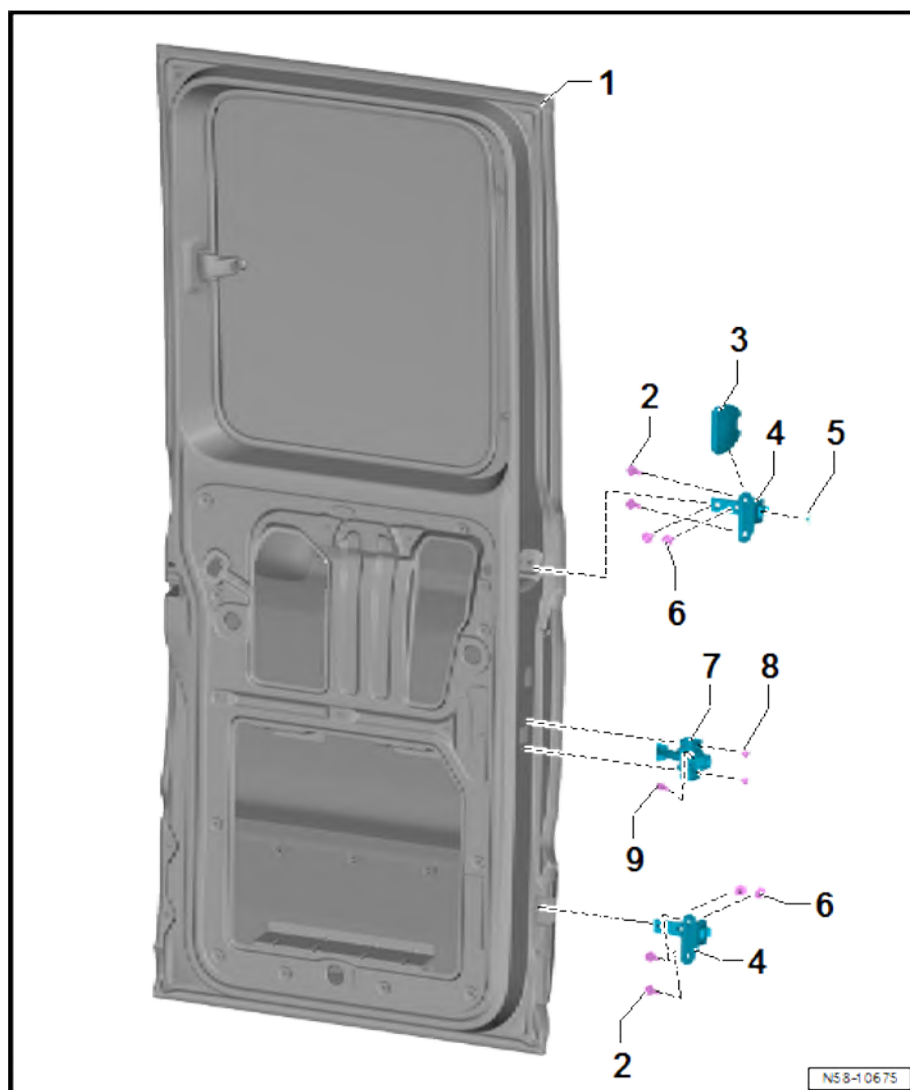
- ☐ Removing and installing
⇒ [page 111](#)

8 - Bolts

- ☐ Qty. 2
- ☐ 9 Nm

9 - Bolt

- ☐ 30 Nm

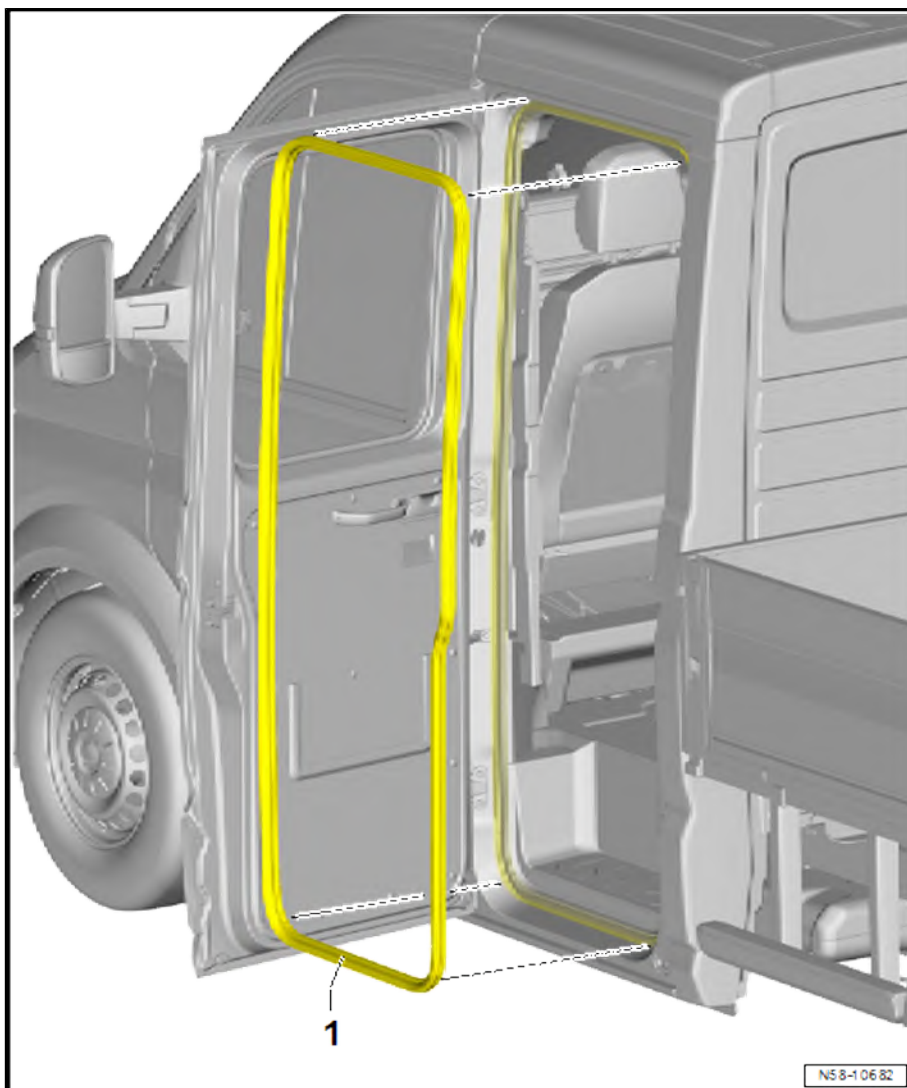




1.2 Assembly overview - door seals

1 - Door seal

- ☐ Renew after removal
- ☐ Removing and installing
⇒ [page 113](#)



1.3 Removing and installing door

Special tools and workshop equipment required

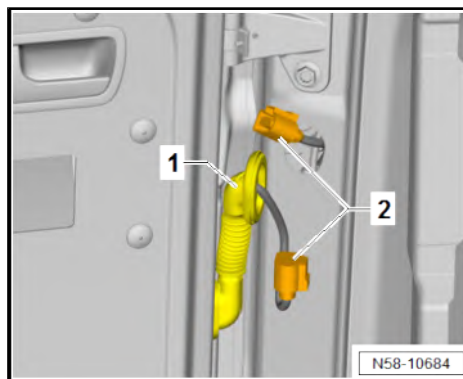
- ◆ Torque wrench - V.A.G 1331-



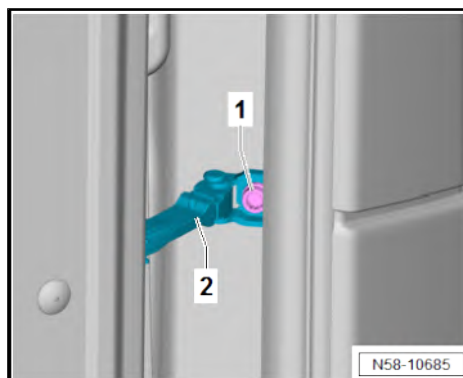


Removing

- Pull bellows -1- off B-pillar.
- Disconnect electrical connectors -2-.



- Unscrew bolt -1-.
- Insert door arrester -2- into door.



- Loosen bolts -2-.
- The following work step must be performed by a second mechanic.
- Have a second mechanic secure door -1-.
- Unscrew bolts -2-.
- Remove door -1- together with second mechanic.

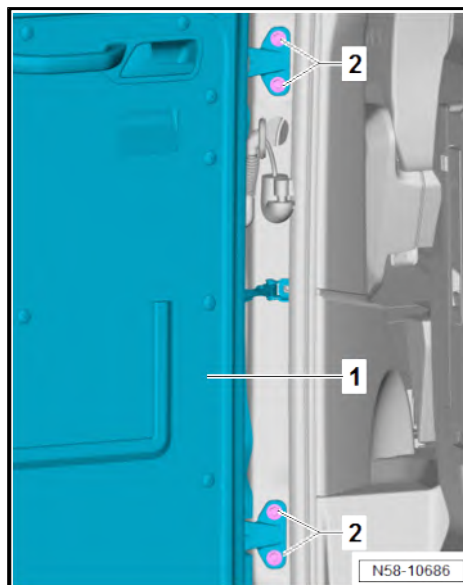
Installing

Install in reverse order of removal, observing the following:

- Use setting gauge - 3371- to check gaps and height difference of door ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .
- If necessary, adjust door ⇒ [page 106](#) .

Specified torques

- ♦ ⇒ [“1.1 Assembly overview - door”, page 104](#)



1.4 Adjusting door

Special tools and workshop equipment required



◆ Setting gauge - 3371-



◆ Torque wrench - V.A.G 1331-



Procedure

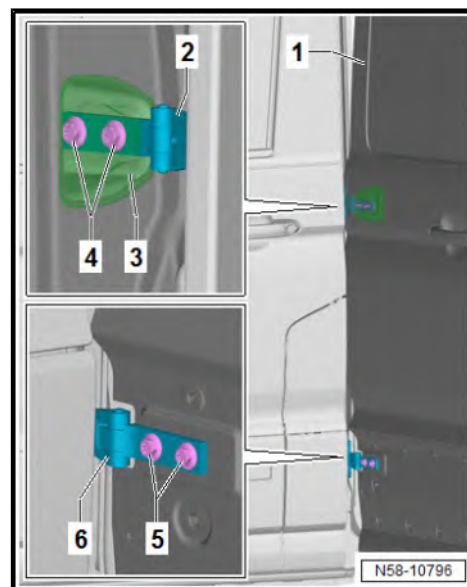
- Vehicle must be standing on all wheels.

Adjusting top door hinge

- Remove rear cab door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Pull off foam section -3-.
- Loosen bolts -4- on hinge -2-.
- Use setting gauge - 3371- to check and adjust panel gaps of door -1- ⇒ Body Repairs; Rep. gr. 00 ; Technical data; Panel gaps/shutlines; Front body .

Adjusting bottom door hinge

- Remove rear door covers ⇒ [page 253](#) .
- Loosen bolts -5- on hinge -6-.
- Use setting gauge - 3371- to check and adjust panel gaps of door -1- ⇒ Body Repairs; Rep. gr. 00 ; Technical data; Panel gaps/shutlines; Front body .
- Following assembly or adjustment work, perform corrosion protection measures on hinges and bolts.



Specified torques

- ◆ ⇒ [“1.1 Assembly overview - door”, page 104](#)



1.5 Adjusting striker pin

⇒ [“1.5.1 Adjusting striker pin, double cab”, page 108](#)

⇒ [“1.5.2 Adjusting striker pin, sliding door”, page 108](#)

⇒ [“1.5.3 Adjusting striker pin, wing door”, page 109](#)

1.5.1 Adjusting striker pin, double cab

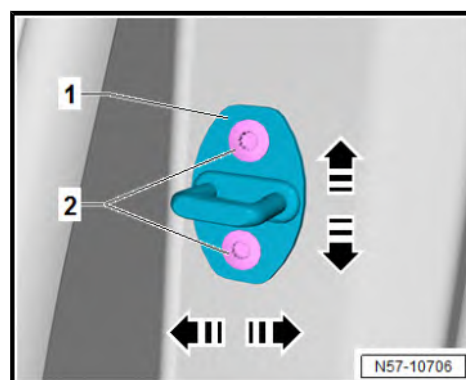
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Procedure

- Vehicle must be standing on all wheels.
- Loosen bolts -2-.
- Adjust striker pin -1- by moving it in the respective -direction of arrow-. Hold it in the correct position, and tighten it to specified torque ⇒ [page 85](#) .
- Use setting gauge - 3371- to check gaps and height difference of door ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front . Repeat adjustment procedure as necessary.



Gaps and height difference of door outside the tolerance range

- Adjust door ⇒ [page 77](#) .

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)

1.5.2 Adjusting striker pin, sliding door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



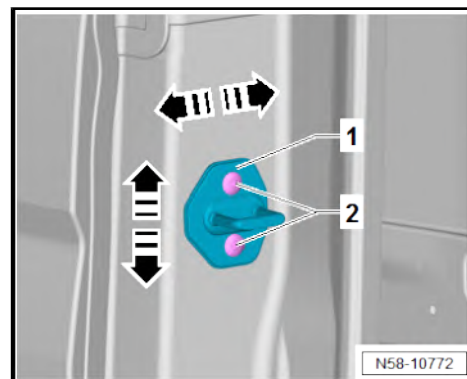


Procedure

- Loosen bolts -2-.
- Adjust striker pin -1- by moving it in the respective -direction of arrow-. Hold it in this position.
- Tighten bolts -2- to specified torque.

Specified torques

Component	Specified torque
Bolts -2-	25 Nm



1.5.3 Adjusting striker pin, wing door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-

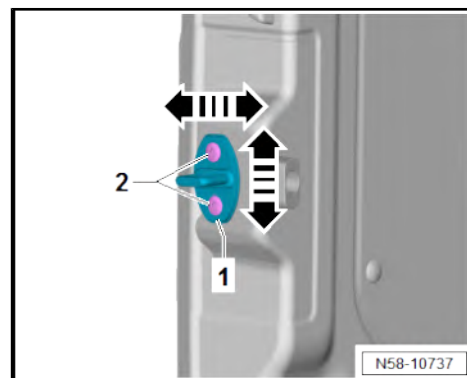


Procedure

- Vehicle must be standing on all wheels.

Adjusting striker pin, wing door

- Loosen bolts -2-.
- Adjust striker pin -1- by moving it in the respective -direction of arrow-.
- Adjust striker pin -1- in such a way that the recess of the door lock with rotary latch is centred relative to the striker pin -1- when it engages.
- Close wing door.
- Check the adjustment.

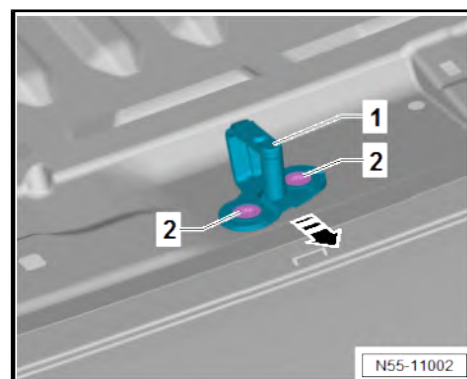


Adjusting striker pin, lock carrier

- Loosen bolts -2-.
- Push striker pin -1- to stop in -direction of arrow-. When doing this, tighten bolts -2- to specified torque.
- Close wing door.
- Check the adjustment.

Specified torques

- ◆ ➤ [“2.1 Assembly overview - door handle and door lock”, page 114](#)





1.6 Removing and installing door hinges

⇒ [“1.6.1 Removing and installing door hinges, top”, page 110](#)

⇒ [“1.6.2 Removing and installing door hinges, bottom”, page 110](#)

1.6.1 Removing and installing door hinges, top

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-

Removing

- Remove rear cab door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Remove door ⇒ [page 105](#) .
- Pull off foam section -2-.
- Unscrew bolts -3-. When doing this, remove upper door hinge -1-.

Installing

Install in reverse order of removal, observing the following:

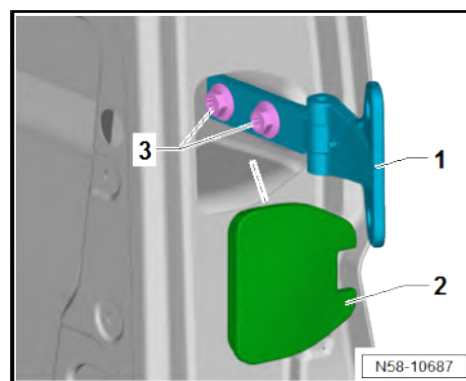
Specified torques

- ◆ ⇒ [“1.1 Assembly overview - door”, page 104](#)

1.6.2 Removing and installing door hinges, bottom

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





Removing

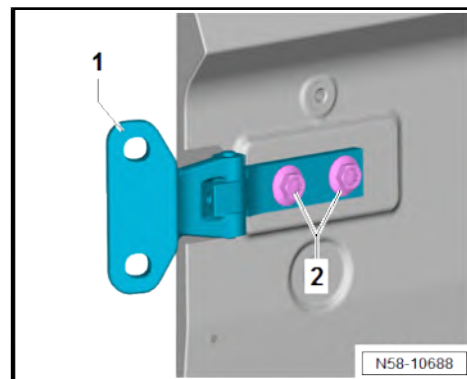
- Remove rear door covers ➔ [page 253](#) .
- Remove door ➔ [page 105](#) .
- Unscrew bolts -2-. When doing this, remove lower door hinge -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“1.1 Assembly overview - door”, page 104](#)



1.7 Removing and installing door arrester

➔ [“1.7.1 Removing and installing door arrester, double cab”, page 111](#)

➔ [“1.7.2 Removing and installing door arrester, sliding door”, page 112](#)

➔ [“1.7.3 Removing and installing door arrester, wing door”, page 112](#)

➔ [“1.7.4 Removing and installing detent mechanism”, page 113](#)

1.7.1 Removing and installing door arrester, double cab

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

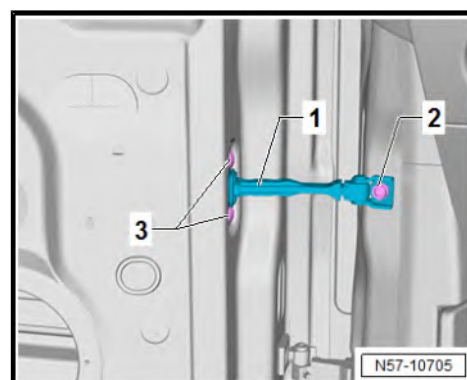
- Remove rear cab door trim ➔ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Unscrew bolt -2-
- Unscrew bolts -3-. When doing this, remove door arrester -1- through door opener.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“1.1 Assembly overview - door”, page 104](#)





1.7.2 Removing and installing door arrester, sliding door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

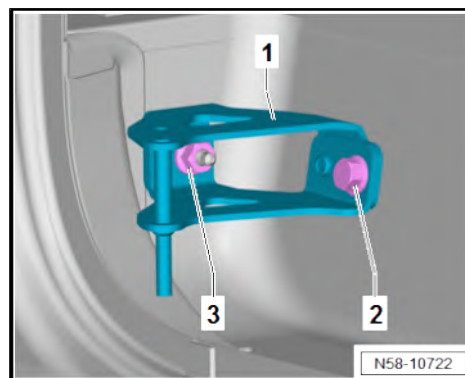
- Remove sliding door step ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing step .
- Unscrew bolt -2-.
- Unscrew nut -3-. When doing this, remove door arrester -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“3.1.2 Assembly overview - sliding door, attachments”, page 147](#)



1.7.3 Removing and installing door arrester, wing door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

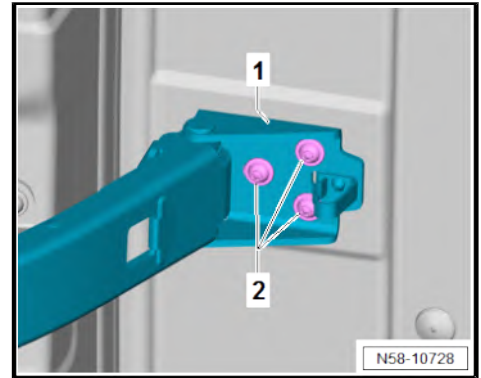
- Unscrew bolts -2-. When doing this, remove door arrester -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“4.1 Assembly overview - wing door”, page 162](#)



1.7.4 Removing and installing detent mechanism

Removing

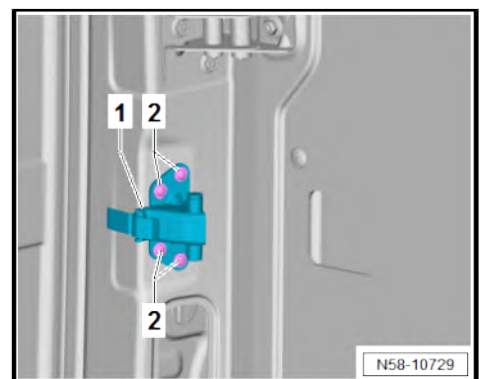
- Unscrew bolts -2-. When doing this, remove detent mechanism -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“4.1 Assembly overview - wing door”, page 162](#)



1.8 Removing and installing inner door seal

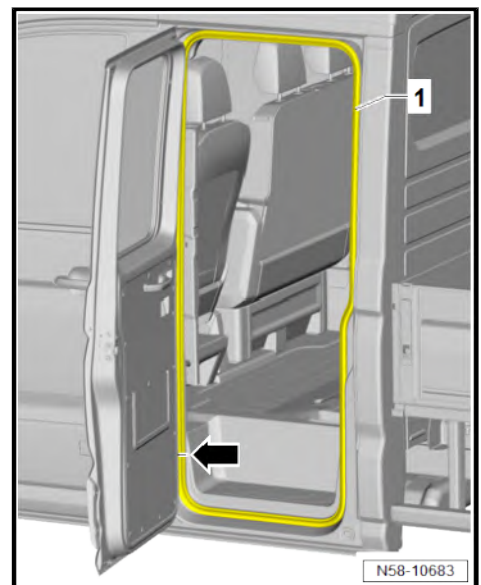
Removing

- Remove step on relevant side ⇒ General body repairs, interior; Rep. gr. 70 ; Interior trims; Removing and installing step .
- Pull inner door seal -1- off body flange.

Installing

Install in reverse order of removal, observing the following:

- Fit new inner door seal -1- with vulcanisation point -arrow- at height of lower door hinge.
- Vulcanisation point -arrow- must not be kinked.
- Install inner door seal -1- all round.





2 Door components

⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)

⇒ [“2.2 Removing and installing door components”, page 120](#)

⇒ [“2.3 Removing and installing striker pin”, page 128](#)

⇒ [“2.4 Removing and installing door handle”, page 130](#)

⇒ [“2.5 Removing and installing door handle bracket”, page 131](#)

⇒ [“2.6 Removing and installing door lock”, page 134](#)

⇒ [“2.7 Removing and installing cap on door handle”, page 141](#)

⇒ [“2.8 Removing and installing lock cylinder”, page 143](#)

2.1 Assembly overview - door handle and door lock

⇒ [“2.1.1 Assembly overview - door handle and door lock, double cab”, page 114](#)

⇒ [“2.1.2 Assembly overview - door handle and door lock, sliding door”, page 116](#)

⇒ [“2.1.3 Assembly overview - door handle and door lock, left”, page 117](#)

⇒ [“2.1.4 Assembly overview - door handle and door lock, right”, page 119](#)

2.1.1 Assembly overview - door handle and door lock, double cab



1 - Cover

- ☐ Removing and installing
⇒ [page 143](#)

2 - Door handle

- ☐ Removing and installing
⇒ [page 130](#)

3 - Bolt

- ☐ 2.5 Nm

4 - Carrier

- ☐ Front
- ☐ Removing and installing
⇒ [page 130](#)

5 - Mounting bracket

- ☐ Removing and installing
⇒ [page 131](#)

6 - Bowden cable

- ☐ From door lock to mounting bracket.

7 - Bowden cable

- ☐ From inner door handle to door lock

8 - Door lock

- ☐ Removing and installing
⇒ [page 134](#)

9 - Striker

- ☐ Removing and installing
⇒ [page 128](#)
- ☐ Adjusting ⇒ [page 108](#)

10 - Bolts

- ☐ Qty. 2
- ☐ 25 Nm

11 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

12 - Bolt

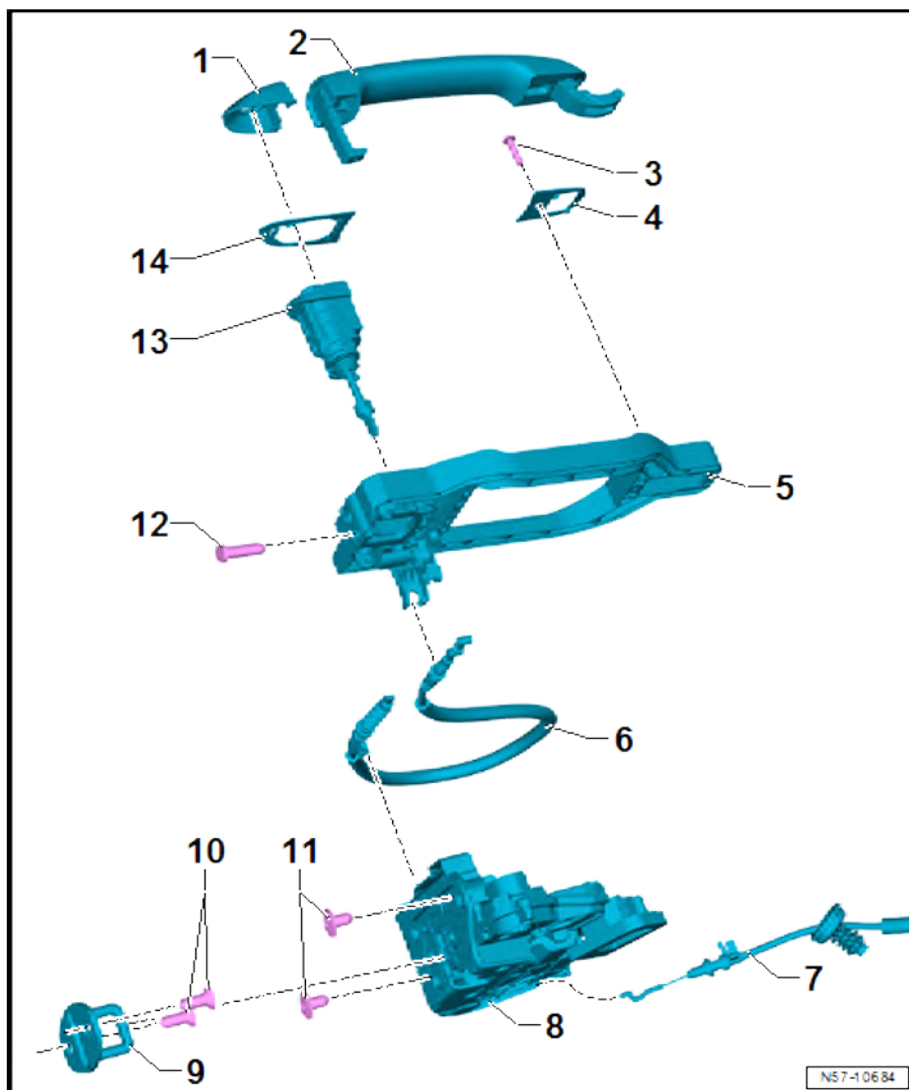
- ☐ 2.5 Nm

13 - Lock cylinder

- ☐ Removing and installing ⇒ [page 143](#)
- ☐ Removing and installing cap on door handle ⇒ [page 141](#)

14 - Carrier

- ☐ Rear
- ☐ Removing and installing ⇒ [page 130](#)



Volkswagen Technical Site: <https://vwts.ru>

2.1.2 Assembly overview - door handle and door lock, sliding door

1 - Interior door handle

- ☐ Removing and installing
⇒ [page 120](#)

2 - Bolts

- ☐ Qty. 3
- ☐ 8 Nm

3 - Stop

- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 134](#)

4 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

5 - Bowden cable

- ☐ From stop to lower door lock

6 - Door lock

- ☐ Removing and installing
⇒ [page 134](#)

7 - Bolts

- ☐ Renew after removal
- ☐ Qty. 3
- ☐ 8 Nm +90°

8 - Bowden cable

- ☐ From door lock to Bowden cable coupling

9 - Bowden cable

- ☐ From Bowden cable coupling to bottom roller guide

10 - Bowden cable coupling

- ☐ Removing and installing ⇒ [page 120](#)

11 - Bolt

- ☐ 3 Nm

12 - Bowden cable

- ☐ From Bowden cable coupling to interior door handle

13 - Bowden cable

- ☐ From mounting bracket to Bowden cable coupling

14 - Mounting bracket

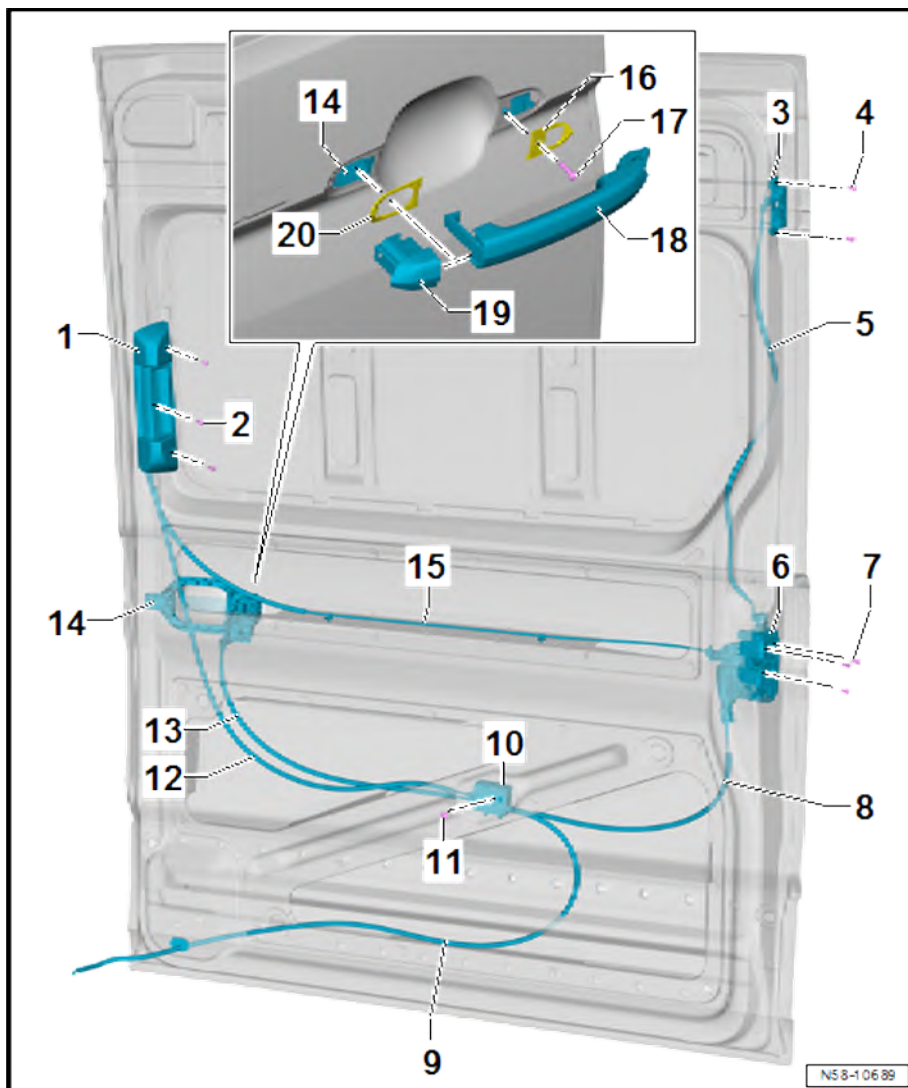
- ☐ Removing and installing ⇒ [page 131](#)

15 - Bowden cable

- ☐ From interior door handle to bottom door lock

16 - Carrier

- ☐ Right
- ☐ Removing and installing ⇒ [page 130](#)





17 - Bolt

- ☐ 2.5 Nm

18 - Door handle

- ☐ Removing and installing ⇒ [page 130](#)

19 - Cover

- ☐ Removing and installing ⇒ [page 141](#)

20 - Left base

- ☐ Removing and installing ⇒ [page 130](#)

2.1.3 Assembly overview - door handle and door lock, left

1 - Door lock

- ☐ Top
- ☐ Removing and installing
⇒ [page 134](#)

2 - End plate

- ☐ Top
- ☐ Removing and installing
⇒ [page 169](#)

3 - Bolts

- ☐ Qty. 2
- ☐ 30 Nm

4 - Guide pin

- ☐ Removing and installing
⇒ [page 169](#)
- ☐ 20 Nm

5 - Guide wedge

- ☐ Removing and installing
⇒ [page 169](#)

6 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

7 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

8 - Bowden cable

- ☐ From handle to upper door lock

9 - Operating mechanism

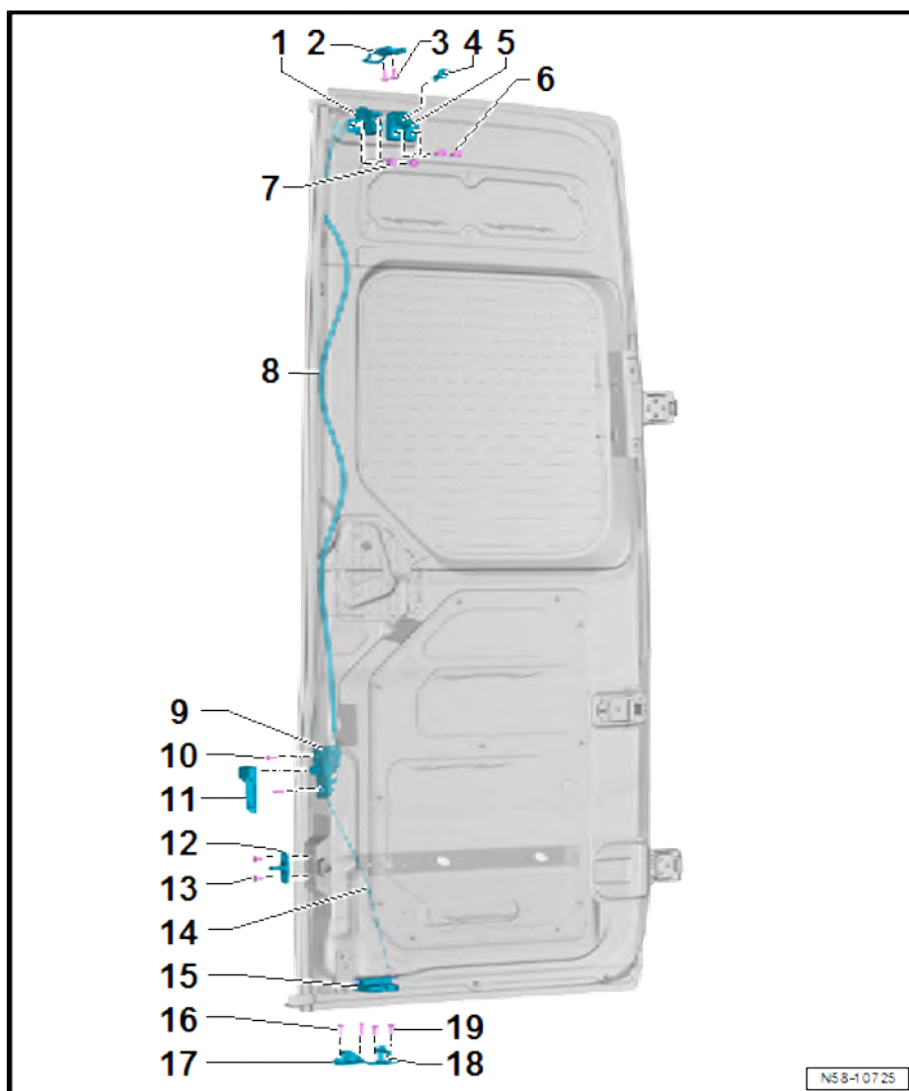
- ☐ Removing and installing
⇒ [page 134](#)

10 - Bolts

- ☐ Qty. 2
- ☐ 2 Nm

11 - Lever

- ☐ Removing and installing ⇒ [page 134](#)



N58-10725



12 - Striker

- ☐ Removing and installing ⇒ [page 128](#)
- ☐ Adjusting ⇒ [page 108](#)

13 - Bolts

- ☐ Qty. 2
- ☐ 25 Nm

14 - Bowden cable

- ☐ From handle to lower door lock

15 - Door lock

- ☐ Bottom
- ☐ Removing and installing ⇒ [page 134](#)

16 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

17 - Guide

- ☐ Removing and installing ⇒ [page 169](#)

18 - Striker

- ☐ Bottom
- ☐ Removing and installing ⇒ [page 128](#)
- ☐ Adjusting ⇒ [page 108](#)

19 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm



2.1.4 Assembly overview - door handle and door lock, right

1 - Guide wedge

- ☐ Removing and installing
⇒ [page 169](#)

2 - Guide pin

- ☐ Removing and installing
⇒ [page 169](#)
- ☐ 20 Nm

3 - End plate

- ☐ Top
- ☐ Removing and installing
⇒ [page 169](#)

4 - Bolts

- ☐ Qty. 2
- ☐ 30 Nm

5 - Door lock

- ☐ Top
- ☐ Removing and installing
⇒ [page 134](#)

6 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

7 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

8 - Bowden cable

- ☐ From top door lock to
bottom door lock

9 - Interior door handle

- ☐ Removing and installing
⇒ [page 134](#)

10 - Mounting bracket

- ☐ Removing and installing ⇒ [page 131](#)

11 - Bolt

- ☐ Qty. 4
- ☐ 1.5 Nm

12 - Bowden cable

- ☐ From interior door handle to door lock

13 - Bowden cable

- ☐ From mounting bracket to lower door lock

14 - Bolts

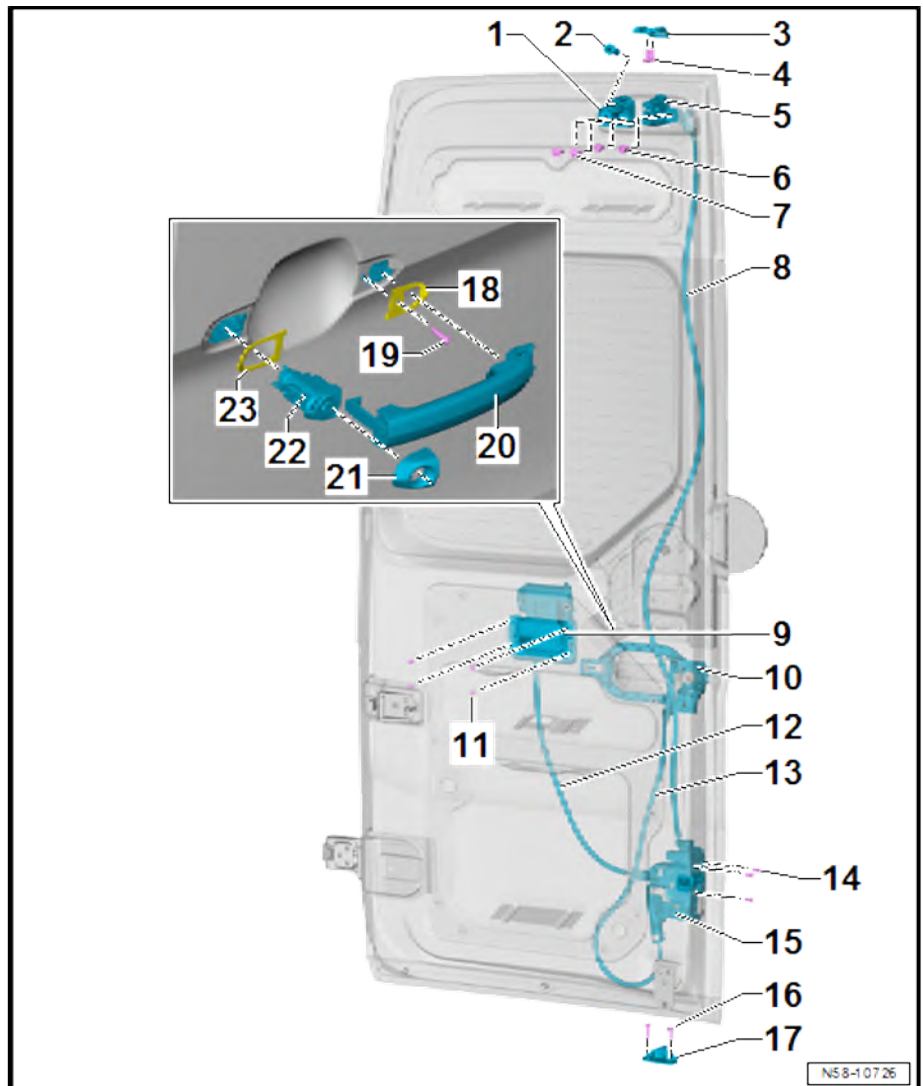
- ☐ Qty. 2
- ☐ 20 Nm

15 - Door lock

- ☐ Removing and installing ⇒ [page 134](#)

16 - Bolts

- ☐ Qty. 2



N5 8-1 07 26



- ☐ 8 Nm

17 - Guide

- ☐ Removing and installing ⇒ [page 169](#)

18 - Carrier

- ☐ Right
- ☐ Removing and installing ⇒ [page 130](#)

19 - Bolt

- ☐ 2.5 Nm

20 - Door handle

- ☐ Removing and installing ⇒ [page 130](#)

21 - Cover

- ☐ On door handle
- ☐ Removing and installing ⇒ [page 143](#)

22 - Lock cylinder

- ☐ Removing and installing ⇒ [page 143](#)

23 - Carrier

- ☐ Left
- ☐ Removing and installing ⇒ [page 130](#)

2.2 Removing and installing door components

⇒ [“2.2.1 Removing and installing roller guide, top”, page 120](#)

⇒ [“2.2.2 Removing and installing roller guide, bottom”, page 121](#)

⇒ [“2.2.3 Removing and installing roller guide, middle”, page 122](#)

⇒ [“2.2.4 Removing and installing contact strip in right sliding door F352”, page 123](#)

⇒ [“2.2.5 Removing and installing Bowden cable coupling”, page 124](#)

⇒ [“2.2.6 Removing and installing locking pin”, page 125](#)

⇒ [“2.2.7 Removing and installing stop”, page 125](#)

⇒ [“2.2.8 Removing and installing interior door handle”, page 126](#)

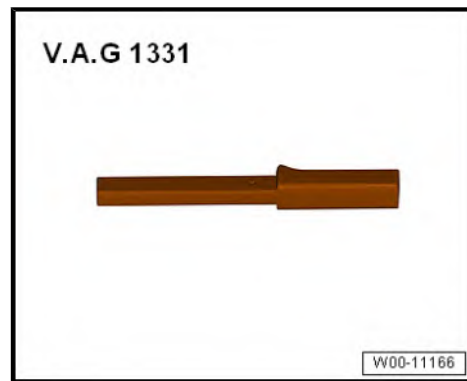
⇒ [“2.2.9 Removing and installing catch plate”, page 127](#)

2.2.1 Removing and installing roller guide, top

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331-



Removing

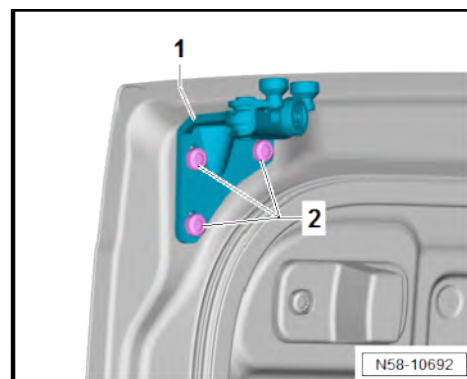
- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure sliding door.
- Unscrew bolts -2-. When doing this, push upper roller guide -1- out of guide rail.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“3.1 Assembly overview – sliding door”, page 145](#)



2.2.2 Removing and installing roller guide, bottom

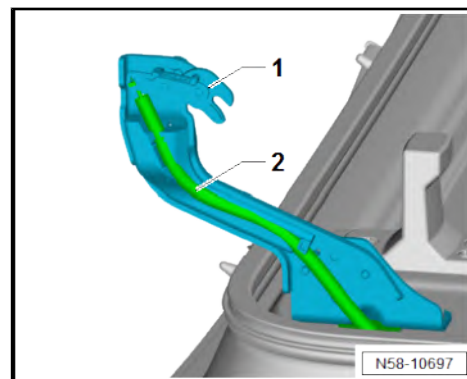
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

- Detach Bowden cable -2- from lower roller guide -1-.
- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure sliding door.





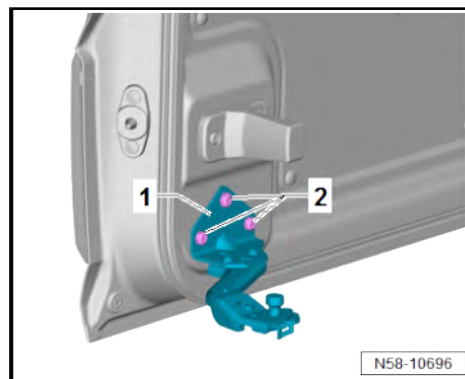
- Unscrew bolts -2-. When doing this, push lower roller guide -1- out of guide rail.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ ["3.1 Assembly overview – sliding door", page 145](#)



2.2.3 Removing and installing roller guide, middle

Special tools and workshop equipment required

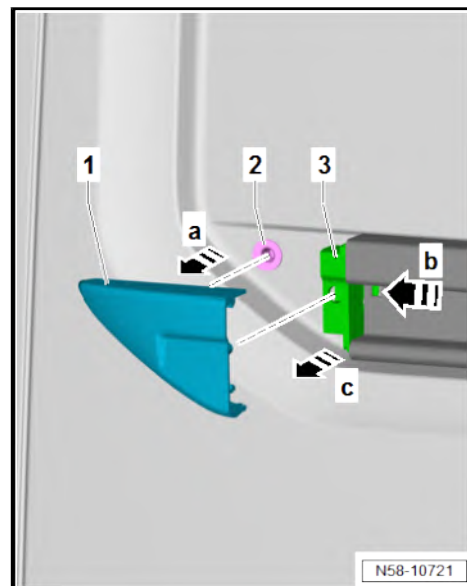
- ♦ Removal wedge - 3409-
- ♦ Torque wrench - V.A.G 1331-





Removing

- Use removal wedge - 3409- to unclip cap -1- in -direction of arrow a- from clip -2-.
- Use removal wedge - 3409- to release locking lug in -direction of arrow b-. When doing this, pull out mounting -3- in -direction of arrow c-.
- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure sliding door.



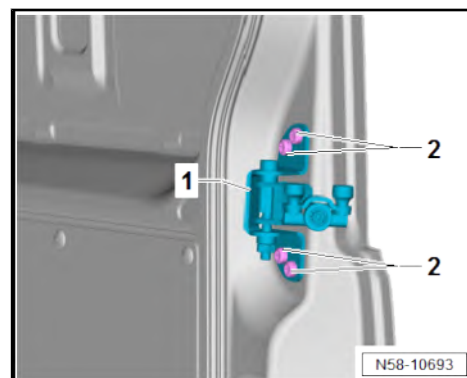
- Unscrew bolts -2-. When doing this, push roller guide -1- out of guide rail.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [“3.1 Assembly overview – sliding door”, page 145](#)



2.2.4 Removing and installing contact strip in right sliding door - F352-

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-





Removing

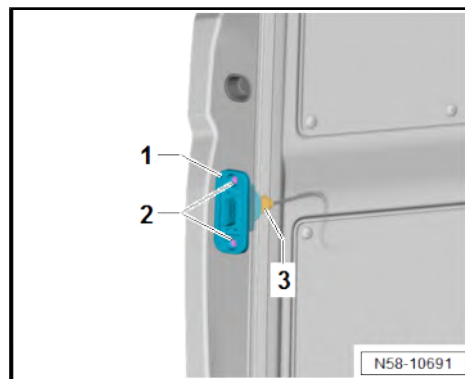
- Unscrew bolts -2-.
- Remove contact carrier -1- from sliding door until electrical connector -3- can be disconnected.
- Disconnect electrical connector -3-. When doing this, remove contact carrier -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

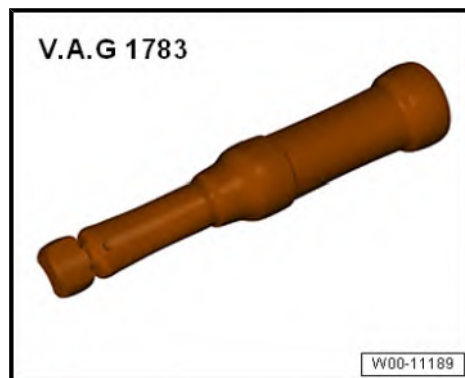
- ♦ ⇒ [“3.1 Assembly overview – sliding door”, page 145](#)



2.2.5 Removing and installing Bowden cable coupling

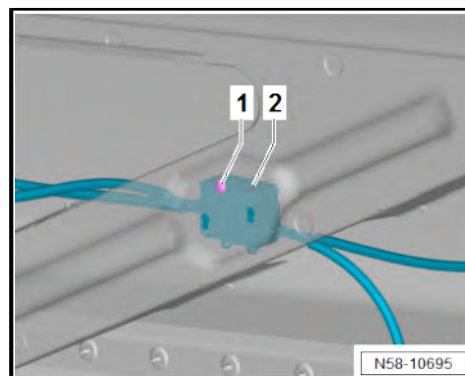
Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-

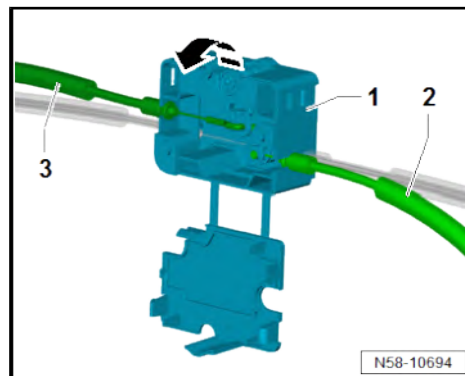


Removing

- Remove sliding door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Door trims, rear; Removing and installing rear door trim .
- Unscrew bolt -1-
- Guide Bowden cable coupling -2- out of sliding door.



- Open front cap on Bowden cable coupling -1- in -direction of arrow-.
- Unhook Bowden cables -2- and -3- from Bowden cable coupling -1-.





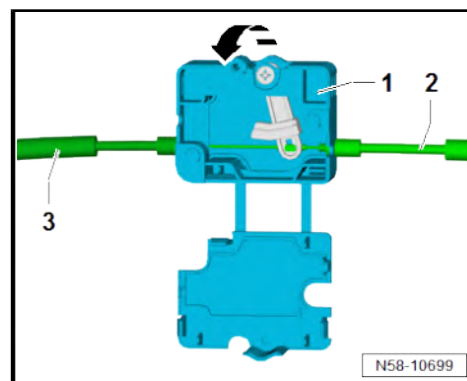
- Open rear cap on Bowden cable coupling -1- in -direction of arrow-.
- Unhook Bowden cables -2- and -3- from Bowden cable coupling -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)



2.2.6 Removing and installing locking pin

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

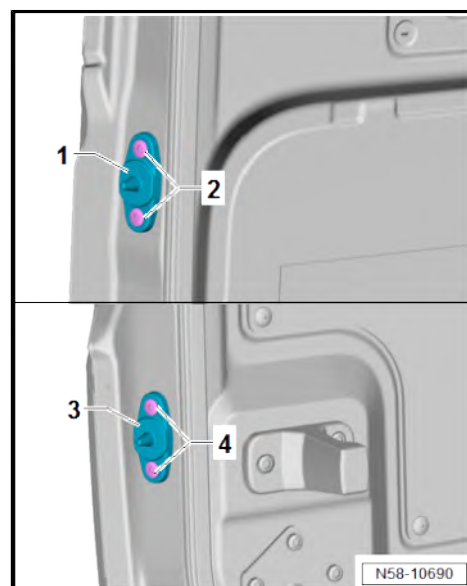
- Unscrew bolts -2-. When doing this, remove upper locking pin -1-.
- Unscrew bolts -4-. When doing this, remove lower locking pin -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“3.1 Assembly overview – sliding door”, page 145](#)



2.2.7 Removing and installing stop

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331-



Removing

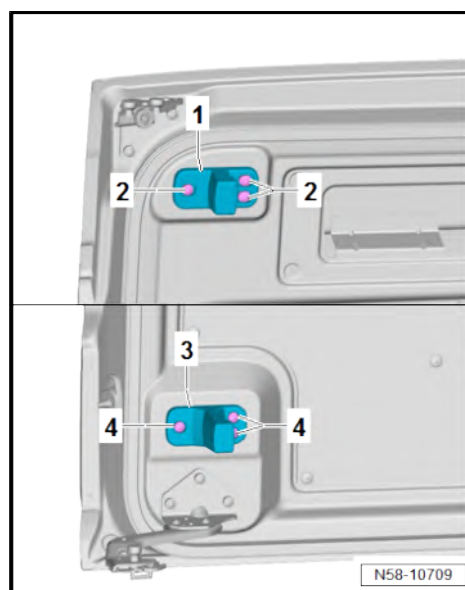
- Unscrew bolts -2-. When doing this, remove upper stop -1-.
- Unscrew bolts -4-. When doing this, remove lower stop -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["3.1 Assembly overview – sliding door", page 145](#)



2.2.8 Removing and installing interior door handle

Special tools and workshop equipment required

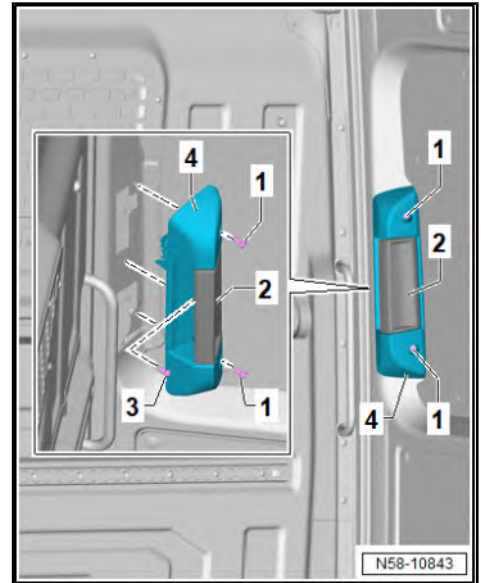
- ◆ Torque wrench - V.A.G 1783-



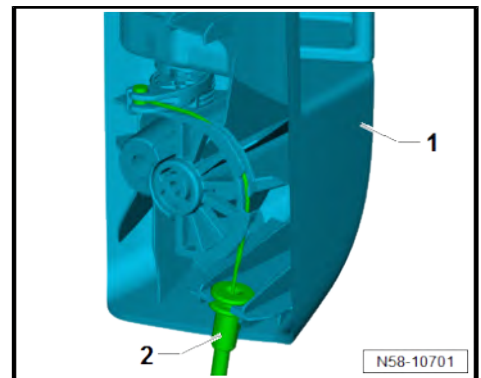


Removing

- Unscrew bolts -1-.
- Pull handle flap -2-. Hold it in this position.
- Unscrew bolt -3-.
- Push interior door handle -4- upwards out of sliding door until Bowden cables can be unhooked.



- Unhook lower Bowden cable -2- from interior door handle -1-.



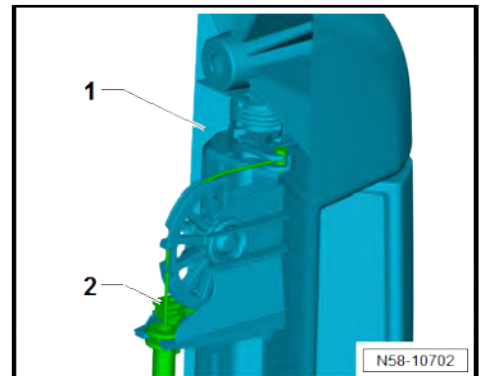
- Unhook upper Bowden cable -2- from interior door handle -1-. When doing this, remove interior door handle -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [“3.1 Assembly overview – sliding door”, page 145](#)



2.2.9 Removing and installing catch plate

Special tools and workshop equipment required



◆ Removal wedge - 3409-



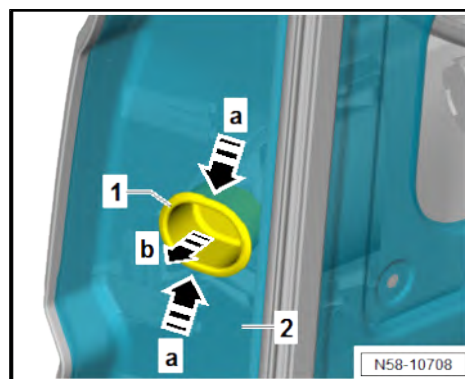
Removing

- Use removal wedge - 3409- to lever out catch plate -1- in -directions of arrow a-. When doing this, remove it in -direction of arrow b- from sliding door -2-.

Installing

Install in reverse order of removal, observing the following:

- Insert catch plate -1- in sliding door.
- Catch plate -1- must be felt to engage.



2.3 Removing and installing striker pin

⇒ [“2.3.1 Removing and installing striker pin, double cab”, page 128](#)

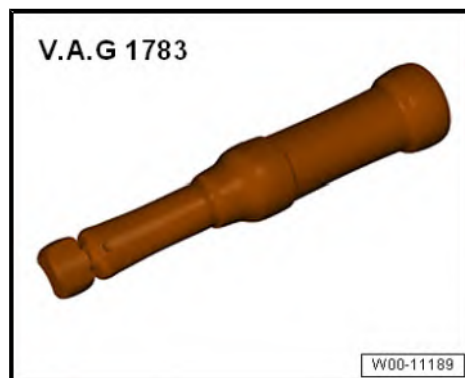
⇒ [“2.3.2 Removing and installing striker pin, sliding door”, page 129](#)

⇒ [“2.3.3 Removing and installing striker pin, wing door”, page 129](#)

2.3.1 Removing and installing striker pin, double cab

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





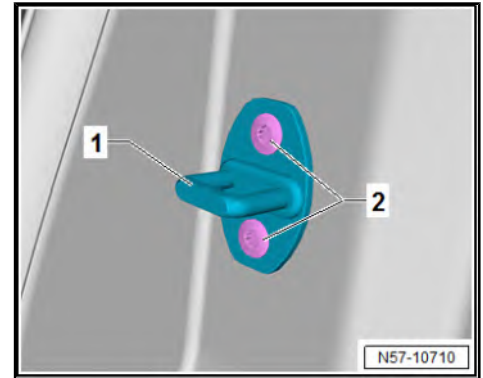
Removing

- Unscrew bolts -2-. When doing this, remove striker pin -1-.

Installing

Install in reverse order of removal, observing the following:

- Adjust striker pin ➤ [page 108](#) .



2.3.2 Removing and installing striker pin, sliding door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



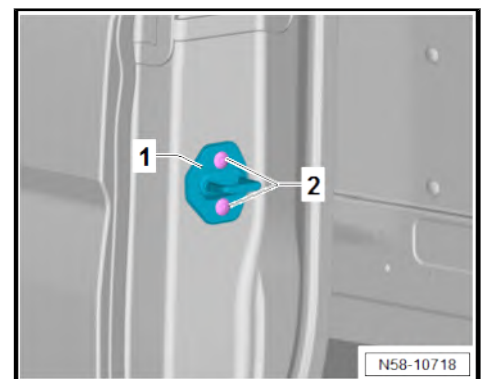
Removing

- Unscrew bolts -2-. When doing this, remove striker pin -1-.

Installing

Install in reverse order of removal, observing the following:

- Adjust striker pin ➤ [page 108](#) .



2.3.3 Removing and installing striker pin, wing door

Special tools and workshop equipment required

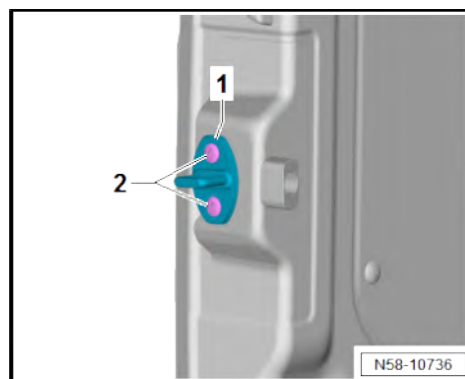


- ◆ Torque wrench - V.A.G 1331-



Removing

- Unscrew bolts -2-. When doing this, remove striker pin -1-.

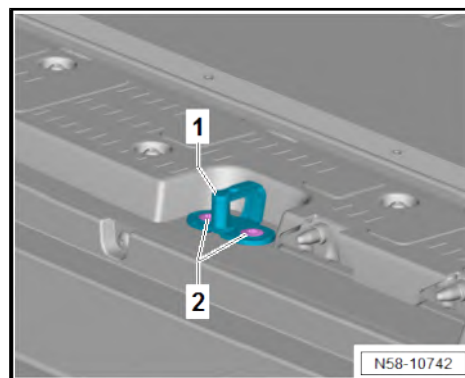


- Unscrew bolts -2-.
- Remove striker pin -1-.

Installing

Install in reverse order of removal, observing the following:

- Adjust striker pin ⇒ [page 108](#) .



2.4 Removing and installing door handle

⇒ [“2.4.1 Removing and installing door handle”, page 130](#)

⇒ [“2.4.2 Removing and installing door handle underlays”, page 131](#)

2.4.1 Removing and installing door handle

Removing

Sliding door

- Removing cap on door handle ⇒ [page 97](#) .

Wing door

- Remove lock cylinder ⇒ [page 95](#) .

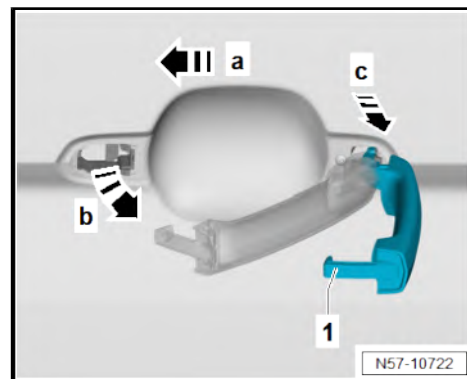


Continued for all vehicles

- Pull door handle -1- in -direction of arrow a- out of mounting in mounting bracket. When doing this, swing it in -direction of arrow b-.
- Remove door handle -1- in -direction of arrow c- from mounting bracket, keeping door handle at right angles to mounting bracket.

Installing

Install in reverse order.



2.4.2 Removing and installing door handle underlays

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



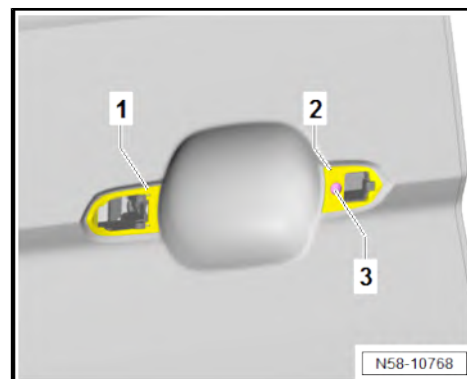
Removing

- Remove door handle ⇒ [page 130](#) .
- Unclip rear underlay -1-.
- Unscrew bolt -3-
- Unclip front underlay -2-.

Installing

Install in reverse order of removal, observing the following:

- ◆ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)



2.5 Removing and installing door handle bracket

⇒ [“2.5.1 Removing and installing door handle bracket, double cab”, page 131](#)

⇒ [“2.5.2 Removing and installing door handle bracket, sliding door”, page 132](#)

⇒ [“2.5.3 Removing and installing door handle bracket, wing door”, page 133](#)

2.5.1 Removing and installing door handle bracket, double cab

Special tools and workshop equipment required

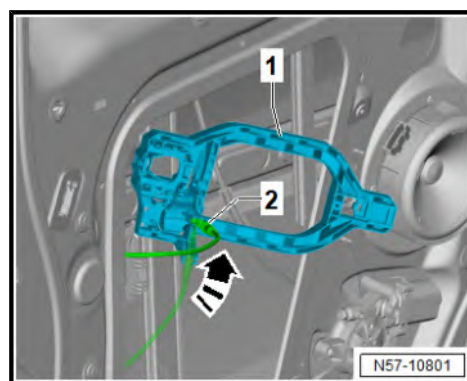


- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove rear cab door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Door window in upper position.
- Remove underlays for door handle ⇒ [page 131](#) .
- If fitted, unclip Bowden cable from clip on outer door panel.
- Remove mounting bracket -1- from door.
- When doing this, pull Bowden cable -2- out of retainer on mounting bracket -1-. When doing this, swing it in -direction of arrow- by approx. 90°.
- Push Bowden cable -2- out of guide on mounting bracket -1-. When doing this, remove mounting bracket -1-.



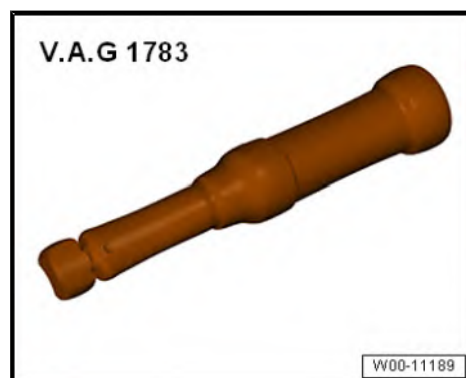
Installing

Install in reverse order.

2.5.2 Removing and installing door handle bracket, sliding door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

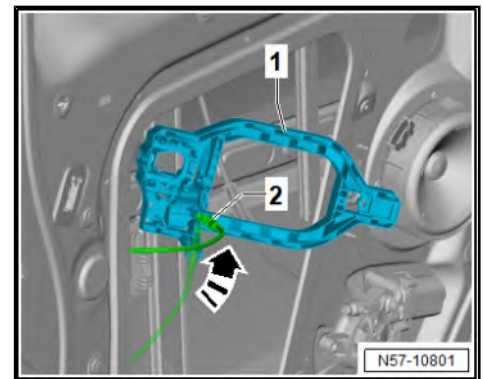
- Remove sliding door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Door trims, rear; Removing and installing rear door trim .
- Remove underlays for door handle ⇒ [page 131](#) .



- If fitted, unclip Bowden cable from clip on outer door panel.
- Remove mounting bracket -1- from door.
- Pull Bowden cable -2- out of retainer. When doing this, swing it in -direction of arrow- by approx. 90°.
- Remove Bowden cable -2- from guide on mounting bracket -1-.

Installing

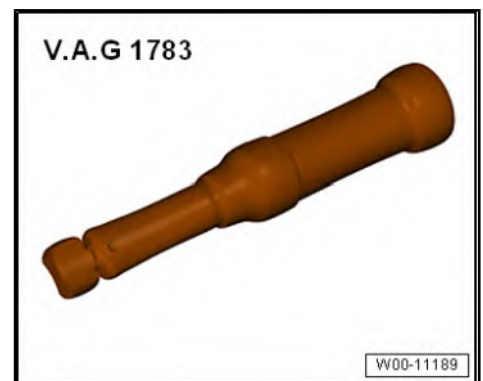
Install in reverse order.



2.5.3 Removing and installing door handle bracket, wing door

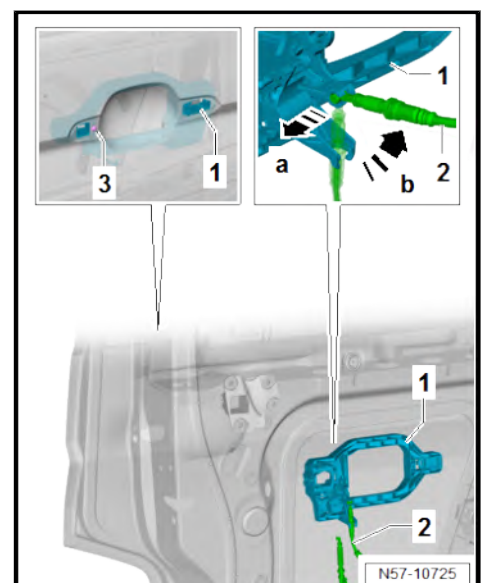
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove door trim for wing door ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Remove bases ⇒ [page 131](#) .
- If fitted, unclip Bowden cable from clip on outer door panel.
- Unscrew bolt -3-
- Remove mounting bracket -1- from door.
- Unclip Bowden cable -2- in -direction of arrow b- from mounting bracket -1-. When doing this, push it in -direction of arrow a-.





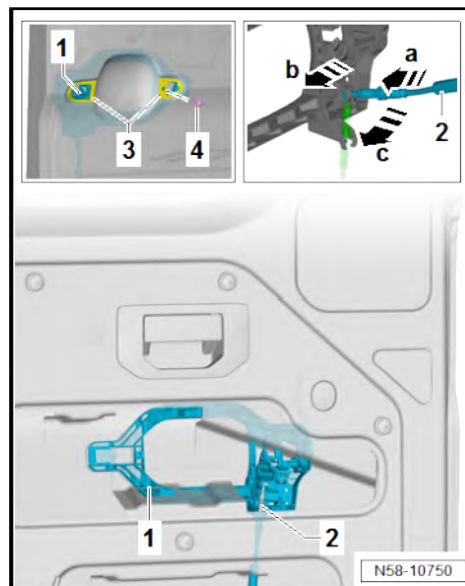
Installing

Install in reverse order of removal, observing the following:

- Insert Bowden cable -2- in -direction of arrow a- into mounting. When doing this, push it in -direction of arrow a-.
- Swing Bowden cable -2- in -direction of arrow c-, and engage it.
- Clip in underlays -3-.
- Tighten bolt -4-.

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)



2.6 Removing and installing door lock

⇒ [“2.6.1 Removing and installing door lock, double cab”, page 134](#)

⇒ [“2.6.2 Removing and installing door lock, sliding door”, page 135](#)

⇒ [“2.6.3 Removing and installing stop”, page 136](#)

⇒ [“2.6.4 Removing and installing door lock, wing door”, page 137](#)

⇒ [“2.6.5 Removing and installing door lock, top”, page 138](#)

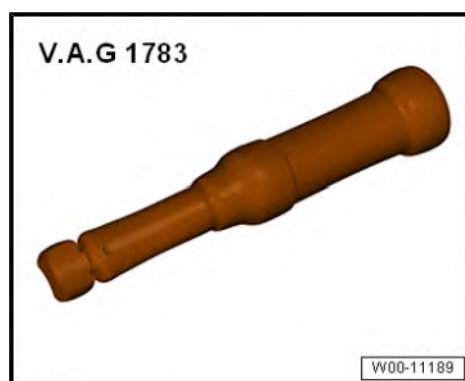
⇒ [“2.6.6 Removing and installing handle and door lock, bottom”, page 139](#)

⇒ [“2.6.7 Removing and installing interior door handle”, page 140](#)

2.6.1 Removing and installing door lock, double cab

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

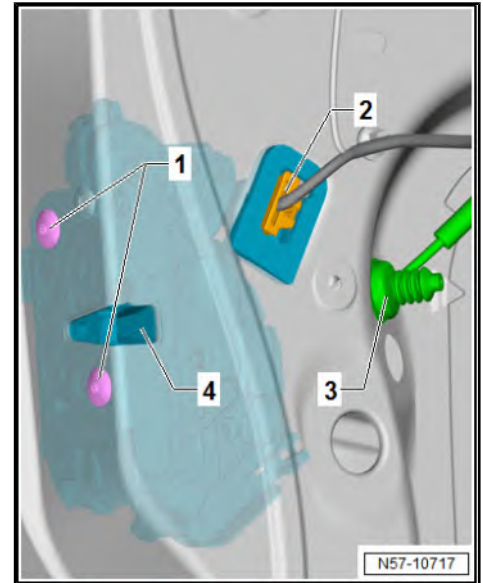
- Remove mounting bracket ⇒ [page 131](#) .
- Disconnect electrical connector -2-.
- Push grommet -3- together with Bowden cable into door.
- Unscrew bolts -1-. When doing this, remove door lock -4- together with Bowden cables.

Installing

Install in reverse order of removal, observing the following:

Specified torques

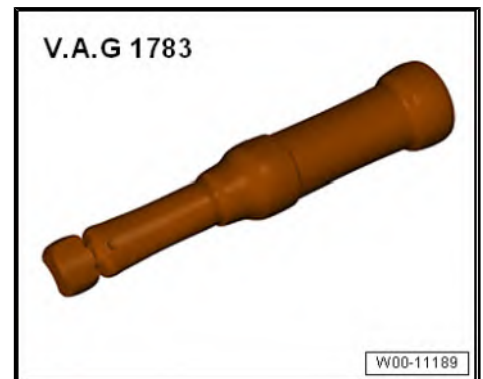
- ◆ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)



2.6.2 Removing and installing door lock, sliding door

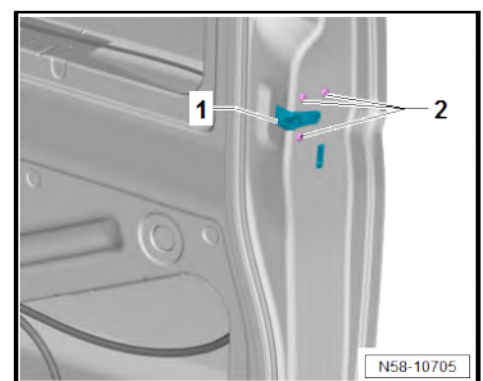
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



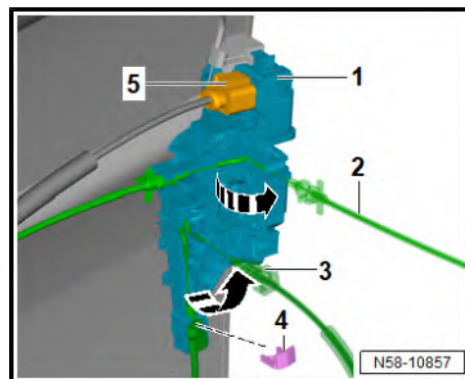
Removing

- Remove sliding door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Door trims, rear; Removing and installing rear door trim .
- Unscrew bolts -2-. When doing this, remove door lock -1-.





- Separate connector -5- from door lock -1-.
- Unhook Bowden cable -2- in direction of -arrow- from door lock -1-.
- Remove securing device -4-.
- Unhook Bowden cable -3- in direction of -arrow- from door lock -1-.



- Unhook Bowden cable -2- from door lock -1-.

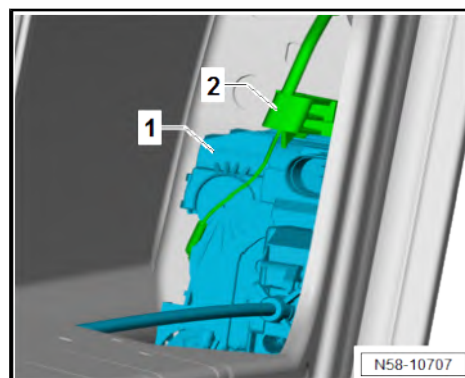
ZD

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ ["2.1 Assembly overview - door handle and door lock", page 114](#)



2.6.3 Removing and installing stop

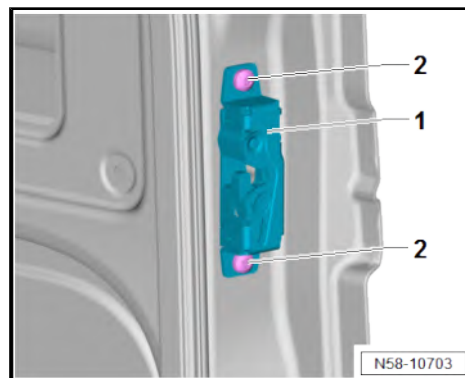
Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-



Removing

- Unscrew bolts -2-. When doing this, remove stop -1-.





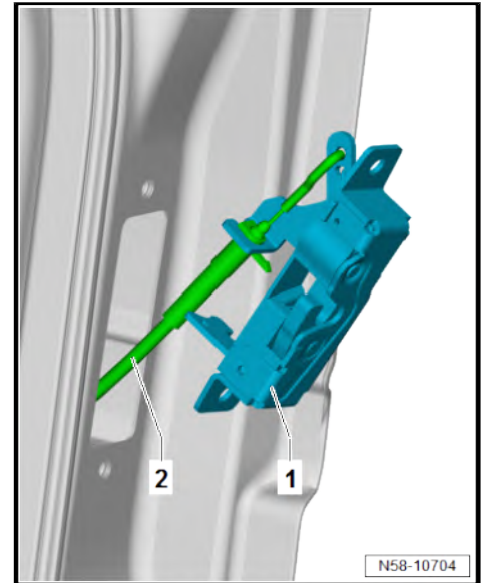
- Guide stop -1- out of sliding door.
- Unhook Bowden cable -2- from stop -1-. When doing this, remove stop -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)



2.6.4 Removing and installing door lock, wing door

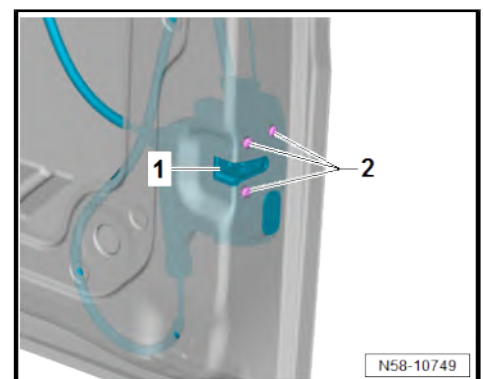
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



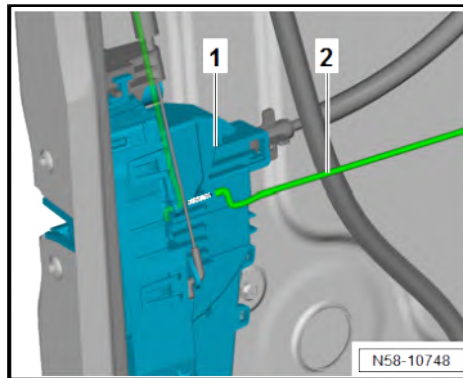
Removing

- Remove trim for wing door ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Unscrew bolts -2-. When doing this, remove door lock -1-.
- Guide door lock -1- out of wing door.

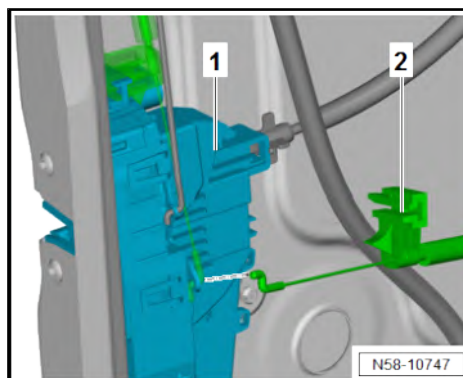




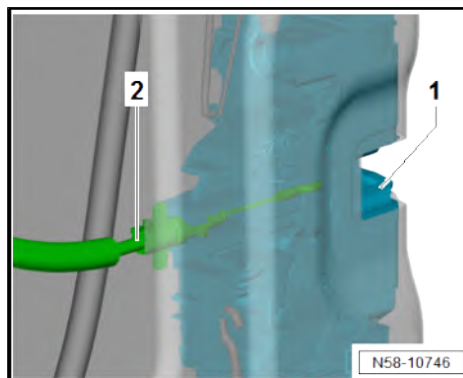
- Detach connecting rod -2- from door lock -1-.



- Swing Bowden cable to top door lock -2- until cable can be guided out of eye in door lock -1-.



- Unhook Bowden cable -2- from door lock -1-.



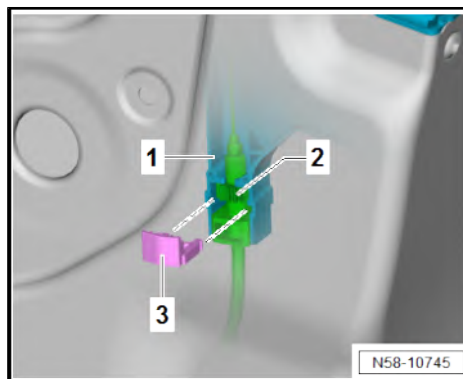
- Unclip clip -3-.
- Unhook Bowden cable -2- from door lock -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)

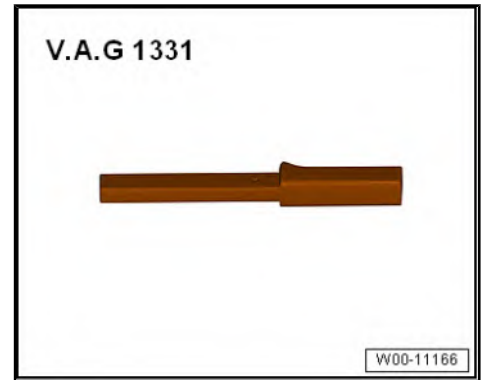


2.6.5 Removing and installing door lock, top

Special tools and workshop equipment required

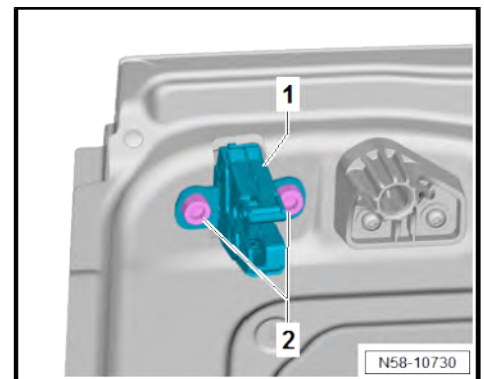


- ◆ Torque wrench - V.A.G 1331-



Removing

- Unscrew bolts -2-.
- Remove door lock -1- until Bowden cable can be unhooked.



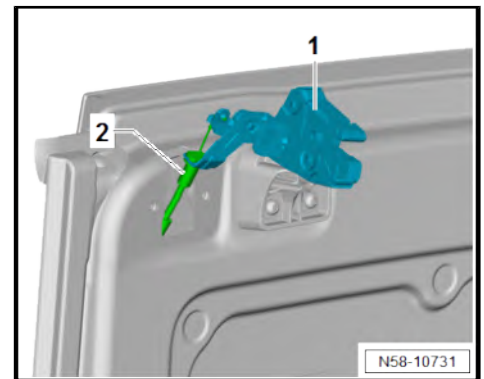
- Unhook Bowden cable -2- from door lock -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["2.1 Assembly overview - door handle and door lock", page 114](#)



2.6.6 Removing and installing handle and door lock, bottom

Special tools and workshop equipment required

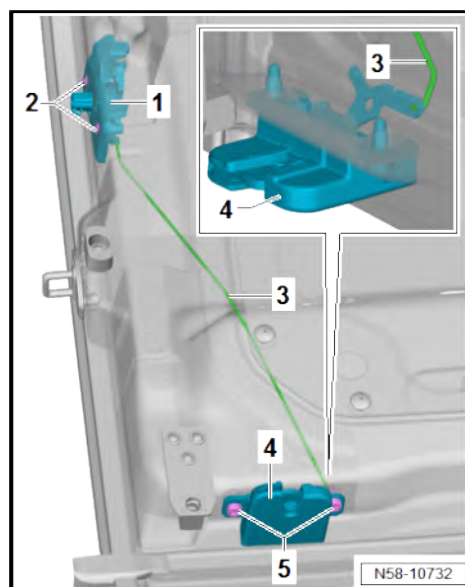
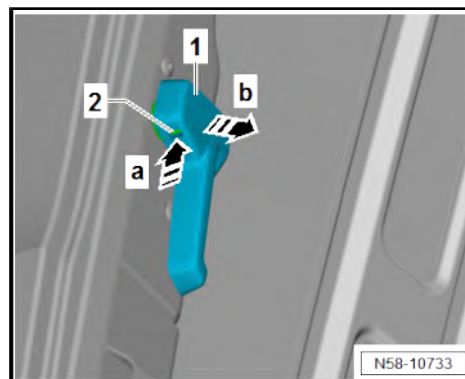
- ◆ Torque wrench - V.A.G 1331-





Removing

- Remove trim for wing door ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Use a small screwdriver to release locking lug -2- in -direction of arrow a-. When doing this, remove lever -1- in -direction of arrow b-.
- Unclip operating rod -3- on inside from door lock -4-.
- Unscrew bolts -5-. When doing this, remove door lock -4-.
- Unscrew bolts -2-. When doing this, remove handle -1-.



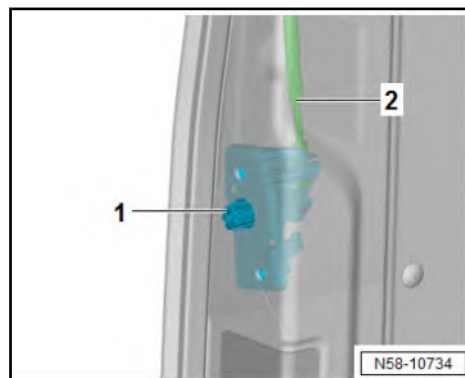
- Detach Bowden cable -2- from handle -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ ["2.1 Assembly overview - door handle and door lock", page 114](#)



2.6.7 Removing and installing interior door handle

Special tools and workshop equipment required

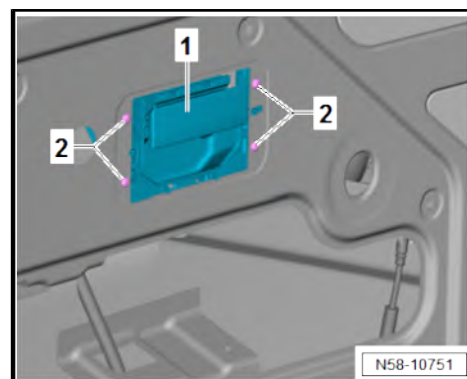


- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove trim for wing door ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Unscrew bolts -2-. When doing this, remove interior door handle -1- from wing door until Bowden cable can be unhooked.



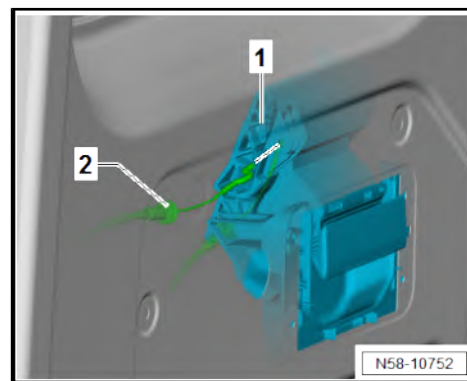
- Detach Bowden cable -2- from interior door handle -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)



2.7 Removing and installing cap on door handle

⇒ [“2.7.1 Removing and installing cap on door handle, double cab”, page 141](#)

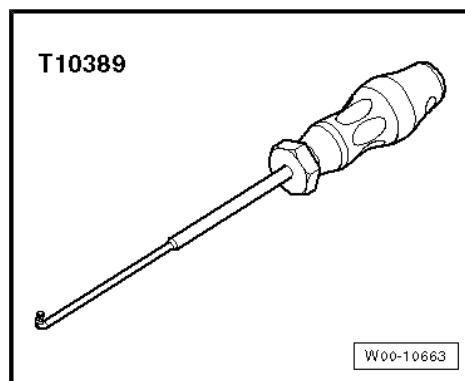
⇒ [“2.7.2 Removing and installing cap on door handle, sliding door”, page 143](#)

2.7.1 Removing and installing cap on door handle, double cab

Special tools and workshop equipment required

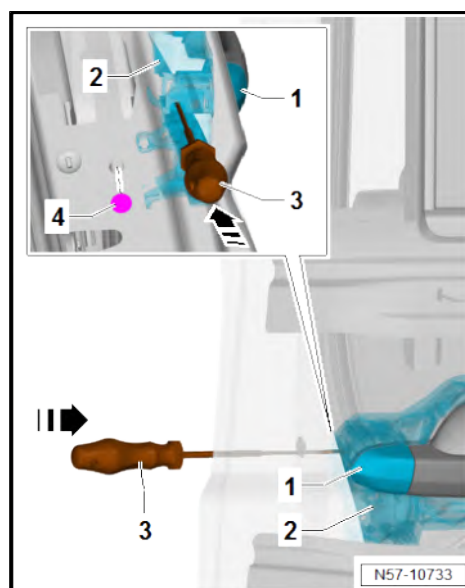


◆ Assembly tool - T10389-



Removing

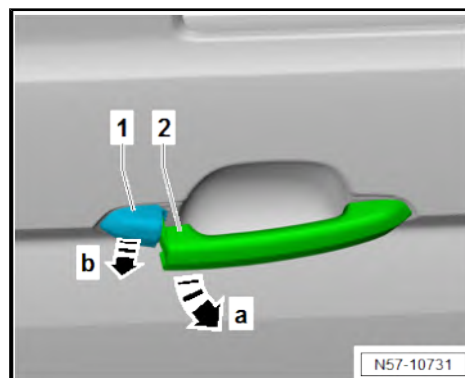
- If fitted, pull off film -4-.
- The following work steps must be performed by a second mechanic.
- Use a screwdriver -3- to release locking hook of mounting bracket -2- through exposed hole in -direction of arrow-.
- Have a second mechanic push cap -1- against outer door panel.



- Pull door handle -2- in direction of -arrow a-. When doing this, hold door handle -2- in this position.
- Remove cap -1- in -direction of arrow b-.

Installing

Install in reverse order.

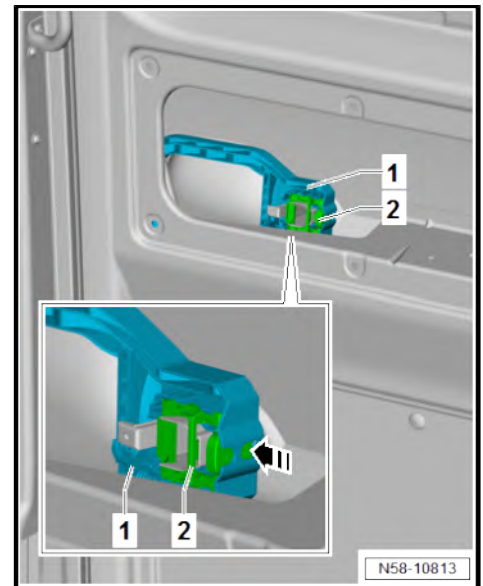




2.7.2 Removing and installing cap on door handle, sliding door

Removing

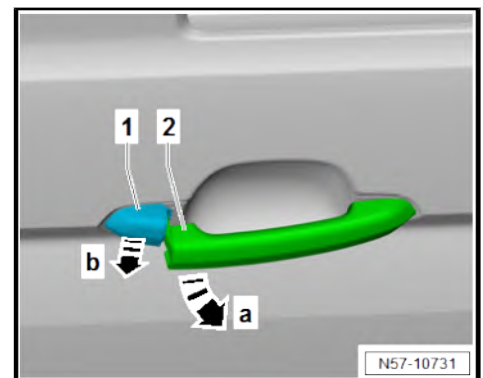
- Remove top sliding door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Use a screwdriver to release locking element -2- in -direction of arrow- from mounting bracket -1-. When doing this, push cap against door.



- Pull door handle -2- in direction of -arrow a-. When doing this, hold door handle -2- in this position.
- Remove cap -1- in -direction of arrow b-.

Installing

Install in reverse order.



2.8 Removing and installing lock cylinder

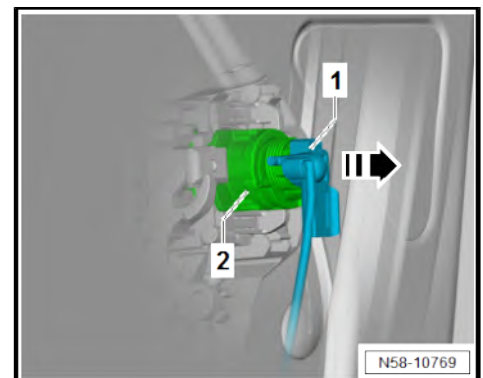
⇒ [“2.8.1 Removing and installing lock cylinder, wing door”, page 143](#)

⇒ [“2.8.2 Removing and installing cap on lock cylinder”, page 144](#)

2.8.1 Removing and installing lock cylinder, wing door

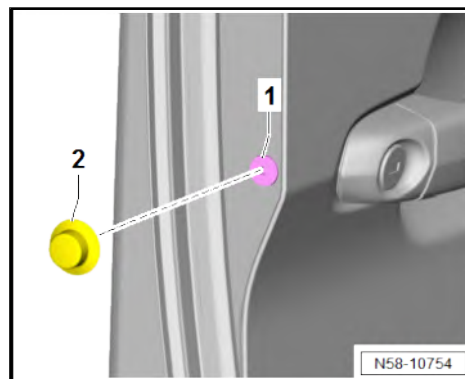
Removing

- Remove trim panel ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trim panels; Removing and installing rear door trim .
- Detach clip -1- from lock cylinder -2- in direction of -arrow-.





- Remove cap -2-.
- Unscrew bolt -1-

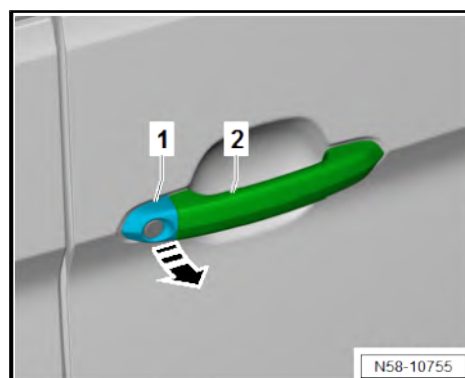


- Pull door handle -2- in -direction of arrow-. When doing this, hold door handle -2- in this position.
- Pull out lock cylinder -1- perpendicular to door in direction of -arrow- from mounting bracket.

Installing

Install in reverse order of removal, observing the following:

- During installation, lock cylinder must be pressed against outside door panel.



Specified torques

- ♦ ⇒ ["2.1 Assembly overview - door handle and door lock", page 114](#)

2.8.2 Removing and installing cap on lock cylinder

Special tools and workshop equipment required

- ♦ Removal wedge - 3409-



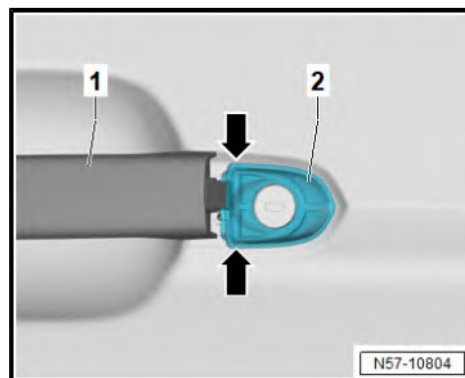
Removing

- Pull door handle -1-. When doing this, hold door handle -1- in this position.
- Use removal wedge - 3409- to lever off lock cylinder cap -arrows- in area marked with -2-.

Installing

Install in reverse order of removal, observing the following:

- Push on lock cylinder cap -2-. When doing this, engage cap in lock cylinder.





3 Sliding door

⇒ [“3.1 Assembly overview – sliding door”, page 145](#)

⇒ [“3.2 Assembly overview - door seals”, page 148](#)

⇒ [“3.3 Removing and installing sliding door”, page 149](#)

⇒ [“3.4 Adjusting sliding door”, page 151](#)

⇒ [“3.5 Removing and installing add-on parts”, page 156](#)

⇒ [“3.6 Removing and installing inner door seal”, page 160](#)

3.1 Assembly overview – sliding door

⇒ [“3.1.1 Assembly overview – sliding door”, page 145](#)

⇒ [“3.1.2 Assembly overview - sliding door, attachments”, page 147](#)

3.1.1 Assembly overview – sliding door

1 - Sliding door

- ☐ Removing and installing
⇒ [page 149](#)
- ☐ Adjusting ⇒ [page 151](#)

2 - Roller guide

- ☐ Top
- ☐ Removing and installing
⇒ [page 120](#)
- ☐ Adjusting ⇒ [page 151](#)

3 - Bolts

- ☐ Qty. 2
- ☐ 30 Nm

4 - Bolts

- ☐ Qty. 3
- ☐ 30 Nm

5 - Stop

- ☐ Top
- ☐ Removing and installing
⇒ [page 120](#)
- ☐ Adjusting ⇒ [page 151](#)

6 - Bolts

- ☐ Qty. 3
- ☐ 30 Nm

7 - Stop

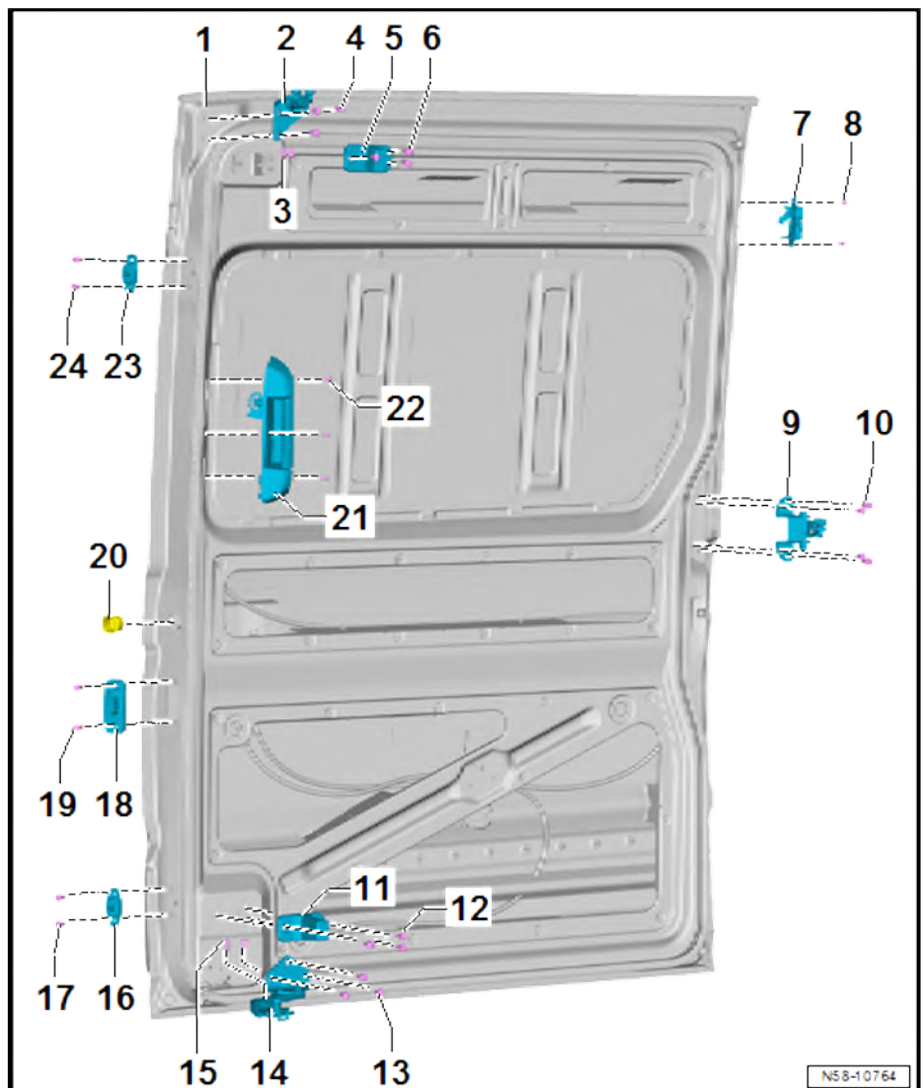
- ☐ Removing and installing
⇒ [page 134](#)

8 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

9 - Roller guide

- ☐ Centre
- ☐ Removing and installing ⇒ [page 120](#)
- ☐ Adjusting ⇒ [page 151](#)



N58-10764



10 - Bolts

- ☐ Qty. 3
- ☐ 30 Nm

11 - Stop

- ☐ Bottom
- ☐ Removing and installing [⇒ page 120](#)
- ☐ Adjusting [⇒ page 151](#)

12 - Bolts

- ☐ Qty. 3
- ☐ 30 Nm

13 - Bolts

- ☐ Qty. 3
- ☐ 30 Nm

14 - Roller guide

- ☐ Bottom
- ☐ Removing and installing [⇒ page 120](#)
- ☐ Adjusting [⇒ page 151](#)

15 - Bolts

- ☐ Qty. 2
- ☐ 12 Nm

16 - Locking peg

- ☐ Bottom
- ☐ Removing and installing [⇒ page 120](#)
- ☐ Adjusting [⇒ page 151](#)

17 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

18 - Contact strip in right sliding door - F352-

- ☐ Removing and installing [⇒ page 120](#)
- ☐ Adjusting [⇒ page 151](#)

19 - Bolts

- ☐ Qty. 2
- ☐ 18 Nm

20 - Catch plate

- ☐ Removing and installing [⇒ page 127](#)

21 - Interior door handle

- ☐ Removing and installing [⇒ page 120](#)

22 - Bolts

- ☐ Qty. 3
- ☐ 8 Nm

23 - Locking peg

- ☐ Top
- ☐ Removing and installing [⇒ page 120](#)
- ☐ Adjusting [⇒ page 151](#)

24 - Bolts

- ☐ Qty. 2

- ☐ 20 Nm

3.1.2 Assembly overview - sliding door, attachments

1 - Guide rail

- ☐ Top
- ☐ Removing and installing
⇒ [page 156](#)

2 - Dished spring

3 - Striker

- ☐ 50 Nm

4 - Nuts

- ☐ Qty. 6 or 8, depending on equipment
- ☐ 7 Nm

5 - Catch plate

- ☐ Top
- ☐ Removing and installing
⇒ [page 156](#)

6 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

7 - Washer

8 - Arrester

- ☐ Removing and installing
⇒ [page 156](#)
- ☐ 30 Nm

9 - Bolts

- ☐ Qty. 2
- ☐ 3 Nm

10 - Rear right door lock unit - VX24-

- ☐ Removing and installing
⇒ [page 156](#)

11 - Catch plate

- ☐ Bottom
- ☐ Removing and installing ⇒ [page 156](#)

12 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

13 - Bolt

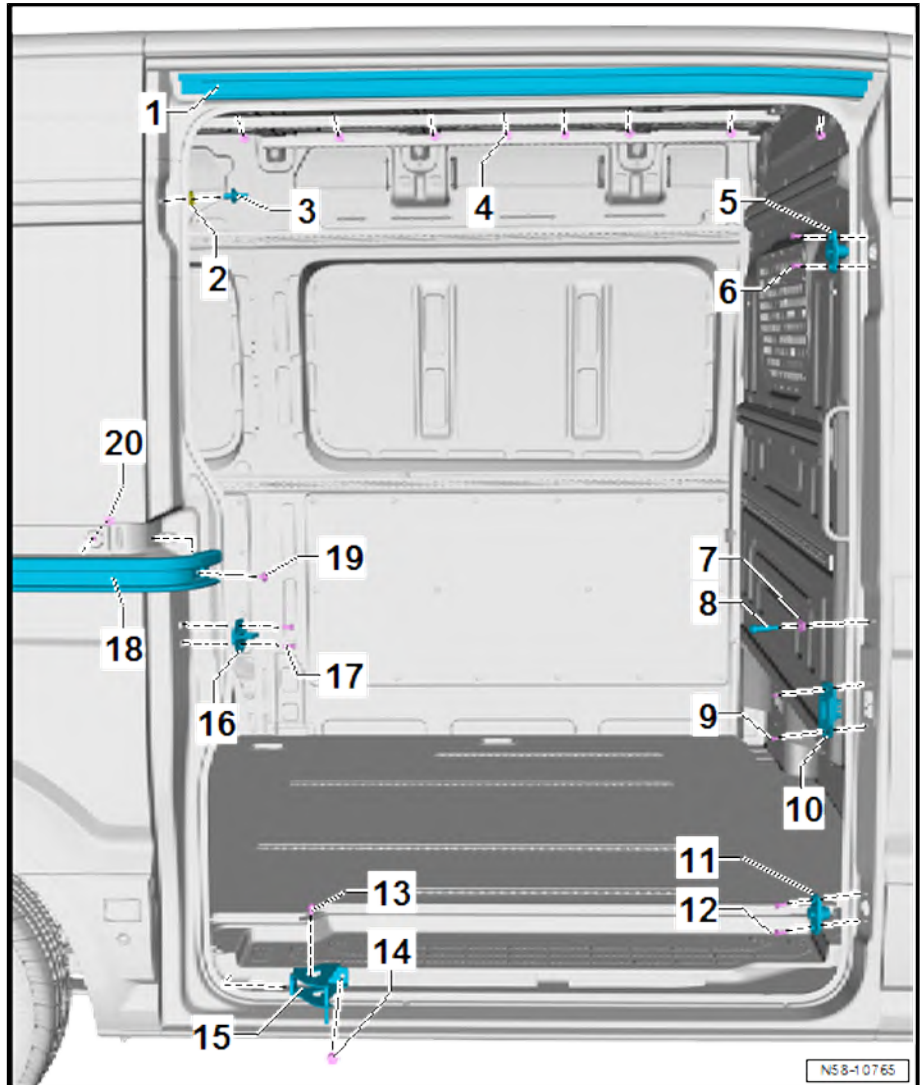
- ☐ 20 Nm

14 - Nut

- ☐ 20 Nm

15 - Door arrester

- ☐ Removing and installing ⇒ [page 111](#)





16 - Striker

- ☐ Removing and installing ⇒ [page 128](#)
- ☐ Adjusting ⇒ [page 108](#)

17 - Bolts

- ☐ Qty. 2
- ☐ 25 Nm

18 - Guide rail

- ☐ Centre
- ☐ Removing and installing ⇒ [page 156](#)

19 - Bolt

- ☐ 9 Nm

20 - Nuts

- ☐ Qty. 7
- ☐ 9 Nm

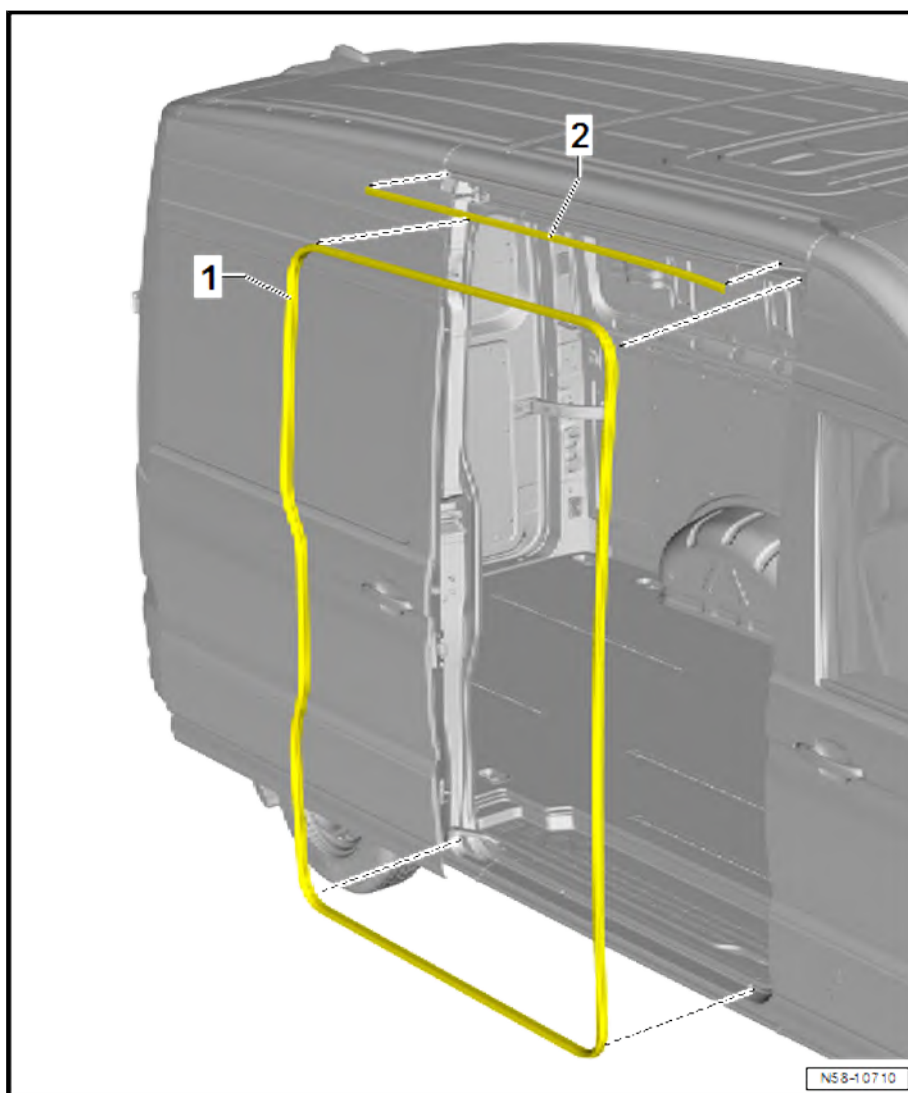
3.2 Assembly overview - door seals

1 - Door seal

- ☐ Renew after removal
- ☐ Removing and installing
⇒ [page 113](#)

2 - Pre-seal

- ☐ Renew after removal
- ☐ Removing and installing
⇒ [page 113](#)

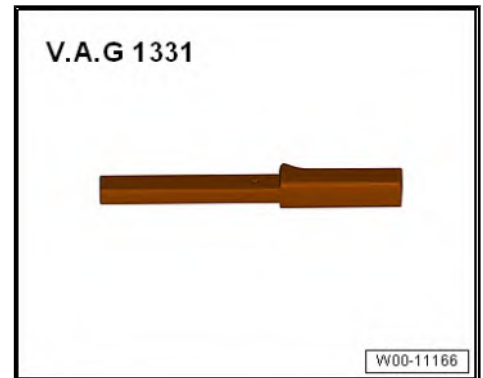




3.3 Removing and installing sliding door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-

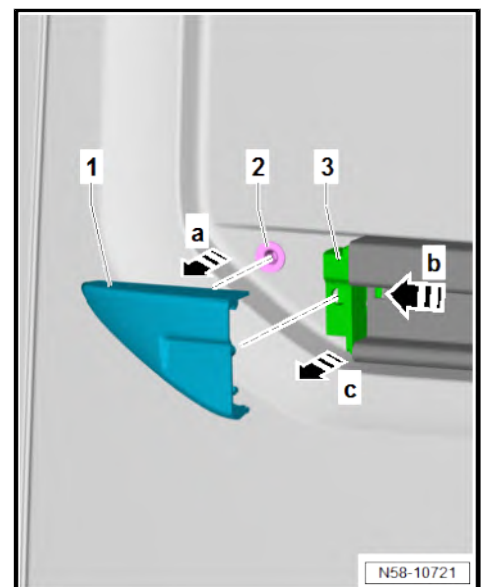


- ◆ Torque wrench - V.A.G 1332-



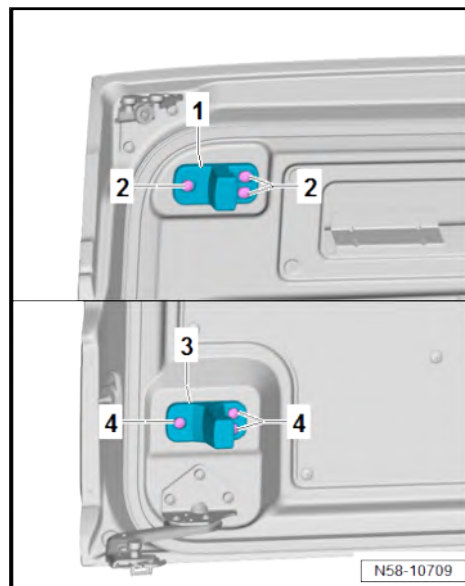
Removing

- Use removal wedge - 3409- to unclip cap -1- in -direction of arrow a- from clip -2-.
- Use removal wedge - 3409- to release locking lug in -direction of arrow b-. When doing this, pull out mounting -3- in -direction of arrow c-.

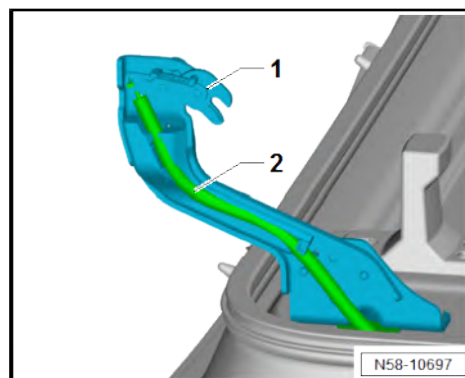




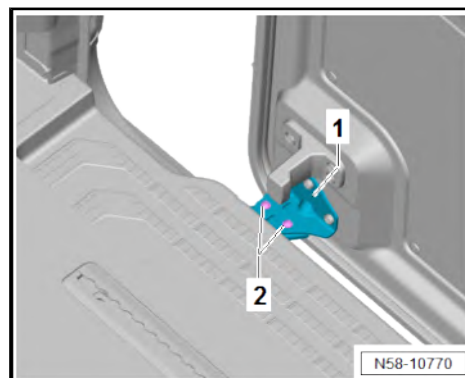
- Unscrew bolts -2-. When doing this, remove stop -1-.
- Unscrew bolts -4-. When doing this, remove stop -3-.



- Detach Bowden cable -1- from bottom roller guide -2-.



- Mark position of bolts -2-.
- The following work steps must be performed by a second mechanic.
- Unscrew bolts -2- from bottom roller guide -1-.



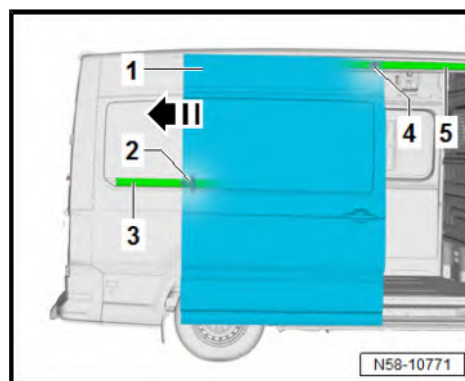
- Push back sliding door -1- in -direction of arrow- until roller guide -2- is pushed out of guide rail -3-.
- Push back sliding door -1- in -direction of arrow- until roller guide -4- is pushed out of guide rail -5-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [“3.1 Assembly overview – sliding door”, page 145](#)





3.4 Adjusting sliding door

Sequence of operations

- ⇒ ["3.4.1 Adjusting locking pegs", page 151](#)
- ⇒ ["3.4.2 Adjusting contact carrier", page 151](#)
- ⇒ ["3.4.3 Adjusting lower roller guide", page 152](#)
- ⇒ ["3.4.4 Adjusting upper roller guide", page 153](#)
- ⇒ ["3.4.5 Adjusting centre roller guide", page 154](#)
- ⇒ ["3.4.6 Adjusting stop", page 155](#)
- ⇒ ["3.4.7 Adjusting adjusting buffer", page 156](#)

3.4.1 Adjusting locking pegs

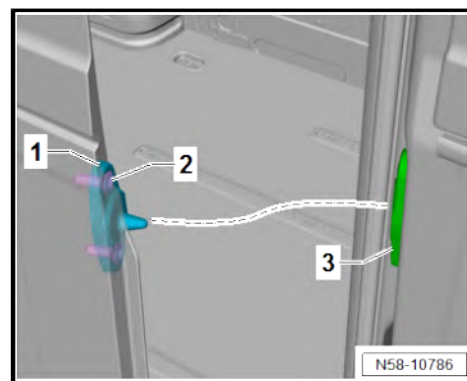
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Procedure

- Vehicle must be standing on all wheels.
- Loosen bolts -2-.
- Close sliding door slowly. When doing this, the locking pin -1- is centred by the catch plate -3- in the B-pillar.
- Open sliding door.
- Tighten bolts -2- to specified torque.



Specified torques

- ◆ ⇒ ["3.1 Assembly overview – sliding door", page 145](#)

3.4.2 Adjusting contact carrier

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



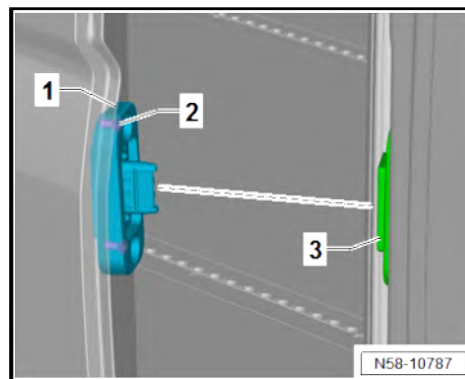


Procedure

- Vehicle must be standing on all wheels.
- Loosen bolts -2-.
- Close sliding door slowly. When doing this, the contact carrier -1- is centred by the contact strip -3- in the B-pillar.
- Open sliding door.
- Tighten bolts -2- to specified torque.

Specified torques

- ♦ ➔ [“3.1 Assembly overview – sliding door”, page 145](#)



3.4.3 Adjusting lower roller guide

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-

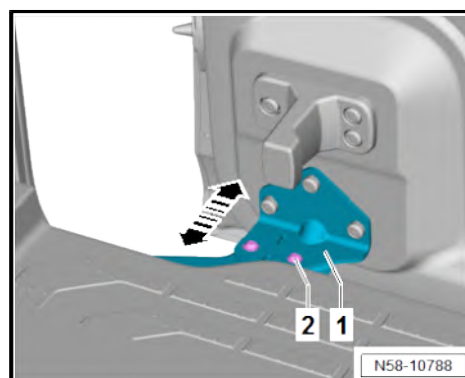


- ♦ Setting gauge - 3371-



Procedure

- Vehicle must be standing on all wheels.
- Loosen bolts -2-.
- Align roller guide -1- by moving it in the respective -direction of arrow-. Hold it in the correct position, and tighten it to specified torque.





- Loosen bolts -2-.
- Align roller guide -1- by moving it in the respective -direction of arrow-. Hold it in the correct position, and tighten it to specified torque.
- Adjust adjustment buffer ⇒ [page 156](#) .
- Use the setting gauge - 3371- to check or adjust the gap dimensions ⇒ Body Repairs; Rep. gr. 00 ; Technical data; Body gaps/shut lines; Front body .

Specified torques

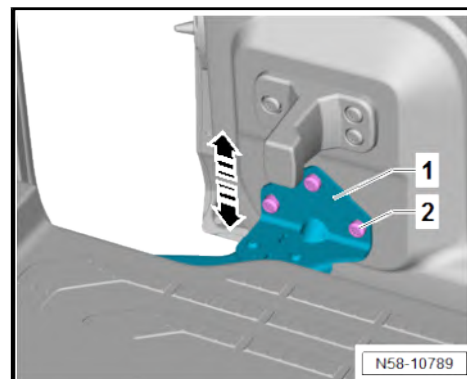
- ◆ ⇒ [“3.1 Assembly overview – sliding door”, page 145](#)

3.4.4 Adjusting upper roller guide

Special tools and workshop equipment required

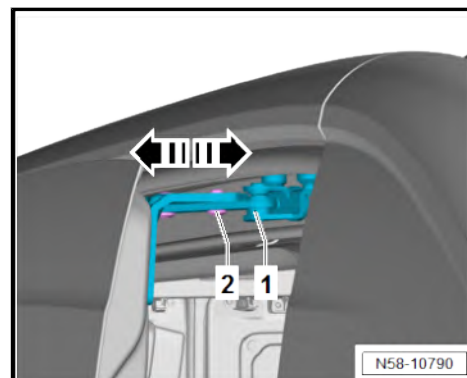
- ◆ Torque wrench - V.A.G 1331-

- ◆ Setting gauge - 3371-



Procedure

- Vehicle must be standing on all wheels.
- Loosen bolts -2-.
- Align roller guide -1- by moving it in the respective -direction of arrow-. Hold it in the correct position, and tighten it to specified torque.

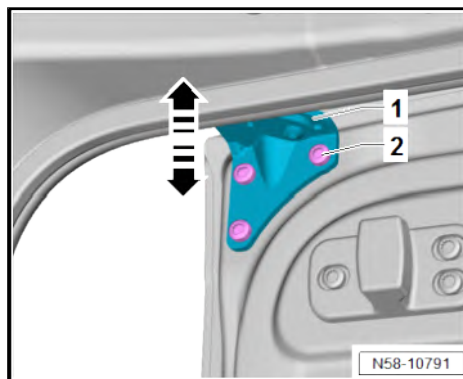




- Loosen bolts -2-.
- Align roller guide -1- by moving it in the respective -direction of arrow-. Hold it in the correct position, and tighten it to specified torque.
- Adjust adjustment buffer ⇒ [page 156](#) .
- Use the setting gauge - 3371- to check or adjust the gap dimensions ⇒ Body Repairs; Rep. gr. 00 ; Technical data; Body gaps/shut lines; Front body .

Specified torques

- ♦ ⇒ [“3.1 Assembly overview – sliding door”, page 145](#)



3.4.5 Adjusting centre roller guide

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-



- ♦ Setting gauge - 3371-

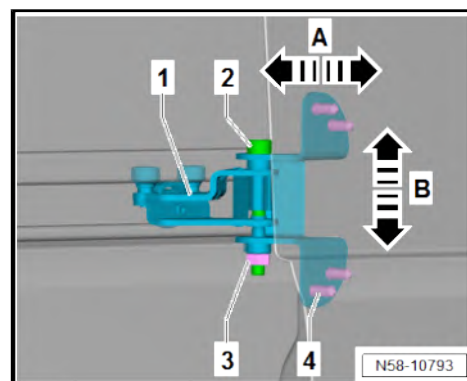


Procedure

- Vehicle must be standing on all wheels.



- Loosen nut -3-.
- Turn hinge pin -2-. When doing this, adjust roller guide -1- in -directions of arrow B-.
- Tighten nut -3-.
- Loosen bolts -4-.
- Align sliding door in direction of -arrow A-.
- Tighten bolts -4-.
- Adjust adjustment buffer ➤ [page 156](#) .
- Use the setting gauge - 3371- to check or adjust the gap dimensions ➤ Body Repairs; Rep. gr. 00 ; Technical data; Body gaps/shut lines; Front body .



Specified torques

- ◆ ➤ ["3.1 Assembly overview – sliding door", page 145](#)

3.4.6 Adjusting stop

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-

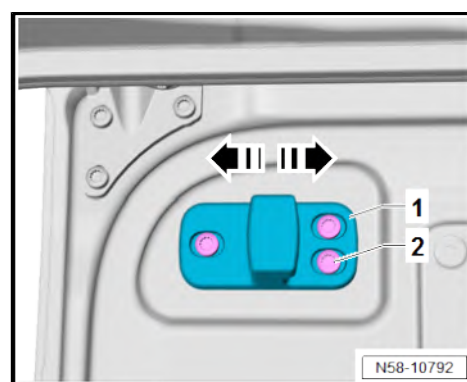


Procedure

- Vehicle must be standing on all wheels.
- Loosen bolts -2-.
- Open sliding door until stop -1- comes into contact with C-pillar.
- Move stop -1- in direction of -arrow- on C-pillar.
- Tighten bolts -2-.

Specified torques

- ◆ ➤ ["3.1 Assembly overview – sliding door", page 145](#)

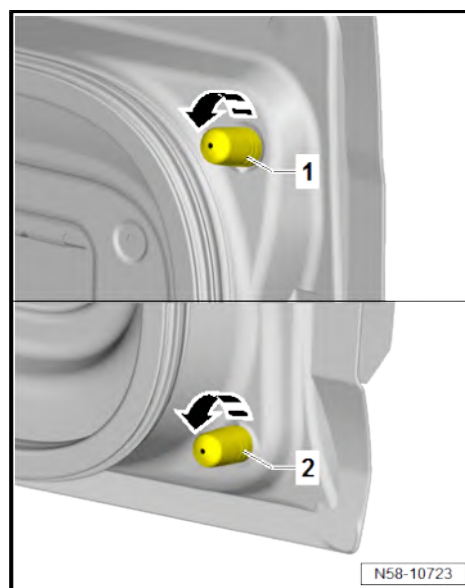




3.4.7 Adjusting adjusting buffer

Procedure

- Vehicle must be standing on all wheels.
- Screw in upper adjusting buffer -1- and lower adjusting buffer -2- as far as stop.
- Once sliding door has been adjusted, unscrew adjusting buffers -1- and -2- in direction of -arrow- until they are in light contact with body.



3.5 Removing and installing add-on parts

⇒ ["3.5.1 Removing and installing guide rail, top", page 156](#)

⇒ ["3.5.2 Removing and installing detent segment", page 157](#)

⇒ ["3.5.3 Removing and installing guide rail, middle", page 157](#)

⇒ ["3.5.4 Removing and installing catch plate", page 158](#)

⇒ ["3.5.5 Removing and installing arrester", page 159](#)

⇒ ["3.5.6 Removing and installing rear right door lock unit VX24", page 159](#)

3.5.1 Removing and installing guide rail, top

Special tools and workshop equipment required

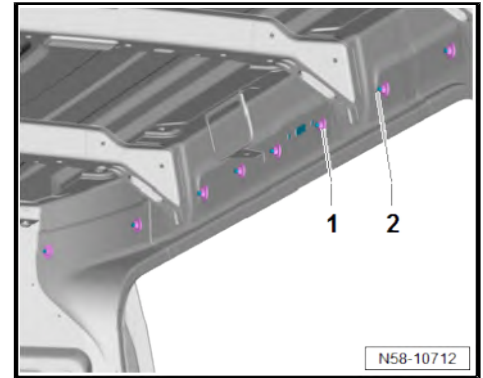
- ◆ Torque wrench - V.A.G 1783-





Removing

- Remove sliding door ➤ [page 149](#) .
- Unscrew nuts -1- at top guide rail -2-.



- Guide out guide rail -1- through roof aperture in -direction of arrow-.

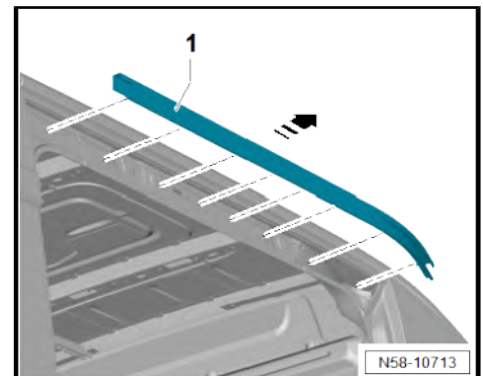
Installing

Install in reverse order of removal, observing the following:

- Insert self-adhesive zinc film - D 381 ZN1 A2- between top guide rail -1- and roof aperture.

Specified torques

- ◆ ➤ [“3.1 Assembly overview – sliding door”, page 145](#)



3.5.2 Removing and installing detent segment

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

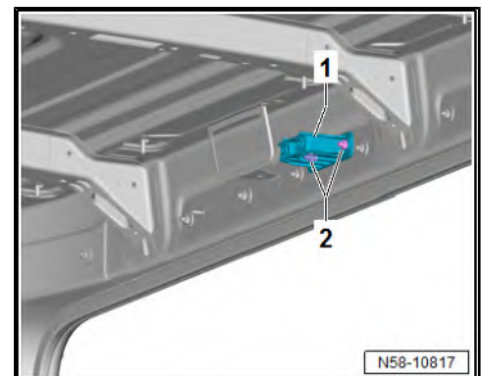
- Unscrew nuts -2-. When doing this, remove detent segment -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -2-	7 Nm



3.5.3 Removing and installing guide rail, middle

Special tools and workshop equipment required

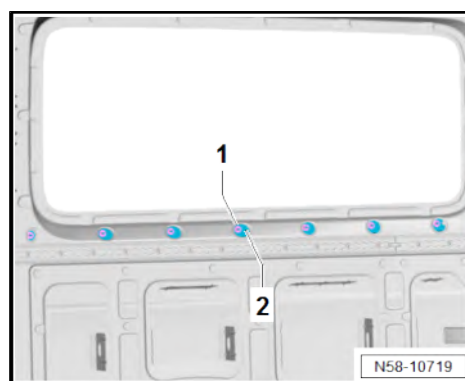


- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove sliding door ➔ [page 149](#) .
- Remove side panel gap cover ➔ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Overview of fitting locations - trims, interior .
- Unscrew nuts -1- at centre guide rail -2-.



- Unscrew bolt -2- When doing this, remove centre guide rail -1-.

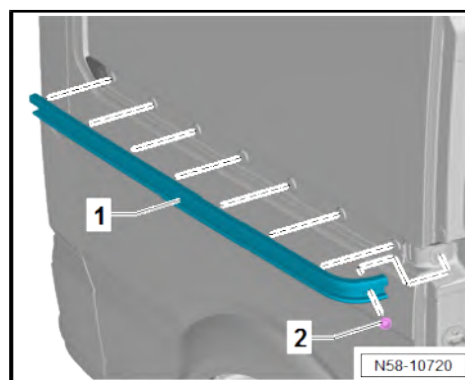
Installing

Install in reverse order of removal, observing the following:

- Insert self-adhesive zinc film - D 381 ZN1 A2- between centre guide rail -1- and side panel.

Specified torques

- ◆ ➔ [“3.1 Assembly overview – sliding door”, page 145](#)



3.5.4 Removing and installing catch plate

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





Removing

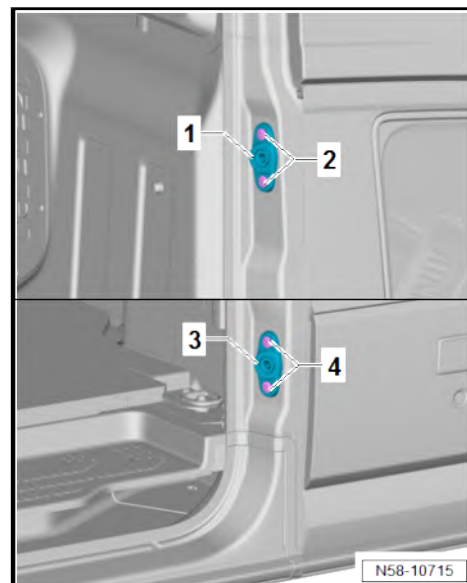
- Unscrew bolts -2-.
- Remove upper catch plate -1- from B-pillar.
- Unscrew bolts -4-.
- Remove lower catch plate -3- from B-pillar.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“3.1.2 Assembly overview - sliding door, attachments”, page 147](#)



3.5.5 Removing and installing arrester

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

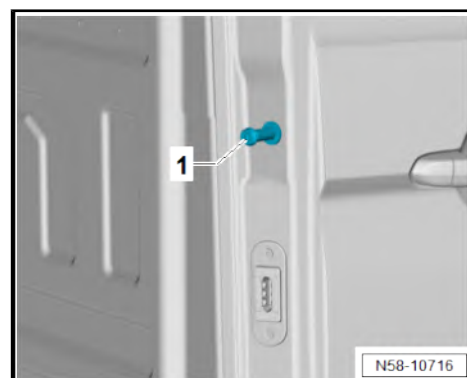
- Unscrew catch plate -1- from B-pillar.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“3.1.2 Assembly overview - sliding door, attachments”, page 147](#)



3.5.6 Removing and installing rear right door lock unit - VX24-

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

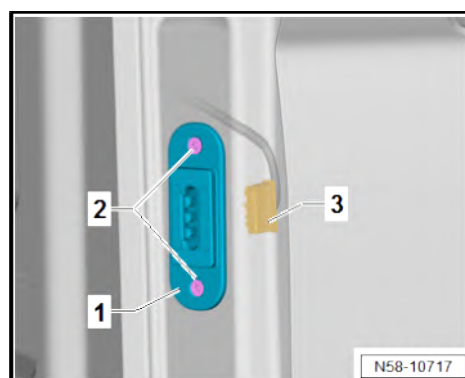
- Unscrew bolts -2-.
- Remove contact strip -1- from B-pillar.
- Separate connector -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“3.1.1 Assembly overview – sliding door”, page 145](#)



3.6 Removing and installing inner door seal

⇒ [“3.6.1 Removing and installing inner door seal”, page 160](#)

⇒ [“3.6.2 Removing and installing door seal”, page 160](#)

3.6.1 Removing and installing inner door seal

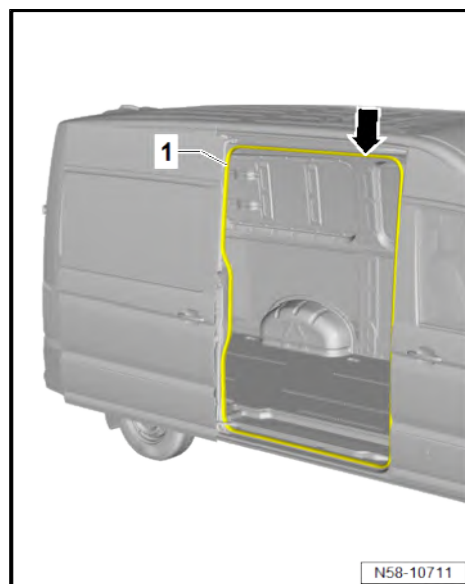
Removing

- Remove bottom roller guide ⇒ [page 120](#) .
- Pull inner door seal -1- off body flange.

Installing

Install in reverse order of removal, observing the following:

- Fit new inner door seal -1-.
- The vulcanisation point -arrow- must be at a distance of approx. 200 mm from the C-pillar.
- Vulcanisation point -arrow- must not be kinked.
- Install new inner door seal -1- all round.



3.6.2 Removing and installing door seal

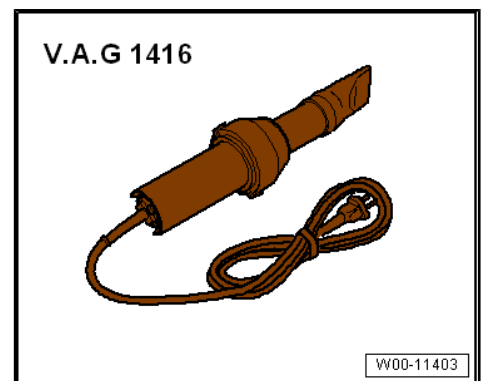
Special tools and workshop equipment required



- ◆ Roller for door insulation foil - 3356-



- ◆ Hot-air blower - V.A.G 1416-



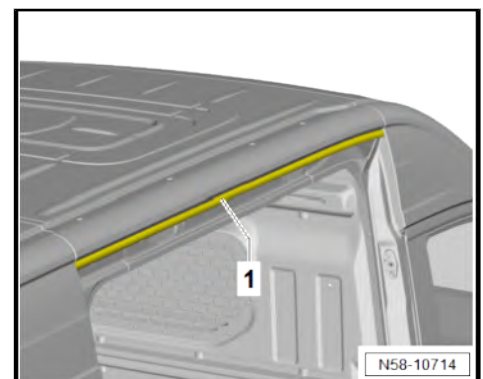
Removing

- Heat pre-seal -1- with hot air blower - V.A.G 1416- . When doing this, pull off pre-seal -1-.

Installing

Install in reverse order of removal, observing the following:

- Clean bonding surface on body using adhesive remover.
- Fit pre-seal -1- on flange of side frame, and position it.
- Press on pre-seal -1- using roller - 3356- .



4 Wing door

⇒ [“4.1 Assembly overview - wing door”, page 162](#)

⇒ [“4.2 Assembly overview - door seals”, page 165](#)

⇒ [“4.3 Removing and installing wing door”, page 165](#)

⇒ [“4.4 Adjusting wing door”, page 166](#)

⇒ [“4.5 Removing and installing hinges”, page 167](#)

⇒ [“4.6 Removing and installing door attachments, wing door”, page 169](#)

⇒ [“4.7 Removing and installing outer door seal”, page 174](#)

⇒ [“4.8 Removing and installing inner door seal”, page 175](#)

4.1 Assembly overview - wing door

⇒ [“4.1.1 Assembly overview - left wing door”, page 162](#)

⇒ [“4.1.2 Assembly overview - right wing door”, page 164](#)

4.1.1 Assembly overview - left wing door

1 - Wing door

- ☐ Removing and installing
⇒ [page 165](#)
- ☐ Adjusting ⇒ [page 108](#)

2 - Hinge

- ☐ Upper and lower
- ☐ Removing and installing
⇒ [page 110](#)

3 - Stop

- ☐ Upper and lower
- ☐ Clipped into hinge

4 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 20 Nm +90°

5 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 20 Nm +90°

6 - Detent mechanism

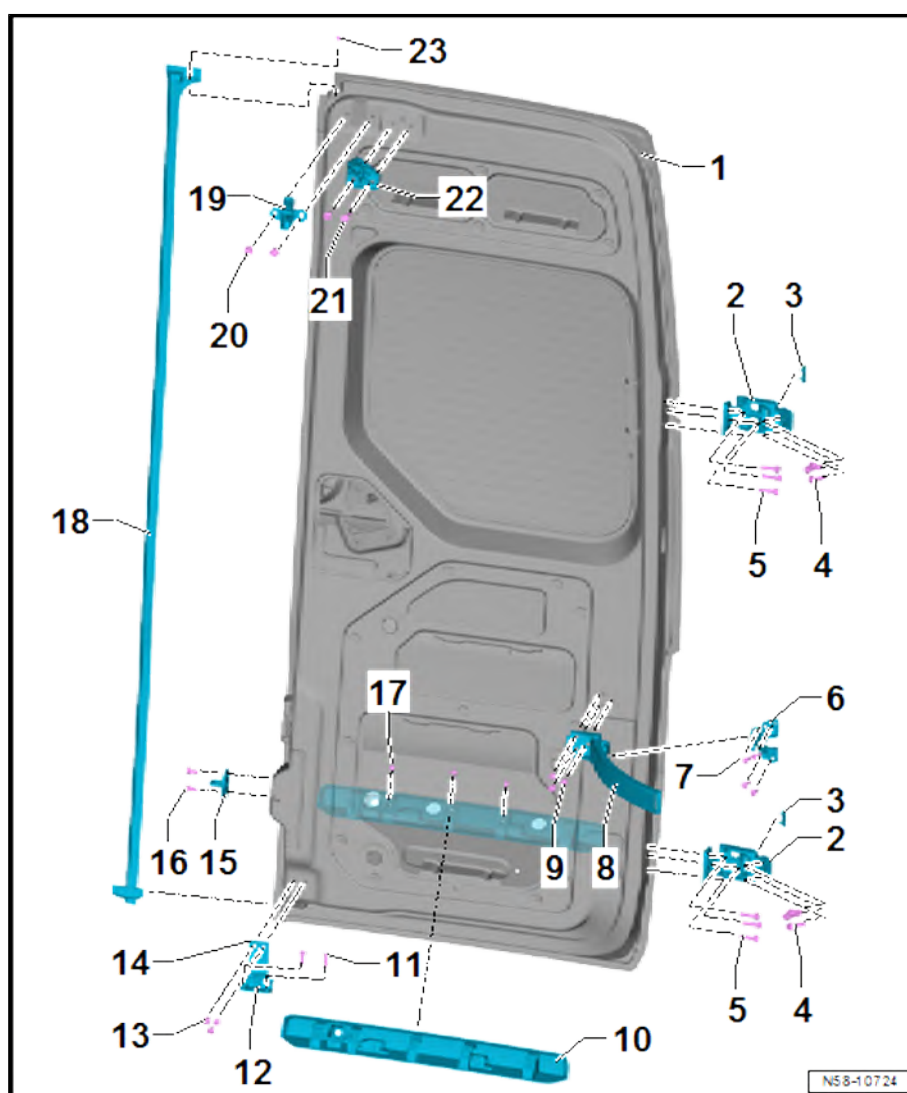
- ☐ Removing and installing
⇒ [page 111](#)

7 - Bolts

- ☐ Qty. 4
- ☐ 8 Nm

8 - Door arrester

- ☐ Various versions with opening angle of 180° or 270°
- ☐ Removing and installing
⇒ [page 111](#)





9 - Bolts

- ☐ Qty. 3
- ☐ 8 Nm

10 - Light strip

- ☐ Removing and installing ⇒ [page 169](#)

11 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

12 - Guide

- ☐ Removing and installing ⇒ [page 169](#)

13 - Bolts

- ☐ Qty. 3
- ☐ 8 Nm

14 - Guide ring

- ☐ Removing and installing ⇒ [page 169](#)

15 - Striker

- ☐ Removing and installing ⇒ [page 128](#)

16 - Bolts

- ☐ Qty. 2
- ☐ 25 Nm

17 - Bolts

- ☐ Qty. 3
- ☐ 2 Nm

18 - Inner door seal

- ☐ Removing and installing

19 - Lock

- ☐ Top
- ☐ Removing and installing ⇒ [page 134](#)

20 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

21 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

22 - Guide wedge

- ☐ Removing and installing ⇒ [page 169](#)

23 - Clip

4.1.2 Assembly overview - right wing door

1 - Right rear wing door

- ☐ Removing and installing
⇒ [page 165](#)
- ☐ Adjusting ⇒ [page 108](#)

2 - Lock

- ☐ Top
- ☐ Removing and installing
⇒ [page 134](#)

3 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

4 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

5 - Guide wedge

- ☐ Removing and installing
⇒ [page 169](#)

6 - Guide ring

- ☐ Removing and installing
⇒ [page 169](#)

7 - Bolts

- ☐ Qty. 3
- ☐ 8 Nm

8 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

9 - Guide

- ☐ Removing and installing
⇒ [page 169](#)

10 - Bolts

- ☐ Renew after removal
- ☐ Qty. 3
- ☐ 20 Nm +90°

11 - Bolts

- ☐ Renew after removal
- ☐ 2 each at top and bottom
- ☐ 20 Nm +90°

12 - Hinge

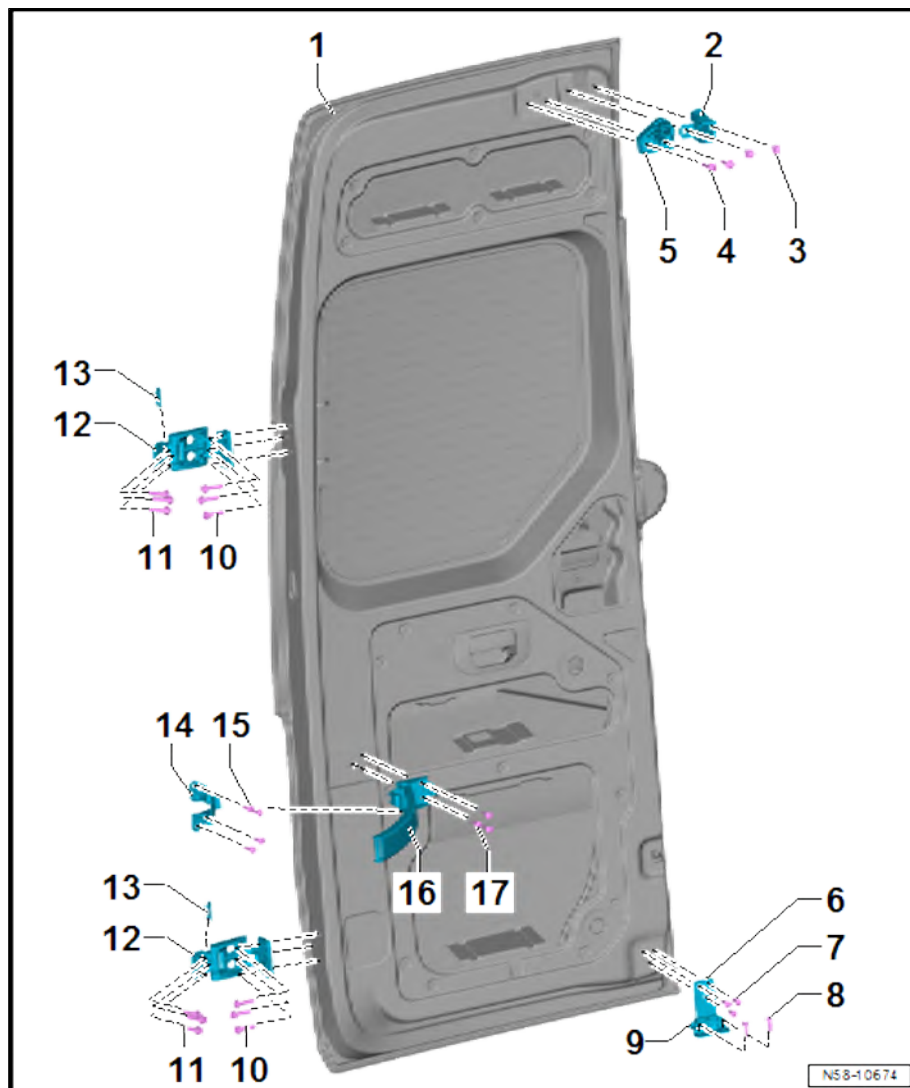
- ☐ Upper and lower
- ☐ Removing and installing ⇒ [page 110](#)

13 - Stop

- ☐ Upper and lower
- ☐ Clipped into hinge

14 - Detent mechanism

- ☐ Removing and installing ⇒ [page 111](#)





15 - Bolts

- ☐ Qty. 4
- ☐ 8 Nm

16 - Door arrester

- ☐ Various versions with opening angle of 180° or 270°
- ☐ Removing and installing ⇒ [page 111](#)

17 - Bolts

- ☐ Qty. 3
- ☐ 8 Nm

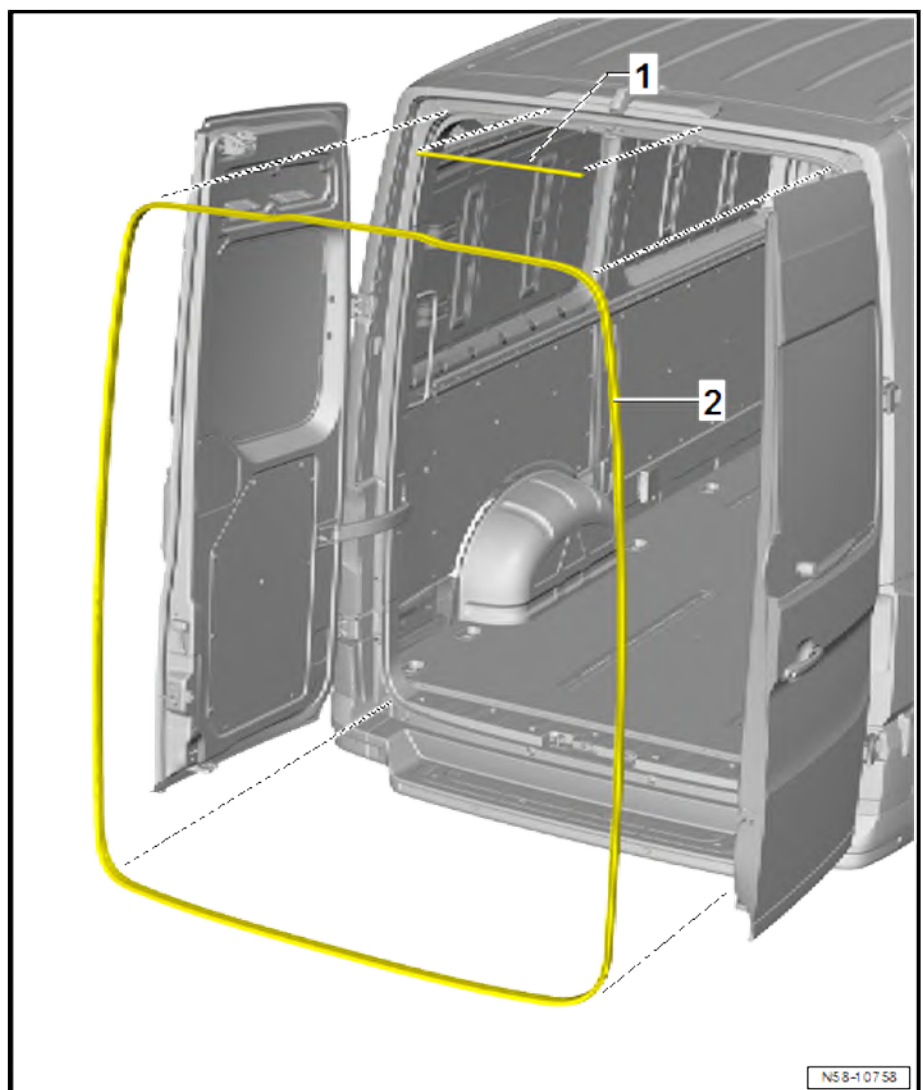
4.2 Assembly overview - door seals

1 - Pre-seal

- ☐ Renew after removal
- ☐ Renewing ⇒ [page 113](#)

2 - Inner door seal

- ☐ Renew after removal
- ☐ Removing and installing
⇒ [page 175](#)



4.3 Removing and installing wing door

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331-



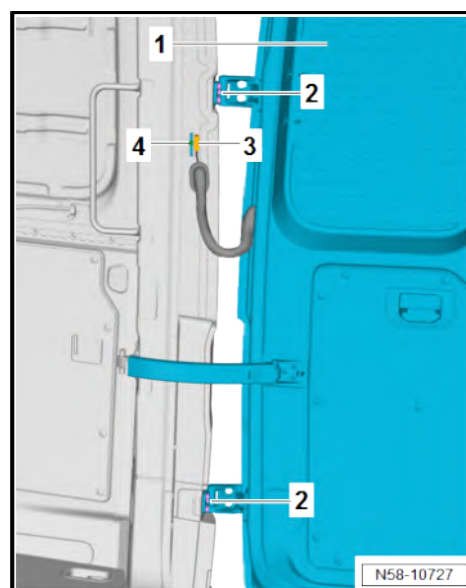
Removing

- Unhook door arrester.
- Disconnect connectors -3-.
- Disconnect hose coupling -4-.
- Loosen bolts -2- at top and bottom.
- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure wing door -1-.
- Unscrew bolts -2-.
- Remove rear wing door -1-.

Installing

Install in reverse order of removal, observing the following:

If fitted, the Trailer Assist camera must be calibrated following accident repairs or upon removal and reinstallation ⇒ Communication; Rep. gr. 91 ; Reversing camera; Calibrating reversing camera .



Specified torques

- ◆ ⇒ [“4.1 Assembly overview - wing door”, page 162](#)

4.4 Adjusting wing door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





◆ Setting gauge - 3371-

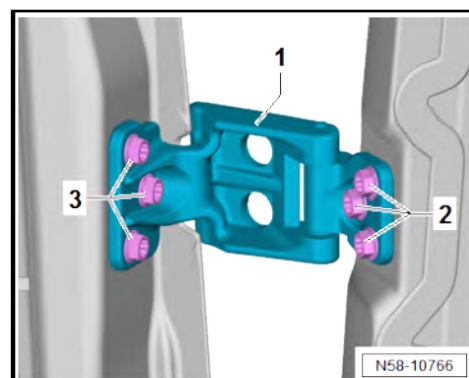


Procedure

- Vehicle must be standing on all wheels.

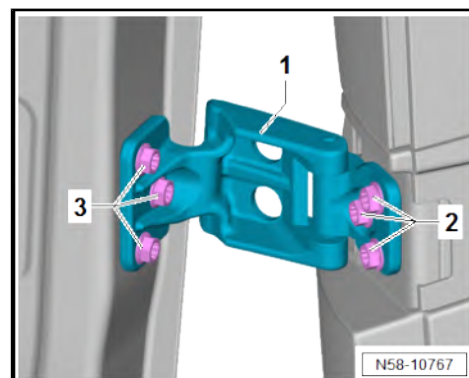
Adjusting top door hinge

- Loosen bolts -2- on hinge -1-.
- Loosen bolts -3- on hinge -1-.
- Adjust door hinge -1-.



Adjusting bottom door hinge

- Loosen bolts -2- on hinge -1-.
- Loosen bolts -3- on hinge -1-.
- Adjust door hinge -1-.
- Following assembly or adjustment work, perform corrosion protection measures on hinges and bolts.
- Use the setting gauge - 3371- to check or adjust the gap dimensions ➔ Body Repairs; Rep. gr. 00 ; Technical data; Body gaps/shut lines; Front body .



Specified torques

- ◆ ➔ [“4.1 Assembly overview - wing door”, page 162](#)

4.5 Removing and installing hinges

➔ [“4.5.1 Removing and installing hinges, bottom”, page 167](#)

➔ [“4.5.2 Removing and installing hinges, top”, page 168](#)

4.5.1 Removing and installing hinges, bottom

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331-



Removing

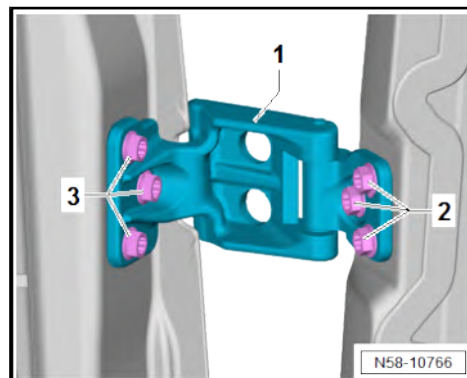
- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure wing door.
- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Pull off hinge -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“4.1 Assembly overview - wing door”, page 162](#)



4.5.2 Removing and installing hinges, top

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure wing door.



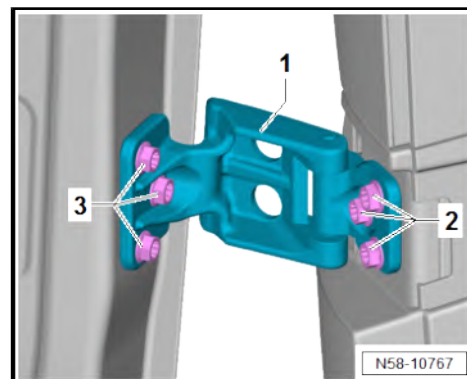
- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Pull off hinge -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“4.1 Assembly overview - wing door”, page 162](#)



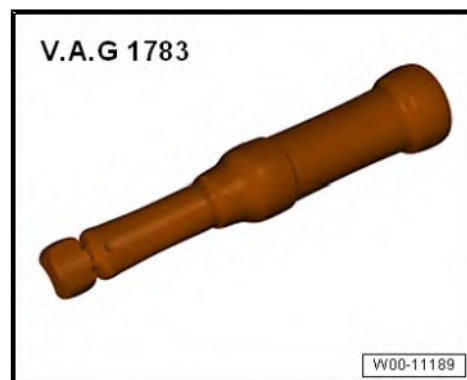
4.6 Removing and installing door attachments, wing door

- ⇒ [“4.6.1 Removing and installing light strip”, page 169](#)
- ⇒ [“4.6.2 Removing and installing guide wedge”, page 170](#)
- ⇒ [“4.6.3 Removing and installing locking element and upper centring pin”, page 170](#)
- ⇒ [“4.6.4 Removing and installing guide ring”, page 171](#)
- ⇒ [“4.6.5 Removing and installing guide”, page 171](#)
- ⇒ [“4.6.6 Removing and installing stop”, page 172](#)
- ⇒ [“4.6.7 Removing and installing magnetic stop”, page 173](#)
- ⇒ [“4.6.8 Removing magnetic stop bracket”, page 173](#)

4.6.1 Removing and installing light strip

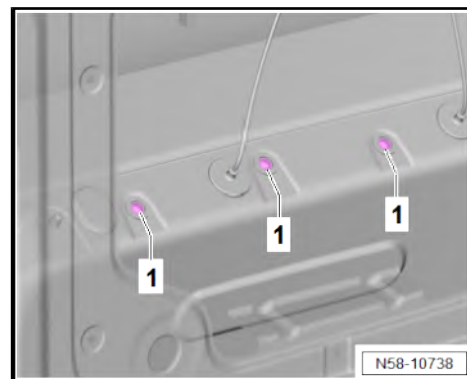
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove trim for wing door ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Remove number plate lights ⇒ Electrical system; Rep. gr. 94 ; Number plate light; Removing and installing number plate light -X4/-X5-, wing door .
- Unscrew bolts -1-.





- Remove electrical connector -2- from light strip -1-.
- Remove light strip -1-.

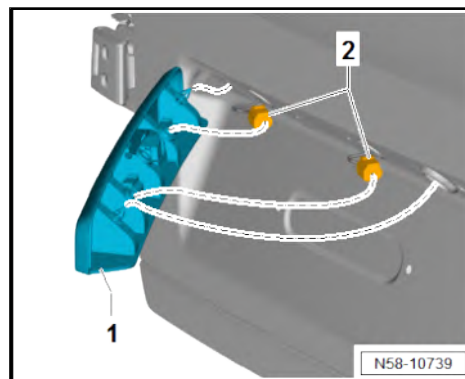
Installing

Install in reverse order of removal, observing the following:

- If present, calibrate reversing camera ⇒ Communication; Rep. gr. 91 ; Reversing camera system; Calibrating reversing camera system .

Specified torques

- ♦ ⇒ [“4.1 Assembly overview - wing door”, page 162](#)



4.6.2 Removing and installing guide wedge

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-



Removing

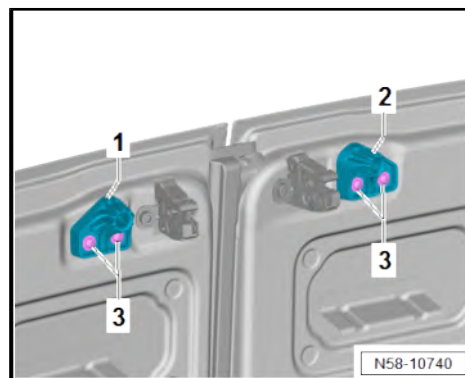
- Unscrew bolts -3-. When doing this, remove guide wedges -1- and -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [“4.1 Assembly overview - wing door”, page 162](#)



4.6.3 Removing and installing locking element and upper centring pin

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-





Removing

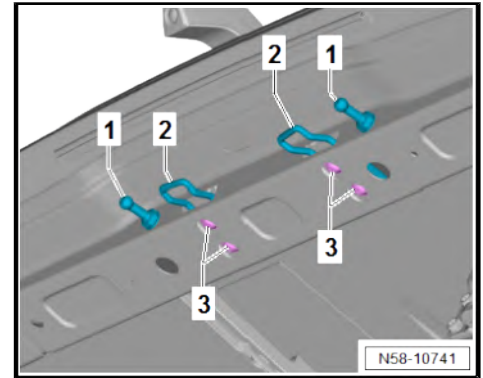
- Remove moulded headliner in passenger compartment ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .
- Unscrew bolts -3-.
- Remove upper end plate -2- on left or right.
- Unscrew guide pin -1- on left and right.

Installing

Install in reverse order of removal, observing the following:

Specified torques

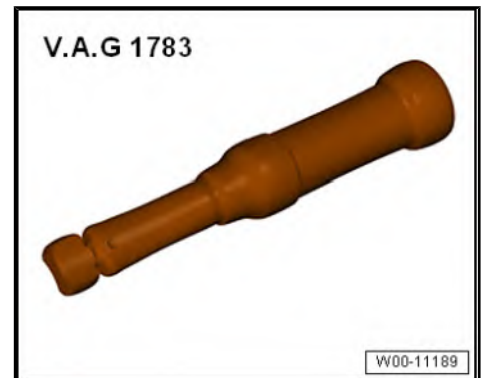
- ◆ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)



4.6.4 Removing and installing guide ring

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

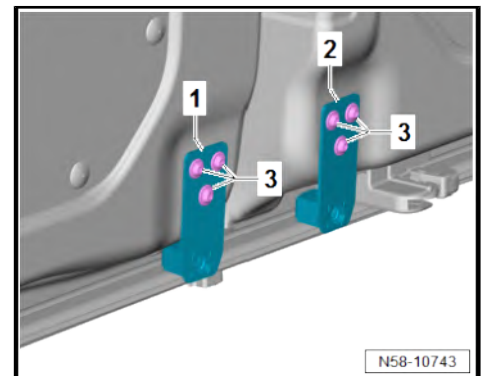
- Unscrew bolts -3-.
- Remove right guide ring -1- or left guide ring -2- from wing door.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“4.1 Assembly overview - wing door”, page 162](#)



4.6.5 Removing and installing guide

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

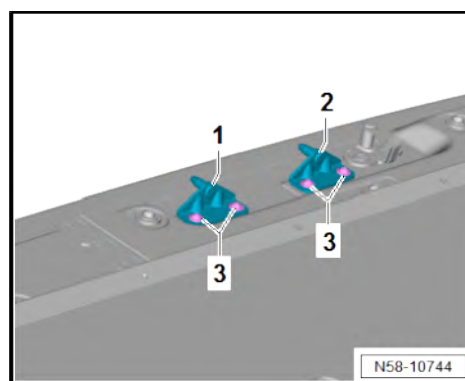
- Unscrew bolts -3-.
- Remove right guide -1- or left guide -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - door handle and door lock”, page 114](#)



4.6.6 Removing and installing stop

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





Removing

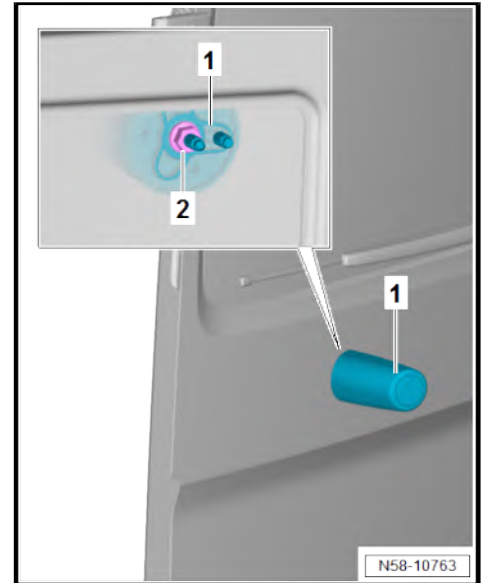
- Remove trim for wing door ⇒ General body repairs, interior; Rep. gr. 70 ; Rear door trims; Removing and installing rear door trim .
- Unscrew nut -2-.
- Remove stop -1- from wing door.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Stop	20 Nm



4.6.7 Removing and installing magnetic stop

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

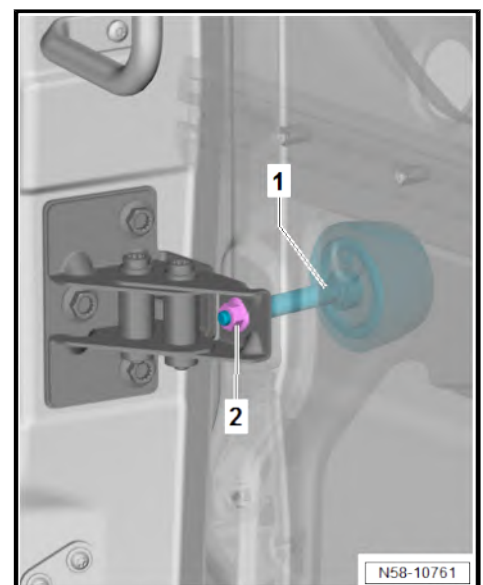
- Unscrew nut -2-.
- Remove magnetic stop -1- from outside.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Magnetic stop	20 Nm



4.6.8 Removing magnetic stop bracket

Special tools and workshop equipment required



◆ Torque wrench - V.A.G 1331-



Removing

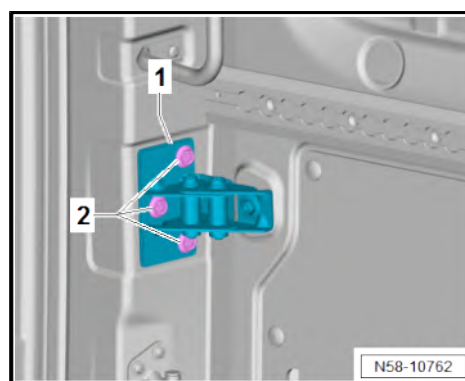
- Remove magnetic stop ⇒ [page 173](#) .
- Unscrew bolts -2-.
- Remove magnetic stop bracket -1- from outside.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Magnetic stop bracket	20 Nm



4.7 Removing and installing outer door seal

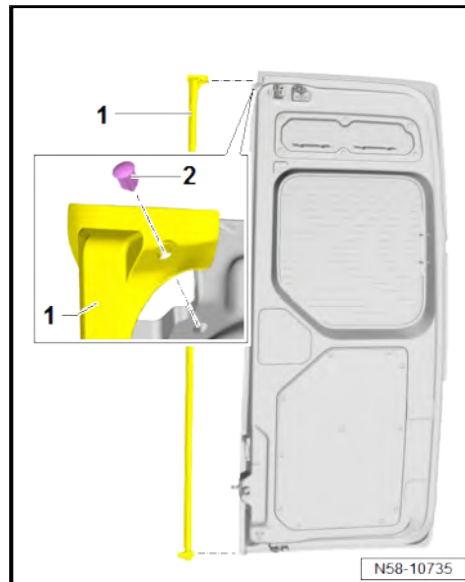
Removing

- Unclip clip -2-.
- Pull outer door seal -1- off wing door.

Installing

Install in reverse order of removal, observing the following:

- Fit moulding of outer door seal -1- at top and bottom and align.
- Secure moulding at top and press in clip -2-.
- Firmly press on outer door seal -1- over whole wing door.





4.8 Removing and installing inner door seal

⇒ ["4.8.1 Removing and installing inner door seal", page 175](#)

⇒ ["4.8.2 Removing and installing door seal", page 175](#)

4.8.1 Removing and installing inner door seal

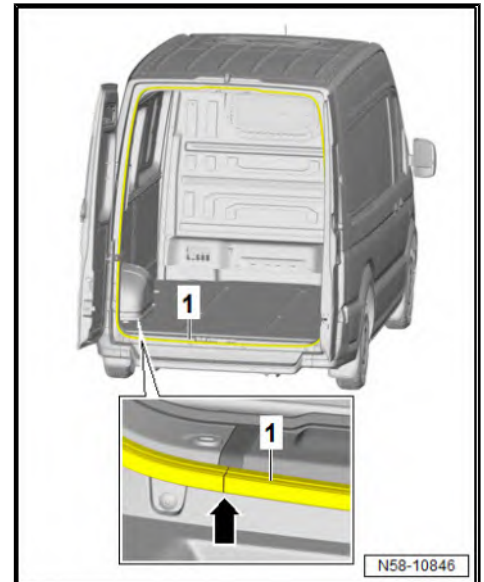
Removing

- Pull inner door seal -1- off body flange.
- Remove lock carrier trim ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing lock carrier trim .

Installing

Install in reverse order of removal, observing the following:

- Fit new inner door seal -1- with vulcanisation point -arrow- being in the area shown.
- Vulcanisation point -arrow- must not be kinked.
- Install inner door seal -1- all round.



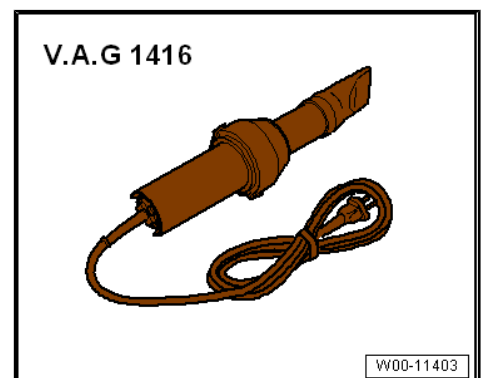
4.8.2 Removing and installing door seal

Special tools and workshop equipment required

- ◆ Roller for door insulation foil - 3356-



- ◆ Hot-air blower - V.A.G 1416-





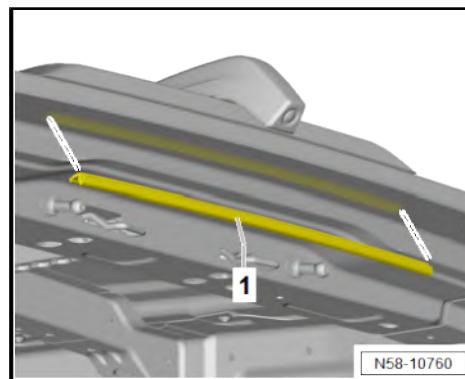
Removing

- Heat pre-seal -1- with hot air blower - V.A.G 1416- . When doing this, pull off pre-seal -1-.

Installing

Install in reverse order of removal, observing the following:

- Clean bonding surface on body using adhesive remover.
- Fit pre-seal -1- to roof frame, and position it.
- Press on pre-seal -1- using roller - 3356- .





63 – Bumpers

1 Bumper, front

⇒ [“1.1 Assembly overview - bumper cover”, page 177](#)

⇒ [“1.2 Removing and installing bumper cover”, page 179](#)

⇒ [“1.3 Removing and installing impact bar”, page 182](#)

⇒ [“1.4 Removing and installing add-on parts”, page 184](#)

1.1 Assembly overview - bumper cover

⇒ [“1.1.1 Assembly overview - bumper cover”, page 177](#)

⇒ [“1.1.2 Assembly overview - bumper cover, e-Crafter or e-TGE”, page 178](#)

⇒ [“1.1.3 Assembly overview - bumper cover attachments”, page 179](#)

1.1.1 Assembly overview - bumper cover

1 - Bumper cover

- ☐ Removing and installing
⇒ [page 179](#)

2 - Bolts

- ☐ Qty. 2
- ☐ 3 Nm

3 - Bolts

- ☐ Qty. 6
- ☐ 2 Nm

4 - Guide

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 184](#)

5 - Bolts

- ☐ Qty. 2
- ☐ 3 Nm

6 - Bolts

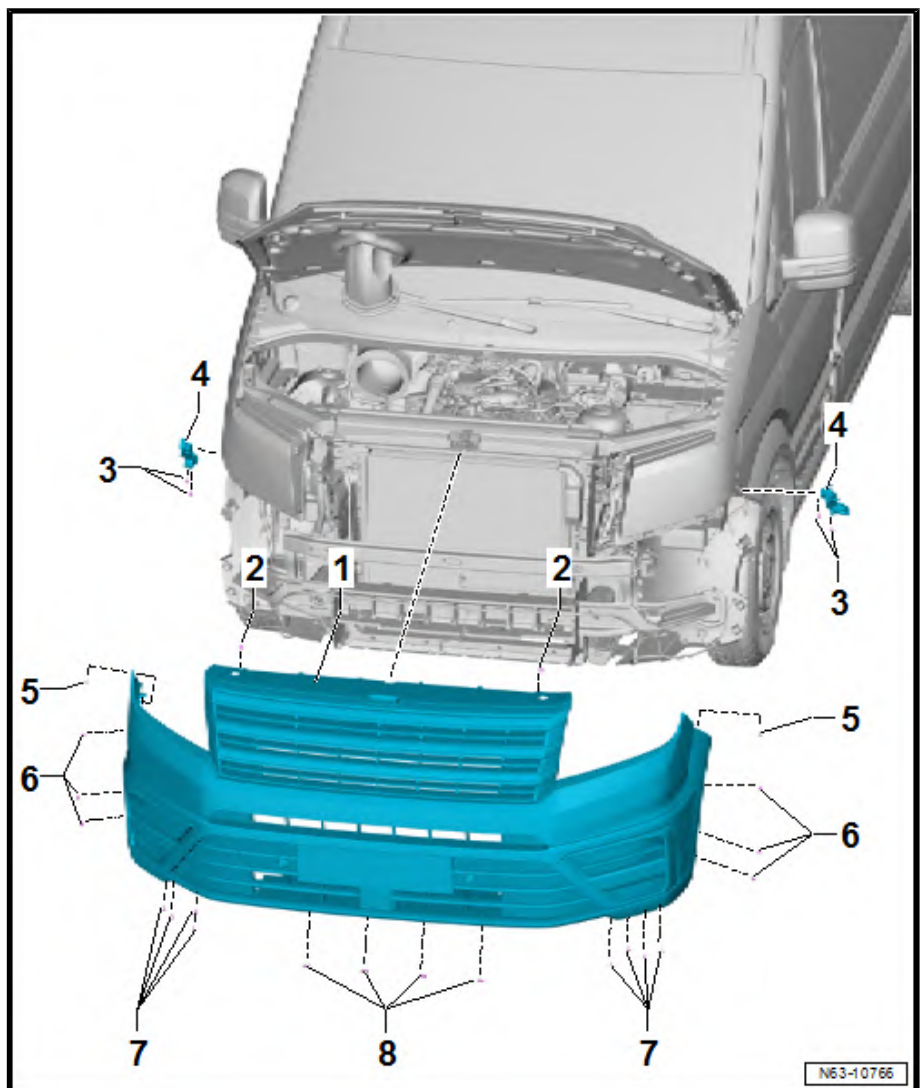
- ☐ Qty. 6
- ☐ 3 Nm

7 - Bolts

- ☐ Qty. 8
- ☐ 3 Nm

8 - Spreader rivet

- ☐ Qty. 4





1.1.2 Assembly overview - bumper cover, e-Crafter or e-TGE

1 - Guide

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 184](#)

2 - Bolts

- ☐ Qty. 6
- ☐ 3 Nm

3 - Bolts

- ☐ Qty. 2
- ☐ Specified torque
⇒ [page 12](#)

4 - Impact bar

- ☐ Removing and installing
⇒ [page 182](#)

5 - Bolts

- ☐ Qty. 2
- ☐ Specified torque
⇒ [page 12](#)

6 - Bolts

- ☐ Qty. 22
- ☐ 3 Nm

7 - Bumper cover

- ☐ Removing and installing
⇒ [page 179](#)

8 - Spreader rivet

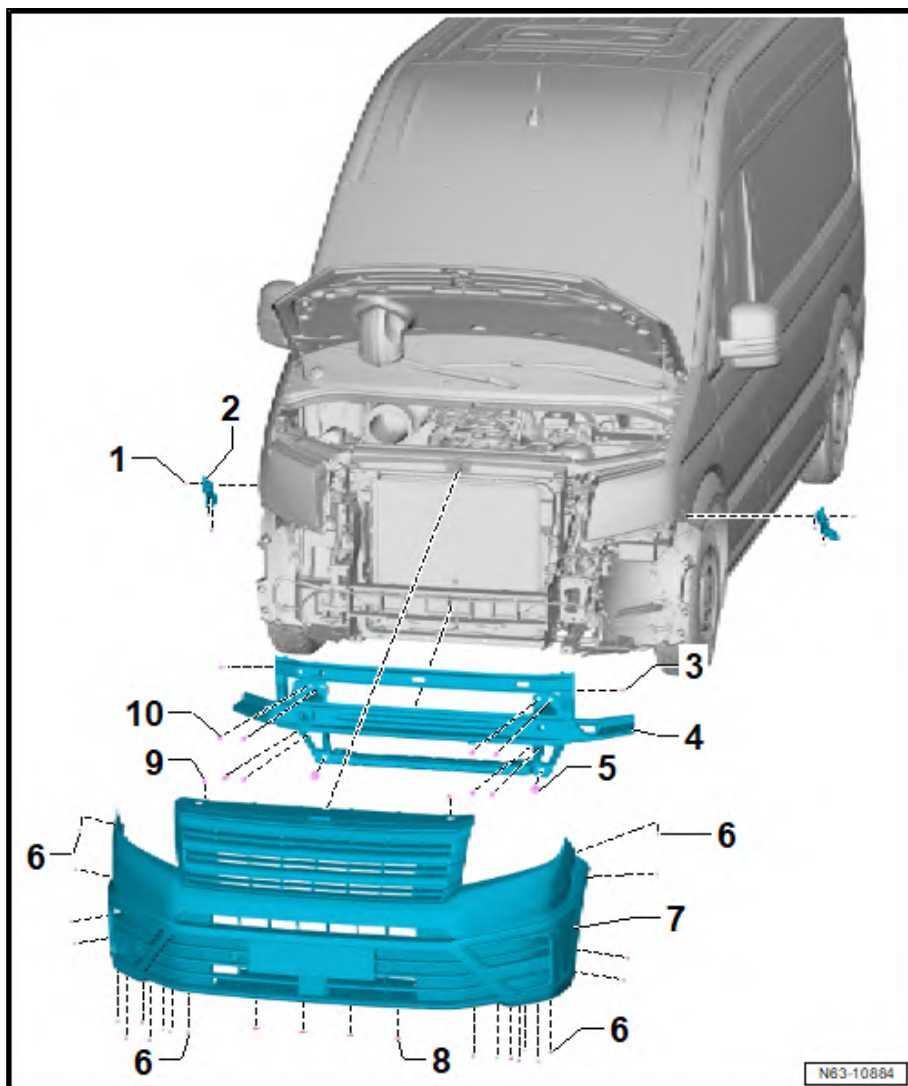
- ☐ Qty. 4

9 - Bolts

- ☐ Qty. 2
- ☐ 3 Nm

10 - Bolts

- ☐ Qty. 8
- ☐ Specified torque ⇒ [page 12](#)





1.1.3 Assembly overview - bumper cover attachments

1 - Bumper cover

- ☐ Removing and installing
⇒ [page 179](#)

2 - Bolts

- ☐ Qty. 2
- ☐ 2 Nm

3 - Support members

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 185](#)

4 - Steps

- ☐ Qty. 2

5 - Covers

- ☐ Qty. 2

6 - Trim

- ☐ Qty. 2
- ☐ For cover

7 - Guide frame

- ☐ Qty. 2
- ☐ Depending on equipment
- ☐ For headlight washer system

8 - Trim

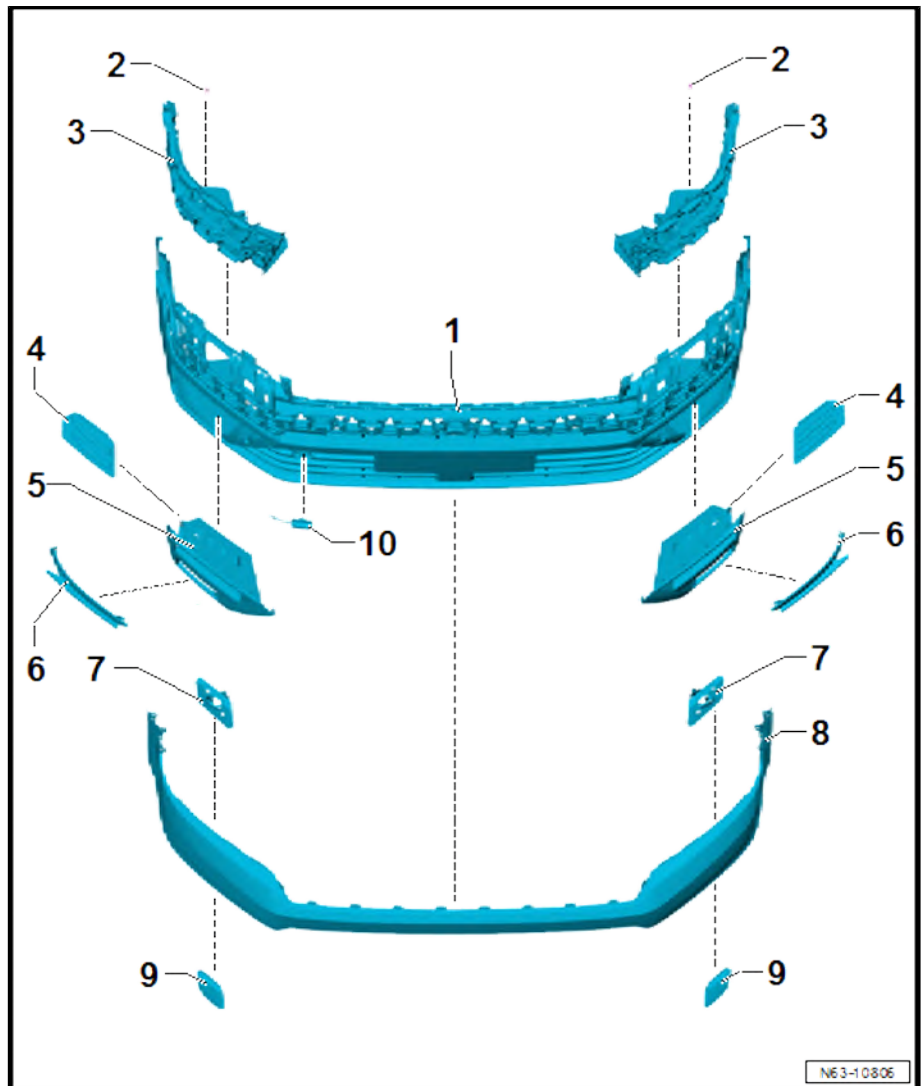
- ☐ For bumper cover
- ☐ Removing and installing
⇒ [page 185](#)

9 - Caps

- ☐ Qty. 2
- ☐ Depending on equipment
- ☐ For headlight washer system

10 - Cover

- ☐ For towing eye.



1.2 Removing and installing bumper cover

Special tools and workshop equipment required



◆ Setting gauge - 3371-

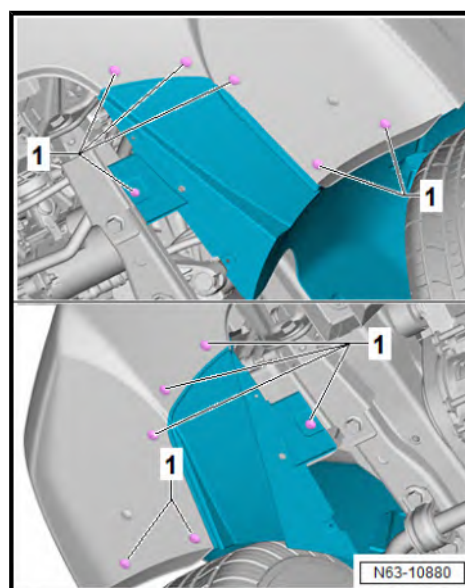


◆ Torque wrench - V.A.G 1783-



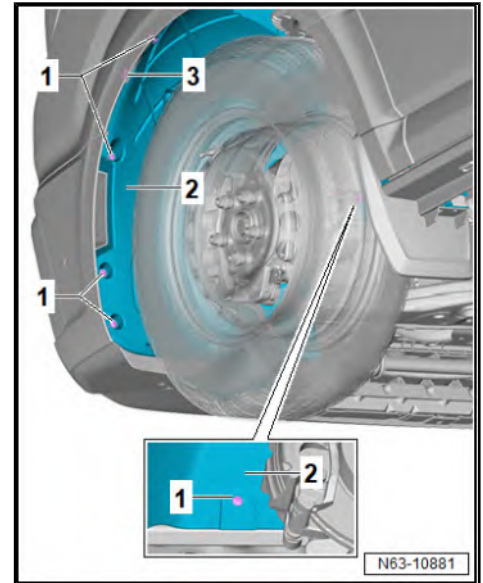
Removing

- If fitted, remove noise insulation ⇒ [page 271](#) .
- Unscrew bolts -1-.

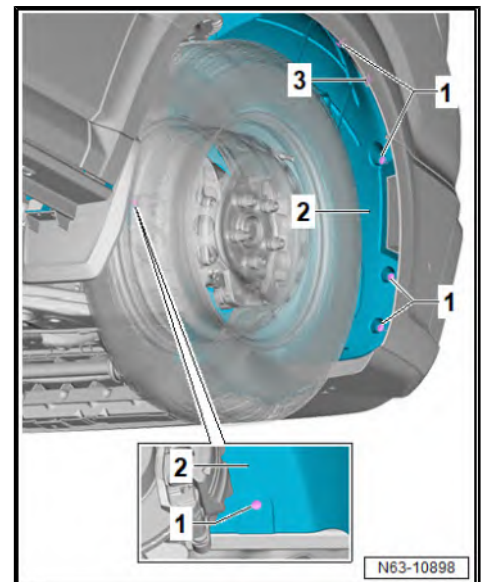




- Unscrew bolts -1-.
- Fold back wheel housing liner -2-. When doing this, unscrew bolt -3-.



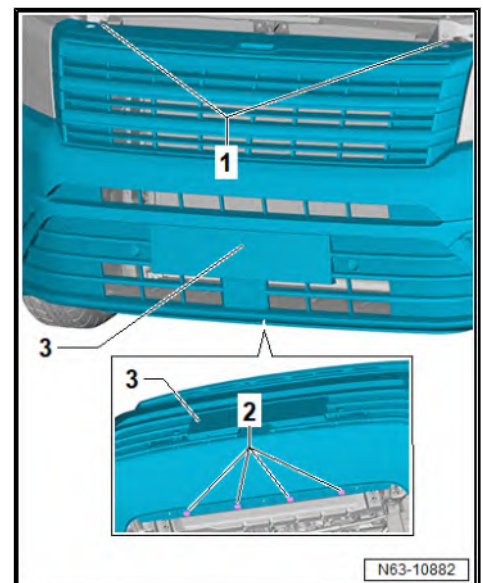
- Unscrew bolts -1-.
- Fold back wheel housing liner -2-. When doing this, unscrew bolt -3-.



- Unscrew bolts -1-.
- Unscrew spreader rivets -2-.
- The following work steps must be performed by a second mechanic.
- Pull bumper cover -3- together with second mechanic in parallel out of fasteners on wings.
- Have a second mechanic secure bumper cover -3-.

Vehicles with headlight washer system

- Place a suitable clean container under washer fluid reservoir.
- Release hose for washer fluid on washer fluid reservoir. When doing this, pull it off washer fluid reservoir.





Vehicles with adaptive cruise control

- Disconnect electrical connector -1-.

Vehicles with parking aid and/or fog lights

- Disconnect electrical connector -2-.

Continued for all vehicles

- Remove bumper cover.

Installing

Install in reverse order of removal, observing the following:

Vehicles with adaptive cruise control

- Calibrate adaptive cruise control ⇒ Running gear, axles, steering; Rep. gr. 44 ; Adaptive cruise control .

Continued for all vehicles

- Use setting gauge - 3371- to check gaps and height difference; adjust as necessary ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .

Specified torques

- ◆ ⇒ [“1.1 Assembly overview - bumper cover”, page 177](#)

1.3 Removing and installing impact bar

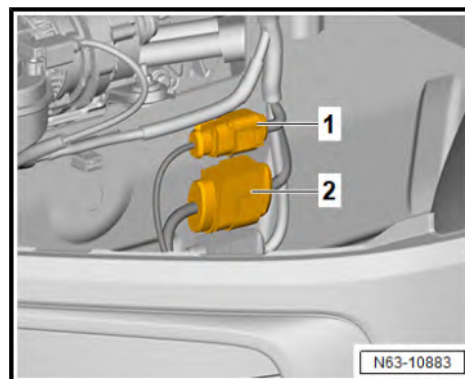
⇒ [“1.3.1 Removing and installing impact bar”, page 182](#)

⇒ [“1.3.2 Removing and installing foam filler piece”, page 184](#)

1.3.1 Removing and installing impact bar

Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



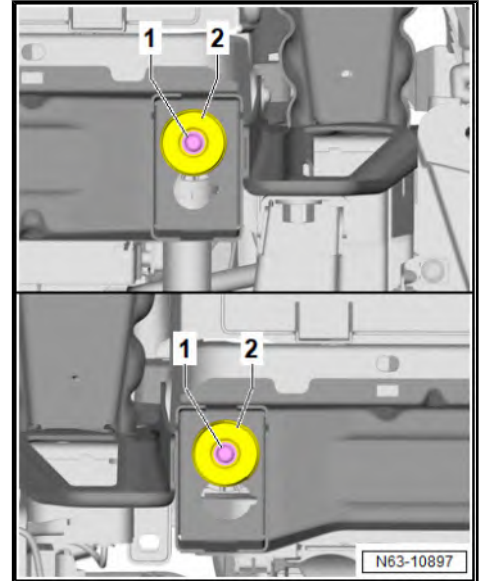
- ◆ Torque wrench - V.A.G 1332-



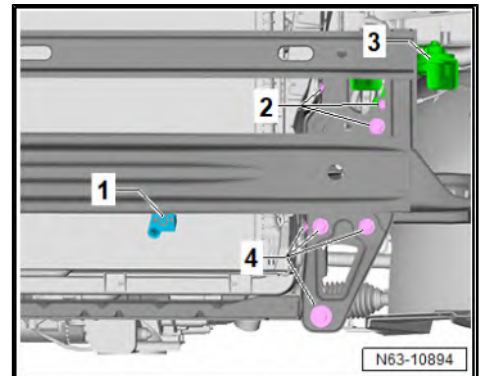


Removing

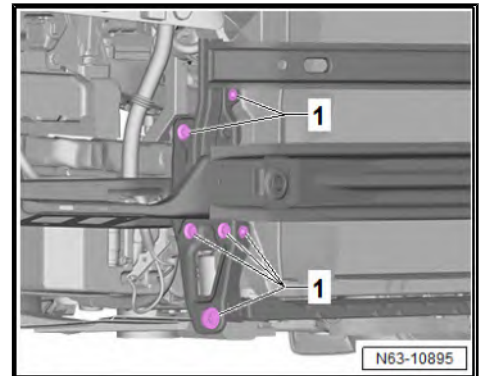
- Remove foam filler piece ➤ [page 182](#) .
- Unscrew bolts -1-. When doing this, remove rubber mounting -2-.



- Use removal wedge - 3409- to release locking lug on retainer -1- for ambient temperature sensor - G17- . When doing this, remove retainer -1- downwards, and lay it aside.
- Unscrew bolt -2- When doing this, remove retainer -3- together with treble horn - H2- and bass horn - H7- , and lay it aside.
- Unscrew bolts -4-.



- Unscrew bolts -1-.
- The following work steps must be performed by a second mechanic.





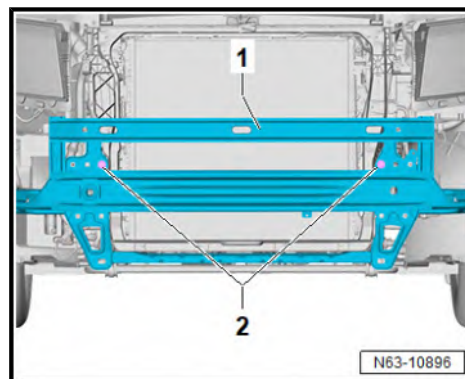
- Have a second mechanic secure impact bar -1-.
- Unscrew bolts -2-. When doing this, have a second mechanic remove impact bar -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["1.1 Assembly overview - lock carrier", page 10](#)
- ◆ Radiator/radiator fan; Assembly overview - radiator/radiator fan ⇒ Rep. gr. 19 ; Radiator/radiator fan; Assembly overview - radiator/radiator fan
- ◆ Horn; Assembly overview - horn ⇒ Electrical system; Rep. gr. 90 ; Horn; Assembly overview - horn



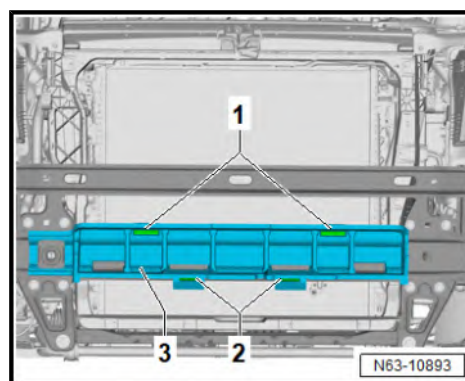
1.3.2 Removing and installing foam filler piece

Removing

- Remove bumper cover ⇒ [page 179](#) .
- Pull fasteners -1- upwards.
- Pull fasteners -2- downwards. When doing this, remove foam filler piece -3-.

Installing

Install in reverse order.



1.4 Removing and installing add-on parts

⇒ ["1.4.1 Removing and installing guide", page 184](#)

⇒ ["1.4.2 Removing and installing reinforcement", page 185](#)

⇒ ["1.4.3 Removing and installing trim", page 186](#)

1.4.1 Removing and installing guide

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

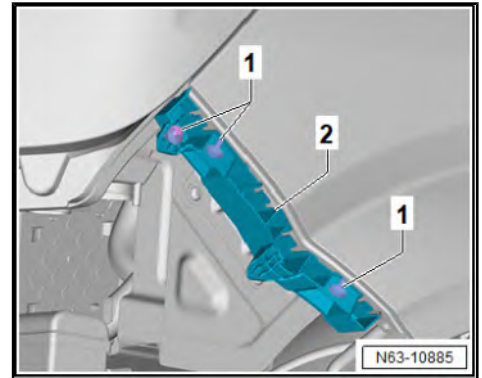
- Remove bumper cover ➔ [page 179](#) .
- Unscrew bolts -1-.
- Unclip guide -2- from fasteners in wing.

Installing

Install in reverse order of removal, observing the following:

Specified torques

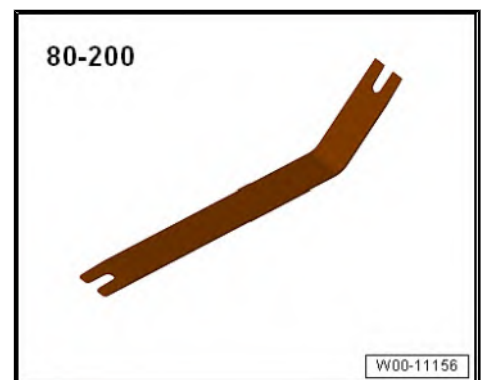
- ◆ ➔ [“1.1 Assembly overview - bumper cover”, page 177](#)



1.4.2 Removing and installing reinforcement

Special tools and workshop equipment required

- ◆ Removal lever - 80-200-



Removing

Vehicles with fog light

- Remove fog light on relevant side ➔ Electrical system; Rep. gr. 94 ; Fog lights; Removing and installing fog light .

Continued for all vehicles

- Remove bumper cover ➔ [page 179](#) .

Vehicles with park assist steering

- Disconnect electrical connector for front left sender for park assist steering on left side of vehicle - G568- or front right sender for park assist steering on right side of vehicle - G569- .

Vehicles with parking aid

- Disconnect electrical connector for front right parking aid sender - G252- or front left parking aid sender - G255- .

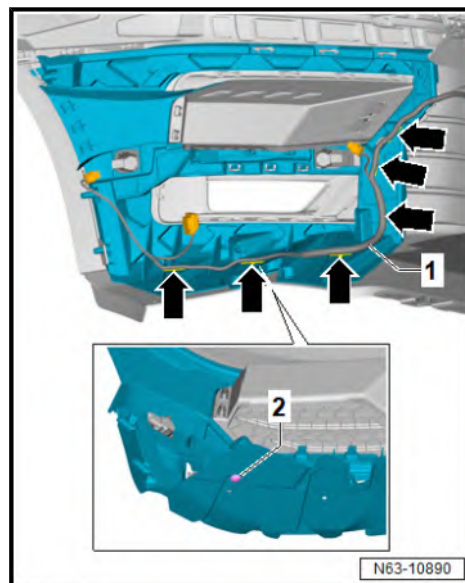


Vehicles with parking aid and/or fog lights

- Use removal lever - 80-200- to unclip wiring harness -1- in area marked with -arrows-.
- Move wiring harness -1- aside.
- Unscrew bolt -2-

Vehicles with headlight washer system, if right support is removed

- Unclip hose for washer fluid of headlight washer system from support.

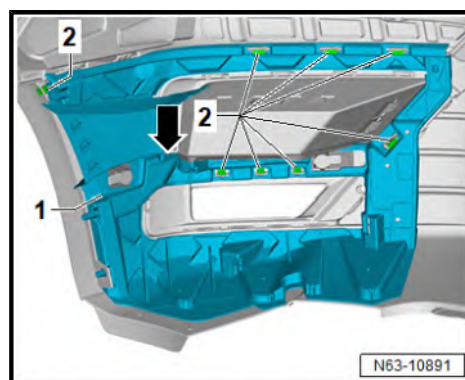


Continued for all vehicles

- Unclip support -1- at points -2-. When doing this, hold support -1- under tension.
- Push support -1- downwards in area marked with -arrow-. When doing this swing it out towards outer side.

Installing

Install in reverse order.



1.4.3 Removing and installing trim

Special tools and workshop equipment required

- ♦ Removal wedge - 3409-



Removing

- Remove supports ⇒ [page 185](#) .



Vehicles with headlight washer system

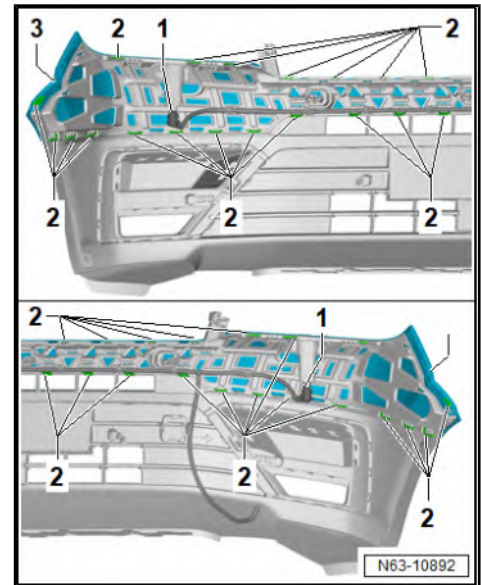
- Remove hoses for washer fluid -1- from pop-up cylinders.

Continued for all vehicles

- The following work steps must be performed by a second mechanic.
- Have a second mechanic secure bumper cover.
- Use removal wedge - 3409- to release fasteners -2-. When doing this, hold trim -3- under tension.
- Remove trim -3-.

Installing

Install in reverse order.



2 Bumper, rear

⇒ [“2.1 Assembly overview - bumper cover”, page 188](#)

⇒ [“2.2 Assembly overview - impact bar”, page 190](#)

⇒ [“2.3 Removing and installing bumper cover”, page 192](#)

⇒ [“2.4 Removing and installing impact bar”, page 196](#)

⇒ [“2.5 Removing and installing add-on parts”, page 197](#)

⇒ [“2.6 Installing holder for parking aid sender”, page 199](#)

2.1 Assembly overview - bumper cover

⇒ [“2.1.1 Assembly overview - bumper cover”, page 188](#)

⇒ [“2.1.2 Assembly overview - bumper cover, with step”, page 189](#)

2.1.1 Assembly overview - bumper cover

1 - Guide profiles

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 197](#)

2 - Bolts

- ☐ Qty. 10
- ☐ 2 Nm

3 - Trim

- ☐ End piece
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 197](#)

4 - Bolts

- ☐ Qty. 2
- ☐ 2 Nm

5 - End pieces

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 192](#)

6 - Bolts

- ☐ Qty. 4
- ☐ 2 Nm

7 - Bolts

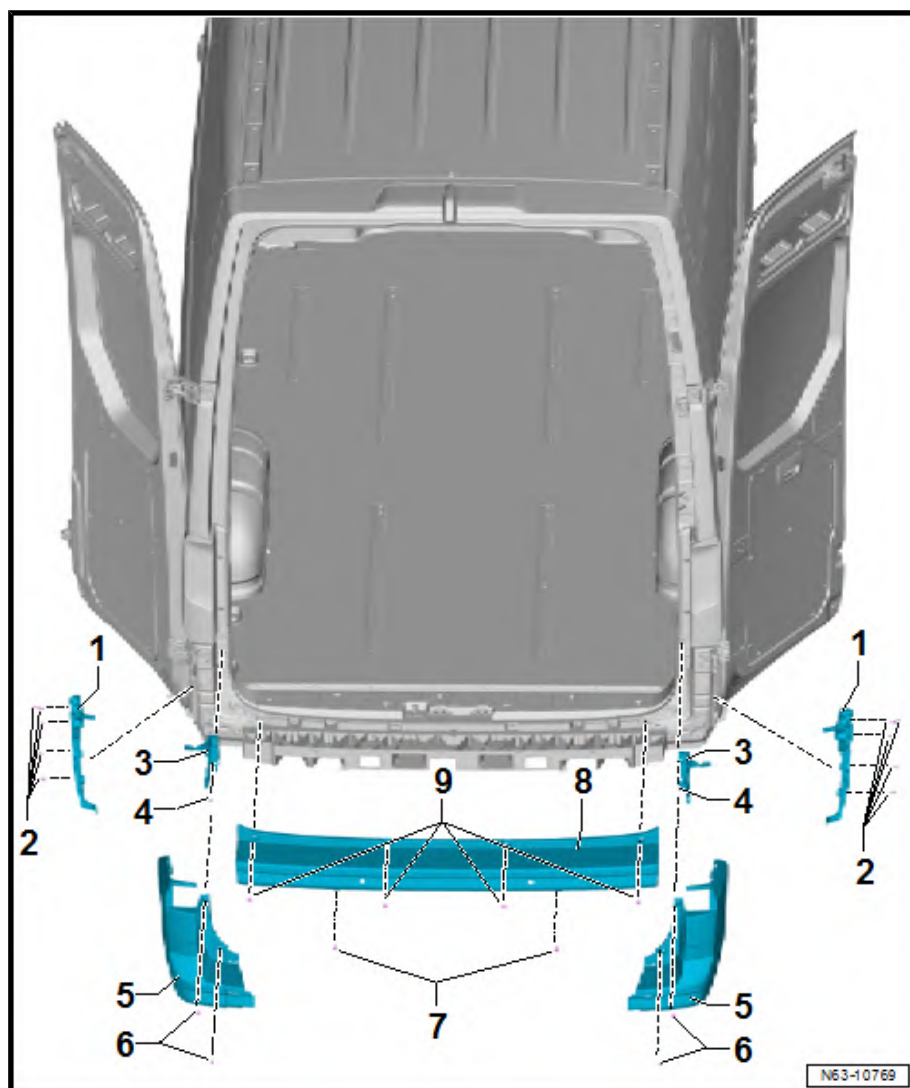
- ☐ Qty. 4
- ☐ 2 Nm

8 - Bumper cover

- ☐ Removing and installing
⇒ [page 192](#)

9 - Bolts

- ☐ Qty. 4
- ☐ 8 Nm





2.1.2 Assembly overview - bumper cover, with step

1 - Guide profiles

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 197](#)

2 - Bolts

- ☐ Qty. 10
- ☐ 2 Nm

3 - Trim

- ☐ End piece
- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 197](#)

4 - Bolts

- ☐ Qty. 2
- ☐ 2 Nm

5 - End pieces

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 192](#)

6 - Bolts

- ☐ Qty. 4
- ☐ 2 Nm

7 - Bolts

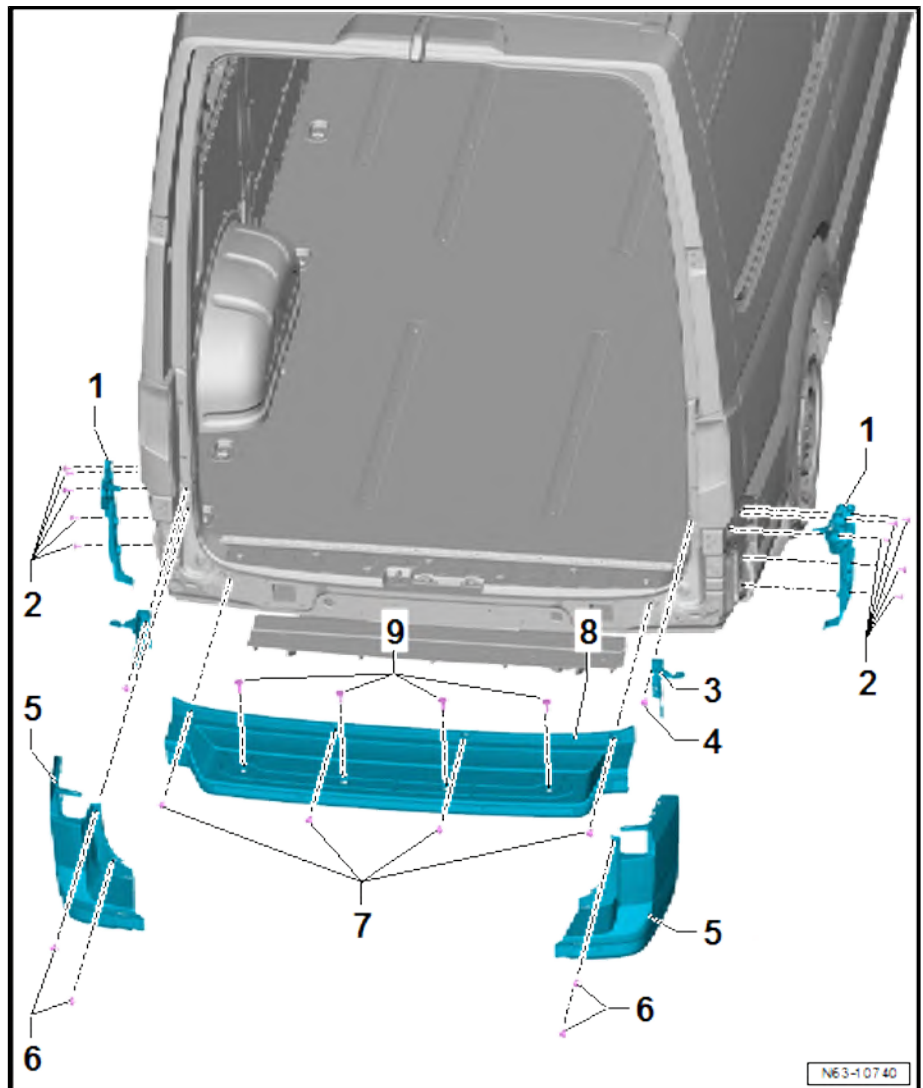
- ☐ Qty. 4
- ☐ 2 Nm

8 - Bumper cover

- ☐ Removing and installing
⇒ [page 192](#)

9 - Bolts

- ☐ Qty. 4
- ☐ 8 Nm





2.2 Assembly overview - impact bar

⇒ [“2.2.1 Assembly overview - impact bar”, page 190](#)

⇒ [“2.2.2 Assembly overview - step”, page 191](#)

⇒ [“2.2.3 Assembly overview – towing eye bracket”, page 192](#)

2.2.1 Assembly overview - impact bar

1 - Impact bar

- ☐ Removing and installing
⇒ [page 196](#)

2 - Bolts

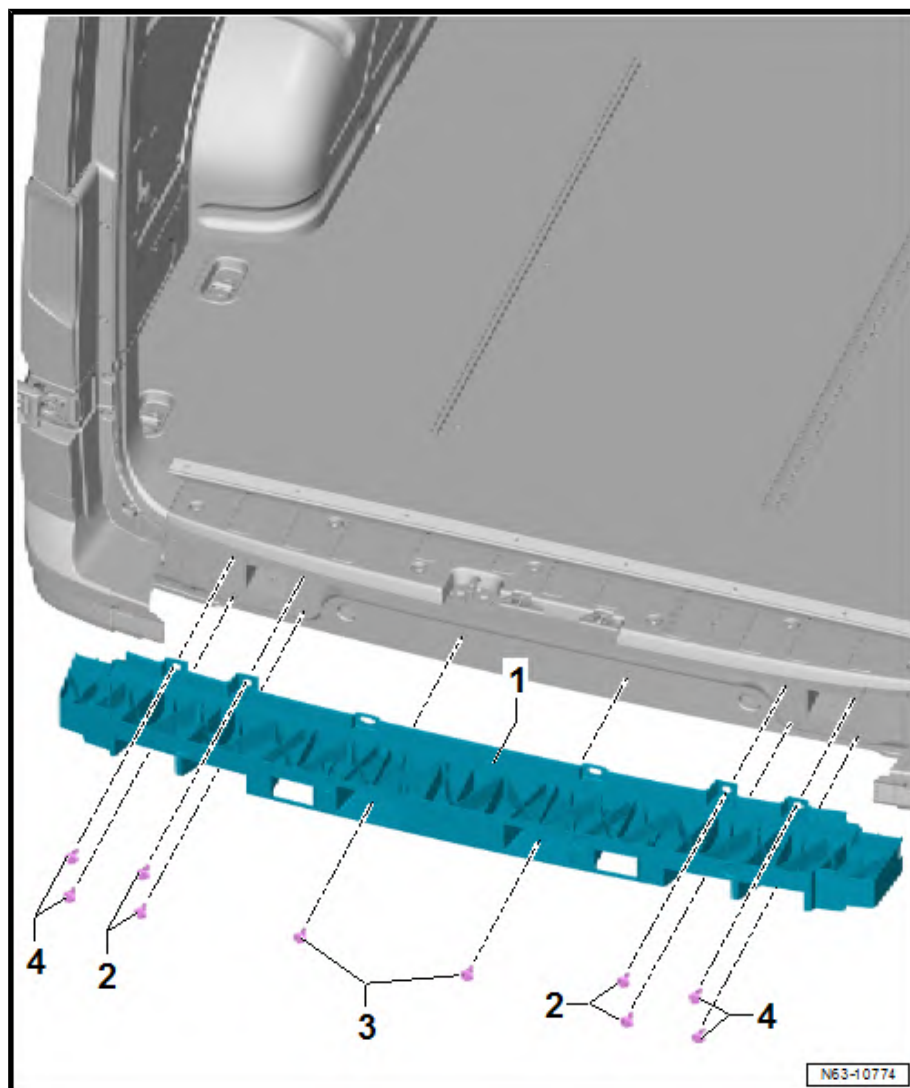
- ☐ Qty. 4
- ☐ For twin tyres
- ☐ 20 Nm

3 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

4 - Bolts

- ☐ Qty. 4
- ☐ For single tyres
- ☐ 20 Nm



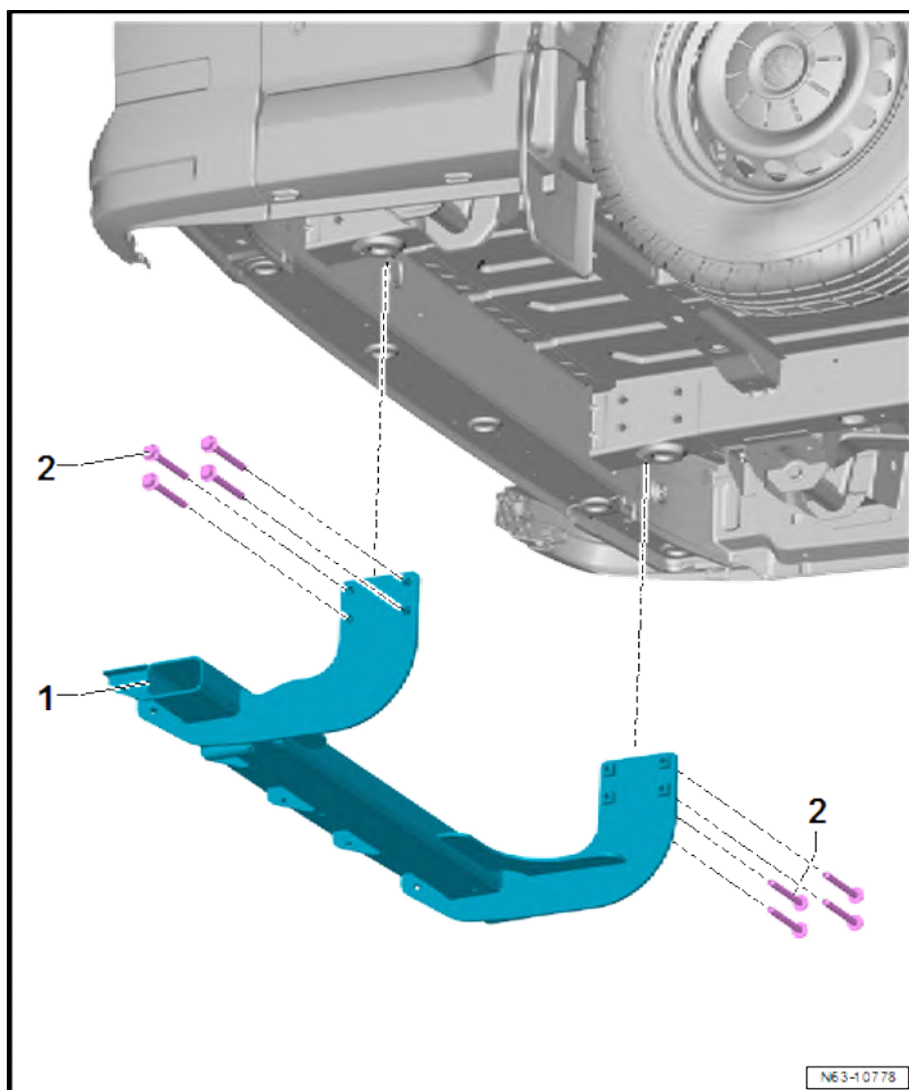
2.2.2 Assembly overview - step

1 - Step

- ☐ Removing and installing
⇒ [page 196](#)

2 - Bolts

- ☐ Renew after removal
- ☐ Qty. 8
- ☐ 70 Nm +270°





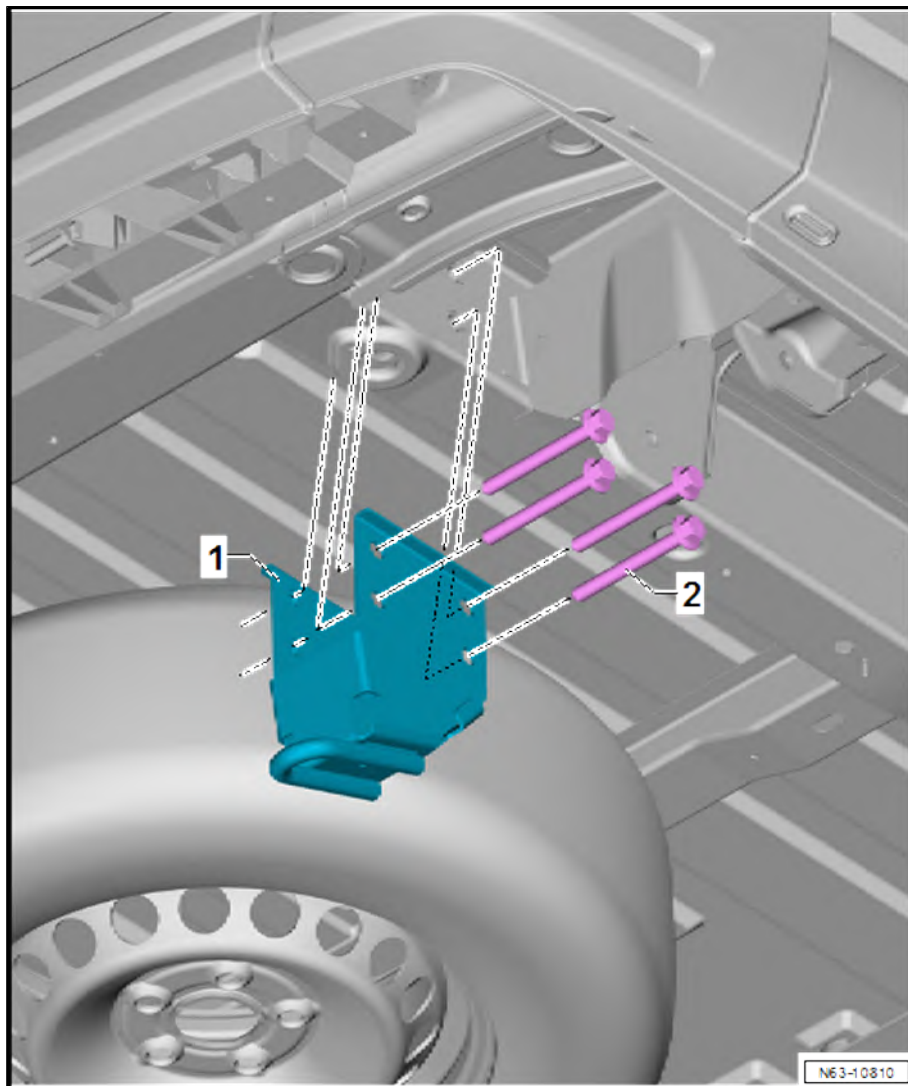
2.2.3 Assembly overview – towing eye bracket

1 - Bracket

- ❑ For towing eye.
- ❑ Removing and installing
⇒ [page 196](#)

2 - Bolts

- ❑ Renew after removal
- ❑ Qty. 4
- ❑ 70 Nm +270°



2.3 Removing and installing bumper cover

⇒ [“2.3.1 Removing and installing bumper cover”, page 192](#)

⇒ [“2.3.2 Removing and installing bumper cover, with step”, page 193](#)

⇒ [“2.3.3 Removing and installing end piece”, page 194](#)

2.3.1 Removing and installing bumper cover

Special tools and workshop equipment required



- ◆ Setting gauge - 3371-



- ◆ Torque wrench - V.A.G 1783-



Removing

Vehicles with towing bracket

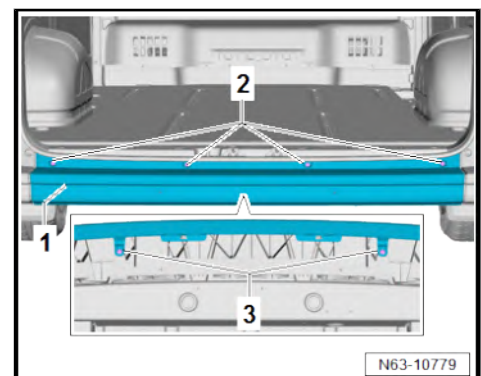
- Remove towing bracket ➔ [page 319](#) .

Continued for all vehicles

- Unscrew bolts -2- and -3-. When doing this, remove bumper cover -1-.

Vehicles with parking aid

- Disconnect electrical connectors on rear left parking aid sender - G203- , rear left centre parking aid sender - G204- , rear right centre parking aid sender - G205- and rear right parking aid sender - G206- .



Installing

Install in reverse order of removal, observing the following:

- Use setting gauge - 3371- to check gaps and height difference; adjust as necessary ➔ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .

Specified torques

- ◆ ➔ [“2.1 Assembly overview - bumper cover”, page 188](#)

2.3.2 Removing and installing bumper cover, with step

Special tools and workshop equipment required



◆ Setting gauge - 3371-



◆ Torque wrench - V.A.G 1783-



Removing

- Unscrew bolts -2- and -3-. When doing this, remove bumper cover -1-.

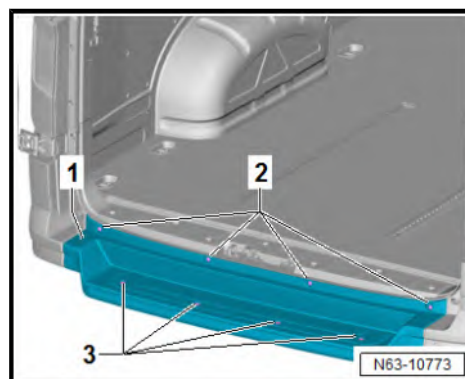
Vehicles with parking aid

- Disconnect electrical connectors on rear left parking aid sender - G203- , rear left centre parking aid sender - G204- , rear right centre parking aid sender - G205- and rear right parking aid sender - G206- .

Installing

Install in reverse order of removal, observing the following:

- Use setting gauge - 3371- to check gaps and height difference; adjust as necessary ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .



Specified torques

- ◆ ⇒ [“2.1 Assembly overview - bumper cover”, page 188](#)

2.3.3 Removing and installing end piece

Special tools and workshop equipment required



◆ Setting gauge - 3371-



◆ Torque wrench - V.A.G 1783-



Removing

- Remove left tail light cluster ⇒ Electrical system; Rep. gr. 94 ; Tail lights; Removing and installing tail light cluster .
- Remove protective side strip of rear side panel ⇒ [page 253](#) .
- Remove centre bumper cover ⇒ [page 192](#) .
- Unscrew bolts -2-.
- Pull end piece -1- out of fasteners of guide.
- Guide out end piece -1- from between D-pillar and wing door.

Vehicles with parking aid

- Disconnect electrical connector on parking aid sender.

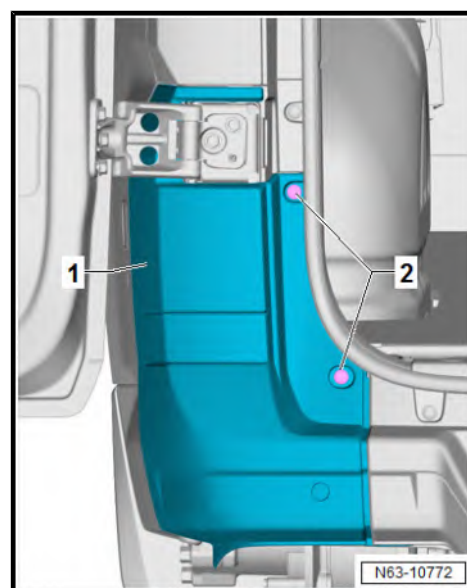
Installing

Install in reverse order of removal, observing the following:

- Use setting gauge - 3371- to check gaps and height difference; adjust as necessary ⇒ Body Repairs; Rep. gr. 00 ; Gaps; Body, front .

Specified torques

- ◆ ⇒ [“2.1 Assembly overview - bumper cover”, page 188](#)





2.4 Removing and installing impact bar

⇒ [“2.4.1 Removing and installing impact bar”, page 196](#)

⇒ [“2.4.2 Removing and installing step”, page 196](#)

⇒ [“2.4.3 Removing and installing towing eye bracket”, page 197](#)

2.4.1 Removing and installing impact bar

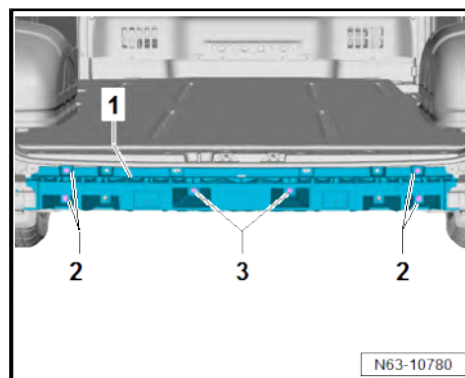
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

- Remove bumper cover ⇒ [page 192](#) .
- Remove left end piece ⇒ [page 194](#) .
- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Remove impact bar -1-.



Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.2 Assembly overview - impact bar”, page 190](#)

2.4.2 Removing and installing step

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-





Removing

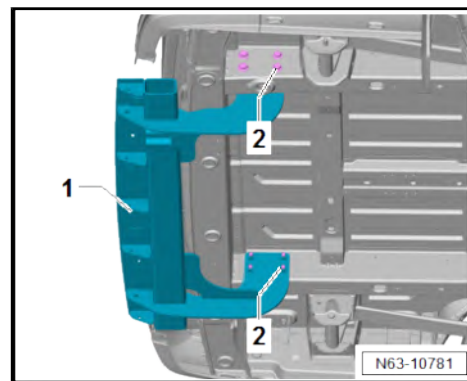
- Remove bumper cover ➔ [page 192](#) .
- The following work step must be performed by a second mechanic.
- Unscrew bolts -2-. When doing this, remove step -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“2.2 Assembly overview - impact bar”, page 190](#)



2.4.3 Removing and installing towing eye bracket

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-



Removing

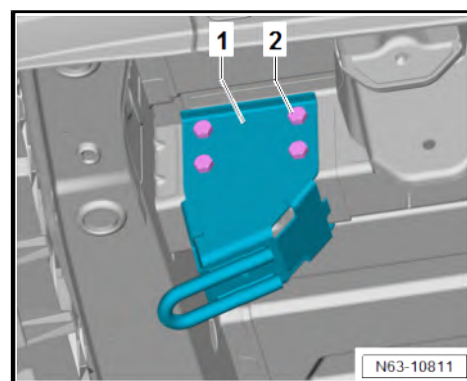
- Unscrew bolts -2-. When doing this, remove towing eye bracket -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“2.2 Assembly overview - impact bar”, page 190](#)



2.5 Removing and installing add-on parts

➔ [“2.5.1 Removing and installing guide bracket”, page 197](#)

➔ [“2.5.2 Removing and installing end piece trim”, page 198](#)

2.5.1 Removing and installing guide bracket

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

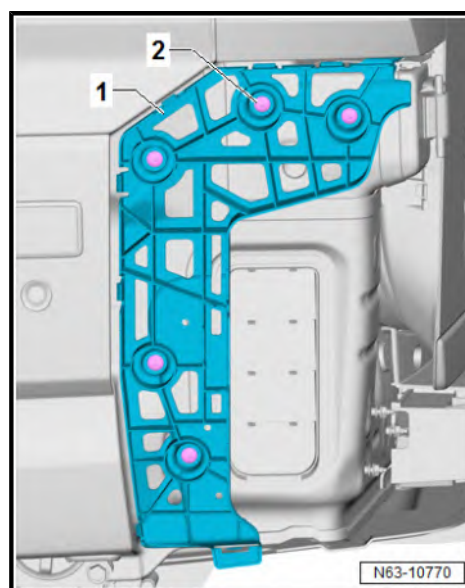
- Remove left end piece ➔ [page 194](#) .
- Unscrew bolts -2-.
- Remove guide bracket -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“2.1 Assembly overview - bumper cover”, page 188](#)



2.5.2 Removing and installing end piece trim

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

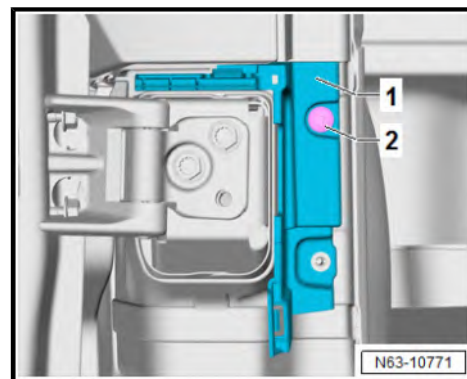
- Remove left end piece ➔ [page 194](#) .
- Unscrew bolt -2-.
- Remove end piece trim -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“2.1 Assembly overview - bumper cover”, page 188](#)



2.6 Installing holder for parking aid sender

Special tools and workshop equipment required

- ◆ Installation tool for PDC bracket - VAS 6614C-



- ◆ Screw-hole punch 18.4 mm diameter - VAS 6614/9-

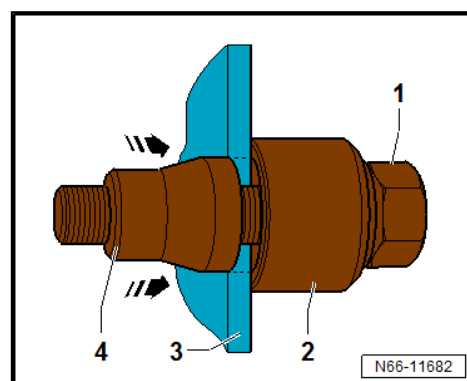
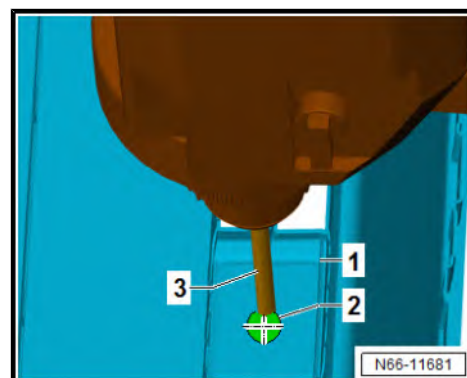
Procedure



Note

Marking -2- for retainers can be found on inside of respective bumper cover -1-.

- Pre-drill hole from inside out with drill bit - VAS 6614/4- -3- from tool kit supplied.
- Fit screw-hole punch 18.4 mm - VAS 6614/9- with bolt head -1- and counter-support -2- from inside of bumper cover -3-.
- Screw on hole punch -4- from outside of bumper cover -3-.
- Press punch -4- against cover.
- Tighten bolt head -1-.
- Punch -4- is pulled through material of bumper cover -3- in direction of -arrow- by turning bolt head -1-.





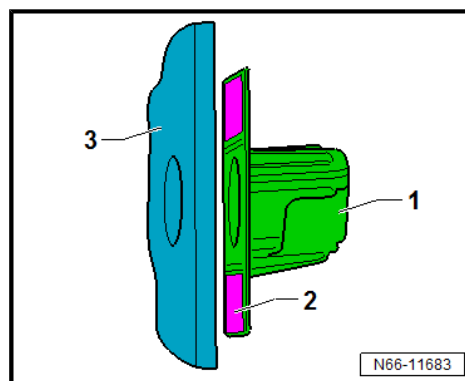
- Thoroughly clean bonding surfaces on bumper cover -3-.
- Pull off backing -2- off double-sided adhesive tape on retainer -1-.



Note

Do not touch the adhesive surface of the bonded punched component once the protective backing has been removed.

- Install retainer -1- in hole punched in bumper cover -3-.
- Press retainer -1- forcefully against bonding surface.
- Do not carry out any further work on the cover or retainers until the minimum curing period of 2 hours has elapsed.





64 – Glazing

1 Repair notes

⇒ [“1.1 Minimum curing periods for bonded windows”, page 201](#)

⇒ [“1.2 Window repair”, page 201](#)

⇒ [“1.3 Installation instructions for bonded windows”, page 202](#)

⇒ [“1.4 Preparing old undamaged windows for fitting”, page 204](#)

⇒ [“1.5 Preparing new windows for installing”, page 204](#)

⇒ [“1.6 Preparing body flange for fitting”, page 205](#)

⇒ [“1.7 Cleaning off excess adhesive”, page 206](#)

1.1 Minimum curing periods for bonded windows

There are special requirements for renewing bonded windows.

One of these standards is, for example, that a freshly bonded windscreen must comply with the safety requirements, even in an accident, following the minimum prescribed curing period.

The vehicle is safe to use only after the minimum curing period has elapsed.

2-component adhesive

The minimum curing time for the 2-component adhesive is 2 hours for all windows.

The operational safety of the vehicle is restored completely after 2 hours.

1-component adhesive

The minimum curing time for the 1-component adhesive is 16 hours for all windows.

The vehicle is definitely safe to operate again after 16 hours if the 1-component adhesive is used.

For 2-component adhesive and 1-component adhesive

Irrespective of which adhesive is used, the vehicle can be moved again half an hour after the window has been installed.

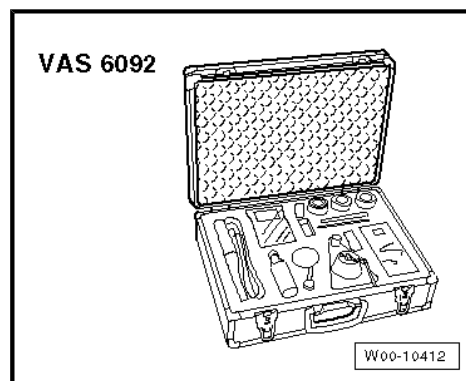
Minimum curing period is the time from bonding the window until vehicle may be used (operational safety). During this time, the vehicle must be standing on a level surface at room temperature (at least 15°C).

1.2 Window repair

Special tools and workshop equipment required



- ◆ Window repair kit - VAS 6092-



Note

- ◆ *Before renewing windscreen, explore possibilities of repairing damage to glass.*
- ◆ *Damaged windscreens can be repaired with the window repair kit - VAS 6092- .*
- ◆ *This damage must remain within a certain size or position.*
- ◆ *Each window repair kit comes with a set of instructions.*

1.3 Installation instructions for bonded windows

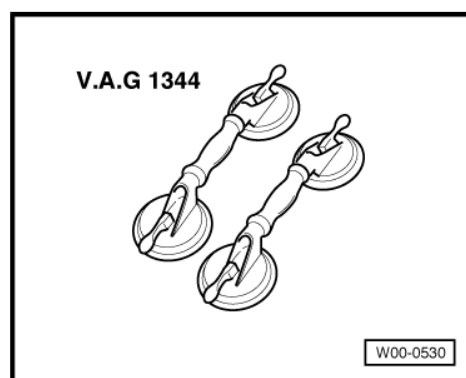
⇒ ["1.3.1 Fitting instructions", page 202](#)

⇒ ["1.3.2 Materials", page 203](#)

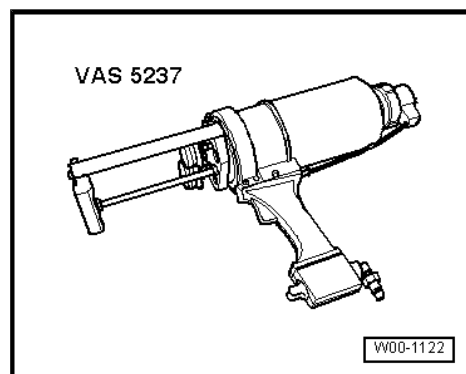
1.3.1 Fitting instructions

Special tools and workshop equipment required

- ◆ Suction holder - V.A.G 1344-



- ◆ Double cartridge gun - VAS 5237-



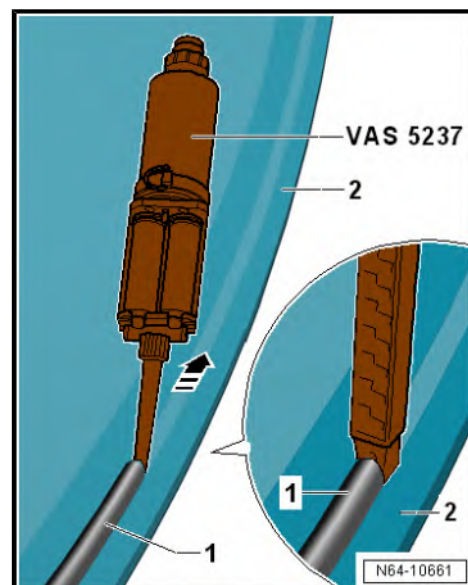


- Apply adhesive sealing material -1-, holding applicator at right angle to window -2- using double cartridge gun - VAS 5237- .



Note

- ◆ *Window must be installed within 10 minutes, or adhesive properties of glass adhesive will be impaired.*
- ◆ *Adhesive bead should not be applied too thickly, and windows should not be pressed too forcibly onto the body flange. Windows no longer have sealing lips and therefore excessive adhesive will be visible in the joint area.*
- ◆ Adhere to adhesive bead dimensions as per assembly overviews.
- ◆ Using 2 suction lifters - V.A.G 1344- , insert glass pane in window aperture, align it centrally, and press it in onto buffer stops.
- ◆ Replace stickers (e.g. for airbag) if there were any.



1.3.2 Materials



Note

For part numbers of the materials, refer to the ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

- ◆ 2-component glass adhesive 1) 2) 3) 6)
- ◆ 1-component glass adhesive 1) 2) 4) 5) 6)
- ◆ Activator 1) 6)
- ◆ Glass and paint primer 1) 6)
- ◆ Cleaning solution 1) 6)
- ◆ Primer applicator 1) 6)
- ◆ Adhesive remover 1) 6)
- ◆ Cutting cord 357 853 999 6) 7
- ◆ Wire reel VAS 6452/1 7
- ◆ Cutting cord 6) 7
- ◆ Buffer stop 6)
- ◆ Cleaning solution 6)
- ◆ Adhesive remover 6)

1) Observe manufacturer's instructions enclosed in the packaging.

2) Observe minimum curing period.

3) Use double cartridge gun - VAS 5237- to apply these materials.

4) Heat according to manufacturer's instructions using cartridge heater - V.A.G 1939- .



5) Small cartridge, 180 ml, for sealing and/or when a 400 ml double cartridge alone is not sufficient.

6) For part numbers of materials, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

7 The cutting cords stated above can be used alternatively to the cutting wires, and vice versa. They are designated for single use.

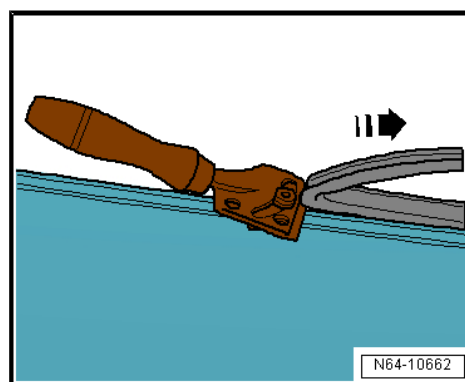
1.4 Preparing old undamaged windows for fitting

Special tools and workshop equipment required

- ♦ Multifunctional stand for vehicle windows - VAS 6447-

Cutting back adhesive bead

- Windscreen is placed on multifunctional stand for vehicle windows - VAS 6447- .
- ♦ If an undamaged window is being re-used, cut back the residual adhesive sealant to 1 to 2 mm shortly before bonding it back in.
- ♦ Do not damage primer and ceramic coating when doing this.
- ♦ Keep bonding surface free of dirt and grease.
- ♦ Do not treat bonding surface with activator, glass/paint primer or cleaning solution immediately after cutting back.
- ♦ The remaining material is required as a base for better adhesion of the new adhesive sealing material to be applied.
- Exception: if bonding is performed later than one day after cutting back adhesive bead, the remaining residual material must be activated with activator .
- Apply activator evenly in one stroke using applicator .



NOTICE

Risk of damage to paintwork from activator.

- Apply activator neatly onto adhesive bead, do not spill.

Allow to air dry for approx. 10 minutes.

1.5 Preparing new windows for installing

Special tools and workshop equipment required

- ♦ Multifunctional stand for vehicle windows - VAS 6447-
- Windscreen is placed on multifunctional stand for vehicle windows - VAS 6447- .

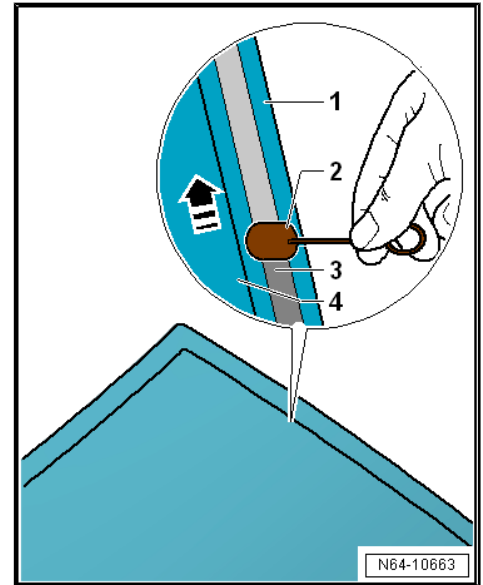


Note

- ♦ *The area on which the adhesive bead is applied is not pre-coated or primed.*
- ♦ *The relevant area is a ceramic coating.*
- ♦ *No windows are precoated.*
- ♦ *The sealing lip is included with a new windscreen.*
- Using cleaning solution , clean edge of window all round to a width of 20 mm.



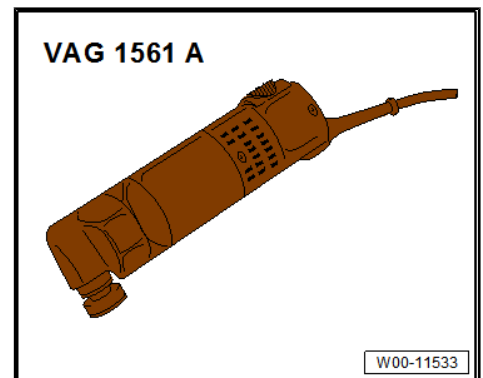
- Then rub edge of window dry using a lint-free cloth.
- Using applicator -2-, apply glass/paint primer -3- evenly in one pass onto ceramic coating -1- of window -4-.
- ◆ Allow to air dry for approx. 10 minutes.



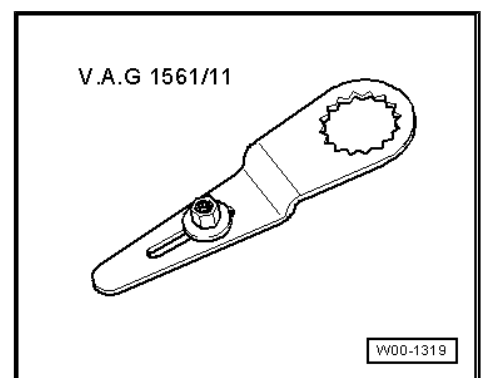
1.6 Preparing body flange for fitting

Special tools and workshop equipment required

- ◆ Electric cutter - V.A.G 1561A-



- ◆ Universal cutter - V.A.G 1561/11-





Procedure



Note

Do not remove adhesive sealant -3- completely under any circumstances.

- Just before reinstalling, cut back remaining adhesive sealant -3- on body flange to 1 ... 2 mm using electric cutter - V.A.G 1561A- -1- and blade, 15 and 25 mm - V.A.G 1561/7+8- -2-.



Note

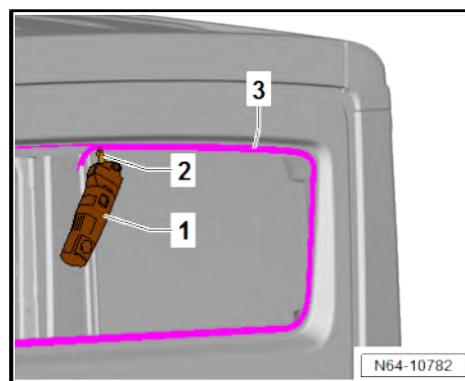
- ♦ *Keep bonding surface free of dirt and grease.*
- ♦ *Do not treat bonding surface with activator, glass/paint primer or cleaning solution immediately after cutting back.*
- ♦ *The remaining material is required as a base for better adhesion of the new adhesive sealing material to be applied.*
- If bonding is performed later than one day after cutting back the adhesive bead, the remaining residual material must be activated with activator .



NOTICE

Risk of damage to paintwork from activator.

- Apply activator neatly onto adhesive bead, do not spill.
- Clean body flange all around using cleaning solution .
- Then rub edge of window dry using a lint-free cloth.
- Apply activator evenly using applicator .



1.7 Cleaning off excess adhesive

- Use adhesive remover as cleaning agent. Safety precautions must be observe for all work procedures.



Note

- ♦ *When cleaning the vehicle interior, do not press freshly installed window outwards.*
- ♦ *Otherwise, the window will be detached.*
- First, clean the painted surface roughly with a dry cloth. Remove residue using adhesive remover .
- Cleaning plastic trim: Allow adhesive sealant to cure (approx. 1 hour) and then peel off.

2 Windscreen

⇒ [“2.1 Assembly overview - windscreen”, page 207](#)

⇒ [“2.2 Removing and installing windscreen”, page 207](#)

2.1 Assembly overview - windscreen

1 - Windscreen

- ☐ Removing and installing
⇒ [page 207](#)

2 - Windscreen seal

- ☐ Is part of windscreen.

3 - Electrical connectors

- ☐ Qty. 2
- ☐ For heated windscreens

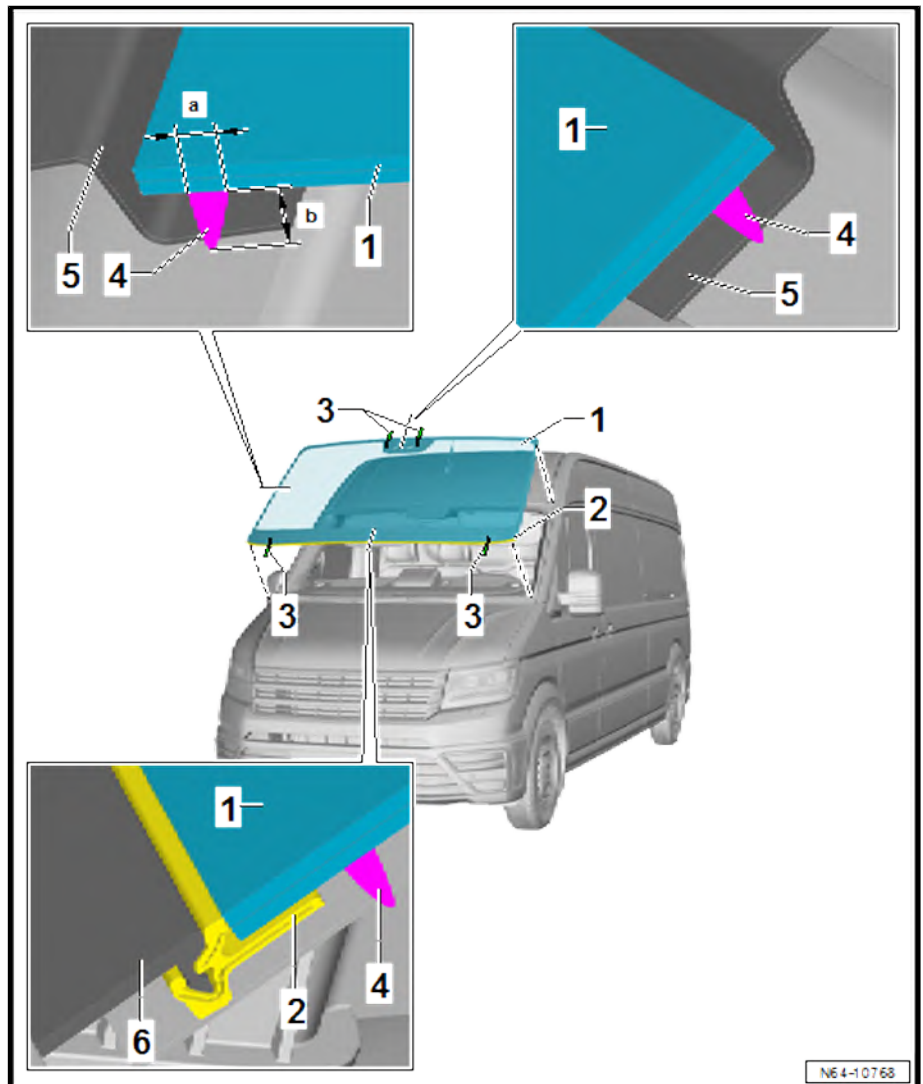
4 - PUR adhesive sealant

- ☐ Cross section of bead:
width = 7 mm
-dimension a-, height =
10 mm -dimension b-
(including precoating,
residual material on win-
dow glass and window
flange).
- ☐ Materials ⇒ [page 203](#)
- ☐ Installation instructions
⇒ [page 202](#) .
- ☐ Minimum curing period
⇒ [page 201](#) .

5 - Body flange

6 - Plenum chamber cover

- ☐ Removing and installing
⇒ [page 39](#)



2.2 Removing and installing windscreen

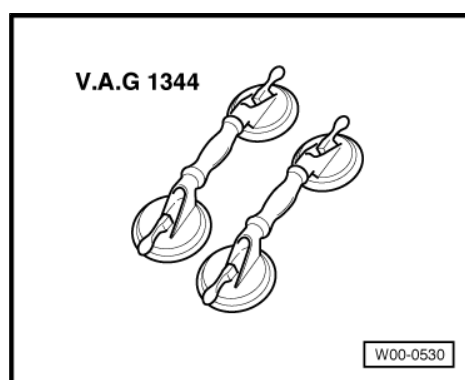
Special tools and workshop equipment required



◆ Setting gauge - 3371-

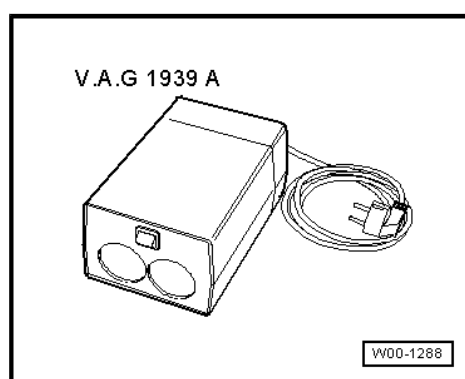


◆ Suction holder - V.A.G 1344-

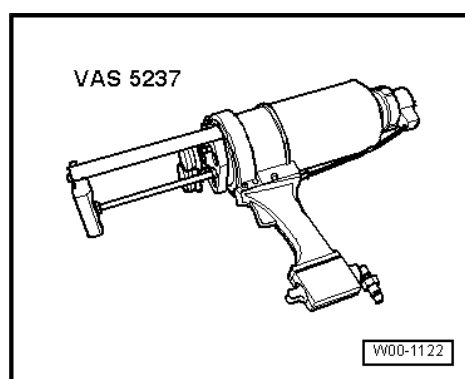


◆ Masking film - V.A.G 1474/7-

◆ Cartridge heating unit - V.A.G 1939A-



◆ Double cartridge gun - VAS 5237-

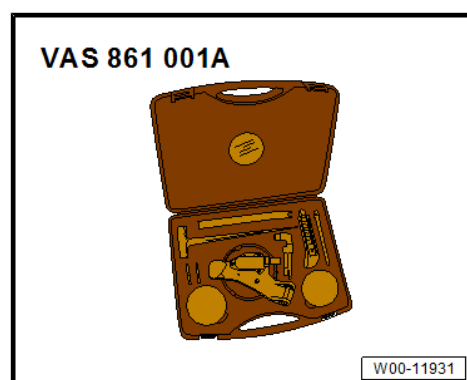




- ◆ Multifunctional stand for vehicle windows - VAS 6447-



- ◆ Windscreen removal kit - VAS 861 001A-



- ◆ Scraper set - VAS 861 005-



- ◆ Commercially available cordless screwdriver

Removing

- Remove A-pillar upper trims on left and right ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing A-pillar trim .
- Remove left and right sun visors ⇒ General body repairs, interior; Rep. gr. 68 ; Equipment; Removing and installing sun visor .
- Remove interior mirror ⇒ General body repairs, interior; Rep. gr. 68 ; Interior mirror; Removing and installing interior mirror .

Vehicles with front camera

- Calibrate front camera ⇒ Running gear, axles, steering; Rep. gr. 44 ; Front camera for driver assist systems; Calibrating front camera for driver assist systems .

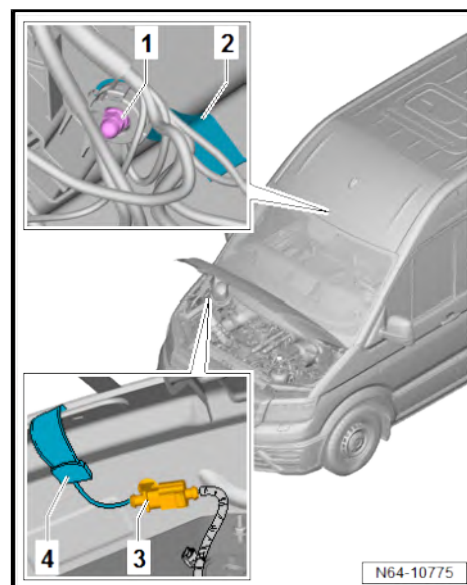


Continued for all vehicles

- Lower moulded headliner in area of windscreen ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .
- Removing plenum chamber cover ⇒ [page 39](#) .

Vehicles with windscreen heater

- Separate electrical connectors -3- in plenum chamber on left and right.
- Secure wiring harness -4- to windscreen on left and right using adhesive tape.
- Unscrew nut -1- at top of roof frame.
- Secure wiring harness -2- to windscreen using adhesive tape.
- Use awl from window removal set - VAS 861 001A- to pierce through at bottom left from vehicle interior towards outside.
- Attach cutting cord from window removal set - VAS 861 001A- to awl from window removal set - VAS 861 001A- .
- Pull awl together with cutting cord into vehicle interior.
- Guide end of cutting cord through relay piece on separating device from window removal set - VAS 861 001A- .
- Guide end of cutting cord through hole in shaft.
- Secure end of cutting cord against being pulled out.
- Position separating device from window removal set - VAS 861 001A- at bottom right on windscreen.
- Place cutting cord all round under windscreen.
- Secure second end of cutting cord on cord hook from window removal set - VAS 861 001A- .
- Attach cord hook on body.
- Slide suitable covering film - V.A.G 1474/7- between windscreen and dash panel.
- Use commercially available cordless screwdriver to start removing windscreen via reel device of separating device.
- Cut out windscreen using cutting cord only until the outermost point of the windscreen has been reached.
- Move separating device to top right.
- Use commercially available cordless screwdriver to continue removing windscreen via reel device of separating device.
- Cut out windscreen using cutting cord only until the outermost point of the windscreen has been reached.
- Move separating device to top left.
- Use commercially available cordless screwdriver to continue removing windscreen via reel device of separating device.
- Cut out windscreen using cutting cord only until the outermost point of the windscreen has been reached.
- Move separating device to bottom left.
- Use commercially available cordless screwdriver to continue removing windscreen via reel device of separating device.
- Secure 2 suction lifters - V.A.G 1344- on windscreen.
- The following work step must be performed by a second mechanic.
- Remove windscreen together with second mechanic.
- Place windscreen on multifunctional stand for vehicle windows - VAS 6447- .





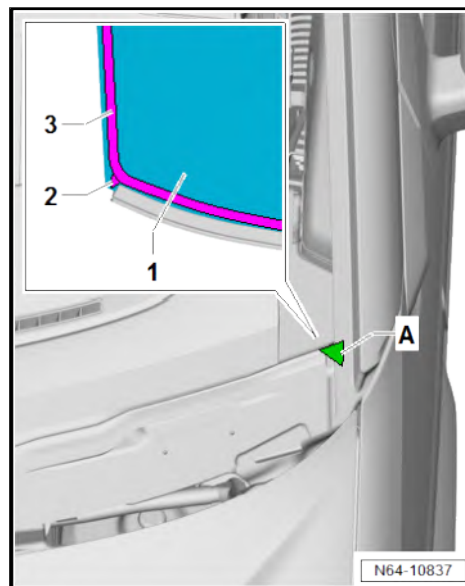
Installing

Install in reverse order of removal, observing the following:

- Preparing body flange for fitting ⇒ [page 205](#)
- Using applicator - D 009 500 25- , apply glass and paint primer to surface -A-.
- Apply adhesive sealant -3- to windscreen -1- as shown in illustration.
- Apply additional adhesive sealant -2- to left and right of windscreen -1-.
- Preparing old undamaged windows for fitting ⇒ [page 204](#)
- Preparing new windows for fitting ⇒ [page 204](#)
- Fitting notes for bonded windows ⇒ [page 202](#) .
- Minimum curing period ⇒ [page 201](#)

Vehicles with front camera

- Calibrate front camera ⇒ Running gear, axles, steering; Rep. gr. 44 ; Front camera for driver assist systems; Calibrating front camera for driver assist systems .



3 Rear windscreen

⇒ [“3.1 Assembly overview - rear window”, page 213](#)

⇒ [“3.2 Removing and installing rear window”, page 216](#)

3.1 Assembly overview - rear window

⇒ [“3.1.1 Assembly overview – rear window, wing door”, page 213](#)

⇒ [“3.1.2 Assembly overview – rear window, double cab”, page 214](#)

⇒ [“3.1.3 Assembly overview - rear window, partition panel”, page 215](#)

3.1.1 Assembly overview – rear window, wing door

1 - Rear windscreen

- ☐ Left
- ☐ Removing and installing
⇒ [page 216](#)

2 - Rear windscreen

- ☐ Right
- ☐ Removing and installing
⇒ [page 216](#)

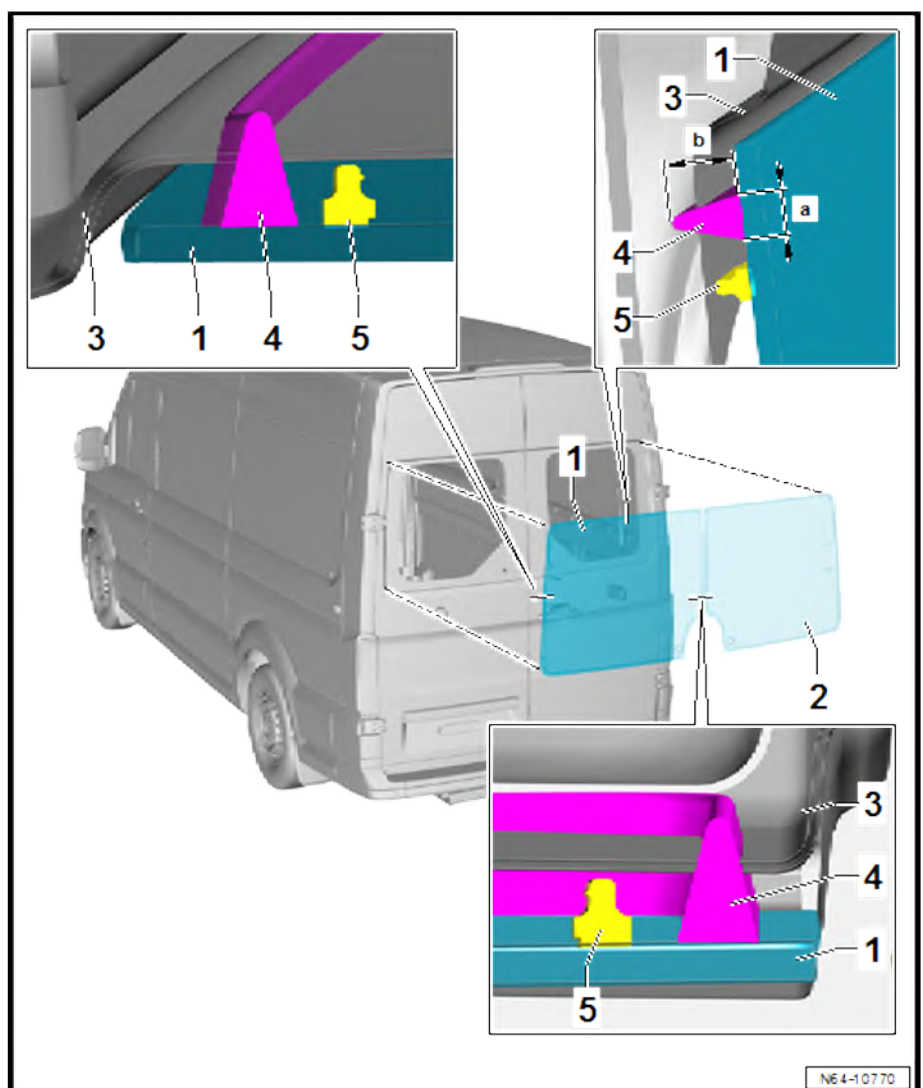
3 - Body flange

4 - PUR adhesive sealant

- ☐ Cross section of bead:
width = 7 mm
-dimension a-, height =
10 mm -dimension b-
(including precoating,
residual material on win-
dow glass and window
flange).
- ☐ Materials ⇒ [page 203](#)
- ☐ Installation instructions
⇒ [page 202](#) .
- ☐ Minimum curing period
⇒ [page 201](#) .

5 - Buffer stop

- ☐ Various number
- ☐ Materials ⇒ [page 203](#)



3.1.2 Assembly overview – rear window, double cab

1 - Rear windscreen

- ❑ Removing and installing
⇒ [page 216](#)

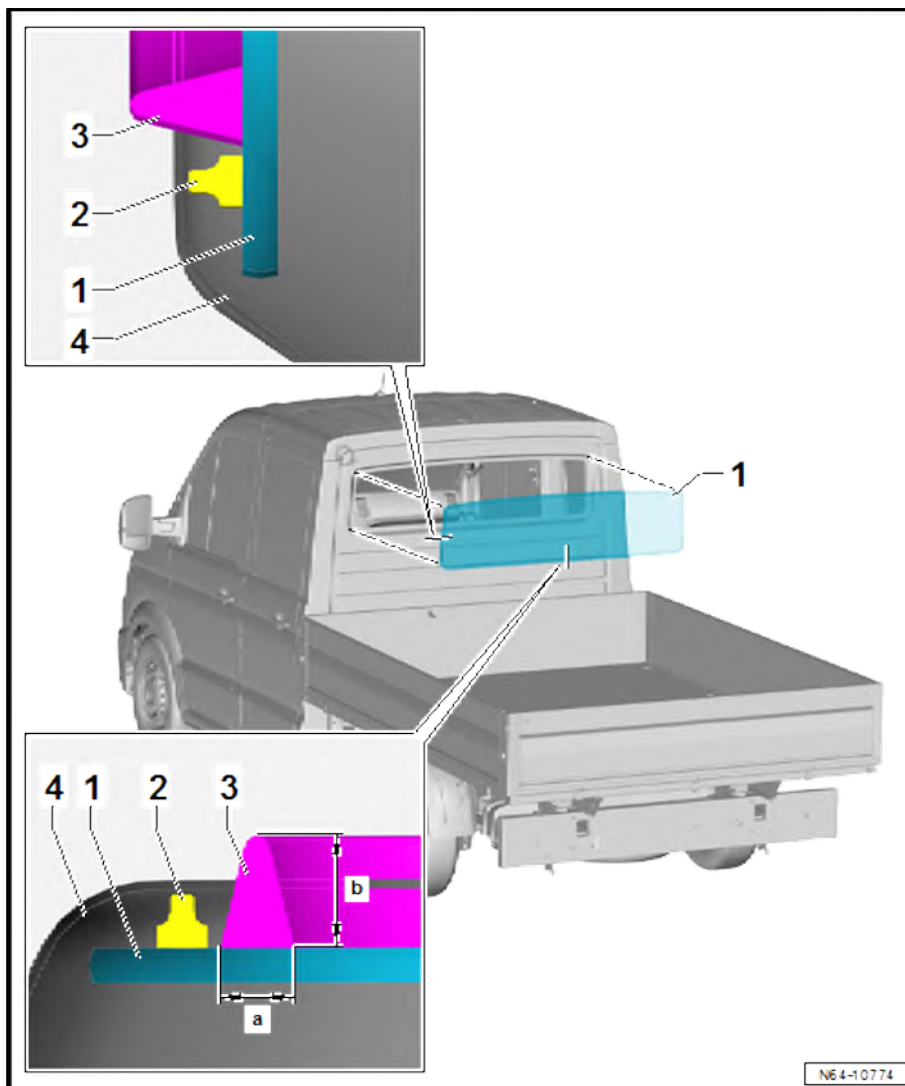
2 - Buffer stop

- ❑ Various number
- ❑ Materials ⇒ [page 203](#)

3 - PUR adhesive sealant

- ❑ Cross section of bead:
width = 7 mm
-dimension a-, height =
10 mm -dimension b-
(including precoat-
ing, residual material on win-
dow glass and window
flange).
- ❑ Materials ⇒ [page 203](#)
- ❑ Installation instructions
⇒ [page 202](#) .
- ❑ Minimum curing period
⇒ [page 201](#) .

4 - Body flange



N64-10774

3.1.3 Assembly overview - rear window, partition panel

1 - Rear windscreen

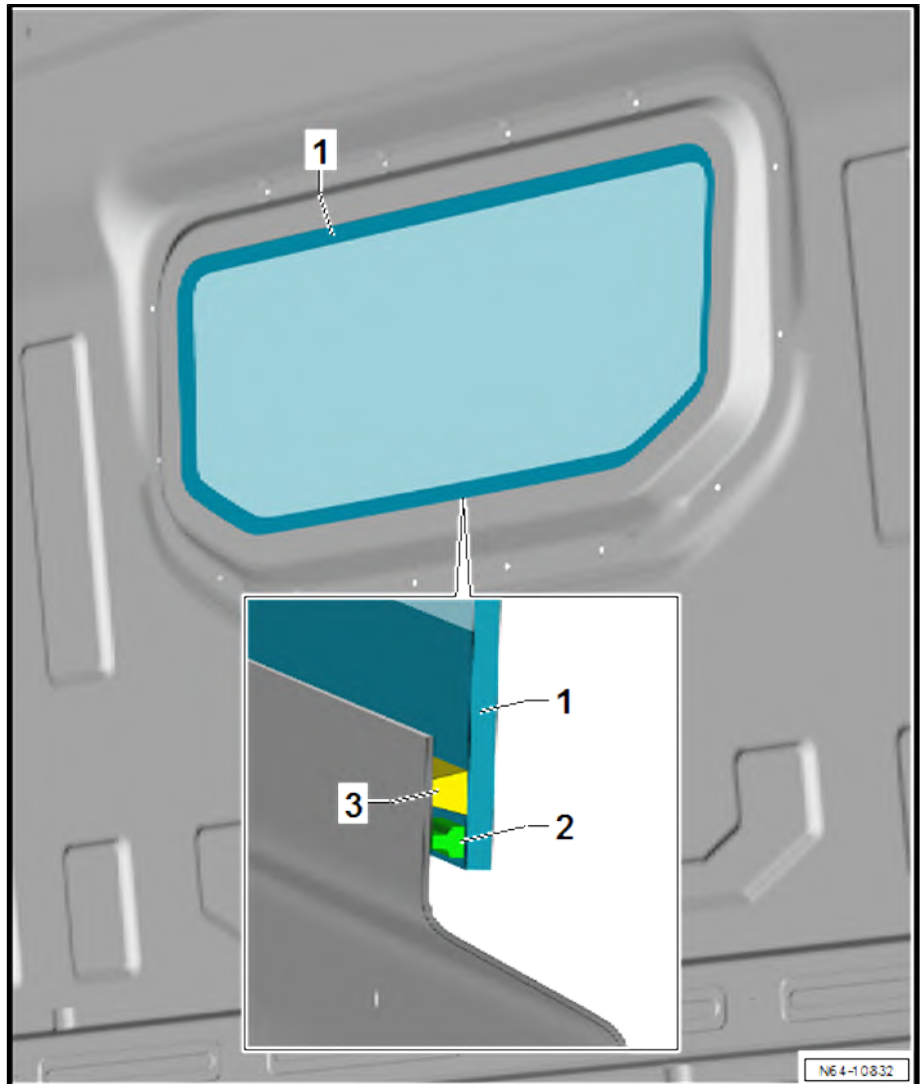
- ☐ Partition panel
- ☐ Removing and installing
⇒ [page 222](#)

2 - Buffer stop

- ☐ Various number
- ☐ Materials ⇒ [page 203](#)

3 - PUR adhesive sealant

- ☐ Cross section of bead:
approx. width = 13 mm,
approx. height = 13 mm
(including precoating,
residual material on win-
dow glass and window
flange)
- ☐ Materials ⇒ [page 203](#)
- ☐ Installation instructions
⇒ [page 202](#) .
- ☐ Minimum curing period
⇒ [page 201](#) .





3.2 Removing and installing rear window

⇒ [“3.2.1 Removing and installing rear window, wing door”, page 216](#)

⇒ [“3.2.2 Removing and installing rear window, double cab”, page 219](#)

⇒ [“3.2.3 Removing and installing rear window, partition panel”, page 222](#)

⇒ [“3.2.4 Installing sliding windows, reworking, vehicles with partition panel”, page 225](#)

⇒ [“3.2.5 Removing broken rear window”, page 226](#)

3.2.1 Removing and installing rear window, wing door

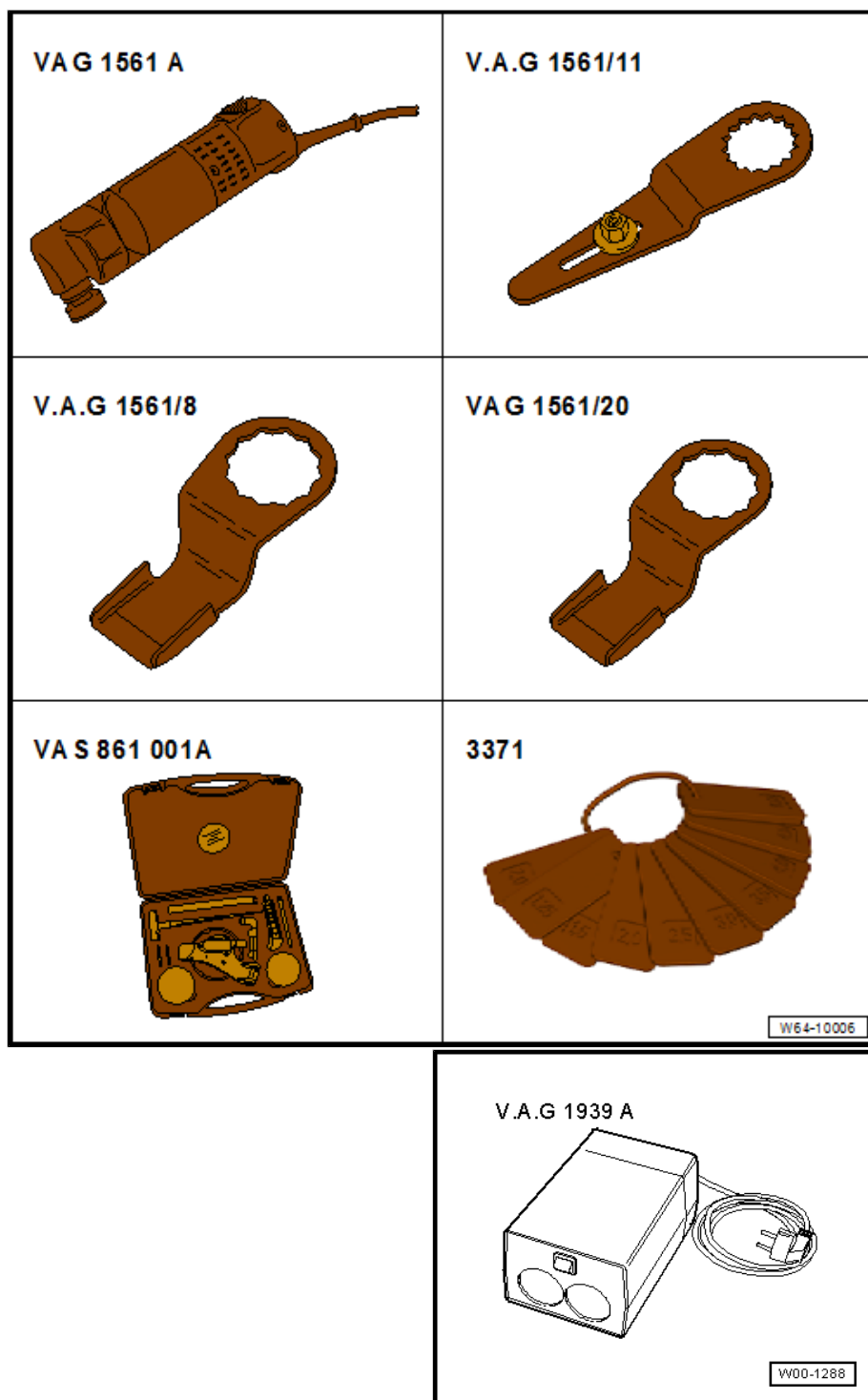
Special tools and workshop equipment required



- ◆ Removal kit for flush bonded windows - VAS 6452-
- ◆ Suction holder - V.A.G 1344-
- ◆ Cutting tool - V.A.G 1351-
- ◆ Windscreen removal kit - V.A.G 1755-
- ◆ Double cartridge gun - VAS 5237-



◆ Hand-cartridge gun - V.A.G 1628-



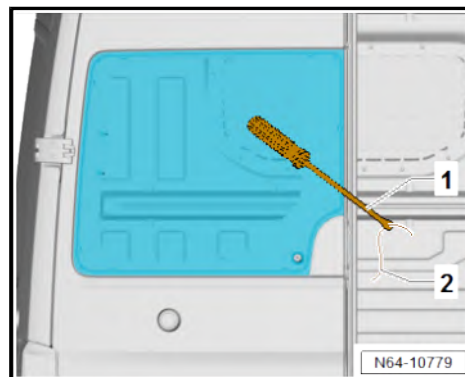
- ◆ Electric cutter - V.A.G 1561A-
- ◆ Universal cutter - V.A.G 1561/11-
- ◆ Blades, 15 and 25 mm - V.A.G 1561/7+8-
- ◆ Scraper (45 mm) - V.A.G 1561/20-
- ◆ Windscreen removal kit - VAS 861 001A-
- ◆ Setting gauge - 3371-



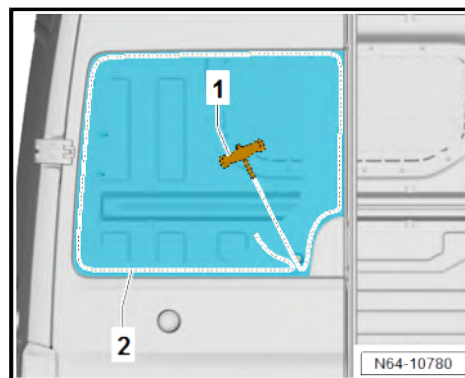
◆ Cartridge heating unit - V.A.G 1939A-

Removing

- Remove wiper arm ⇒ Electrical system; Rep. gr. 92 ; Rear window wiper system; Removing and installing wiper arm .
- Remove badge cap ⇒ [page 311](#) .
- If fitted, separate connectors for window heating.
- Use awl - V.A.G 1474/2- -1- to poke through adhesive sealant from inside out.
- Pull cutting cord -2- through adhesive sealant to inside using awl - V.A.G 1474/2- -1-.



- Secure cutting cord -2- with pull handle - V.A.G 1351/1- -1- to prevent it from being pulled out.
- Lay cutting cord -2- around rear window and guide second end of cutting cord inwards.
- Ensure that cutting cord -2- lies under rear window in corners.





- Place reel device - V.A.G 1654 A- -1- in indicated position.
- Secure cutting cord -3- on inside with pull handle - V.A.G 1351/1- -2- to prevent it from being pulled out.
- Secure other end of cutting cord to reel device - V.A.G 1654 A- -1-.
- Cut rear window free by actuating reel device - V.A.G 1654 A- -1-.
- Press cutting cord -3- against rear window with wedge - V.A.G 1474/5- while cutting.



Note

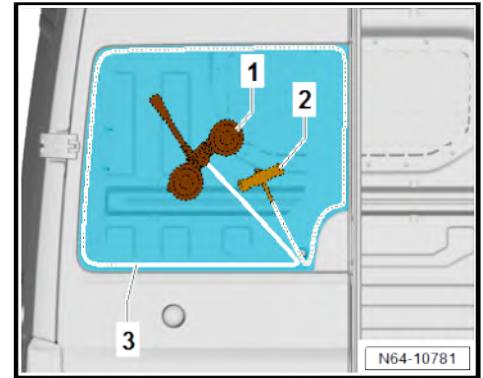
Further dismantling requires the assistance of a second mechanic.

With aid of second mechanic, hold rear window using suction lifter - V.A.G 1344- from outside.

Installing

Install in reverse order of removal, observing the following:

- Preparing old undamaged windows for fitting ⇒ [page 204](#)
- Preparing new windows for fitting ⇒ [page 204](#)
- Preparing body flange for fitting ⇒ [page 205](#)
- Fitting notes for bonded windows ⇒ [page 202](#) .
- Minimum curing period ⇒ [page 201](#)



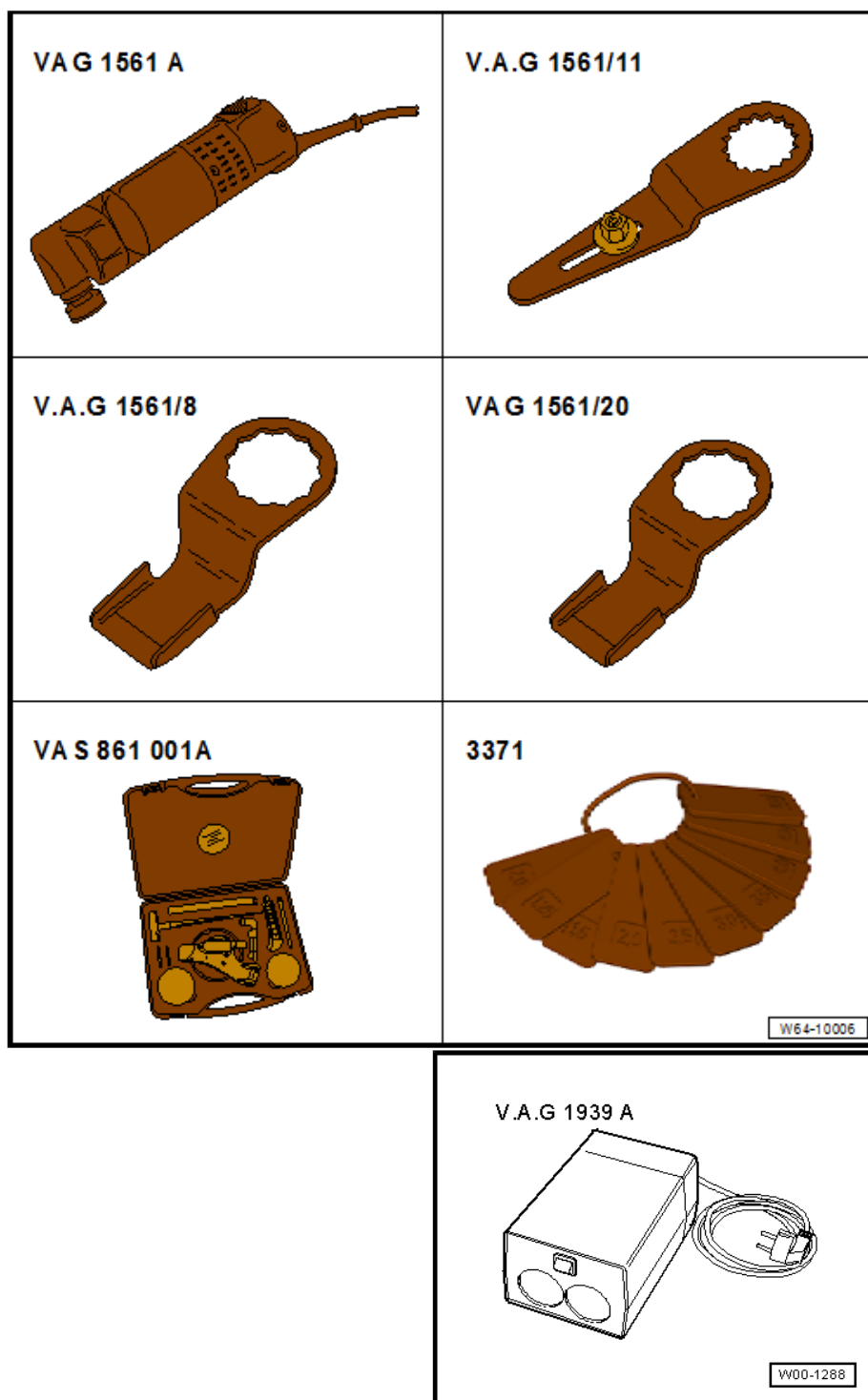
3.2.2 Removing and installing rear window, double cab



**Special tools and workshop
equipment required**



- ◆ Removal kit for flush bonded windows - VAS 6452-
- ◆ Suction holder - V.A.G 1344-
- ◆ Cutting tool - V.A.G 1351-
- ◆ Windscreen removal kit - V.A.G 1755-
- ◆ Double cartridge gun - VAS 5237-
- ◆ Hand-cartridge gun - V.A.G 1628-

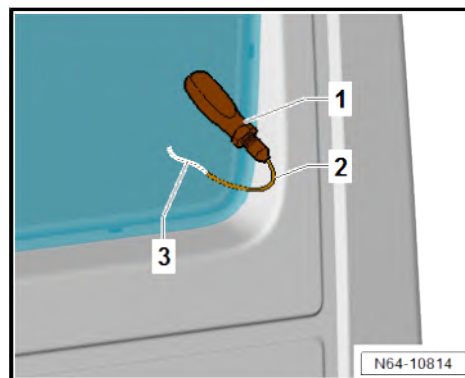


- ◆ Electric cutter - V.A.G 1561A-
- ◆ Universal cutter - V.A.G 1561/11-
- ◆ Blades, 15 and 25 mm - V.A.G 1561/7+8-
- ◆ Scraper (45 mm) - V.A.G 1561/20-
- ◆ Windscreen removal kit - VAS 861 001A-
- ◆ Setting gauge - 3371-
- ◆ Cartridge heating unit - V.A.G 1939A-

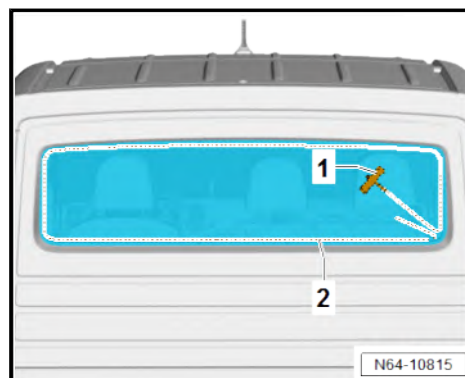


Removing

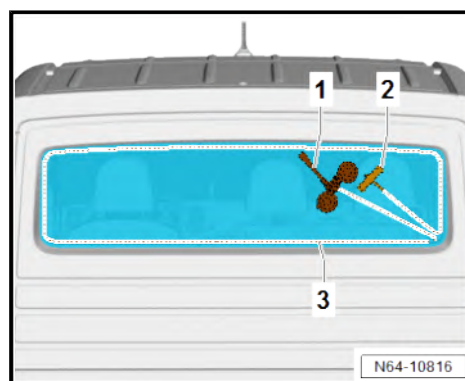
- Take trimming needle -2- from windscreen removal kit - V.A.G 1755- and use handle -1- to force it through adhesive material.
- Unscrew handle -1- and thread cutting cord -3- through eye of needle.
- Pull trimming needle -2- and cutting cord -3- through to inside.
- Pull other end of cutting cord through adhesive sealant as well.



- Place cutting cord -2- around rear window.
- Secure one end of cutting cord to pull handle - V.A.G 1351/1- -1- to counterhold.



- Secure other end of cutting cord to reel device - V.A.G 1654 - -1-.
- Secure inside end of cutting cord against falling out using Pull toggle - V.A.G 1351/1- -2-.
- Cut rear window free by actuating reel device - V.A.G 1654 - -1-.
- Press cutting cord -3- against window with wedge - V.A.G 1474/5- while cutting.



Installing

Install in reverse order of removal, observing the following:

- Preparing old undamaged windows for fitting ⇒ [page 204](#)
- Preparing new windows for fitting ⇒ [page 204](#)
- Preparing body flange for fitting ⇒ [page 205](#)
- Fitting notes for bonded windows ⇒ [page 202](#) .
- Minimum curing period ⇒ [page 201](#)

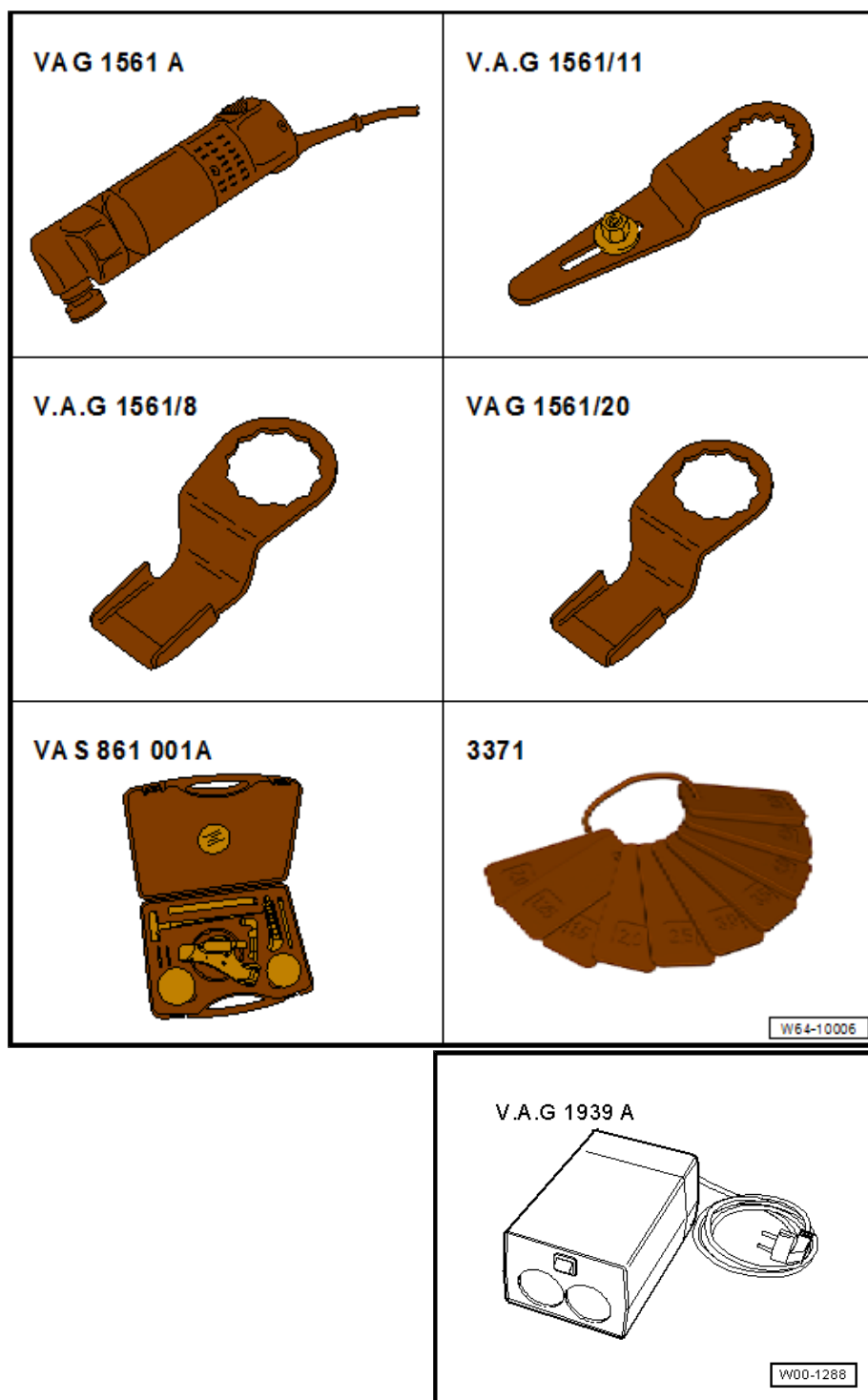
3.2.3 Removing and installing rear window, partition panel



Special tools and workshop equipment required



- ◆ Removal kit for flush bonded windows - VAS 6452-
- ◆ Suction holder - V.A.G 1344-
- ◆ Cutting tool - V.A.G 1351-
- ◆ Windscreen removal kit - V.A.G 1755-
- ◆ Double cartridge gun - VAS 5237-
- ◆ Hand-cartridge gun - V.A.G 1628-

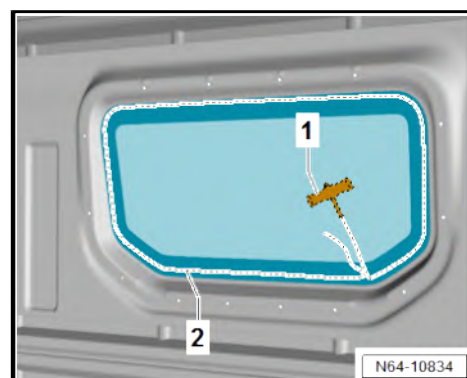
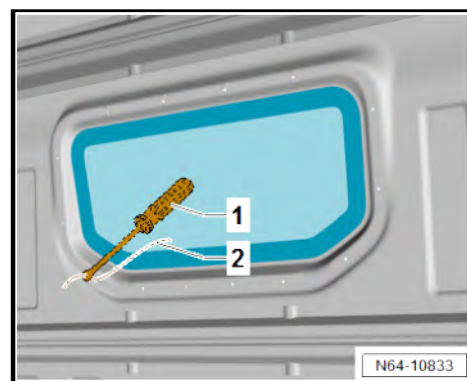


- ◆ Electric cutter - V.A.G 1561A-
- ◆ Universal cutter - V.A.G 1561/11-
- ◆ Blades, 15 and 25 mm - V.A.G 1561/7+8-
- ◆ Scraper (45 mm) - V.A.G 1561/20-
- ◆ Windscreen removal kit - VAS 861 001A-
- ◆ Setting gauge - 3371-
- ◆ Cartridge heating unit - V.A.G 1939A-

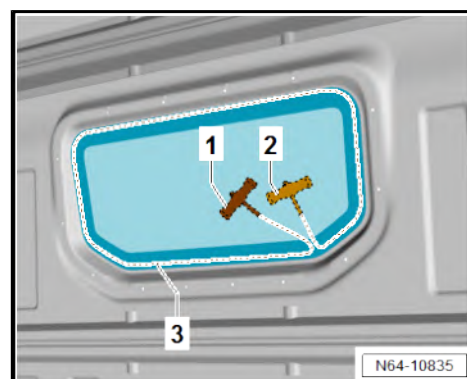


Removing

- Remove window frame for partition panel ⇒ General body repairs, interior; Rep. gr. 68 ; Equipment; Removing and installing partition panel .
- If applicable, unscrew bolts for additional connection to sliding window partition panel ⇒ [page 225](#) .
- Use awl - V.A.G 1474/2- -1- to poke through adhesive sealant from inside out.
- Pull cutting cord -2- through adhesive sealant to inside using awl - V.A.G 1474/2- -1-.
- Secure cutting cord -2- with pull handle - V.A.G 1351/1- -1- to prevent it from being pulled out.
- Place cutting cord -2- around rear window.
- Ensure that cutting cord -2- lies under rear window in corners.



- Secure cutting cord -3- with pull handle - V.A.G 1351/1- -2- to prevent it from being pulled out.
- Secure other end of cutting cord to pull handle - V.A.G 1351/1- -1-.
- Cut rear window free by pulling at pull handle - V.A.G 1351/1- -1-.



Installing

Install in reverse order of removal, observing the following:

- Preparing old undamaged windows for fitting ⇒ [page 204](#)
- Preparing new windows for fitting ⇒ [page 204](#)
- Preparing body flange for fitting ⇒ [page 205](#)
- Fitting notes for bonded windows ⇒ [page 202](#) .
- Minimum curing period ⇒ [page 201](#)

3.2.4 Installing sliding windows, reworking, vehicles with partition panel

Special tools and workshop equipment required

- ◆ Hand drill



- ◆ Torque wrench - V.A.G 1783-

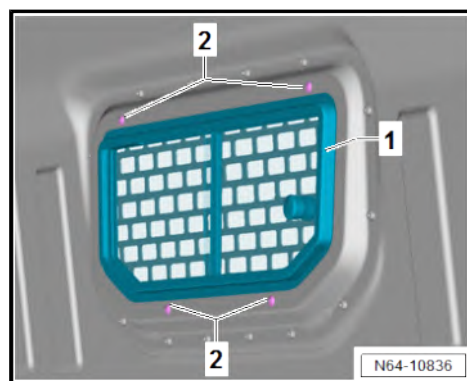


Procedure

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear safety goggles.
 - Wear protective gloves.
-
- Drill 5.0 mm holes for threaded connections -2- in partition panel and sliding window frame -1- using drill .
 - Maximum drilling depth: 13 mm
 - Drill 6.0 mm holes in partition panel (panel only) using drill .
 - Using M6 thread chaser, cut threads in sliding window frame -1-.
 - Moisten treads of bolts with locking fluid for bolts ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
 - Tighten additional bolted connections -2- to specified torque.



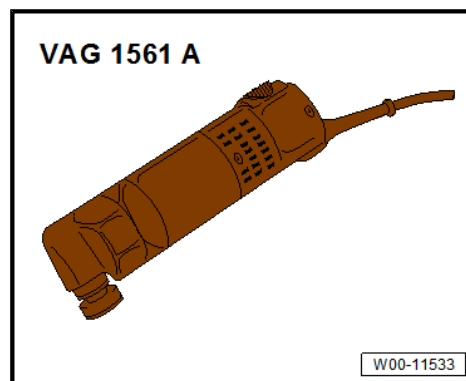
Specified torques

Component	Specified torque
Additional bolted connections -2-	2.5 Nm

3.2.5 Removing broken rear window

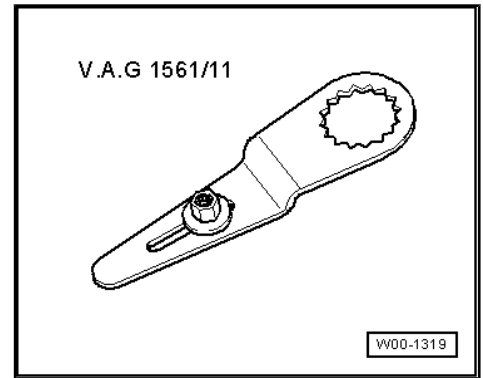
Special tools and workshop equipment required

- ◆ Electric cutter - V.A.G 1561A-





◆ Universal cutter - V.A.G 1561/11-



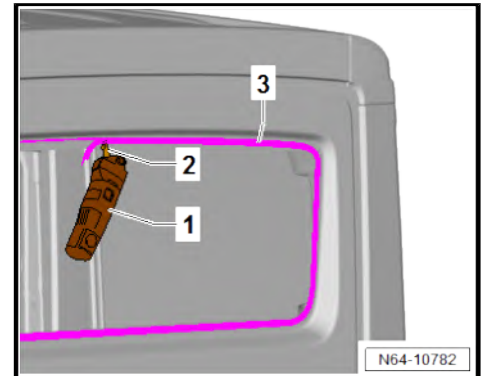
Removing

⚠ CAUTION

**Risk of injury to hands and eyes caused by glass splinters.
Risk of cuts.**

- Wear safety goggles.
- Wear protective gloves.

- Protect body and interior from glass splinters.
- Remove remaining pieces of glass up to adhesive sealing material.
- If fitted, pull off connectors for window heating.
- Apply textile reinforced adhesive tape all around body flange.
- Cut back adhesive sealant -3- in window opening using electric cutter - V.A.G 1561 A- -1- and universal blade - V.A.G 1561/11- -2-.



4 Side window

⇒ [“4.1 Assembly overview - front side window”, page 228](#)

⇒ [“4.2 Assembly overview - rear side windows”, page 230](#)

⇒ [“4.3 Removing and installing front side window”, page 232](#)

⇒ [“4.4 Removing and installing rear side window”, page 237](#)

4.1 Assembly overview - front side window

⇒ [“4.1.1 Assembly overview - front side window”, page 228](#)

⇒ [“4.1.2 Assembly overview - sliding window, sliding door”, page 229](#)

4.1.1 Assembly overview - front side window

1 - Side window

- ☐ Front
- ☐ Removing and installing
⇒ [page 232](#)

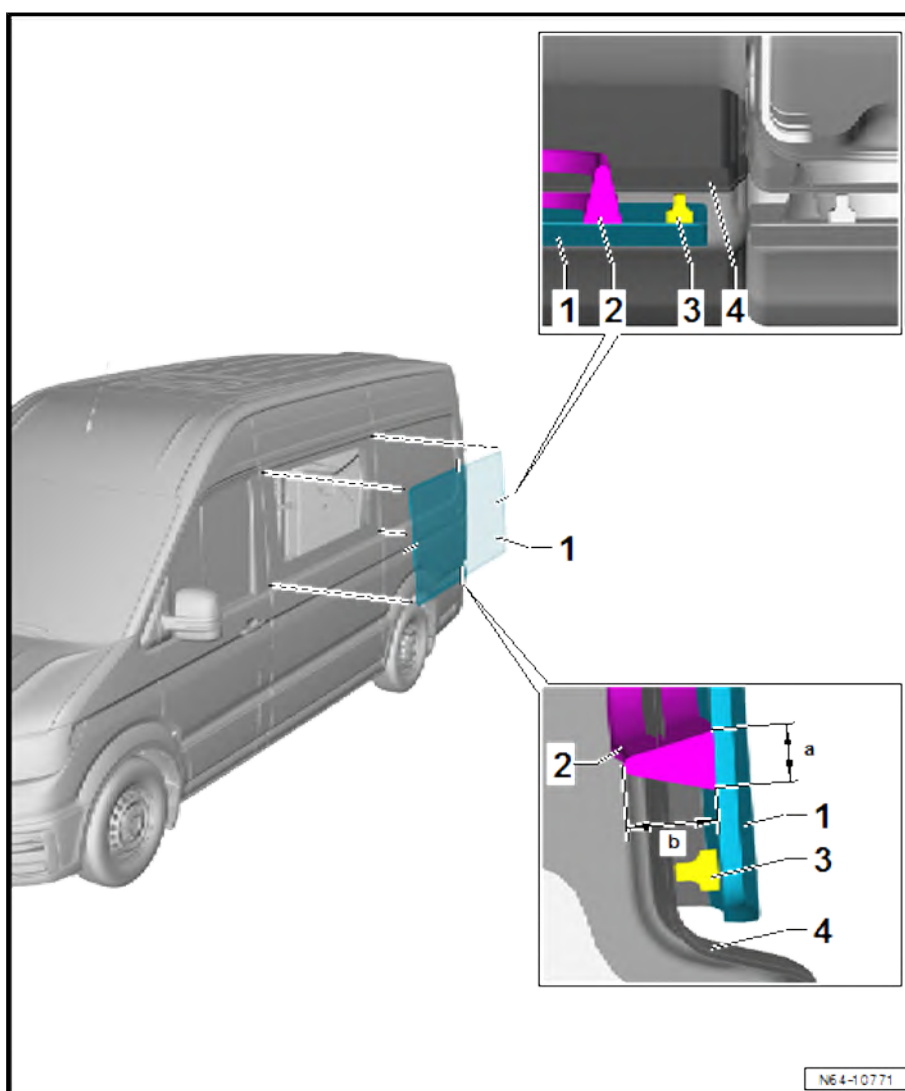
2 - PUR adhesive sealant

- ☐ Cross section of bead:
width = 7 mm
-dimension a-, height =
10 mm -dimension b-
(including precoating,
residual material on win-
dow glass and window
flange).
- ☐ Materials ⇒ [page 203](#)
- ☐ Installation instructions
⇒ [page 202](#) .
- ☐ Minimum curing period
⇒ [page 201](#) .

3 - Buffer stop

- ☐ Various number
- ☐ Materials ⇒ [page 203](#)

4 - Body flange





4.1.2 Assembly overview - sliding window, sliding door

1 - Sliding window

- ❑ If damaged, sliding window can only be replaced as complete item
- ❑ Removing and installing
⇒ [page 232](#)

2 - Cover strip

- ❑ Attached to side window
- ❑ Cannot be removed without being damaged

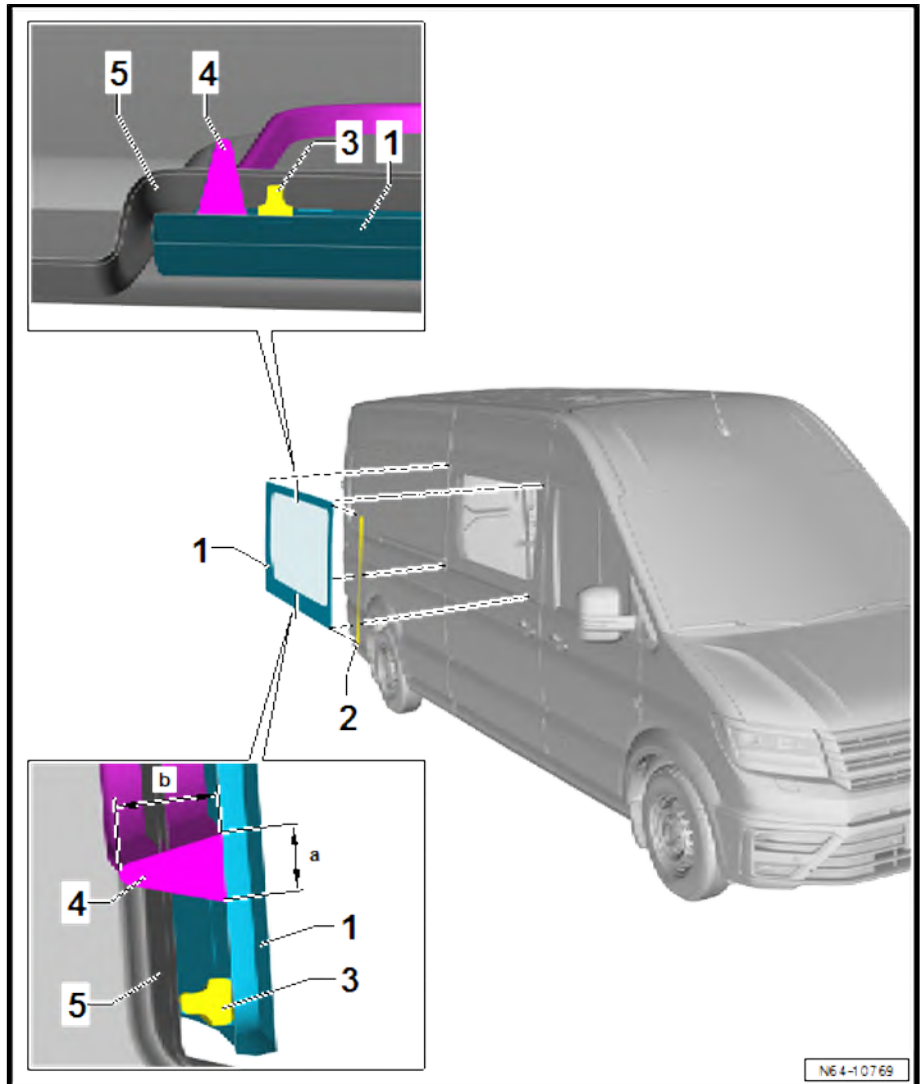
3 - Buffer stop

- ❑ Various number
- ❑ Materials ⇒ [page 203](#)

4 - PUR adhesive sealant

- ❑ Cross section of bead:
width = 7 mm
-dimension a-, height =
10 mm -dimension b-
(including precoating,
residual material on win-
dow glass and window
flange).
- ❑ Materials ⇒ [page 203](#)
- ❑ Installation instructions
⇒ [page 202](#) .
- ❑ Minimum curing period
⇒ [page 201](#) .

5 - Body flange



N6 4-1 07 69

4.2 Assembly overview - rear side windows

⇒ ["4.2.1 Assembly overview - front and rear side window", page 230](#)

⇒ ["4.2.2 Assembly overview - rear side windows", page 231](#)

4.2.1 Assembly overview - front and rear side window

1 - Side window

- ☐ Rear
- ☐ Removing and installing
⇒ [page 237](#)

2 - Side window

- ☐ Front
- ☐ Removing and installing
⇒ [page 232](#)

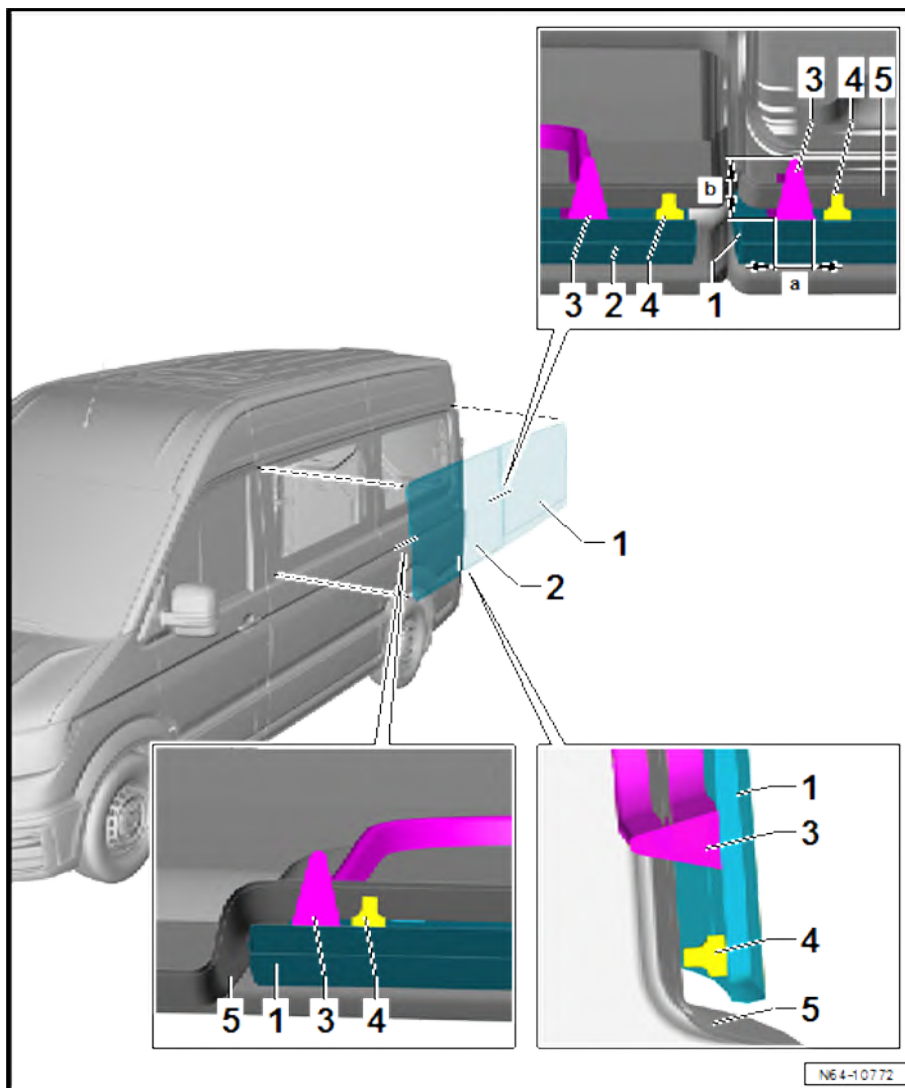
3 - PUR adhesive sealant

- ☐ Cross section of bead:
width = 7 mm
-dimension a-, height =
10 mm -dimension b-
(including precoat-
ing, residual material on win-
dow glass and window
flange).
- ☐ Materials ⇒ [page 203](#)
- ☐ Installation instructions
⇒ [page 202](#) .
- ☐ Minimum curing period
⇒ [page 201](#) .

4 - Buffer stop

- ☐ Various number
- ☐ Materials ⇒ [page 203](#)

5 - Body flange



4.2.2 Assembly overview - rear side windows

1 - Side window

- ☐ Rear
- ☐ Removing and installing
⇒ [page 237](#)

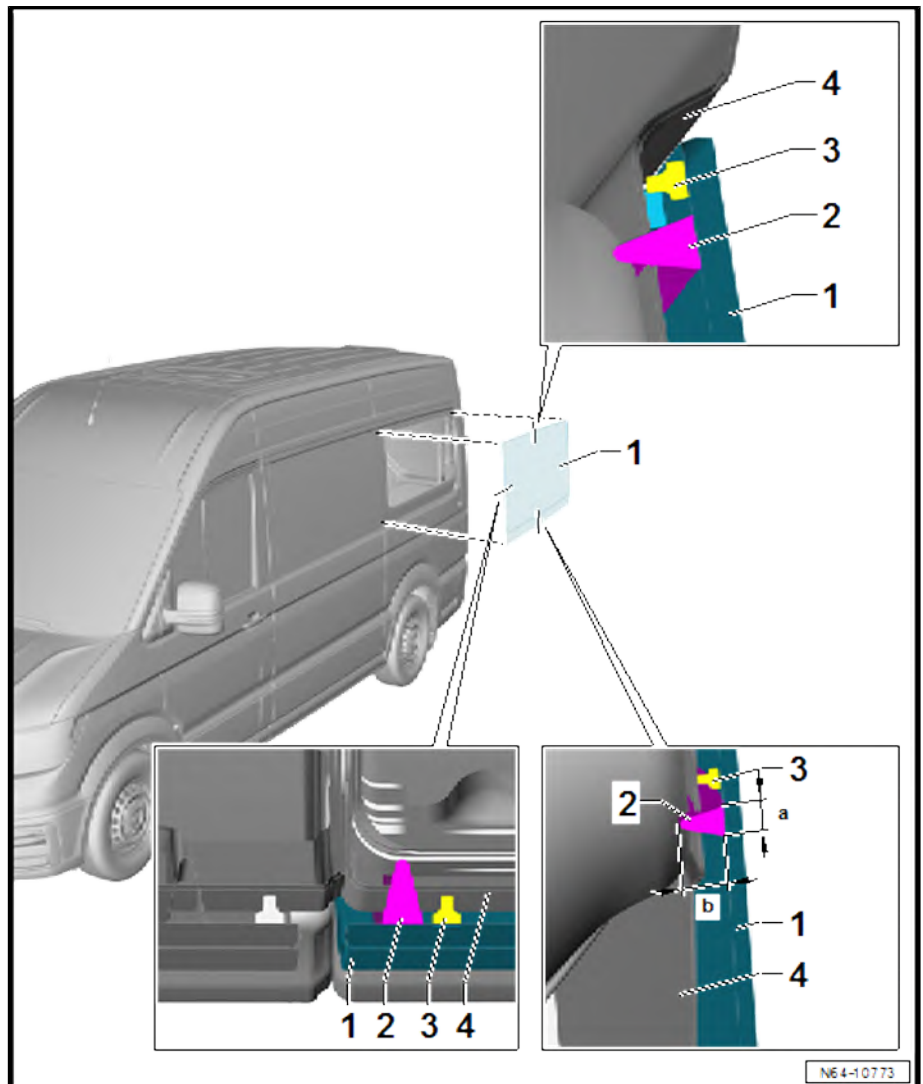
2 - PUR adhesive sealant

- ☐ Cross section of bead:
width = 7 mm
-dimension a-, height =
10 mm -dimension b-
(including precoating,
residual material on win-
dow glass and window
flange).
- ☐ Materials ⇒ [page 203](#)
- ☐ Installation instructions
⇒ [page 202](#) .
- ☐ Minimum curing period
⇒ [page 201](#) .

3 - Buffer stop

- ☐ Various number
- ☐ Materials ⇒ [page 203](#)

4 - Body flange





4.3 Removing and installing front side window

⇒ ["4.3.1 Removing and installing front side window", page 232](#)

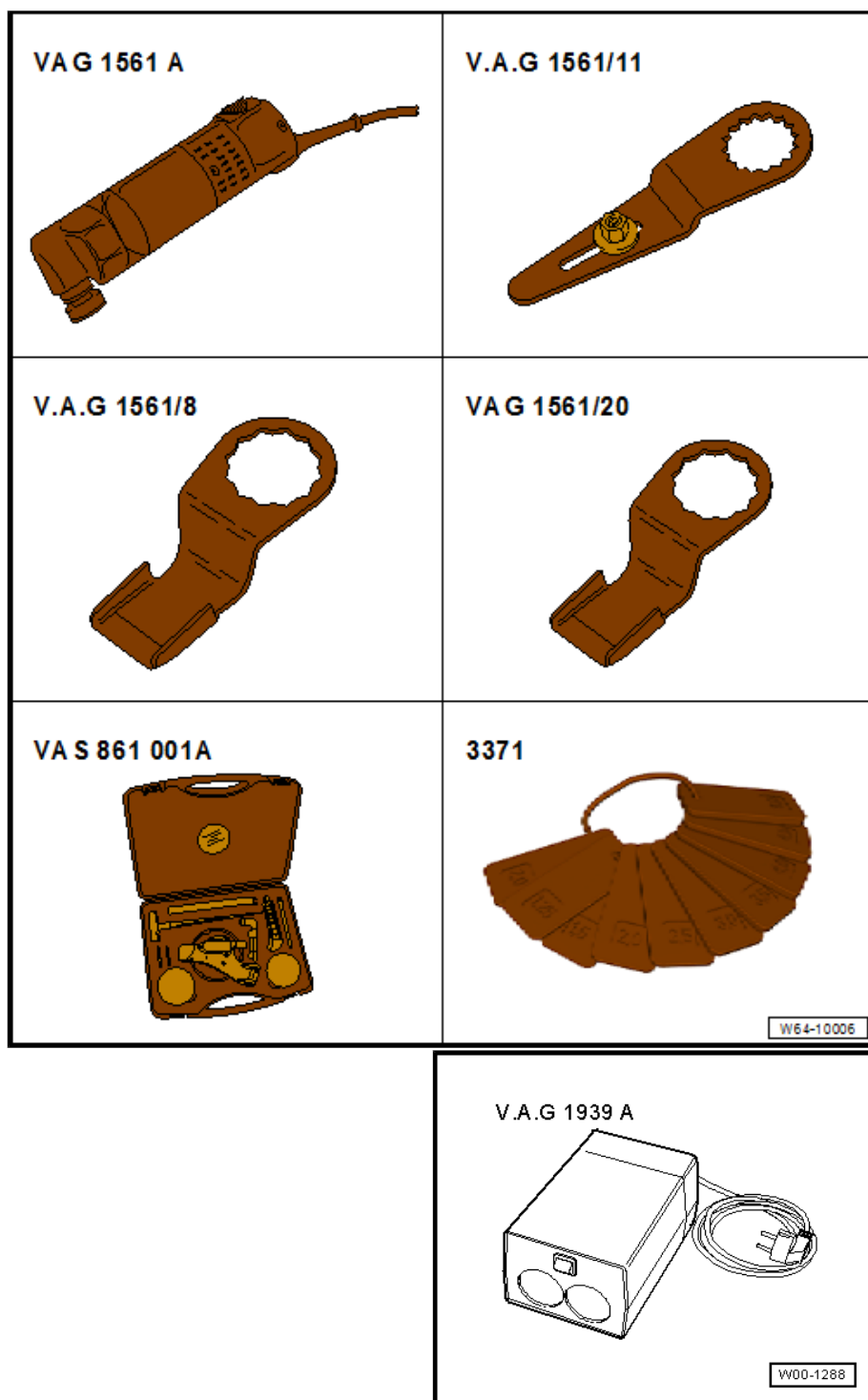
⇒ ["4.3.2 Removing and installing front side window, sliding window", page 234](#)

4.3.1 Removing and installing front side window

Special tools and workshop equipment required



- ◆ Removal kit for flush bonded windows - VAS 6452-
- ◆ Suction holder - V.A.G 1344-
- ◆ Cutting tool - V.A.G 1351-
- ◆ Windscreen removal kit - V.A.G 1755-
- ◆ Double cartridge gun - VAS 5237-
- ◆ Hand-cartridge gun - V.A.G 1628-

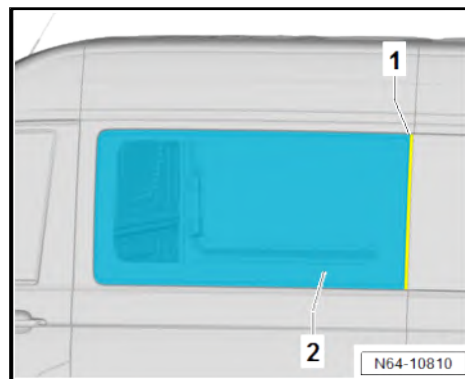


- ◆ Electric cutter - V.A.G 1561A-
- ◆ Universal cutter - V.A.G 1561/11-
- ◆ Blades, 15 and 25 mm - V.A.G 1561/7+8-
- ◆ Scraper (45 mm) - V.A.G 1561/20-
- ◆ Windscreen removal kit - VAS 861 001A-
- ◆ Setting gauge - 3371-
- ◆ Cartridge heating unit - V.A.G 1939A-

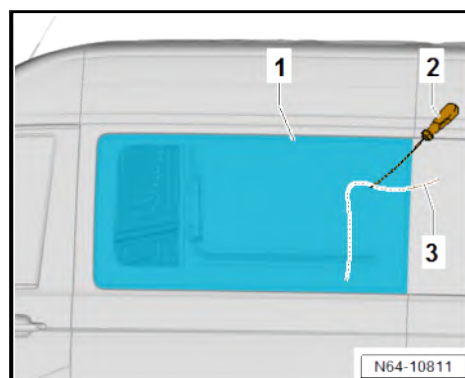


Removing

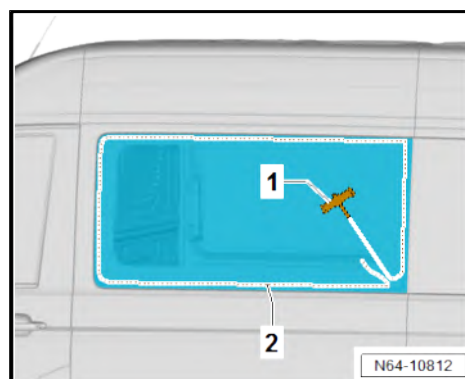
- Pull off cover strip -1- from front side window -2-.



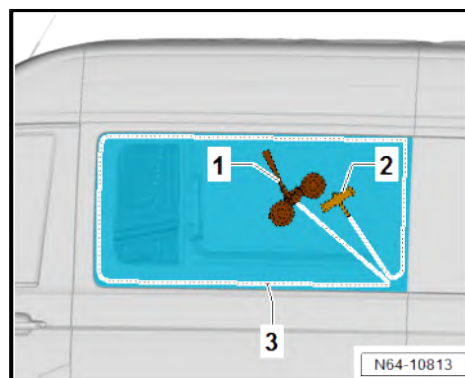
- Force end of cutting cord -3- through adhesive sealant of sliding window -1- towards inside using awl - V.A.G 1474/2- -2-.
- Pull other end of cutting cord through adhesive sealant as well.



- Place cutting cord -2- around side window.
- Secure one end of cutting cord to pull handle - V.A.G 1351/1- -1- to counterhold.



- Secure other end of cutting cord to reel device - V.A.G 1654 - -1-.
- Secure inside end of cutting cord against falling out using Pull toggle - V.A.G 1351/1- -2-.
- Cut side window free by moving reel device - V.A.G 1654 - -1- to required position.
- Press cutting cord -3- against window with wedge - V.A.G 1474/5- while cutting.



Installing

Install in reverse order of removal, observing the following:

- Preparing old undamaged windows for fitting ⇒ [page 204](#)
- Preparing new windows for fitting ⇒ [page 204](#)
- Preparing body flange for fitting ⇒ [page 205](#)
- Fitting notes for bonded windows ⇒ [page 202](#) .
- Minimum curing period ⇒ [page 201](#)

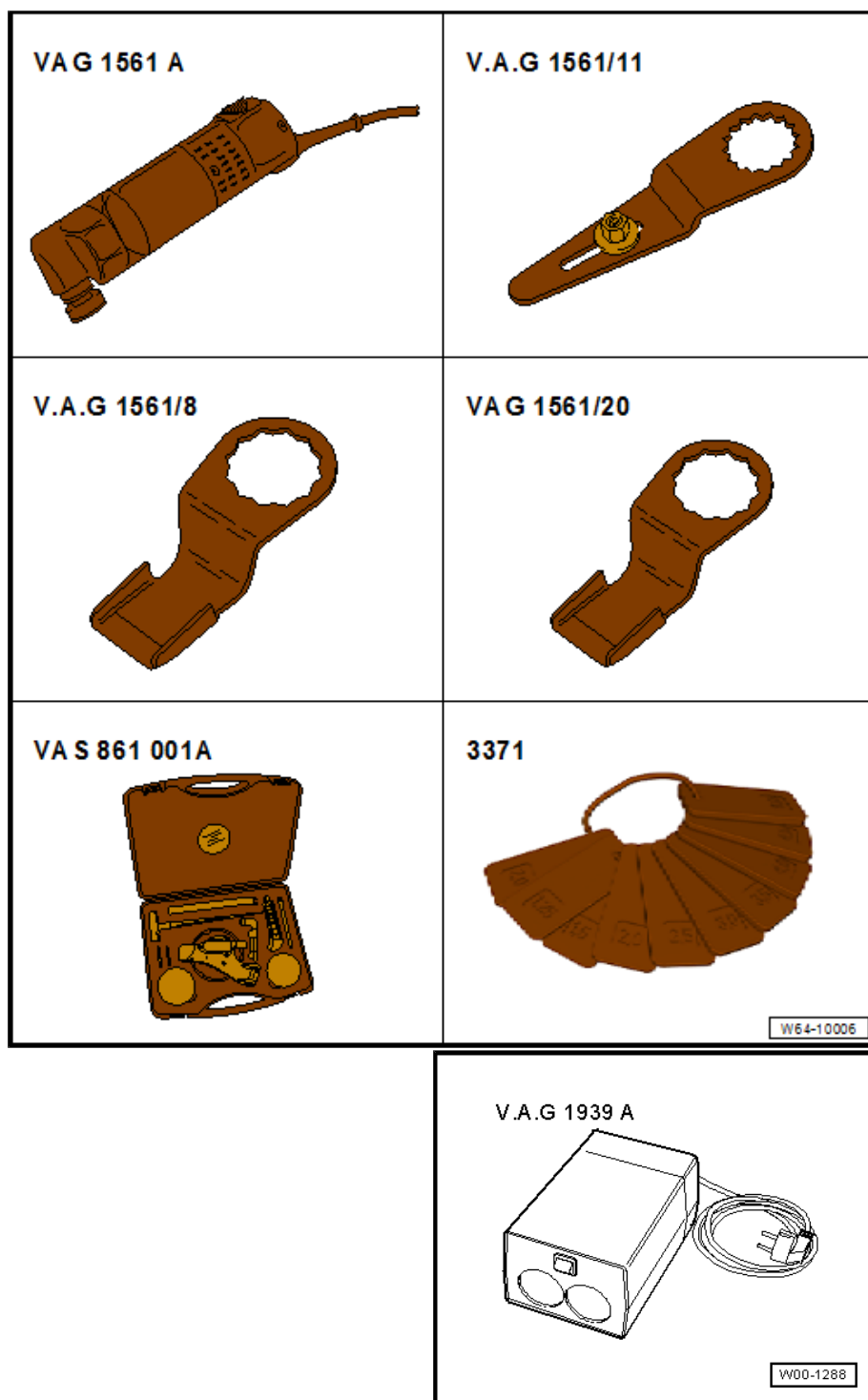
4.3.2 Removing and installing front side window, sliding window



Special tools and workshop equipment required



- ◆ Removal kit for flush bonded windows - VAS 6452-
- ◆ Suction holder - V.A.G 1344-
- ◆ Cutting tool - V.A.G 1351-
- ◆ Windscreen removal kit - V.A.G 1755-
- ◆ Double cartridge gun - VAS 5237-
- ◆ Hand-cartridge gun - V.A.G 1628-

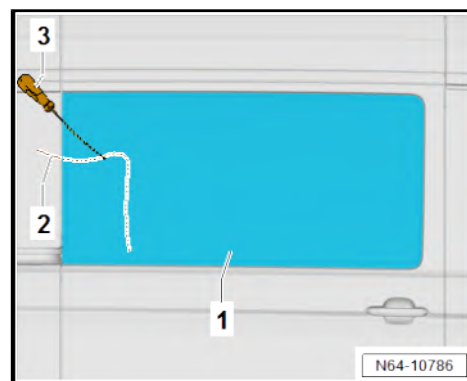


- ◆ Electric cutter - V.A.G 1561A-
- ◆ Universal cutter - V.A.G 1561/11-
- ◆ Blades, 15 and 25 mm - V.A.G 1561/7+8-
- ◆ Scraper (45 mm) - V.A.G 1561/20-
- ◆ Windscreen removal kit - VAS 861 001A-
- ◆ Setting gauge - 3371-
- ◆ Cartridge heating unit - V.A.G 1939A-

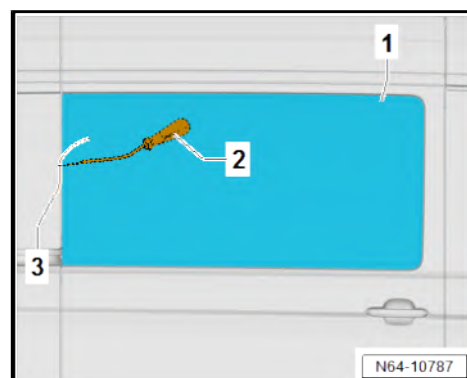


Removing

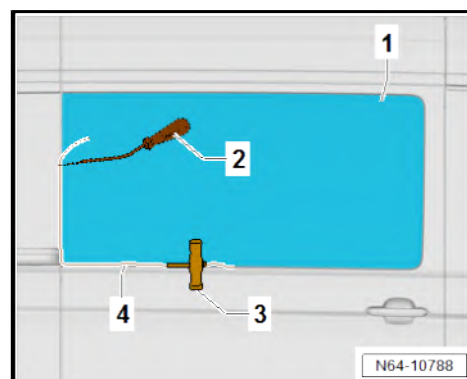
- Force end of cutting cord -2- through adhesive sealant of sliding window -1- towards inside using awl - V.A.G 1474/2- -3-.



- Secure end of cutting cord -3- to cutting tool - V.A.G 1351- -2-.
- Have a second mechanic fit and counterhold cutting tool - V.A.G 1351- on inside of sliding window -1-.



- Secure outer cutting cord -4- on pull handle - V.A.G 1351/1- -3-.
- Cut sliding window -1- free continuously from outside using pull handle - V.A.G 1351/1- -3-.
- Have cutting tool - V.A.G 1351- -2- used from inside accordingly by a second mechanic.



Installing

Install in reverse order of removal, observing the following:

- Preparing old undamaged windows for fitting ⇒ [page 204](#)
- Preparing new windows for fitting ⇒ [page 204](#)
- Preparing body flange for fitting ⇒ [page 205](#)
- Fitting notes for bonded windows ⇒ [page 202](#) .
- Minimum curing period ⇒ [page 201](#)

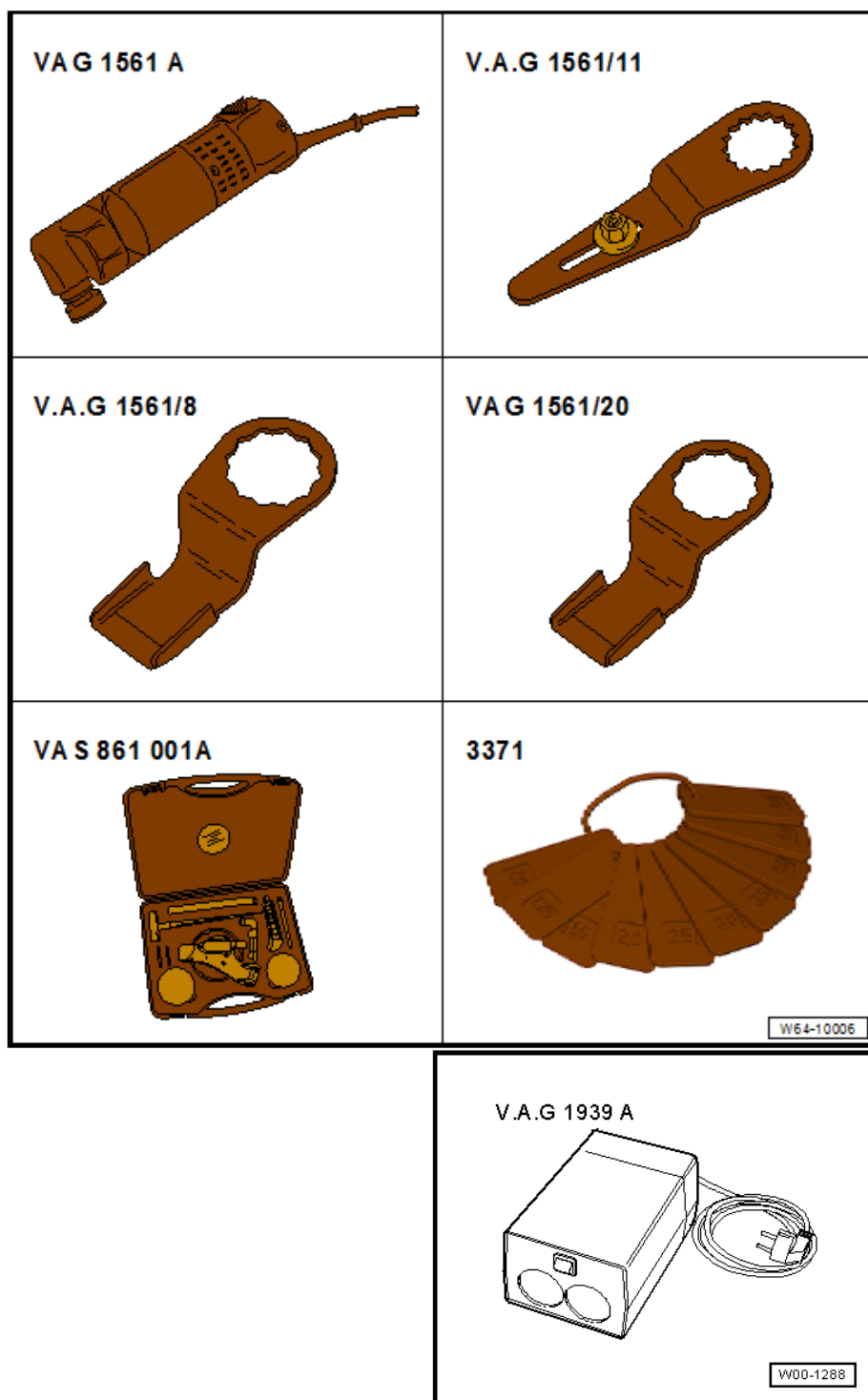
4.4 Removing and installing rear side window



**Special tools and workshop
equipment required**



- ◆ Removal kit for flush bonded windows - VAS 6452-
- ◆ Suction holder - V.A.G 1344-
- ◆ Cutting tool - V.A.G 1351-
- ◆ Windscreen removal kit - V.A.G 1755-
- ◆ Double cartridge gun - VAS 5237-
- ◆ Hand-cartridge gun - V.A.G 1628-



- ◆ Electric cutter - V.A.G 1561A-
- ◆ Universal cutter - V.A.G 1561/11-
- ◆ Blades, 15 and 25 mm - V.A.G 1561/7+8-
- ◆ Scraper (45 mm) - V.A.G 1561/20-
- ◆ Windscreen removal kit - VAS 861 001A-
- ◆ Setting gauge - 3371-
- ◆ Cartridge heating unit - V.A.G 1939A-



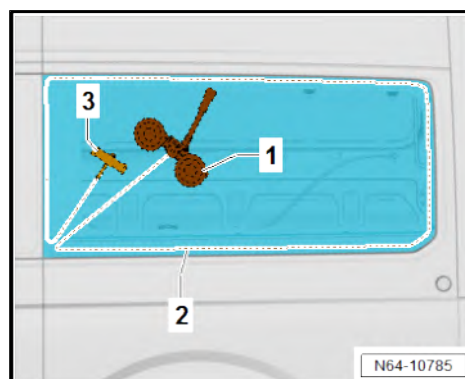
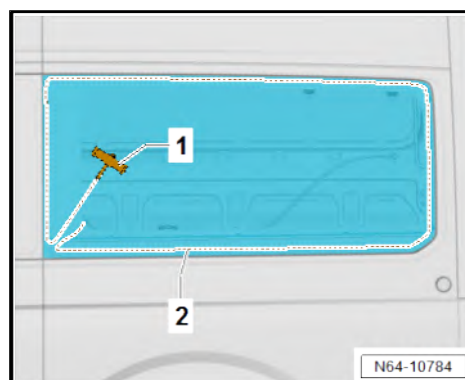
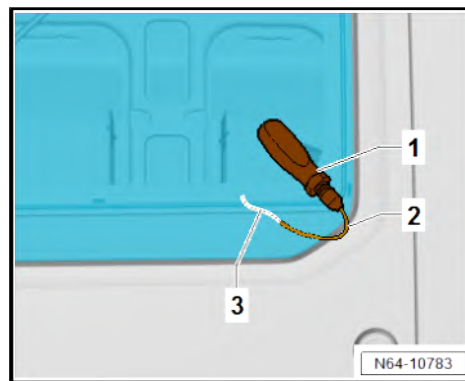
Removing

- Remove gap cover of window frame ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior .
- Take trimming needle -2- from windscreen removal kit - V.A.G 1755- and use handle -1- to force it through adhesive material of adhesive bead.
- Unscrew handle -1- and thread cutting cord -3- through eye of needle.
- Pull trimming needle -2- and cutting cord -3- through to inside.
- Pull other end of cutting cord through adhesive sealant as well.
- Place cutting cord -2- around side window.
- Secure one end of cutting cord to pull handle - V.A.G 1351/1- -1- to counterhold.
- Secure other end of cutting cord to reel device - V.A.G 1654 - -1-.
- Secure inside end of cutting cord against falling out using Pull toggle - V.A.G 1351/1- -3-.
- Cut side window free by moving reel device - V.A.G 1654 - -1- to required position.
- Press cutting cord -2- against window with wedge - V.A.G 1474/5- while cutting.

Installing

Install in reverse order of removal, observing the following:

- Preparing old undamaged windows for fitting ⇒ [page 204](#)
- Preparing new windows for fitting ⇒ [page 204](#)
- Preparing body flange for fitting ⇒ [page 205](#)
- Fitting notes for bonded windows ⇒ [page 202](#) .
- Minimum curing period ⇒ [page 201](#)





5 Door windows

⇒ ["5.1 Assembly overview - rear door window", page 241](#)

⇒ ["5.2 Removing and installing front door window", page 241](#)

⇒ ["5.3 Removing and installing fixed front door window", page 242](#)

⇒ ["5.4 Removing and installing rear door window", page 243](#)

5.1 Assembly overview - rear door window

1 - Cab door

- ☐ Removing and installing
⇒ [page 105](#)

2 - Sealing frame

- ☐ Removing and installing
⇒ [page 243](#)

3 - Vent window

- ☐ Removing and installing
⇒ [page 243](#)

4 - Nuts

- ☐ Qty. 2
- ☐ 8 Nm

5 - Plugs

- ☐ Qty. 2

6 - Bolt

- ☐ With or without washer,
depending on equip-
ment level
- ☐ 4 Nm

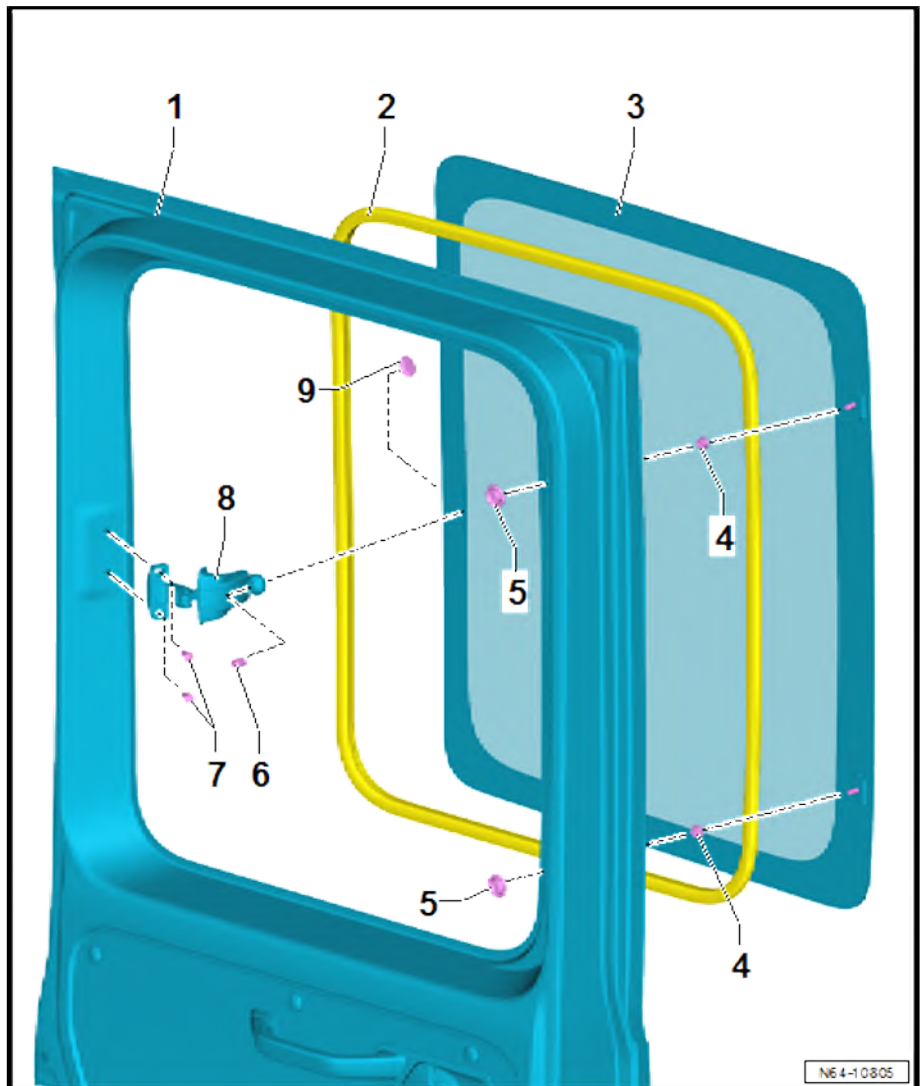
7 - Bolts

- ☐ Qty. 2
- ☐ 8 Nm

8 - Tilting arm

- ☐ Removing and installing
⇒ [page 243](#)

9 - Cap



5.2 Removing and installing front door window

Removing

- Remove front door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .
- Remove protective film ⇒ [page 88](#) .



- Lower door window -1- until retaining hooks are accessible through opening -2- in window regulator.

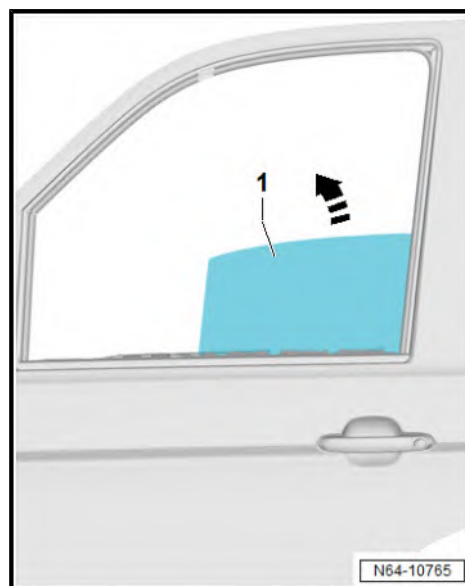
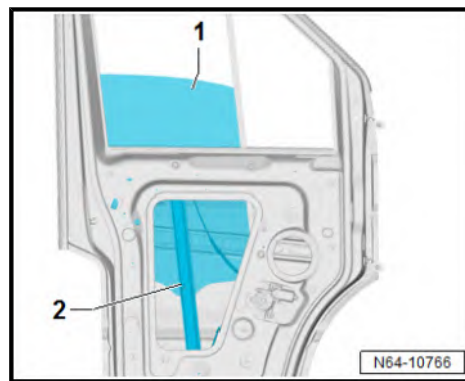


Note

- ◆ *If this part of the work cannot be done due to a fault in the window regulator motor, the window regulator motor can be removed.*
- ◆ *The window can then be pushed to required working position.*
- Use a small screwdriver to carefully release locking devices through opening -2- in window regulator. While doing this, lift door window -1- slightly to prevent it from falling back in locking device.
- Lower door window -1- downwards into door.
- Remove window regulators ⇒ [page 87](#) .
- Remove fixed front door window ⇒ [page 242](#) .
- Lift door window -1- through opening in door, and guide it upwards.
- Swing door window -1- upwards in -direction of arrow- out of door aperture.

Installing

Install in reverse order.



5.3 Removing and installing fixed front door window

Special tools and workshop equipment required

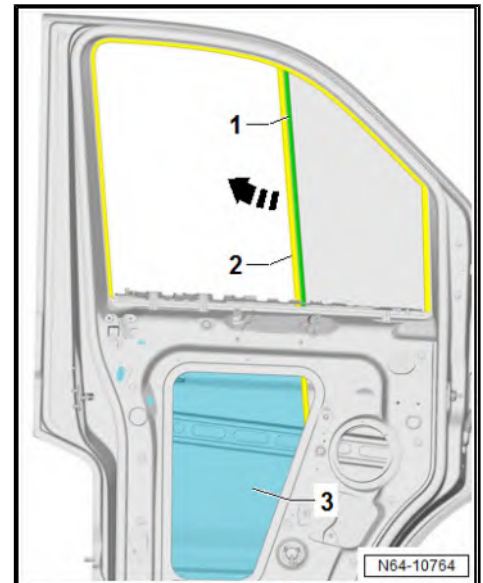
- ◆ Torque wrench - V.A.G 1783-





Removing

- Remove inner window slot seal ➤ [page 97](#) .
- Remove outer window slot seal ➤ [page 98](#) .
- Remove door window ➤ [page 241](#) .
- Pull front door window seal -2- in -direction of arrow- out of middle frame -1-.



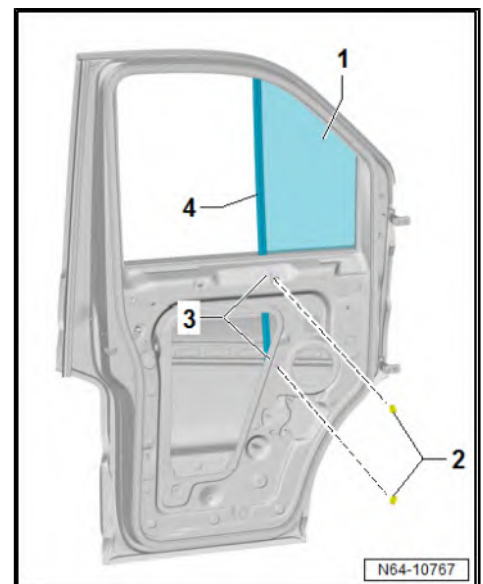
- Unclip sealing cap -2- from door.
- Unscrew bolts -3-.
- Guide guide rail -4- downwards, and swing it upwards out of door aperture.
- Pull fixed door window at front -1- out of window channel.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ ["2.4 Assembly overview - window channels and window slot \(aperture\) seals"](#), [page 86](#)



5.4 Removing and installing rear door window

➤ ["5.4.1 Removing and installing vent window"](#), [page 243](#)

➤ ["5.4.2 Removing and installing sealing frame"](#), [page 245](#)

➤ ["5.4.3 Removing and installing stay"](#), [page 245](#)

5.4.1 Removing and installing vent window

Special tools and workshop equipment required

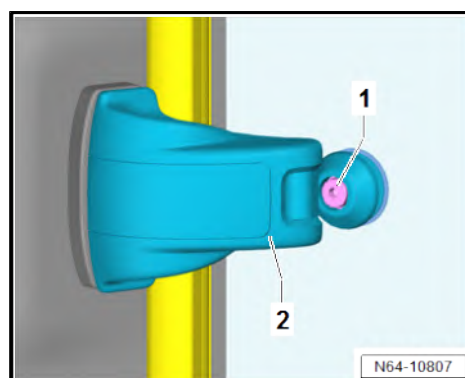


- ◆ Torque wrench - V.A.G 1783-



Removing

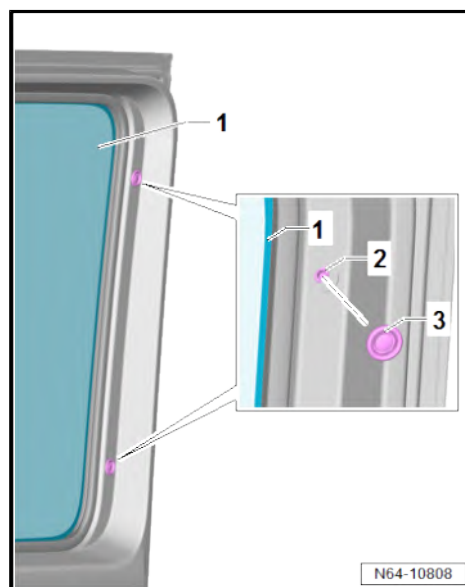
- Unscrew bolt -1- from window mechanism -2-.



- Unclip plug -3- from cab door.
- Unscrew nut -2-.
- Remove vent window -1- from exterior.

Installing

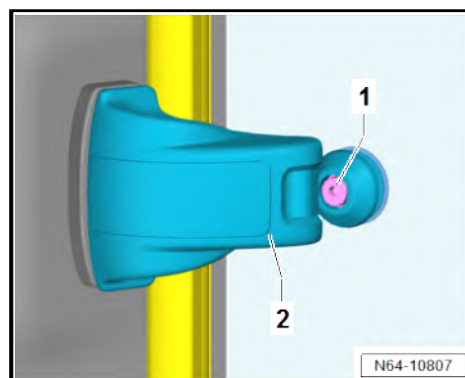
Install in reverse order of removal, observing the following:



- With vent window open, push tilt arm -2- in direction of travel and tighten bolt -1-.

Specified torques

- ◆ ⇒ [“5.1 Assembly overview - rear door window”, page 241](#)

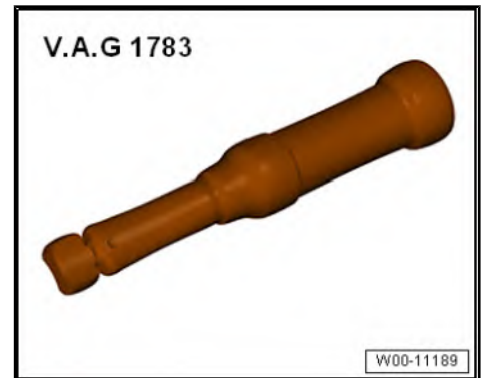




5.4.2 Removing and installing sealing frame

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



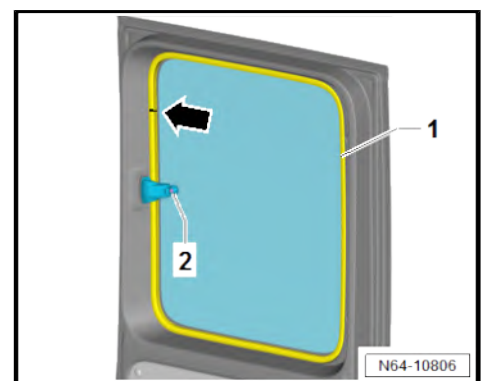
Removing

- Unscrew bolt -2-.
- Pull sealing frame -1- off body flange.

Installing

Install in reverse order of removal, observing the following:

- Starting at tilt arm, install sealing frame -1- at vulcanisation point -arrow-.



Note

Vulcanisation point -arrow- must not be kinked.

Specified torques

- ◆ ⇒ [“5.1 Assembly overview - rear door window”, page 241](#)

5.4.3 Removing and installing stay

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

- Unscrew bolt -2-.
- Unscrew bolts -3-.
- Remove stay -1- from cab door.

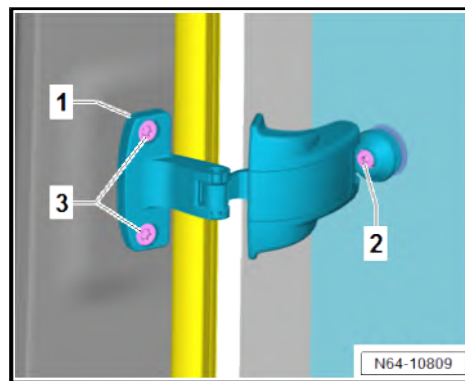
Installing

Install in reverse order of removal, observing the following:

- Pull stay -1- inwards and tighten bolts on cab door -3-.
- With vent window open, push tilt arm -1- in direction of travel and tighten bolt -2-.

Specified torques

- ♦ ⇒ [“5.1 Assembly overview - rear door window”, page 241](#)





66 – Exterior equipment

1 Radiator grille and front trim

⇒ [“1.1 Assembly overview - radiator grille”, page 247](#)

⇒ [“1.2 Removing and installing radiator grille”, page 248](#)

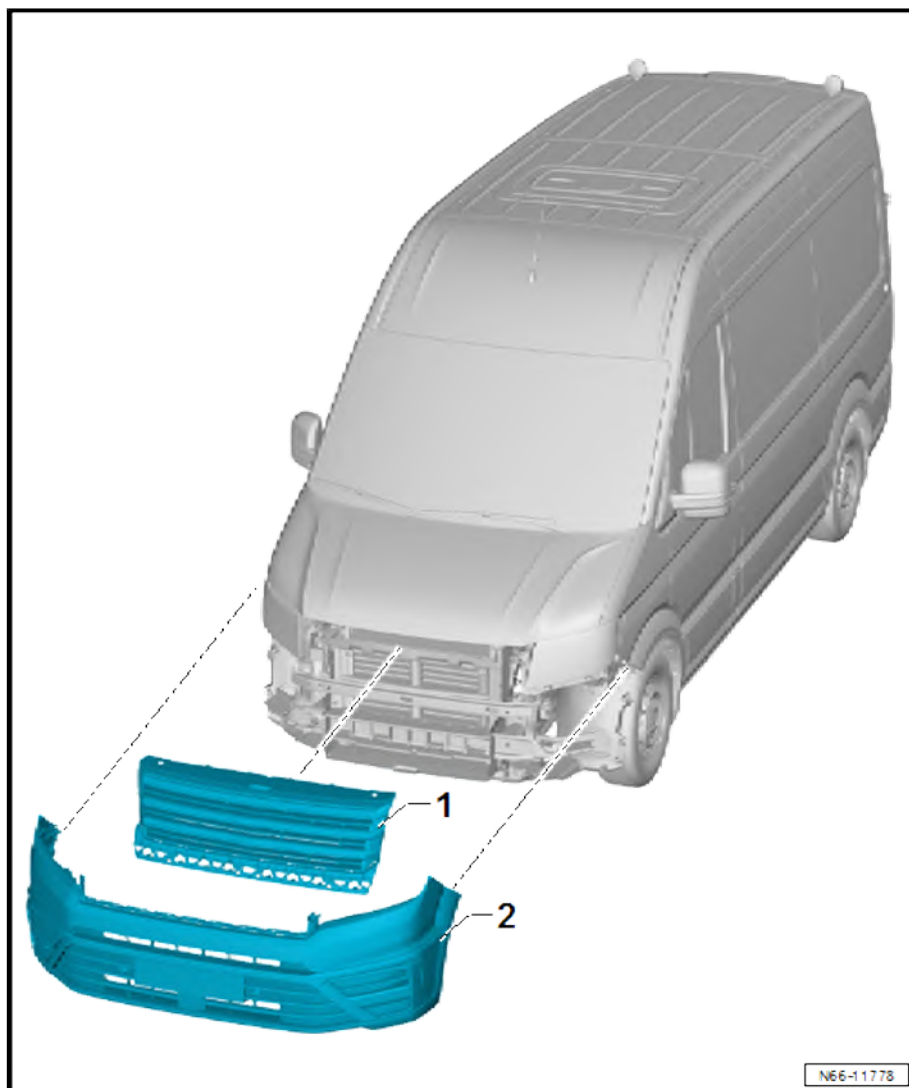
1.1 Assembly overview - radiator grille

1 - Radiator grille

- ☐ Removing and installing
⇒ [page 248](#)

2 - Bumper cover

- ☐ Removing and installing
⇒ [page 179](#)





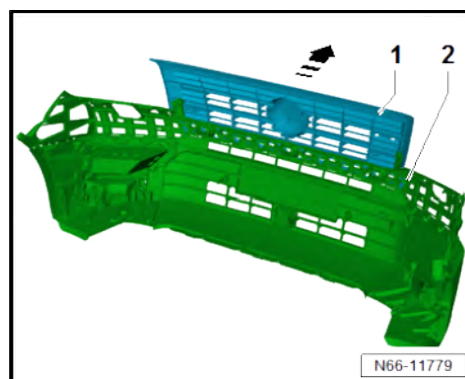
1.2 Removing and installing radiator grille

Removing

- Remove trim ➔ [page 184](#) .
- Release fasteners. When doing this, remove radiator grille
-1- in -direction of arrow- from bumper cover -2-.

Installing

Install in reverse order.





2 Mouldings, trims, extensions

⇒ [“2.1 Assembly overview - covers”, page 249](#)

⇒ [“2.2 Assembly overview - wheel arch covers”, page 251](#)

⇒ [“2.3 Assembly overview - heat shield”, page 252](#)

⇒ [“2.4 Removing and installing cover”, page 253](#)

⇒ [“2.5 Removing and installing trim strip”, page 264](#)

⇒ [“2.6 Removing and installing B-pillar trim”, page 265](#)

⇒ [“2.8 Removing and installing underbody heat shield”, page 268](#)

2.1 Assembly overview - covers

⇒ [“2.1.1 Assembly overview - covers”, page 249](#)

⇒ [“2.1.2 Assembly overview - covers, double cab”, page 250](#)

2.1.1 Assembly overview - covers

1 - Protective strip

- ☐ Wings
- ☐ Removing and installing
⇒ [page 253](#)

2 - Protective strip

- ☐ Door
- ☐ Removing and installing
⇒ [page 253](#)

3 - Trim

- ☐ B-pillar
- ☐ Removing and installing
⇒ [page 265](#)

4 - Bolts

- ☐ Qty. 2
- ☐ 4.5 Nm

5 - Protective strip

- ☐ B-pillar
- ☐ Removing and installing
⇒ [page 253](#)

6 - Protective strip

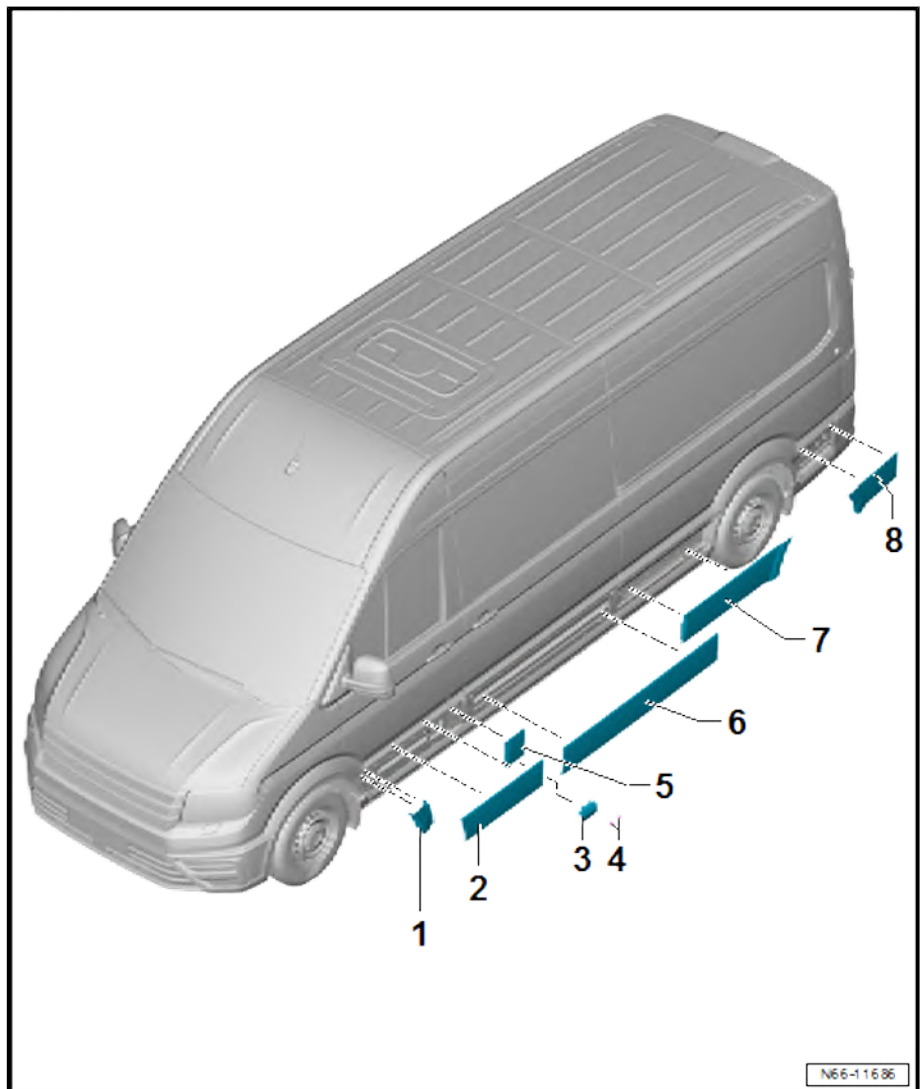
- ☐ Sliding door
- ☐ Removing and installing
⇒ [page 253](#)

7 - Protective strip

- ☐ Front side panel
- ☐ Removing and installing
⇒ [page 253](#)

8 - Protective strip

- ☐ Rear side panel
- ☐ Removing and installing
⇒ [page 253](#)





2.1.2 Assembly overview - covers, double cab

1 - Protective strip

- ☐ Wings
- ☐ Removing and installing
⇒ [page 253](#)

2 - Protective strip

- ☐ Rear door
- ☐ Removing and installing
⇒ [page 253](#)

3 - Trim

- ☐ B-pillar
- ☐ Removing and installing
⇒ [page 265](#)

4 - Bolts

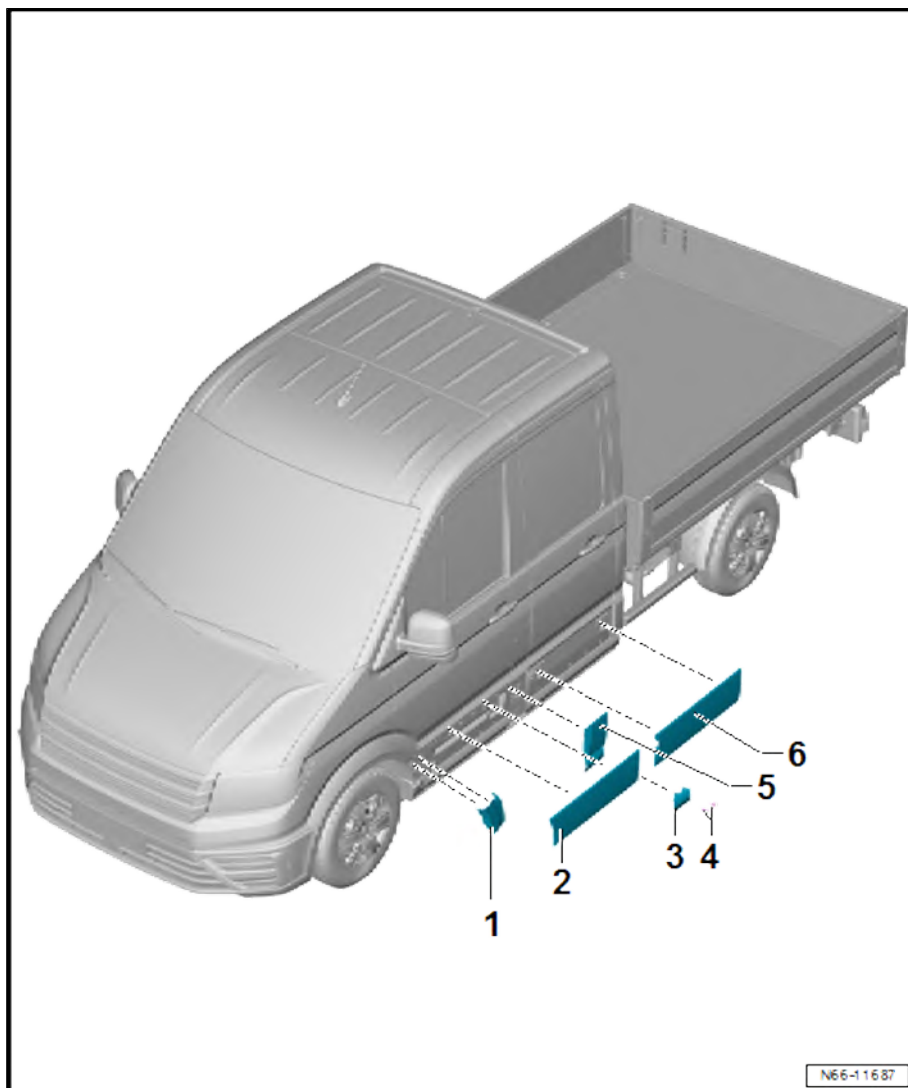
- ☐ Qty. 2
- ☐ 4.5 Nm

5 - Protective strip

- ☐ B-pillar
- ☐ Removing and installing
⇒ [page 253](#)

6 - Protective strip

- ☐ Rear door
- ☐ Removing and installing
⇒ [page 253](#)





2.2 Assembly overview - wheel arch covers

1 - Wheel arch cover

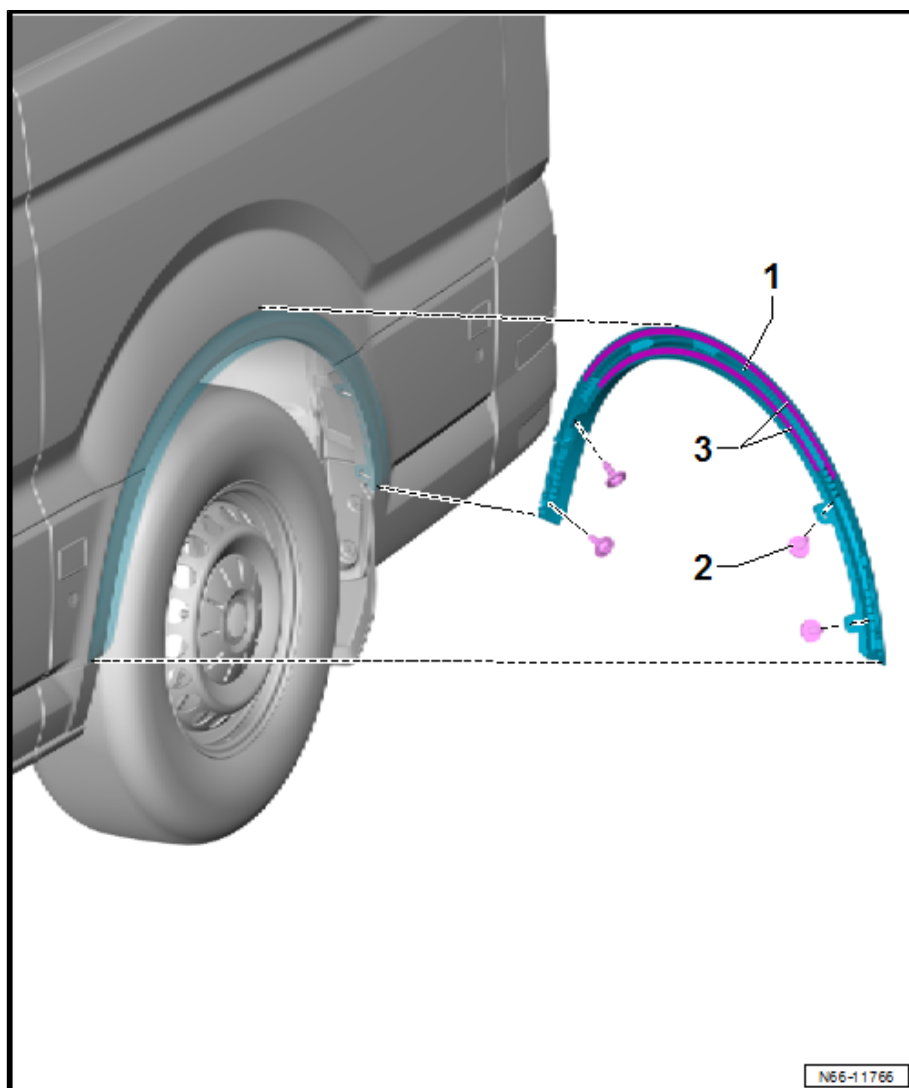
- ☐ Removing and installing
⇒ [page 266](#)

2 - Bolts

- ☐ Qty. 4
- ☐ 1.5 Nm

3 - Adhesive tape

- ☐ Double-sided



Volkswagen Technical Site: <https://vwts.ru>



2.3 Assembly overview - heat shield

1 - Heat shield

- ☐ Front
- ☐ Removing and installing
⇒ [page 268](#)

2 - Clamping washers

- ☐ Qty. 3

3 - Clamping washers

- ☐ Qty. 2

4 - Heat shield

- ☐ Front, centre
- ☐ Removing and installing
⇒ [page 268](#)

5 - Clamping washer

6 - Heat shield

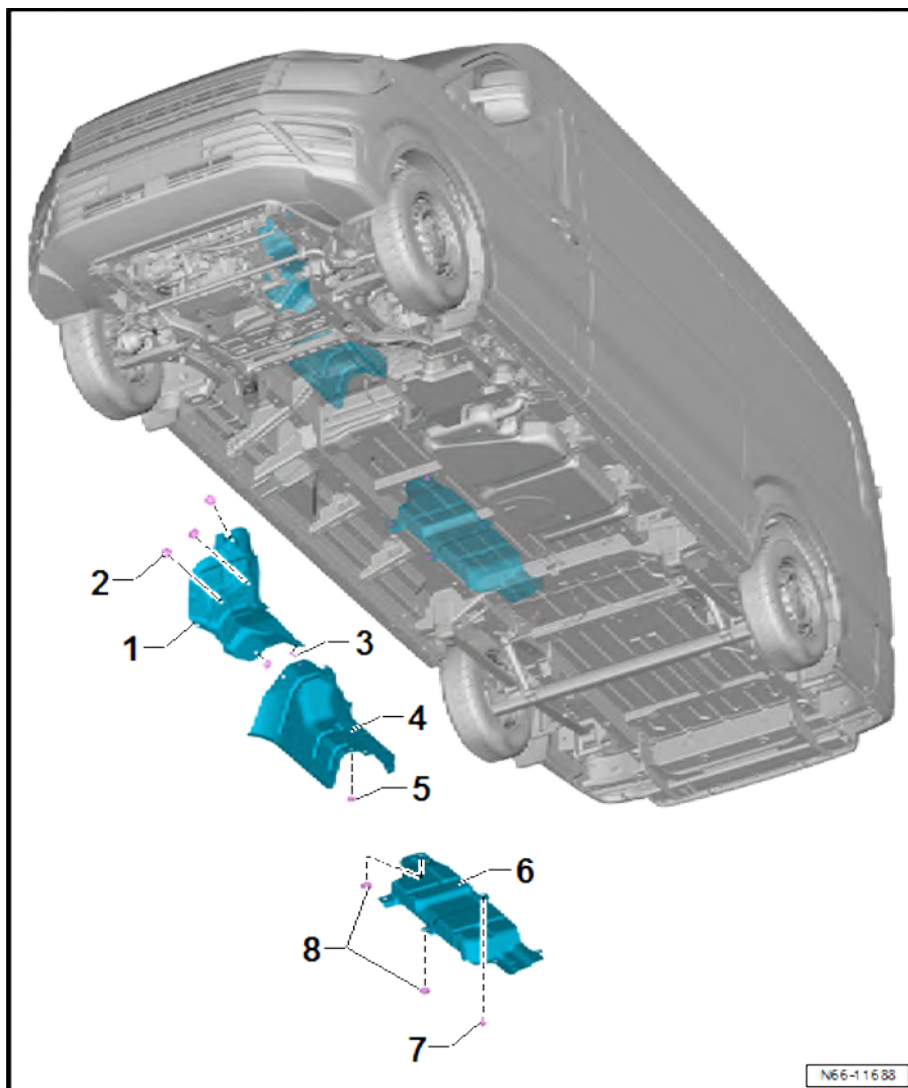
- ☐ Centre
- ☐ Removing and installing
⇒ [page 268](#)

7 - Bolt

- ☐ 12 Nm

8 - Clamping washers

- ☐ Qty. 2





2.4 Removing and installing cover

⇒ [“2.4.1 Fitting instructions”, page 253](#)

⇒ [“2.4.2 Removing and installing cover, wing”, page 254](#)

⇒ [“2.4.3 Removing and installing cover, front door”, page 254](#)

⇒ [“2.4.4 Removing and installing cover, B-pillar”, page 256](#)

⇒ [“2.4.5 Removing and installing cover, sliding door”, page 256](#)

⇒ [“2.4.6 Removing and installing cover, side panel”, page 257](#)

⇒ [“2.4.7 Removing and installing cover, double cab”, page 258](#)

⇒ [“2.4.8 Removing and installing cover, rear”, page 259](#)

⇒ [“2.4.9 Removing and installing cover, D-pillar”, page 260](#)

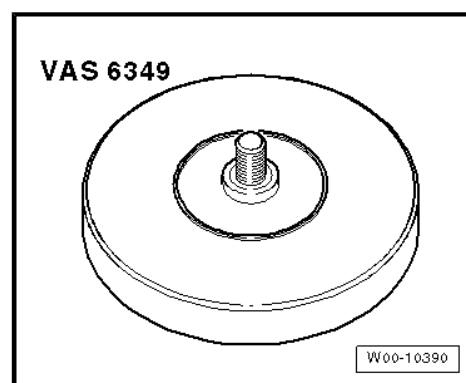
⇒ [“2.4.10 Removing and installing cover, rear side panel”, page 261](#)

⇒ [“2.4.11 Installing sender retainer for parking aid and park assist steering”, page 262](#)

2.4.1 Fitting instructions

Special tools and workshop equipment required

- ◆ Adhesive strip remover - VAS 6349-



When removing cover, adhesive tape may become damaged.

If using new adhesive tape as prescribed in ⇒ Electronic parts catalogue (ETKA) and ⇒ webMANTIS for MAN TGE, the trim strip can be reused.

- Check clips for damage and renew if necessary.

If a cover is installed for the first time on a new vehicle or on newly painted surfaces, the procedure in ⇒ Workshop manual “Paint”; ⇒ Paintwork repairs; ⇒ Paint finish; ⇒ Mouldings & film should be observed.

If a cover is removed and reattached, use only adhesive remover to remove the adhesive residue.

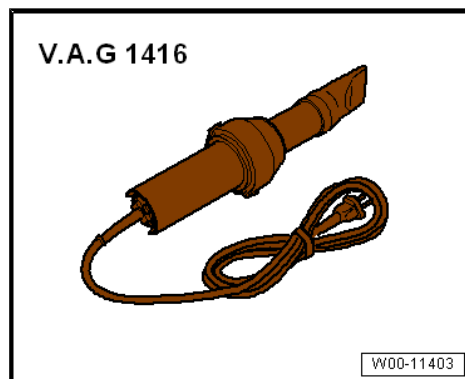
- Remove existing adhesive residue of adhesive tape using adhesive strip remover - VAS 6349- .
- Bonding surfaces must be free of dust and grease.
- Install cover immediately after cleaning.
- Do not remove protective foil until immediately before installation.
- The working temperature is approx. 21°C.



2.4.2 Removing and installing cover, wing

Special tools and workshop equipment required

- ♦ Hot-air blower - V.A.G 1416-



Removing

- Unscrew bolts -3-.
- Heat up cover -1- in area of bonding surface -4- using hot air blower - V.A.G 1416- .
- Pull off cover -1-.
- Pull out cover -1- from catches -2-.

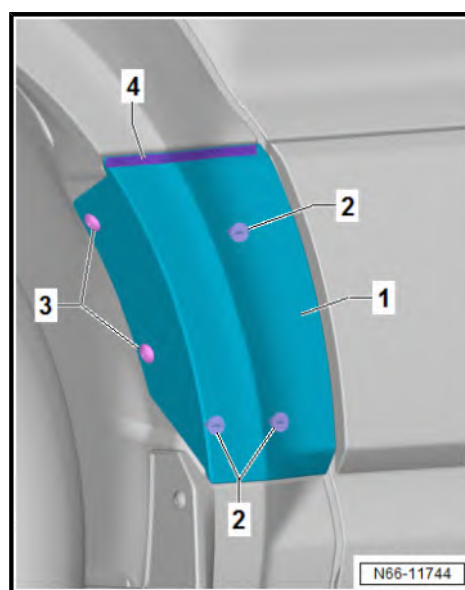
Installing

Install in reverse order of removal, observing the following:

Follow installation instructions ➔ [page 253](#) .

Specified torques

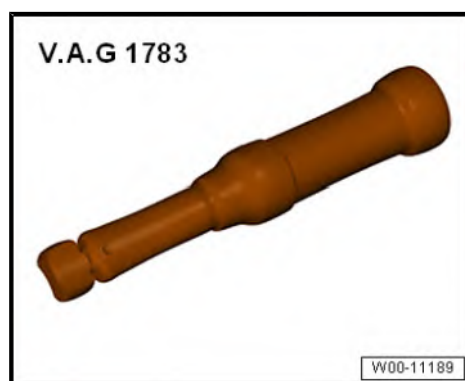
Component	Specified torque
Bolts for cover	1.5 Nm



2.4.3 Removing and installing cover, front door

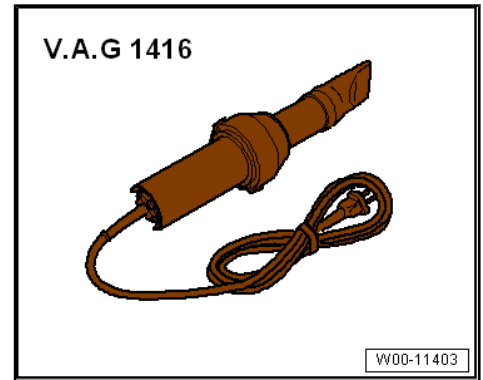
Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-



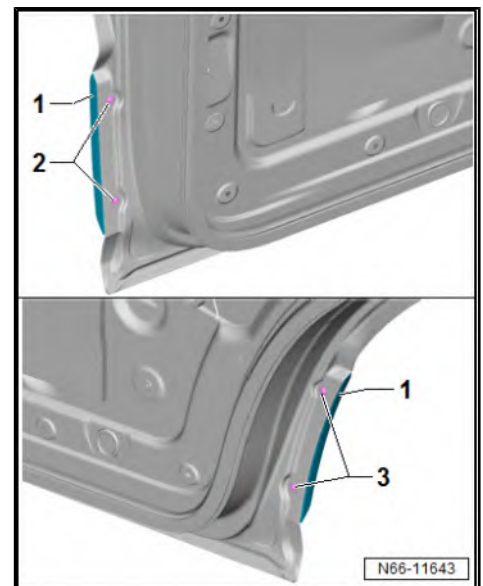


◆ Hot-air blower - V.A.G 1416-

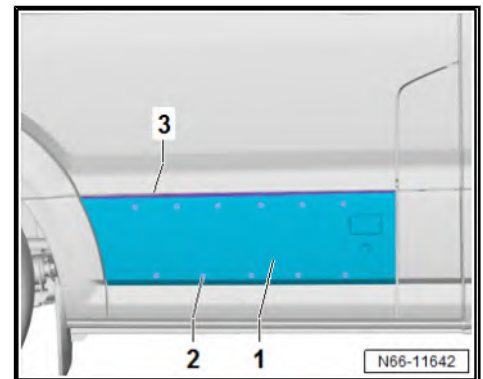


Removing

- Unscrew bolt -2- on inside at front from cover -1-.
- Unscrew bolts -3- on inside at rear from cover -1-.



- Heat up cover -1- in area of bonding surface -3- using hot air blower - V.A.G 1416- .
- Pull off cover -1-.
- Pull out cover -1- from catches -2-.



- If present, separate electrical connector -3- for front side left parking aid sender 2 - G957- -4-.
- Separate electrical connector -2- for front left side marker light bulb - M33- -1-.

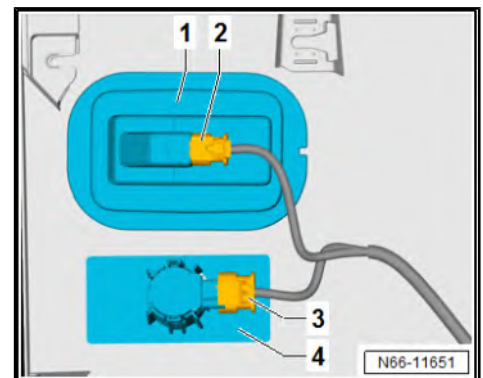
Installing

Install in reverse order of removal, observing the following:

Follow installation instructions ➔ [page 253](#) .

Specified torques

Component	Specified torque
Bolts for cover	1.5 Nm





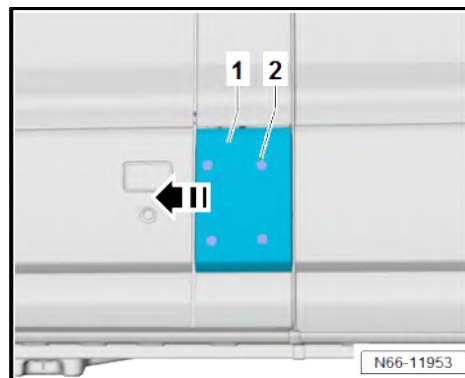
2.4.4 Removing and installing cover, B-pillar

Removing

- Detach cover -1- in direction of -arrow- from fasteners -2-.
- While doing this, remove cover -1-.

Installing

Install in reverse order.



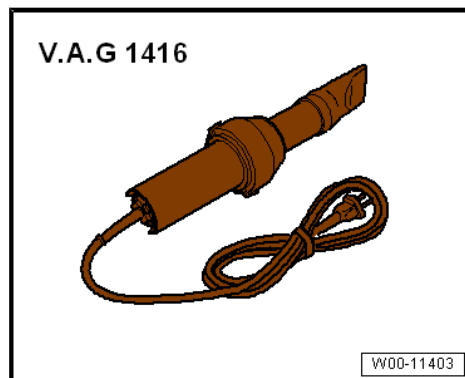
2.4.5 Removing and installing cover, sliding door

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



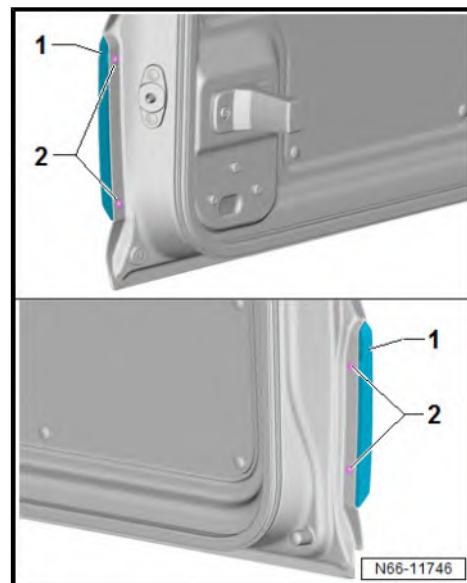
- ◆ Hot-air blower - V.A.G 1416-





Removing

- Unscrew bolts -2- on inside at rear from cover -1-.
- Unscrew bolt -2- on inside at front from cover -1-.



- Heat up cover -1- in area of bonding surface -3- using hot air blower - V.A.G 1416- .
- Pull off cover -1-.
- Pull out cover -1- from catches -2-.

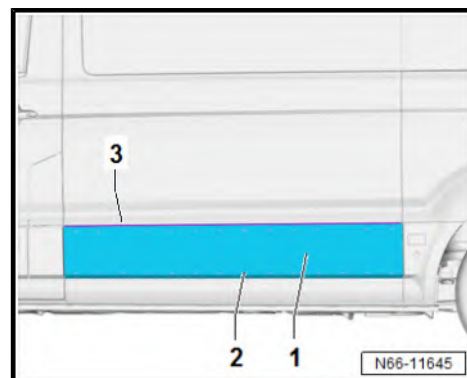
Installing

Install in reverse order of removal, observing the following:

Follow installation instructions ➔ [page 253](#) .

Specified torques

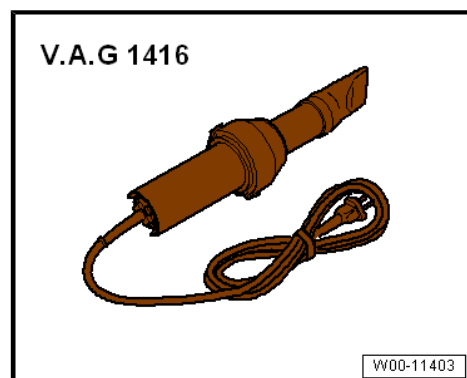
Component	Specified torque
Bolts for cover	1.5 Nm



2.4.6 Removing and installing cover, side panel

Special tools and workshop equipment required

- ◆ Hot-air blower - V.A.G 1416-





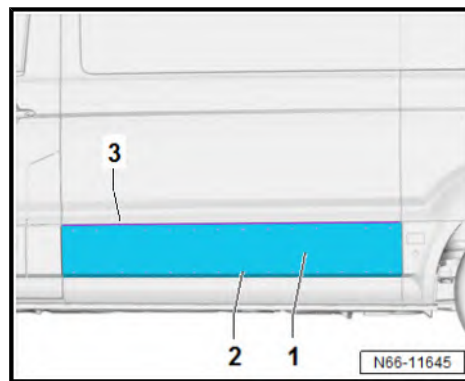
Removing

- Heat up cover -1- in area of bonding surface -3- using hot air blower - V.A.G 1416- .
- Pull off cover -1-.
- Pull out cover -1- from catches -2-.

Installing

Install in reverse order of removal, observing the following:

Follow installation instructions ➔ [page 253](#) .



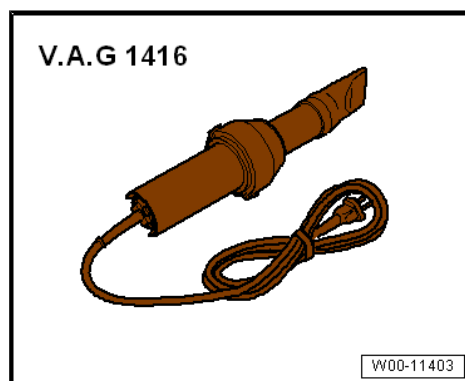
2.4.7 Removing and installing cover, double cab

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



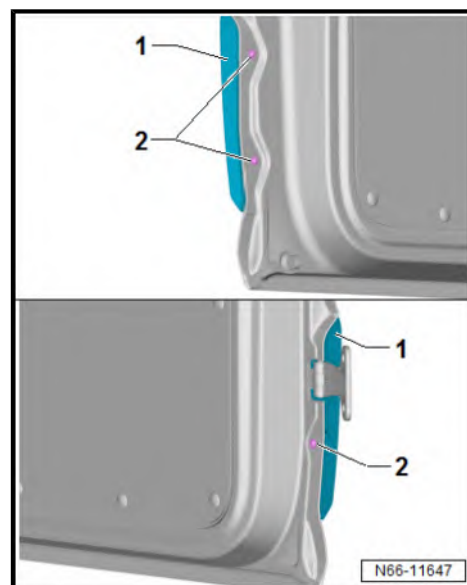
- ◆ Hot-air blower - V.A.G 1416-





Removing

- Unscrew bolts -2- on inside at rear from cover -1-.
- Unscrew bolt -2- on inside at front from cover -1-.



- Heat up cover -1- in area of bonding surface -3- using hot air blower - V.A.G 1416- .
- Pull off cover -1-.
- Pull out cover -1- from catches -2-.

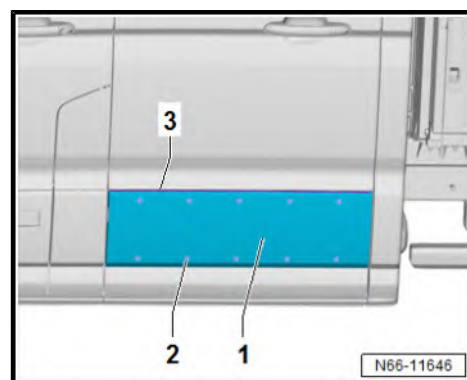
Installing

Install in reverse order of removal, observing the following:

Follow installation instructions ➔ [page 253](#) .

Specified torques

Component	Specified torque
Bolts for cover	1.5 Nm



2.4.8 Removing and installing cover, rear

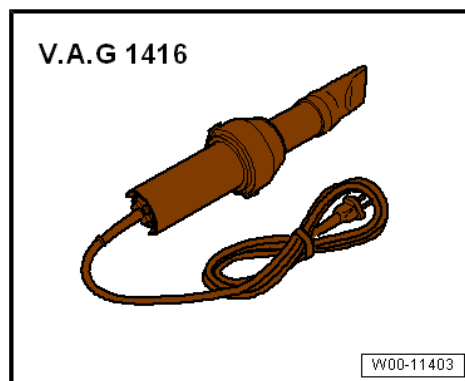
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



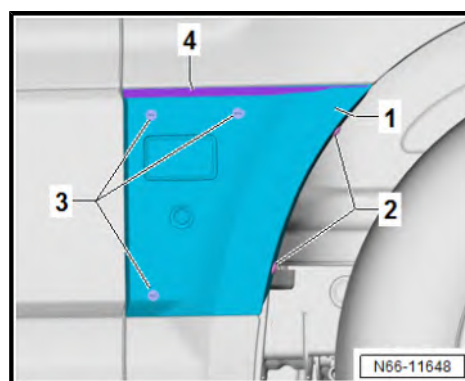


◆ Hot-air blower - V.A.G 1416-



Removing

- Unscrew bolts -2-.
- Heat up cover -1- in area of bonding surface -4- using hot air blower - V.A.G 1416- .
- Pull off cover -1-.
- Pull out cover -1- from catches -3-.



- If present, separate electrical connector -3- for rear side left parking aid sender 2 - G959- -4-.
- Separate electrical connector -2- for rear left side marker light bulb - M37- -1-.

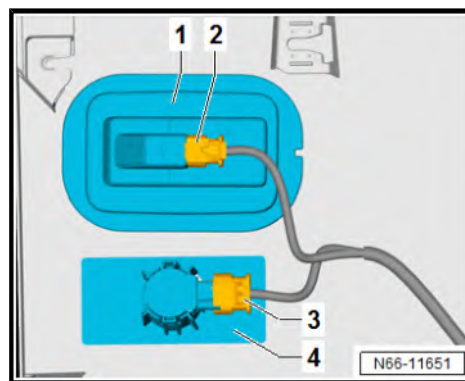
Installing

Install in reverse order of removal, observing the following:

Follow installation instructions ➔ [page 253](#) .

Specified torques

Component	Specified torque
Bolts for cover	1.5 Nm



2.4.9 Removing and installing cover, D-pillar

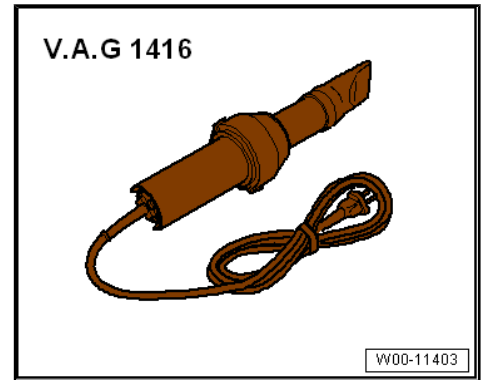
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



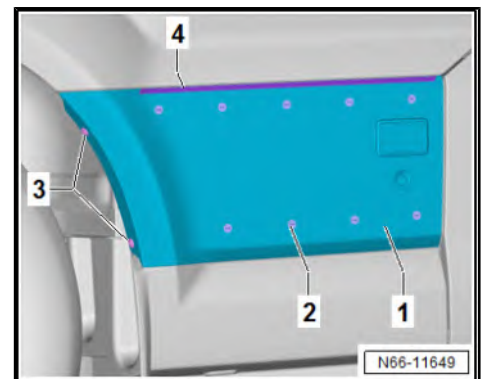


◆ Hot-air blower - V.A.G 1416-

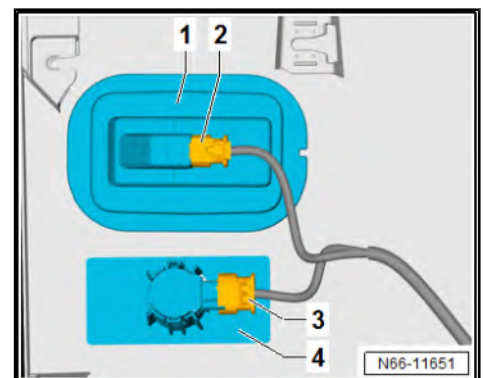


Removing

- Unscrew bolts -3-.
- Heat up cover -1- in area of bonding surface -4- using hot air blower - V.A.G 1416- .
- Pull off cover -1-.
- Pull out cover -1- from catches -2-.



- If present, separate electrical connector -3- for rear side right parking aid sender 2 - G958- -4-.
- Separate electrical connector -2- for rear right side marker light bulb - M38- -1-.



Installing

Install in reverse order of removal, observing the following:

Follow installation instructions ➔ [page 253](#) .

Specified torques

Component	Specified torque
Bolts for cover	1.5 Nm

2.4.10 Removing and installing cover, rear side panel

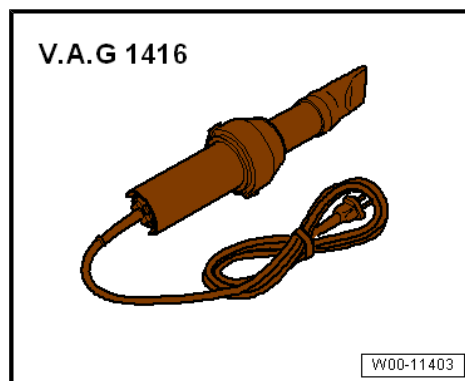
Special tools and workshop equipment required

◆ Torque wrench - V.A.G 1783-



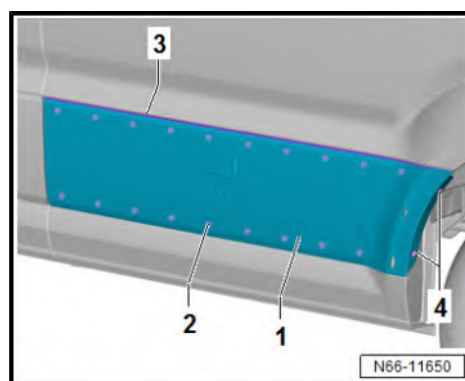


◆ Hot-air blower - V.A.G 1416-



Removing

- Unscrew bolts -4-.
- Heat up cover -1- in area of bonding surface -3- using hot air blower - V.A.G 1416- .
- Pull off cover -1-.
- Pull out cover -1- from catches -2-.



- If present, separate electrical connector -3- for rear side left parking aid sender 2 - G959- -4-.
- Separate electrical connector -2- for rear left side marker light bulb - M37- -1-.

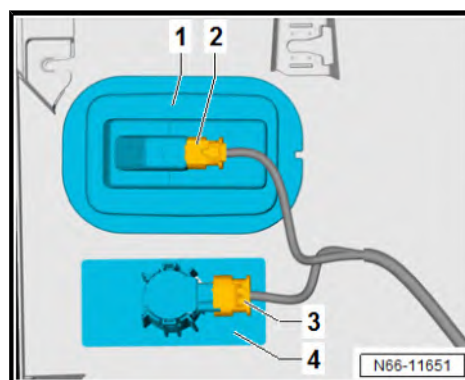
Installing

Install in reverse order of removal, observing the following:

Follow installation instructions ➔ [page 253](#) .

Specified torques

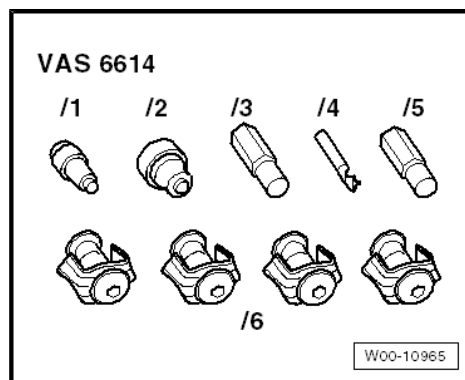
Component	Specified torque
Bolts for cover	1.5 Nm



2.4.11 Installing sender retainer for parking aid and park assist steering

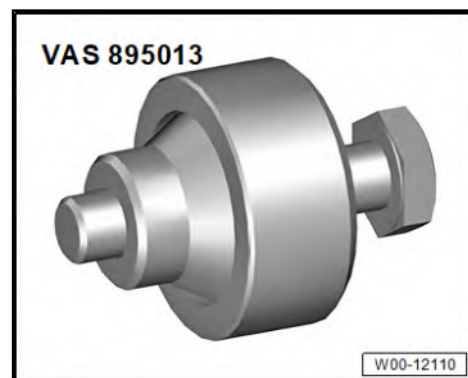
Special tools and workshop equipment required

- ◆ Drill bit - VAS 6614/4- from -VAS 6614-





- ◆ For parking aid holder screw-hole punch Ø 18.4 mm - VAS 6614/9-
- ◆ For park assist steering holder screw-hole punch Ø 28.2mm - VAS 895 013-



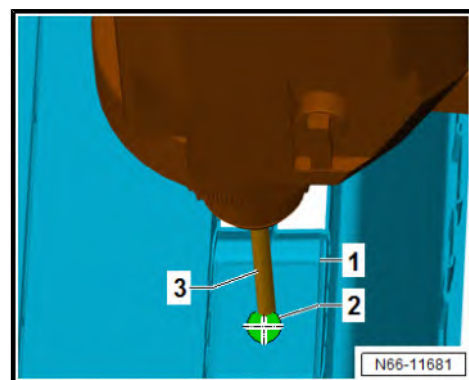
Procedure



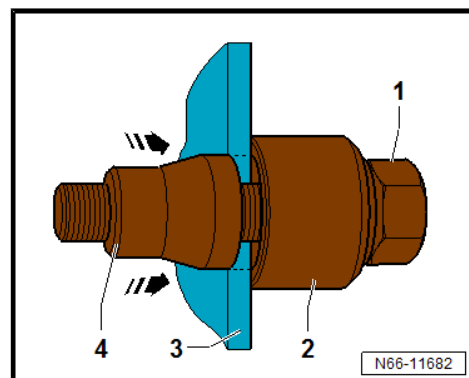
Note

Marking -2- for retainers can be found on inside of respective cover -1-.

- Pre-drill hole from inside out with drill bit - VAS 6614/4- -3- from tool kit supplied.



- Fit screw-hole punch with bolt head -1- and counter support -2- from inside of cover -3-.
- Screw on hole punch -4- from outside of cover -3-.
- Press punch -4- against cover.
- Tighten bolt head -1-.
- Punch -4- is pulled through material of cover -3- in direction of -arrow- by turning bolt head -1-.



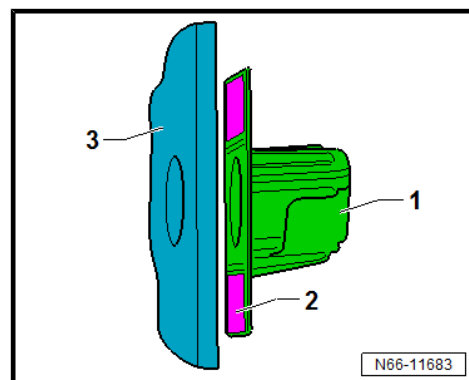
- Thoroughly clean bonding surfaces on cover -3-.
- Pull off backing -2- off double-sided adhesive tape on retainer -1-.



Note

Do not touch the adhesive surface of the bonded punched component once the protective backing has been removed.

- Install retainer -1- in hole punched in cover -3-.
- Press retainer -1- forcefully against bonding surface.
- Do not carry out any further work on the cover or retainers until the minimum curing period of 2 hours has elapsed.





2.5 Removing and installing trim strip

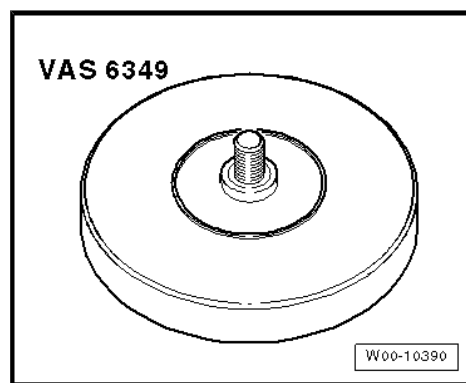
⇒ [“2.5.1 Fitting instructions”, page 264](#)

⇒ [“2.5.2 Removing and installing trim strip, wing door”, page 264](#)

2.5.1 Fitting instructions

Special tools and workshop equipment required

- ◆ Adhesive strip remover - VAS 6349-



When removing cover, adhesive tape may become damaged.

If using new adhesive tape as prescribed in ⇒ Electronic parts catalogue (ETKA) and ⇒ webMANTIS for MAN TGE, the trim strip can be reused.

- Check clips for damage and renew if necessary.

If a cover is installed for the first time on a new vehicle or on newly painted surfaces, the procedure in ⇒ Workshop manual “Paint”; ⇒ Paintwork repairs; ⇒ Paint finish; ⇒ Mouldings & film should be observed.

If a cover is removed and reattached, use only adhesive remover to remove the adhesive residue.

- Remove existing adhesive residue of adhesive tape using adhesive strip remover - VAS 6349- .
- Bonding surfaces must be free of dust and grease.
- Install cover immediately after cleaning.
- Do not remove protective foil until immediately before installation.
- The working temperature is approx. 21°C.

2.5.2 Removing and installing trim strip, wing door

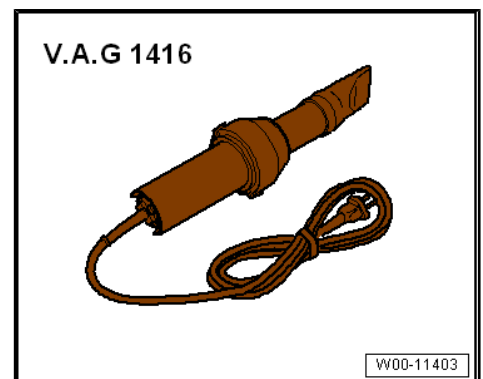
Special tools and workshop equipment required



- ◆ Roller for door insulation foil - 3356-



- ◆ Hot-air blower - V.A.G 1416-

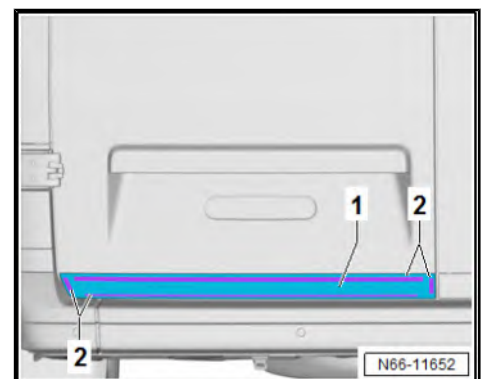


Removing

- Heat up trim strip -1- in area of adhesive surfaces -2- using hot air blower - V.A.G 1416- .
- Pull trim strip -1- off wing door.

Installing

Install in reverse order of removal, observing the following:

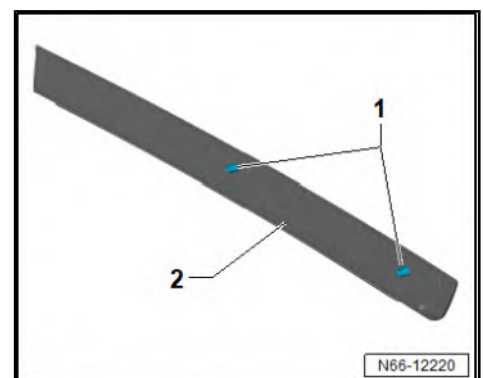


Vehicles as of week 12/2018

- Check for positioning holes in wing door in area of trim strip.
- If holes are available, remove pins -1- on new trim strip -2-.

Continued for all vehicles

- Follow installation instructions ⇒ [page 264](#) .
- Pull protective film off double-sided adhesive tape.
- Fit trim strip -1- on wing door.
- Press on trim strip -1- using roller - 3356- .



2.6 Removing and installing B-pillar trim

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

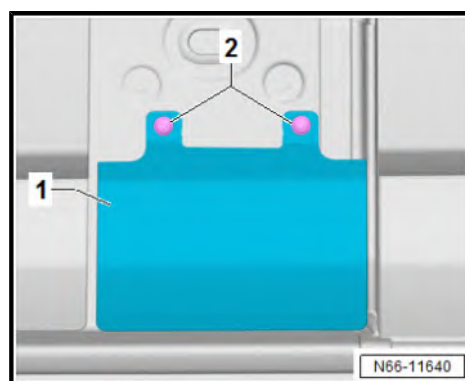
- Remove B-pillar cover ➔ [page 253](#) .
- Unscrew bolts -2-.
- Remove B-pillar trim -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“2.1 Assembly overview - covers”, page 249](#)



2.7 Removing and installing wheel arch covers

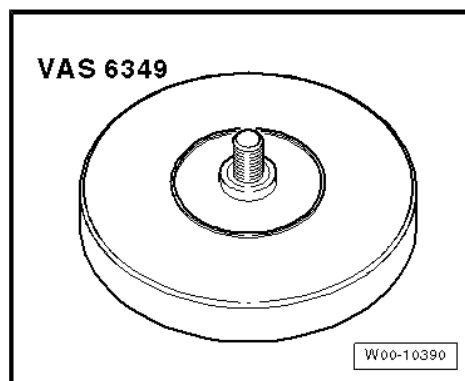
➔ [“2.7.1 Fitting instructions”, page 266](#)

➔ [“2.7.2 Removing and installing rear wheel arch covers”, page 267](#)

2.7.1 Fitting instructions

Special tools and workshop equipment required

- ◆ Adhesive strip remover - VAS 6349-



When removing cover, adhesive tape may become damaged.

If using new adhesive tape as prescribed in ➔ Electronic parts catalogue (ETKA) and ➔ webMANTIS for MAN TGE, the trim strip can be reused.

- Check clips for damage and renew if necessary.

If a cover is installed for the first time on a new vehicle or on newly painted surfaces, the procedure in ➔ Workshop manual “Paint”; ➔ Paintwork repairs; ➔ Paint finish; ➔ Mouldings & film should be observed.



If a cover is removed and reattached, use only adhesive remover to remove the adhesive residue.

- Remove existing adhesive residue of adhesive tape using adhesive strip remover - VAS 6349- .
- Bonding surfaces must be free of dust and grease.
- Install cover immediately after cleaning.
- Do not remove protective foil until immediately before installation.
- The working temperature is approx. 21°C.

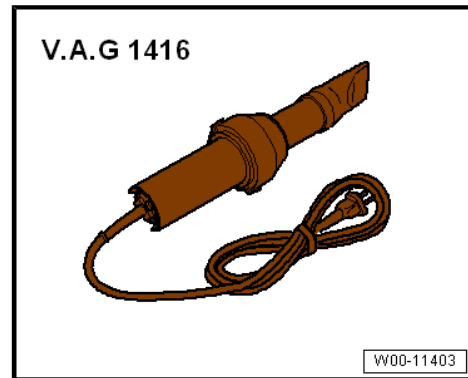
2.7.2 Removing and installing rear wheel arch covers

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



- ◆ Hot-air blower - V.A.G 1416-



Removing

- Unscrew bolts -3-.
- Heat up adhesive tape -2- on wheel arch cover -1- with hot air blower - V.A.G 1416- before removing.
- Remove wheel arch cover -1-.

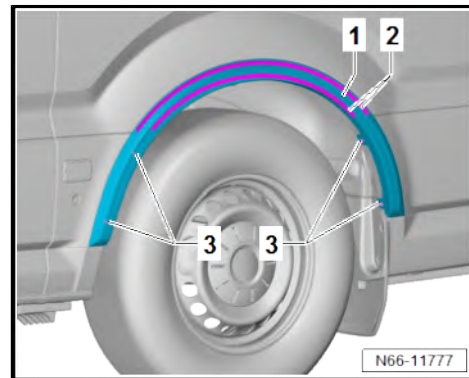
Installing

Install in reverse order of removal, observing the following:

- Note repair instructions ⇒ [page 266](#) .

Specified torques

- ◆ ⇒ [“2.2 Assembly overview - wheel arch covers”, page 251](#)





2.8 Removing and installing underbody heat shield

⇒ ["2.8.1 Removing and installing heat shield, front", page 268](#)

⇒ ["2.8.2 Removing and installing heat shield, front centre", page 268](#)

⇒ ["2.8.3 Removing and installing heat shield, centre", page 268](#)

2.8.1 Removing and installing heat shield, front

Removing

- Remove subframe ⇒ Running gear, axles, steering; Rep. gr. 40 ; Subframe; Removing and installing subframe with steering rack .

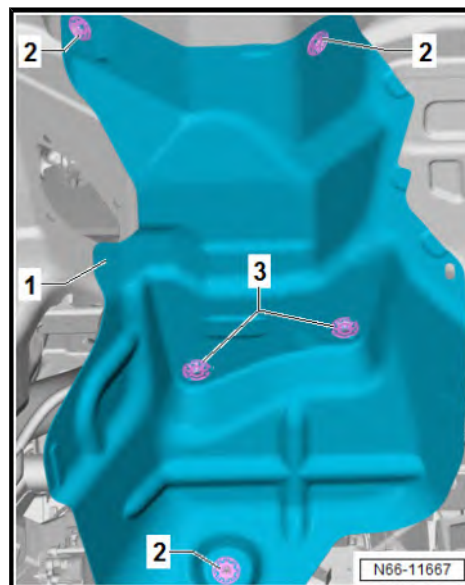
CAUTION

Danger of severe burns from hot exhaust system.
Risk of burns to hands and other parts of body.
– Allow exhaust system to cool down.

- Remove clamping washers -2-.
- Loosen clamping washers -3-.
- Remove front heat shield -1-.

Installing

Install in reverse order.



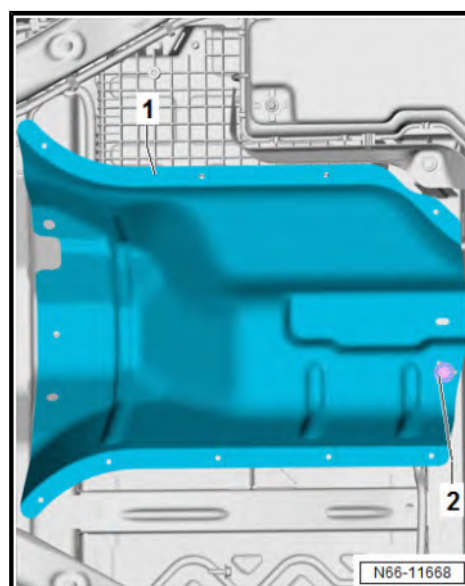
2.8.2 Removing and installing heat shield, front centre

Removing

- Remove front heat shield ⇒ [page 268](#) .
- Loosen lock washer -2-.
- Remove heat shield -1-.

Installing

Install in reverse order.

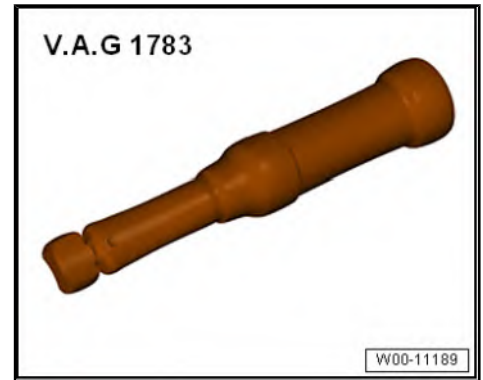


2.8.3 Removing and installing heat shield, centre

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

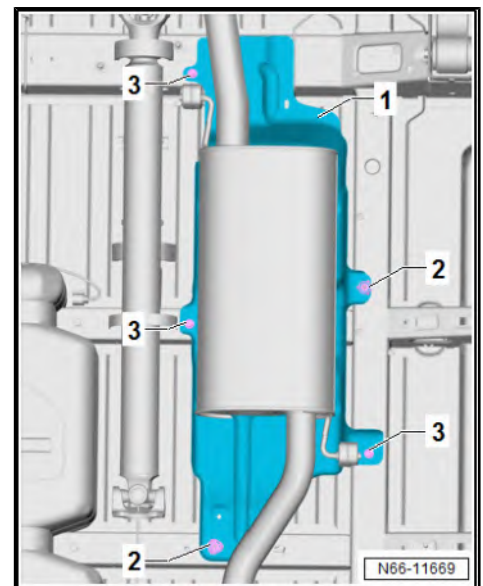
- Remove front silencer ⇒ Rep. gr. 26 ; Exhaust pipes/silencers; Removing and installing front silencer .
- Loosen lock washer -2-.
- Unscrew bolts -3-.
- Remove centre heat shield -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["2.3 Assembly overview - heat shield", page 252](#)





3 Noise insulation

⇒ ["3.1 Assembly overview - noise insulation", page 270](#)

⇒ ["3.2 Removing and installing noise insulation", page 271](#)

3.1 Assembly overview - noise insulation

⇒ ["3.1.1 Assembly overview - noise insulation", page 270](#)

⇒ ["3.1.2 Assembly overview - noise insulation, e-Crafter or e-TGE", page 271](#)

3.1.1 Assembly overview - noise insulation

1 - Noise insulation

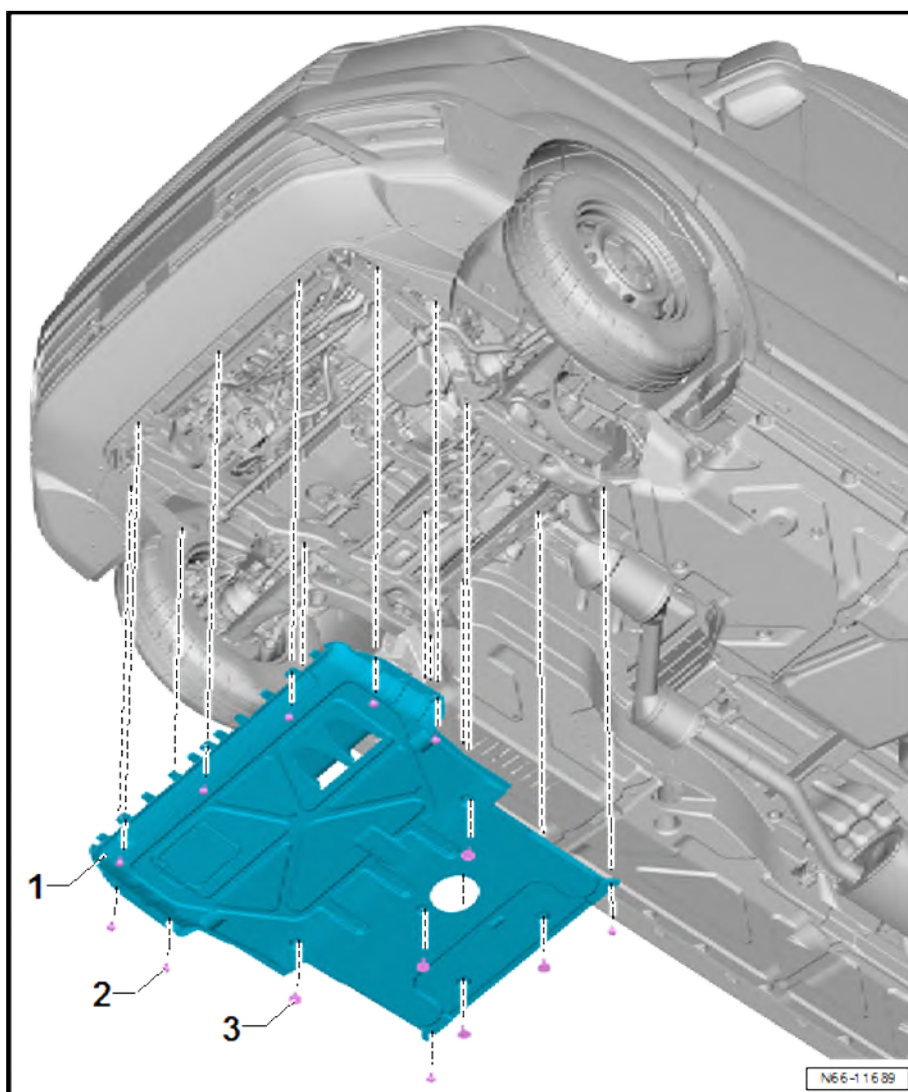
- ☐ Removing and installing
⇒ [page 271](#)

2 - Bolts

- ☐ Qty. 11
- ☐ 2 Nm

3 - Bolts

- ☐ Renew after removal
- ☐ Qty. 5
- ☐ 6 Nm





3.1.2 Assembly overview - noise insulation, e-Crafter or e-TGE

1 - Noise insulation

- ❑ Removing and installing
⇒ [page 271](#)

2 - Bolts

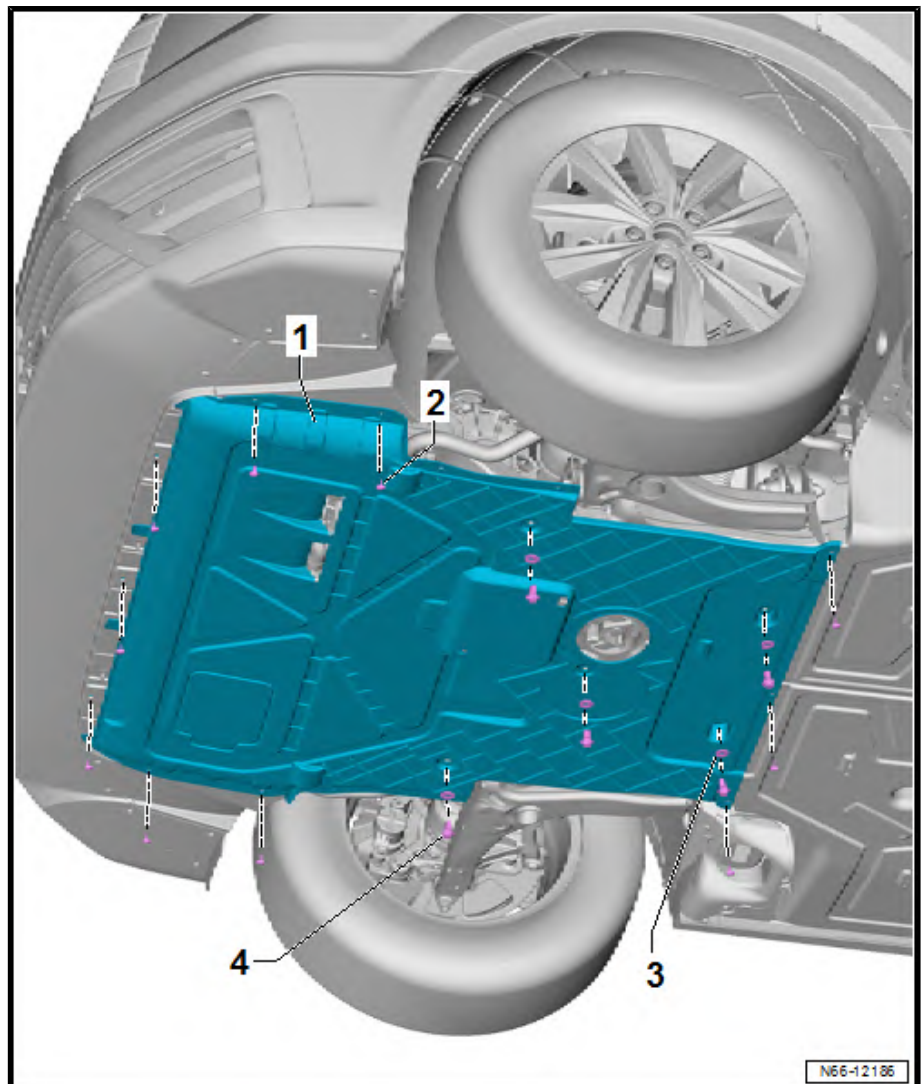
- ❑ Qty. 10
- ❑ 2 Nm

3 - Washers

- ❑ Qty. 5

4 - Bolts

- ❑ Qty. 5
- ❑ 14 Nm



3.2 Removing and installing noise insulation

⇒ [“3.2.1 Removing and installing noise insulation”, page 271](#)

⇒ [“3.2.2 Removing and installing noise insulation, e-Crafter or e-TGE”, page 272](#)

⇒ [“3.2.3 Removing and installing additional noise insulation holder, e-Crafter or e-TGE”, page 273](#)

3.2.1 Removing and installing noise insulation

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

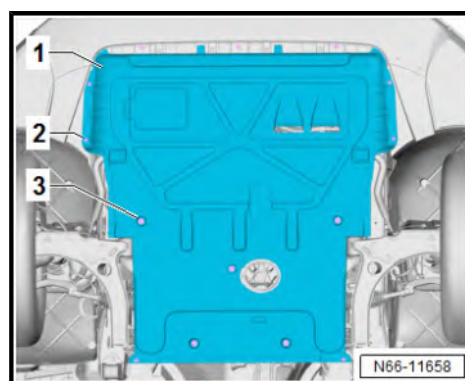
- Unscrew bolts -3-.
- Unscrew bolts -2-.
- Pull noise insulation -1- towards rear out of lock carrier.

Installing

Install in reverse order.

Specified torques

- ◆ ⇒ ["3.1 Assembly overview - noise insulation", page 270](#)



3.2.2 Removing and installing noise insulation, e-Crafter or e-TGE

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-





Removing

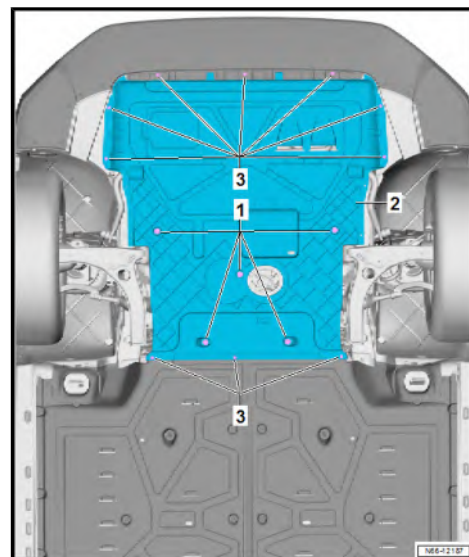
- Unscrew bolts -1- together with washers.
- The following work step must be performed by a second mechanic.
- Have a second mechanic secure noise insulation -2-.
- Unscrew bolts -3-. When doing this, have a second mechanic remove noise insulation -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“3.1 Assembly overview - noise insulation”, page 270](#)



3.2.3 Removing and installing additional noise insulation holder, e-Crafter or e-TGE

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-



Removing

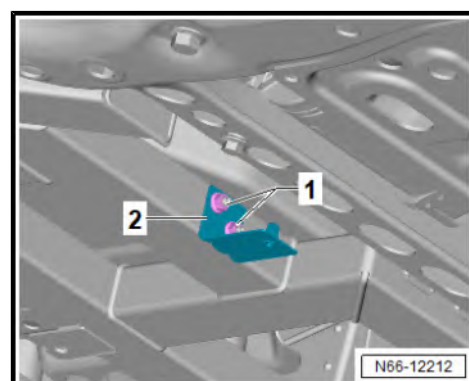
- Remove noise insulation ⇒ [page 271](#) .
- Unscrew nuts -1-.
- Remove additional holder -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -1-	20 Nm



4 Underbody cladding

⇒ [“4.1 Overview of fitting locations - underbody cladding”, page 274](#)

⇒ [“4.2 Assembly overview - underbody cladding”, page 277](#)

⇒ [“4.3 Removing and installing underbody cladding”, page 279](#)

⇒ [“4.4 Removing and installing underbody cladding, e-Crafter or e-TGE”, page 287](#)

4.1 Overview of fitting locations - underbody cladding

⇒ [“4.1.1 Overview of fitting locations - underbody cladding”, page 274](#)

⇒ [“4.1.2 Assembly overview - underbody cladding, e-Crafter or e-TGE”, page 276](#)

4.1.1 Overview of fitting locations - underbody cladding

1 - Noise insulation

- ☐ Removing and installing
⇒ [page 271](#)

2 - Underbody cladding

- ☐ Front left
- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 279](#)

3 - Underbody cladding

- ☐ Left
- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 279](#)

4 - Underbody cladding

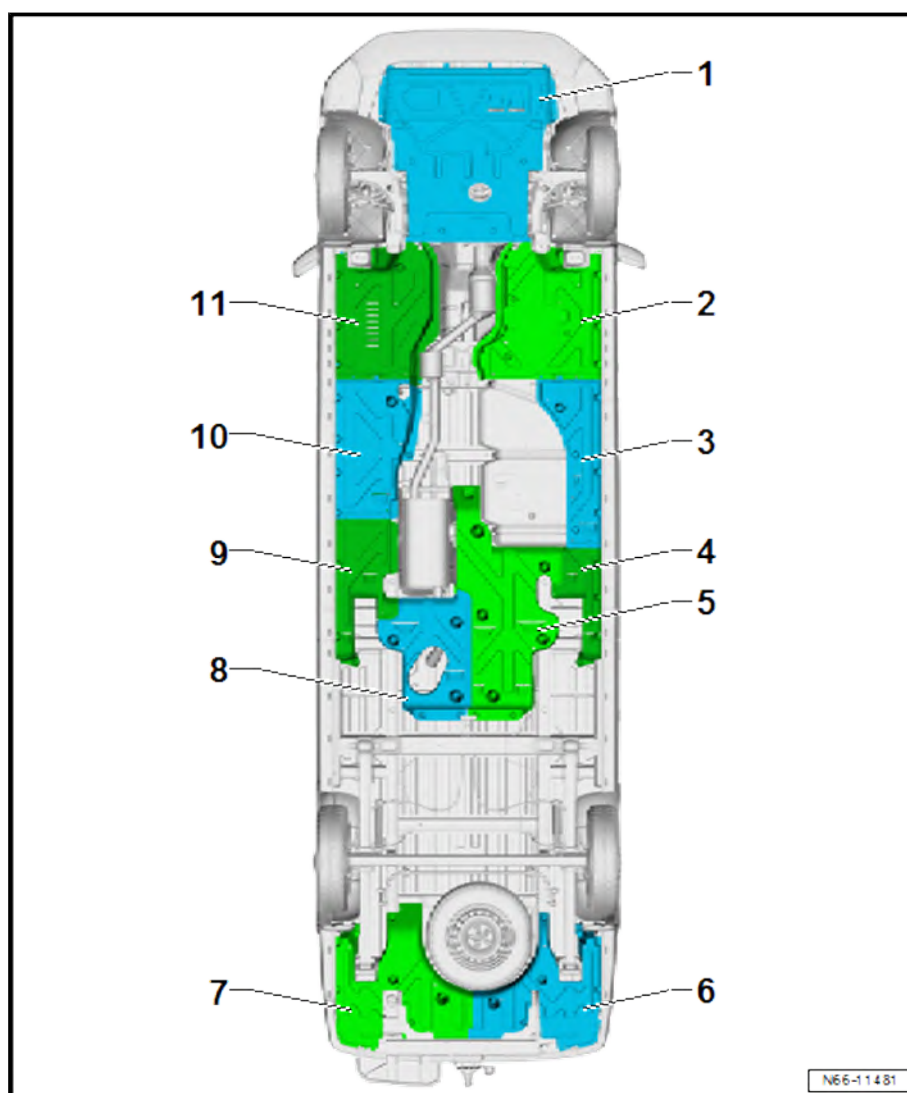
- ☐ Left
- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 279](#)

5 - Underbody cladding

- ☐ Left
- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 279](#)

6 - Underbody cladding

- ☐ Rear left
- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 279](#)





7 - Underbody cladding

- ☐ Rear right
- ☐ Depending on equipment
- ☐ Removing and installing ⇒ [page 279](#)

8 - Underbody cladding

- ☐ Right
- ☐ Depending on equipment
- ☐ Removing and installing ⇒ [page 279](#)

9 - Underbody cladding

- ☐ Right
- ☐ Depending on equipment
- ☐ Removing and installing ⇒ [page 279](#)

10 - Underbody cladding

- ☐ Right
- ☐ Depending on equipment
- ☐ Removing and installing ⇒ [page 279](#)

11 - Underbody cladding

- ☐ Front right
- ☐ Depending on equipment
- ☐ Removing and installing ⇒ [page 279](#)



4.1.2 Assembly overview - underbody cladding, e-Crafter or e-TGE

1 - Plugs

- ☐ Underbody cladding, front left
- ☐ Assembly overview ⇒ [page 277](#)
- ☐ Removing and installing ⇒ [page 287](#)

2 - Underbody cladding

- ☐ Front left
- ☐ Assembly overview ⇒ [page 277](#)
- ☐ Removing and installing ⇒ [page 287](#)

3 - Underbody cladding

- ☐ Rear left
- ☐ Assembly overview ⇒ [page 277](#)
- ☐ Removing and installing ⇒ [page 287](#)

4 - Underbody cladding

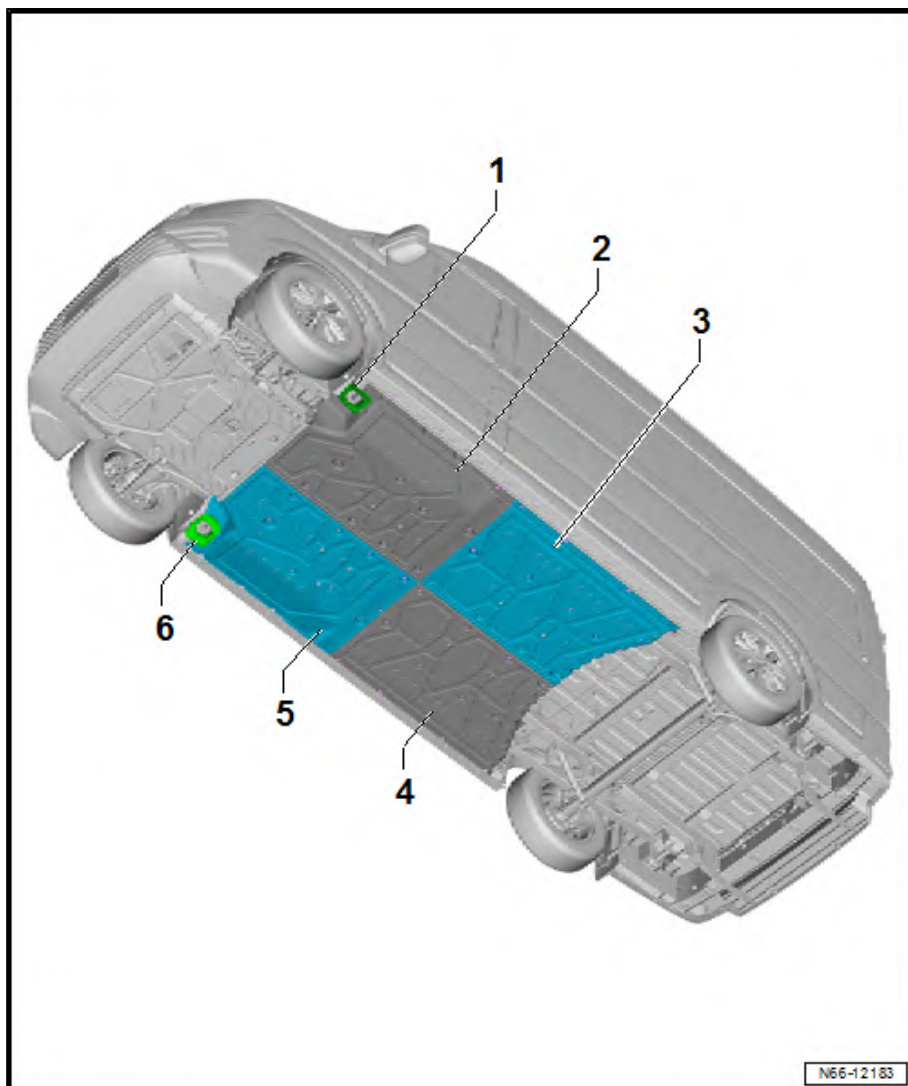
- ☐ Rear right
- ☐ Assembly overview ⇒ [page 277](#)
- ☐ Removing and installing ⇒ [page 287](#)

5 - Underbody cladding

- ☐ Front right
- ☐ Assembly overview ⇒ [page 277](#)
- ☐ Removing and installing ⇒ [page 287](#)

6 - Plugs

- ☐ Underbody cladding, front right
- ☐ Assembly overview ⇒ [page 277](#)
- ☐ Removing and installing ⇒ [page 287](#)





4.2 Assembly overview - underbody cladding

⇒ [“4.2.1 Assembly overview - underbody cladding, e-Crafter or e-TGE, centre of vehicle”, page 277](#)

⇒ [“4.2.2 Assembly overview - underbody cladding, e-Crafter or e-TGE, rear of vehicle”, page 278](#)

4.2.1 Assembly overview - underbody cladding, e-Crafter or e-TGE, centre of vehicle

1 - Plugs

- ☐ Underbody cladding, front right
- ☐ Removing and installing
⇒ [page 287](#)

2 - Underbody cladding

- ☐ Front right
- ☐ Removing and installing
⇒ [page 287](#)

3 - Underbody cladding

- ☐ Front left
- ☐ Removing and installing
⇒ [page 287](#)

4 - Plugs

- ☐ Underbody cladding, front left
- ☐ Removing and installing
⇒ [page 287](#)

5 - Bolts

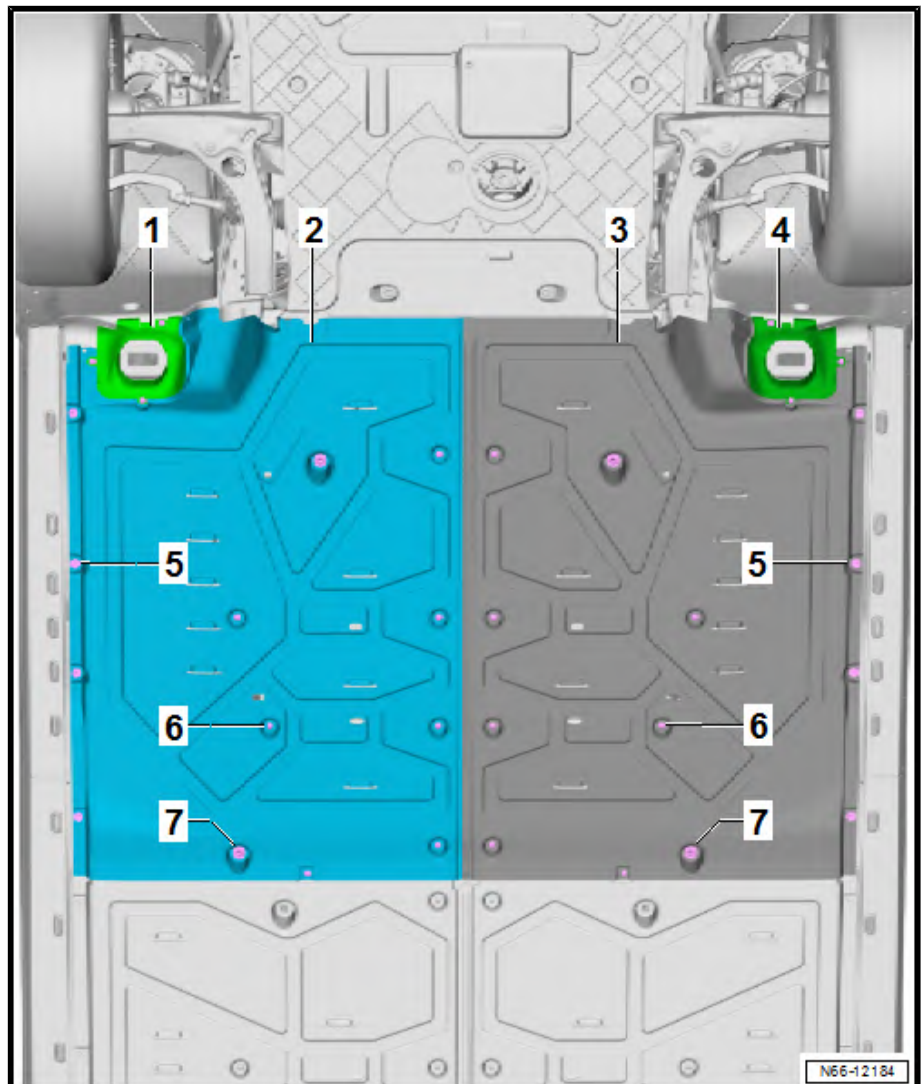
- ☐ Qty. 8
- ☐ 3 Nm

6 - Bolts

- ☐ Qty. 20
- ☐ 2 Nm

7 - Nuts

- ☐ Qty. 4
- ☐ 2 Nm





4.2.2 Assembly overview - underbody cladding, e-Crafter or e-TGE, rear of vehicle

1 - Underbody cladding

- ☐ Rear right
- ☐ Removing and installing
⇒ [page 287](#)

2 - Nuts

- ☐ Qty. 4
- ☐ 2 Nm

3 - Underbody cladding

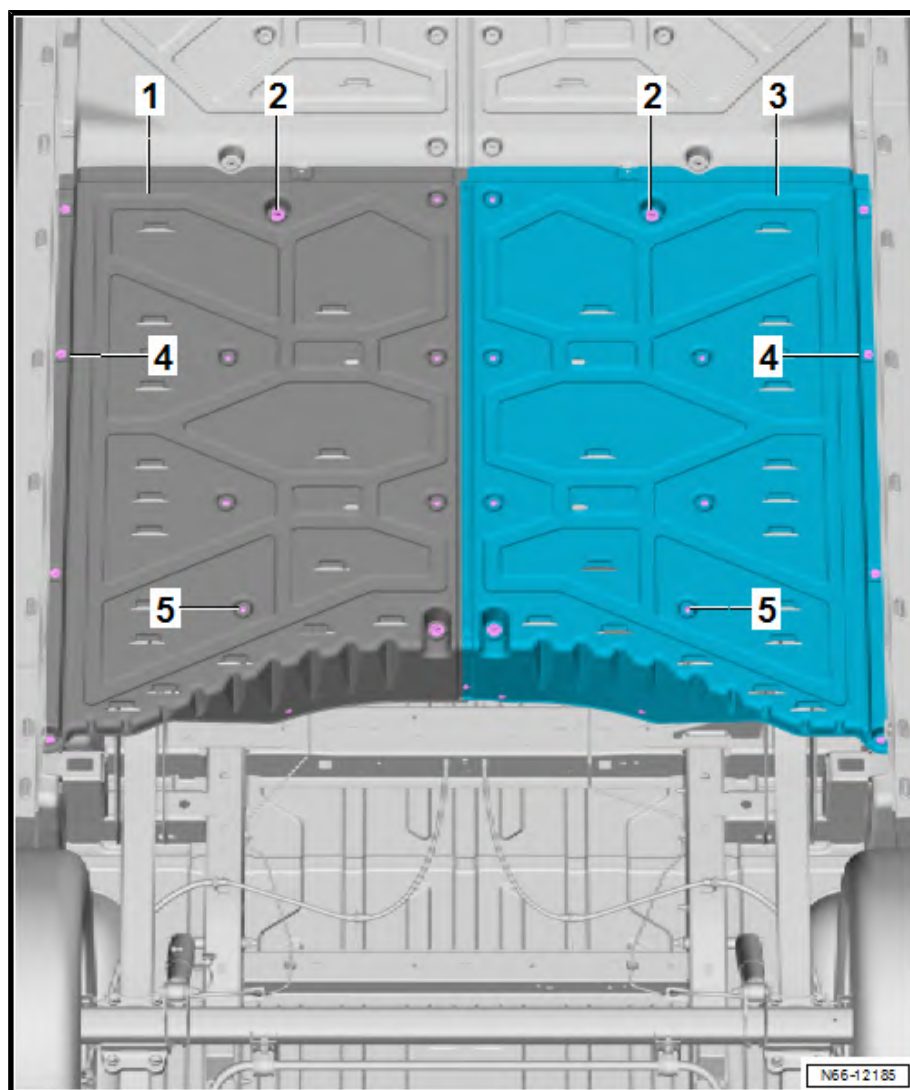
- ☐ Rear left
- ☐ Removing and installing
⇒ [page 287](#)

4 - Bolts

- ☐ Qty. 8
- ☐ 3 Nm

5 - Bolts

- ☐ Qty. 16
- ☐ 2 Nm





4.3 Removing and installing underbody cladding

⇒ [“4.3.1 Removing and installing underbody cladding, rear left”, page 279](#)

⇒ [“4.3.2 Removing and installing underbody cladding, rear right”, page 280](#)

⇒ [“4.3.3 Removing and installing underbody cladding, rear centre”, page 280](#)

⇒ [“4.3.4 Removing and installing underbody cladding, rear left”, page 281](#)

⇒ [“4.3.5 Removing and installing underbody cladding, rear centre right”, page 282](#)

⇒ [“4.3.6 Removing and installing underbody cladding, centre right”, page 282](#)

⇒ [“4.3.7 Removing and installing underbody cladding, front left”, page 283](#)

⇒ [“4.3.8 Removing and installing underbody cladding, front right”, page 284](#)

⇒ [“4.3.9 Removing and installing underbody cladding, centre left”, page 284](#)

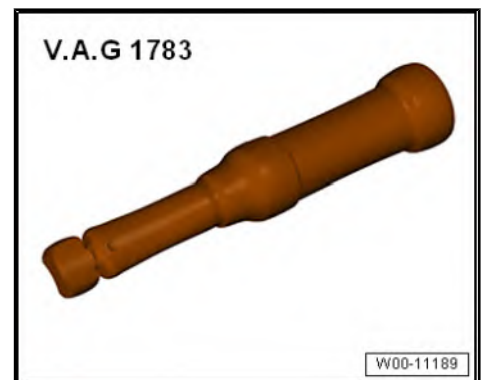
⇒ [“4.3.10 Removing and installing underbody cladding, centre”, page 285](#)

⇒ [“4.3.11 Removing and installing underbody cladding, front”, page 286](#)

4.3.1 Removing and installing underbody cladding, rear left

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

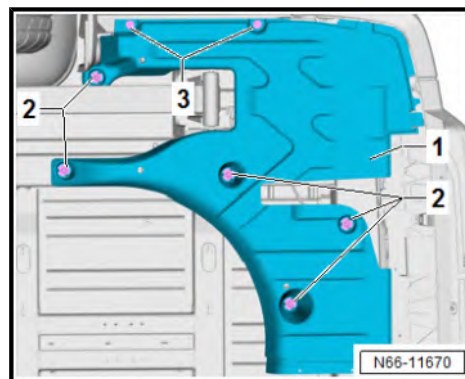
- Unscrew bolts -2-.
- Remove nuts -3-.
- Remove rear left underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -3-	2 Nm
Bolts -2-	3 Nm



4.3.2 Removing and installing underbody cladding, rear right

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

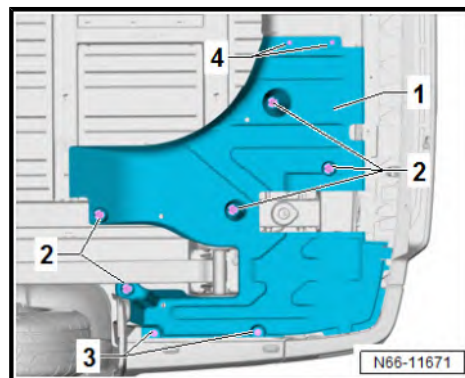
- Unscrew bolts -3-.
- Unscrew bolts -4-.
- Unscrew nuts -2-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -2-	2 Nm
Bolts -3-	2 Nm
Bolts -4-	3 Nm

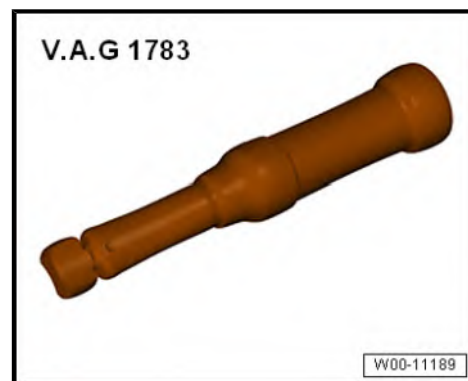


4.3.3 Removing and installing underbody cladding, rear centre

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

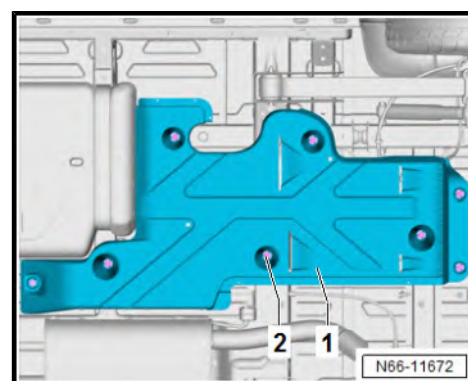
- Unscrew nuts -2-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

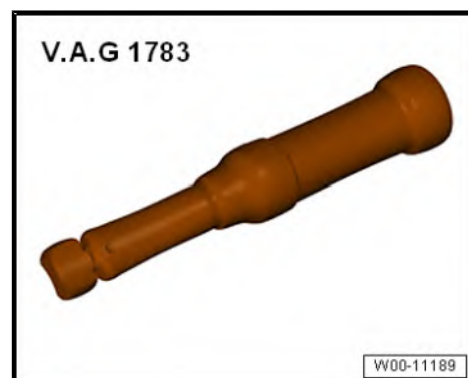
Component	Specified torque
Nuts -2-	2 Nm



4.3.4 Removing and installing underbody cladding, rear left

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

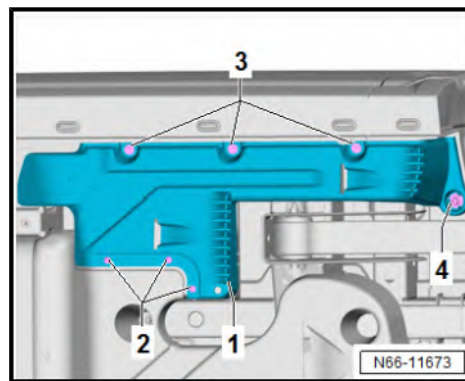
- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Unscrew nut -4-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nut -4-	2 Nm
Bolts -2-	2 Nm
Bolts -3-	3 Nm



4.3.5 Removing and installing underbody cladding, rear centre right

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-



Removing

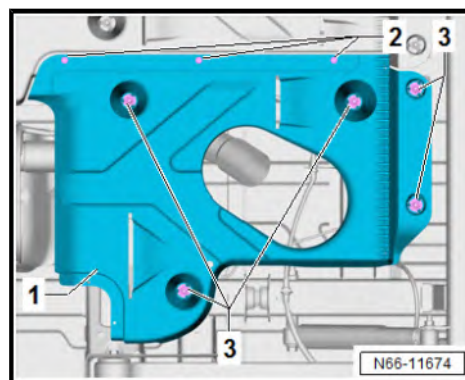
- Unscrew bolts -2-.
- Remove nuts -3-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -3-	2 Nm
Bolts -2-	2 Nm

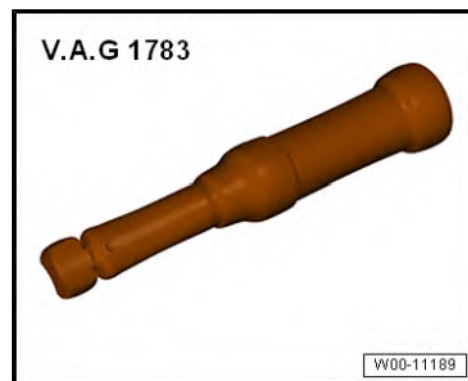


4.3.6 Removing and installing underbody cladding, centre right

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

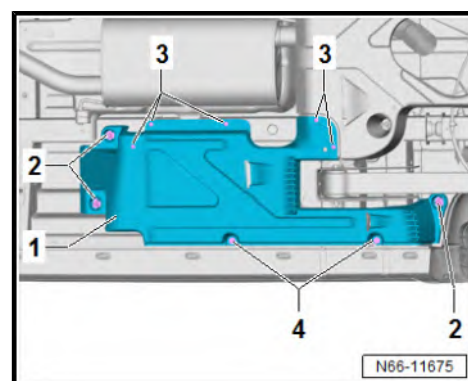
- Unscrew bolts -3-.
- Unscrew bolts -4-.
- Unscrew nuts -2-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -2-	2 Nm
Bolts -3-	2 Nm
Bolts -4-	3 Nm



4.3.7 Removing and installing underbody cladding, front left

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

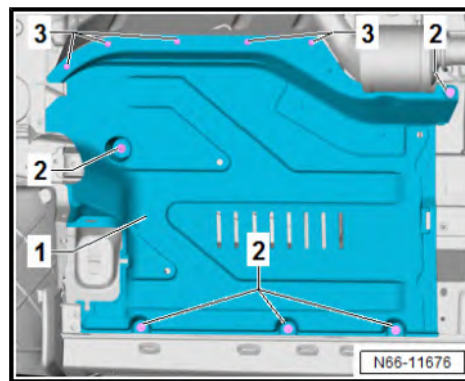
- If fitted, remove mudflap ➔ [page 330](#) .
- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

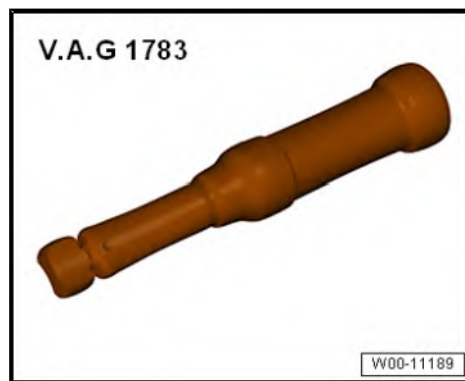
Component	Specified torque
Bolts -2-	3 Nm
Bolts -3-	2 Nm



4.3.8 Removing and installing underbody cladding, front right

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

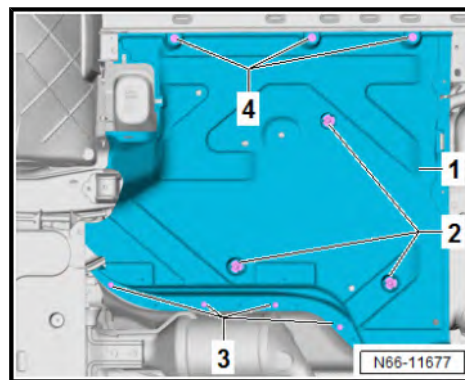
- Unscrew left bolts -3-.
- Unscrew right bolts -4-.
- Unscrew nuts -2-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -2-	2 Nm
Bolts -3-	2 Nm
Bolts -4-	3 Nm

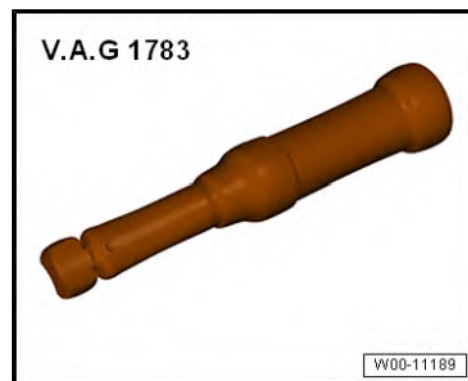


4.3.9 Removing and installing underbody cladding, centre left

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

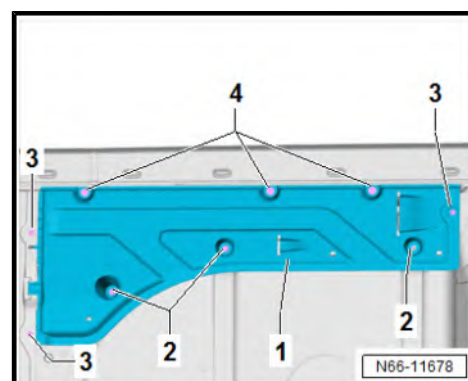
- Unscrew bolts -3-.
- Unscrew bolts -4-.
- Unscrew nuts -2-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nut -2-	2 Nm
Bolts -3-	2 Nm
Bolts -4-	3 Nm



4.3.10 Removing and installing underbody cladding, centre

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

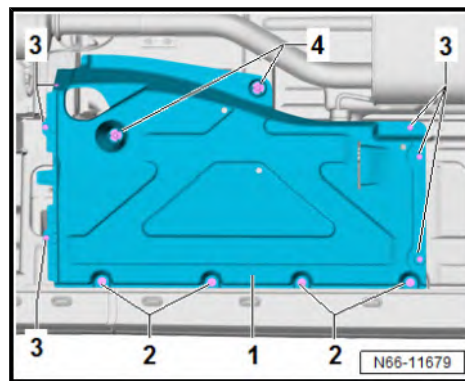
- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Remove nuts -4-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -4-	2 Nm
Bolts -2-	3 Nm
Bolts -3-	2 Nm



4.3.11 Removing and installing underbody cladding, front

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-



Removing

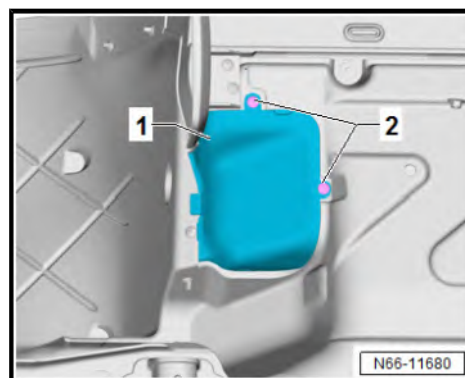
- Unscrew bolts -2-.
- Remove underbody cladding -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -2-	2 Nm





4.4 Removing and installing underbody cladding, e-Crafter or e-TGE

⇒ [“4.4.1 Removing and installing underbody cladding, front left”, page 287](#)

⇒ [“4.4.2 Removing and installing underbody cladding, front right”, page 288](#)

⇒ [“4.4.3 Removing and installing underbody cladding, rear left”, page 288](#)

⇒ [“4.4.4 Removing and installing underbody cladding, rear right”, page 289](#)

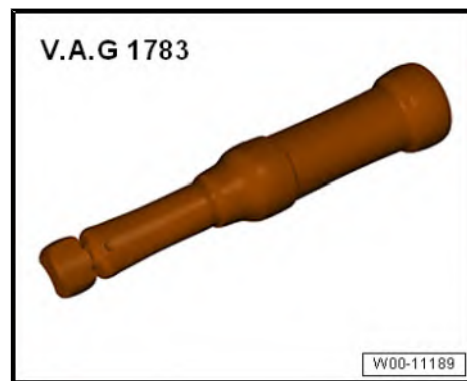
⇒ [“4.4.5 Removing and installing plugs”, page 290](#)

⇒ [“4.4.6 Removing and installing rear underbody cladding bracket”, page 291](#)

4.4.1 Removing and installing underbody cladding, front left

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove plugs ⇒ [page 290](#) .
- Unscrew bolts -1-.
- The following work step must be performed by a second mechanic.
- Have a second mechanic secure underbody cladding -2-.



Note

The underbody cladding is nested within each other. Therefore, the underbody cladding must be swung out from the adjacent ones when removing.

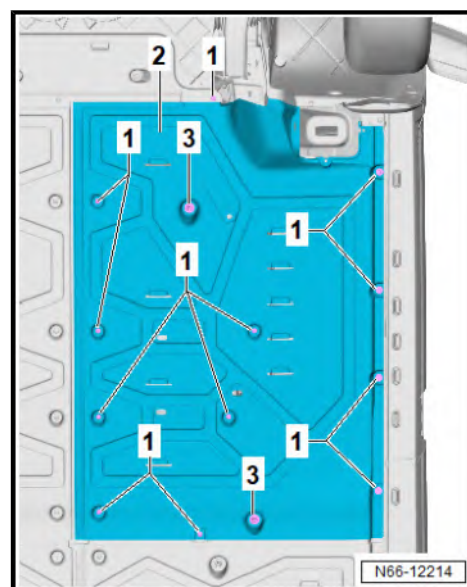
- Unscrew nuts -3-. When doing this, have a second mechanic remove underbody cladding -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“4.2 Assembly overview - underbody cladding”, page 277](#)





4.4.2 Removing and installing underbody cladding, front right

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

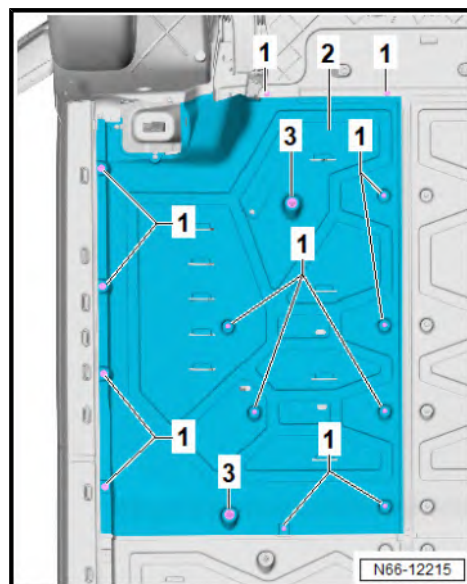
- Remove plugs ⇒ [page 290](#) .
- Unscrew bolts -1-.
- The following work step must be performed by a second mechanic.
- Have a second mechanic secure underbody cladding -2-.



Note

The underbody cladding is nested within each other. Therefore, the underbody cladding must be swung out from the adjacent ones when removing.

- Unscrew nuts -3-. When doing this, have a second mechanic remove underbody cladding -2-.



Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["4.2 Assembly overview - underbody cladding", page 277](#)

4.4.3 Removing and installing underbody cladding, rear left

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

- Unscrew bolts -1-.
- The following work step must be performed by a second mechanic.
- Have a second mechanic secure underbody cladding -2-.



Note

The underbody cladding is nested within each other. Therefore, the underbody cladding must be swung out from the adjacent ones when removing.

- Unscrew nuts -3-. When doing this, have a second mechanic remove underbody cladding -2-.

Installing

Install in reverse order of removal, observing the following:

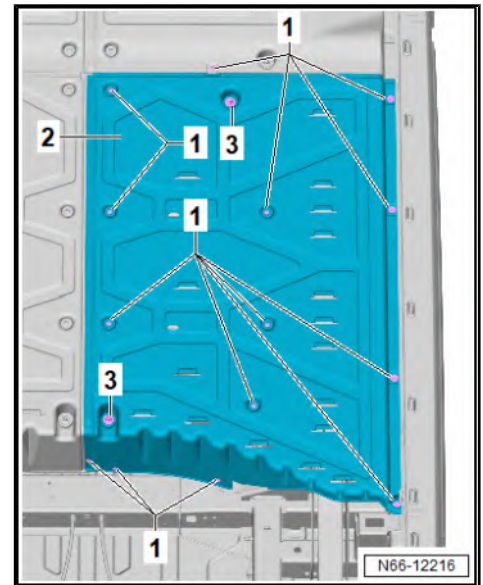
Specified torques

- ◆ ➔ [“4.2 Assembly overview - underbody cladding”, page 277](#)

4.4.4 Removing and installing underbody cladding, rear right

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

- Unscrew bolts -1-.
- The following work step must be performed by a second mechanic.
- Have a second mechanic secure underbody cladding -2-.



Note

The underbody cladding is nested within each other. Therefore, the underbody cladding must be swung out from the adjacent ones when removing.

- Unscrew nuts -3-. When doing this, have a second mechanic remove underbody cladding -2-.

Installing

Install in reverse order of removal, observing the following:

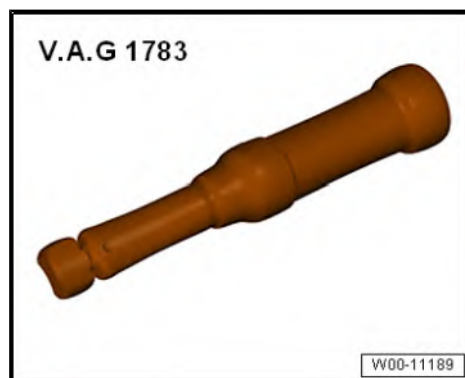
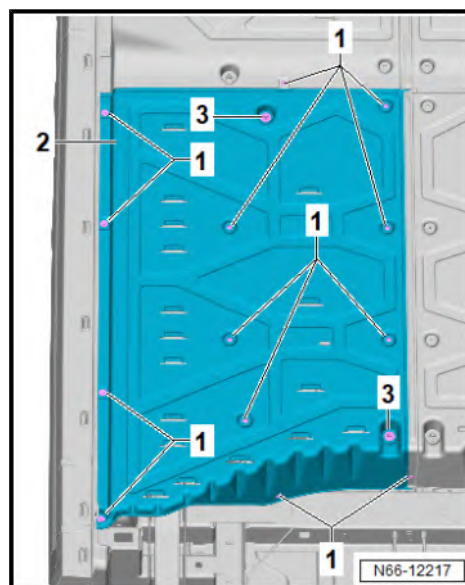
Specified torques

- ♦ ➔ [“4.2 Assembly overview - underbody cladding”, page 277](#)

4.4.5 Removing and installing plugs

Special tools and workshop equipment required

- ♦ Removal wedge - 3409-
- ♦ Torque wrench - V.A.G 1783-





Removing

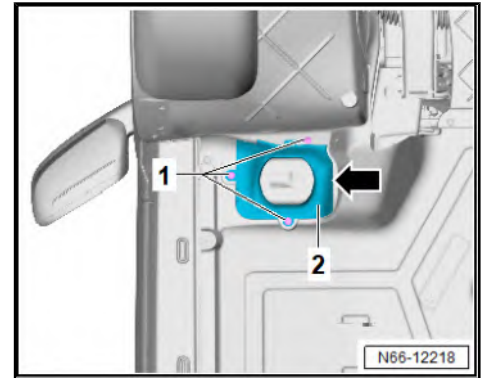
- Unscrew bolts -1-.
- Use removal wedge - 3409- to release locking lug in area marked with -arrow-. When doing this, swing out plug -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

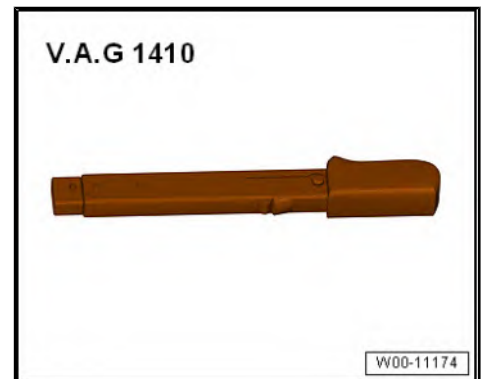
- ◆ ⇒ [“4.2 Assembly overview - underbody cladding”, page 277](#)



4.4.6 Removing and installing rear underbody cladding bracket

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-



Removing

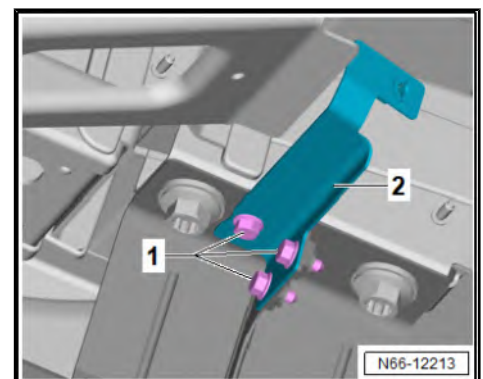
- Remove rear underbody cladding on the relevant side ⇒ [page 287](#) .
- Unscrew bolts -1-. When doing this, remove bracket -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -1-	8 Nm



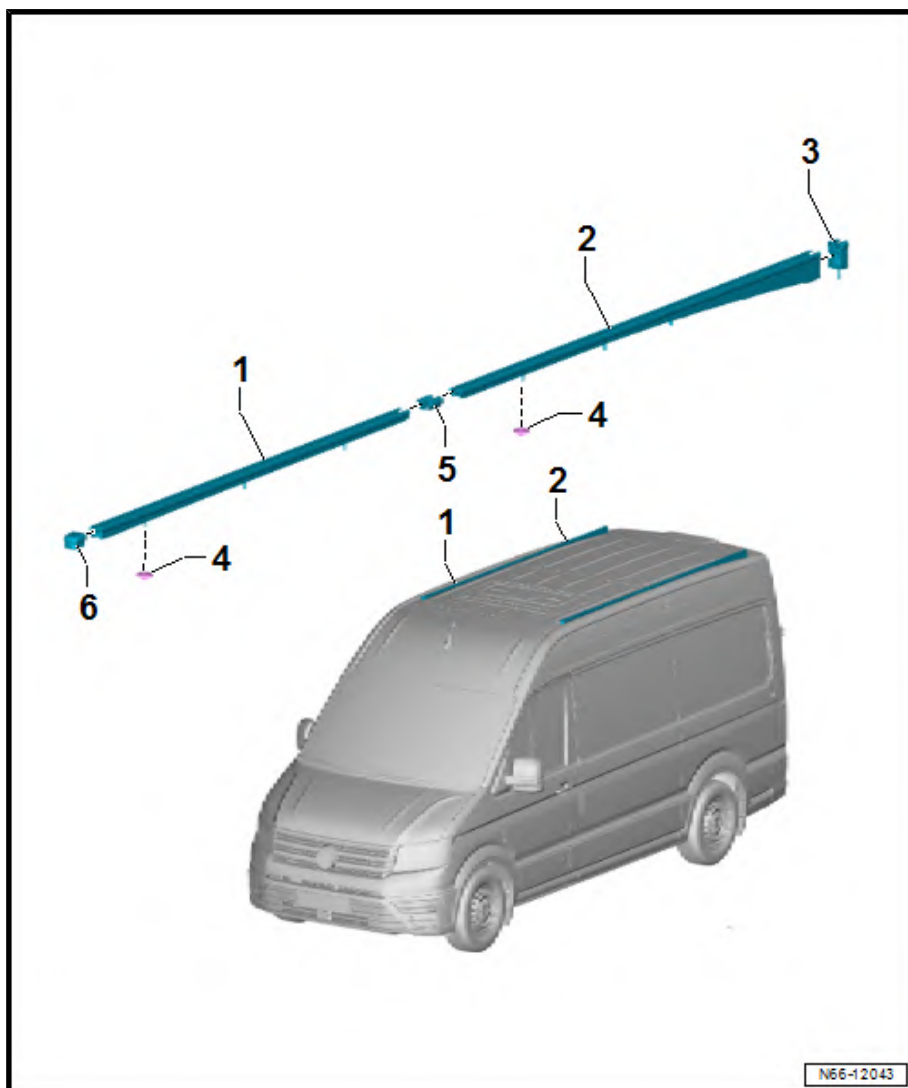


5 Roof rails for roof load carrier

⇒ ["5.1 Assembly overview - roof rails for roof load carrier",
page 292](#)

5.1 Assembly overview - roof rails for roof load carrier

- 1 - C rail
 - ☐ Front
- 2 - C rail
 - ☐ Rear
- 3 - End cap
 - ☐ Rear
- 4 - Nuts
 - ☐ Renew after removal
 - ☐ 5 per guide rail
 - ☐ 8 Nm
- 5 - Connecting piece
- 6 - End cap
 - ☐ Front





6 Exterior mirror

⇒ [“6.1 Assembly overview - exterior mirror”, page 293](#)

⇒ [“6.2 Removing and installing exterior mirror”, page 295](#)

⇒ [“6.3 Removing and installing mirror glass”, page 295](#)

⇒ [“6.4 Removing and installing mirror cover”, page 296](#)

⇒ [“6.5 Removing and installing mirror trim”, page 297](#)

6.1 Assembly overview - exterior mirror

⇒ [“6.1.1 Assembly overview - exterior mirror”, page 293](#)

⇒ [“6.1.2 Assembly overview - exploded mirror, dropside”, page 294](#)

6.1.1 Assembly overview - exterior mirror

1 - Mirror mounting

- ☐ Removing and installing
⇒ [page 295](#)

2 - Dampers

3 - Bolts

- ☐ Quantity may vary depending on mirror mounting
- ☐ 20 Nm

4 - Adjuster unit

5 - Bolt

- ☐ 2 Nm

6 - Mirror glass

- ☐ Removing and installing
⇒ [page 295](#)

7 - Bolts

- ☐ Qty. 4
- ☐ 2 Nm

8 - Mirror frame

- ☐ With auxiliary panoramic mirror
- ☐ Removing and installing
⇒ [page 297](#)

9 - Mirror cover

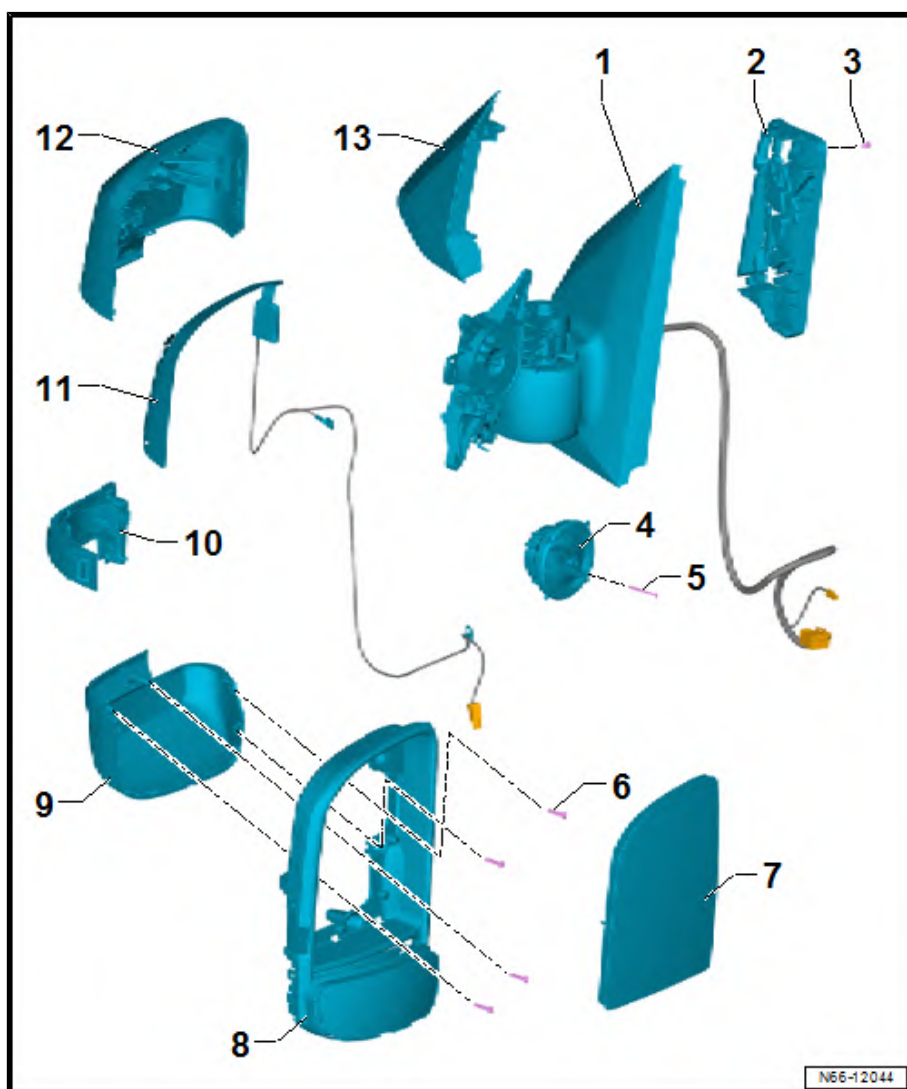
- ☐ Bottom
- ☐ Removing and installing
⇒ [page 296](#)

10 - Aerial module for radio aerial - R248- for FM2 (diversity aerial)

- ☐ Removing and installing ⇒ Communication; Rep. gr. 91 ; Aerial systems; Removing and installing aerial module

11 - Light in exterior mirror

- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 94 ; Lights in exterior mirror; Removing and installing turn signal





12 - Mirror cover

- ☐ Top
- ☐ Removing and installing ⇒ [page 296](#)

13 - Cover

- ☐ Wings
- ☐ Removing and installing ⇒ [page 31](#)

6.1.2 Assembly overview - exploded mirror, dropside

1 - Mirror mounting

- ☐ Removing and installing
⇒ [page 295](#)

2 - Dampers

3 - Bolts

- ☐ Quantity may vary depending on mirror mounting
- ☐ 20 Nm

4 - Cover

5 - Adjuster unit

6 - Bolt

- ☐ 2 Nm

7 - Mirror glass

- ☐ Removing and installing
⇒ [page 295](#)

8 - Bolts

- ☐ Qty. 4
- ☐ 2 Nm

9 - Mirror frame

- ☐ With auxiliary panoramic mirror
- ☐ Removing and installing
⇒ [page 297](#)

10 - Mirror cover

- ☐ Bottom
- ☐ Removing and installing
⇒ [page 296](#)

11 - Aerial module for radio aerial - R248- for FM2 (diversity aerial)

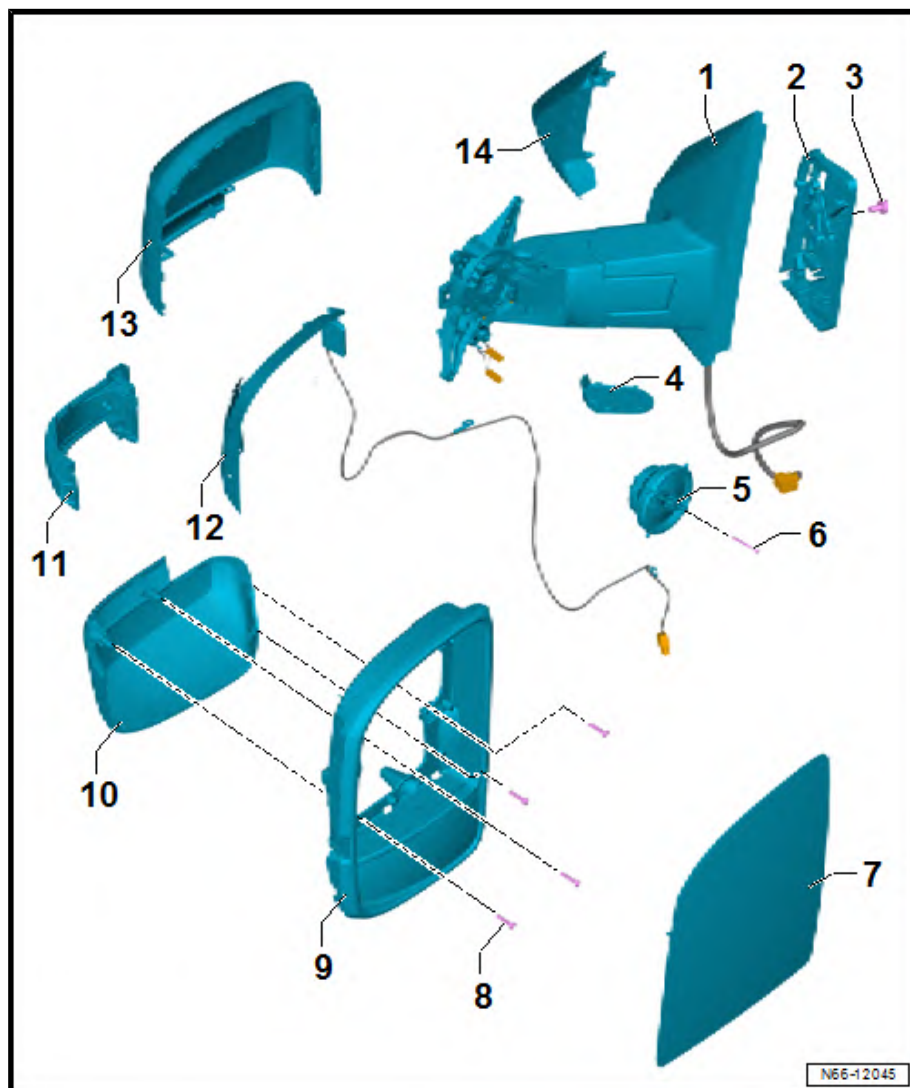
- ☐ Removing and installing ⇒ Communication; Rep. gr. 91 ; Aerial systems; Removing and installing aerial module

12 - Light in exterior mirror

- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 94 ; Lights in exterior mirror; Removing and installing turn signal

13 - Mirror cover

- ☐ Top
- ☐ Removing and installing ⇒ [page 296](#)





14 - Cover

- ☐ Wings
- ☐ Removing and installing ⇒ [page 31](#)

6.2 Removing and installing exterior mirror

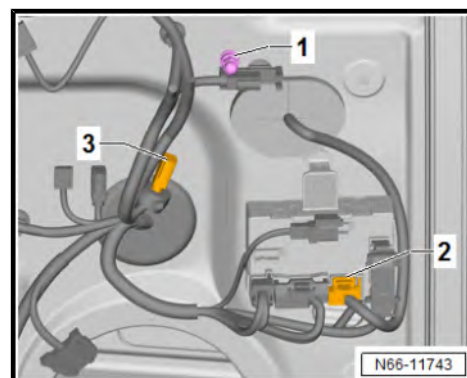
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

- Remove front door trim ⇒ General body repairs, interior; Rep. gr. 70 ; Front door trims; Removing and installing front door trim .
- Disconnect electrical connectors -2- and -3-.
- Unscrew nut -1- and disconnect earth wire.



Vehicles with additional (third) bolt

- Remove plugs -4-.
- Loosen bolt -3-

Continued for all vehicles

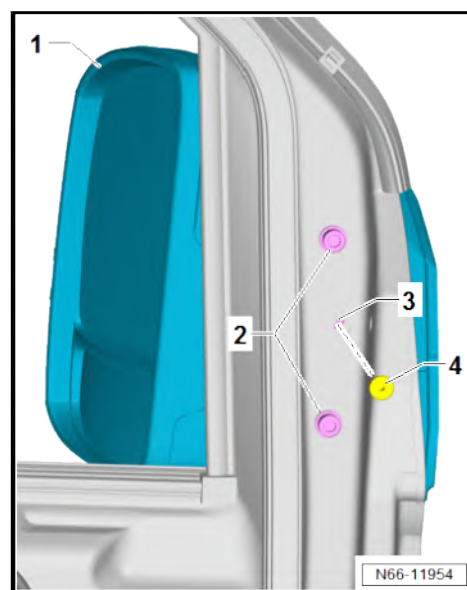
- Unscrew bolts -2-. While doing this, push exterior mirror -1- towards rear out of mirror reinforcement to remove.
- Guide wiring harness for exterior mirror -1- through opening in door.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“6.1 Assembly overview - exterior mirror”, page 293](#)

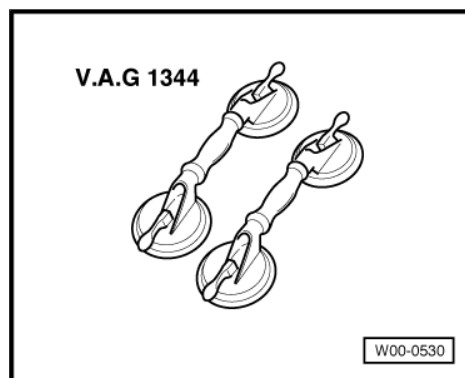


6.3 Removing and installing mirror glass

Special tools and workshop equipment required

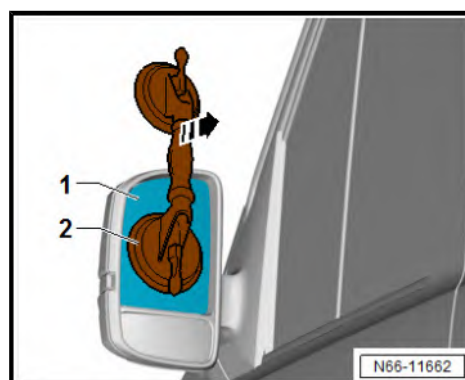


◆ Suction holder - V.A.G 1344-



Removing

- Place suction lifter - V.A.G 1344- on mirror glass -1-.
- Pull off mirror glass -1- from mirror adjustment unit using suction lifter - V.A.G 1344- -2- in direction of -arrow-.

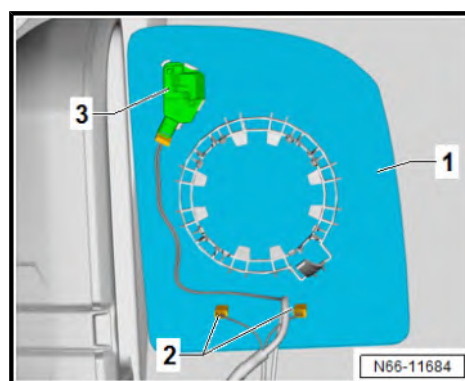


- Swing mirror glass -1- towards side.
- Separate electrical connectors -2-.
- If fitted, separate connector -3-.

Installing

Install in reverse order of removal, observing the following:

- Fit mirror glass.
- When fitting the mirror glass, only press on the centre of the mirror.
- Perform functional check



6.4 Removing and installing mirror cover

⇒ [“6.4.1 Removing and installing mirror cover, top”, page 296](#)

⇒ [“6.4.2 Removing and installing mirror cover, bottom”, page 297](#)

6.4.1 Removing and installing mirror cover, top

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

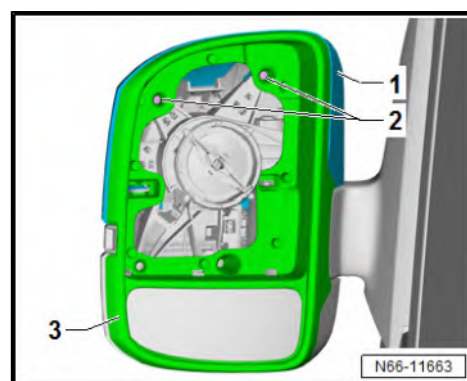
- Remove mirror glass ➤ [page 295](#) .
- Unscrew bolts -2-.
- Remove mirror cover -1- from mirror trim -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [“6.1 Assembly overview - exterior mirror”, page 293](#)



6.4.2 Removing and installing mirror cover, bottom

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

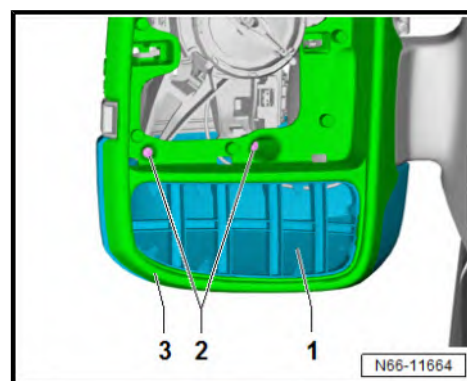
- Remove mirror glass ➤ [page 295](#) .
- Unscrew bolts -2-.
- Remove mirror cover -1- from mirror trim -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [“6.1 Assembly overview - exterior mirror”, page 293](#)



6.5 Removing and installing mirror trim

Special tools and workshop equipment required

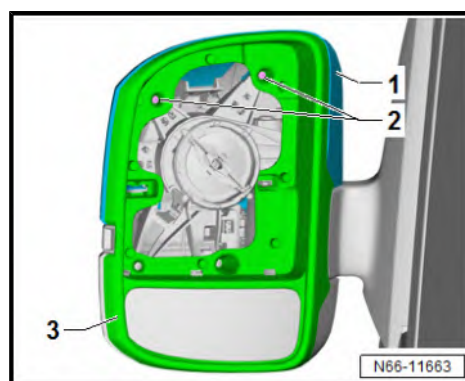


- ◆ Torque wrench - V.A.G 1783-

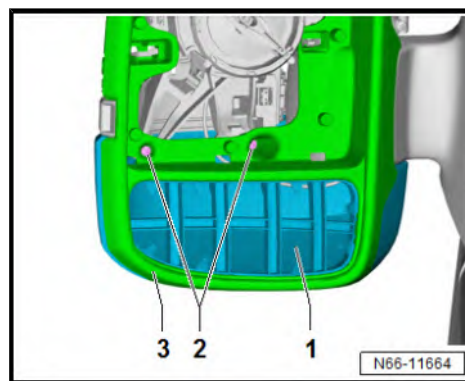


Removing

- Remove mirror glass ➔ [page 295](#) .
- Unscrew bolts -2-.
- Remove top mirror cover -1- from mirror trim -3-.



- Unscrew bolts -2-.
- Remove bottom mirror cover -1- from mirror trim -3-.



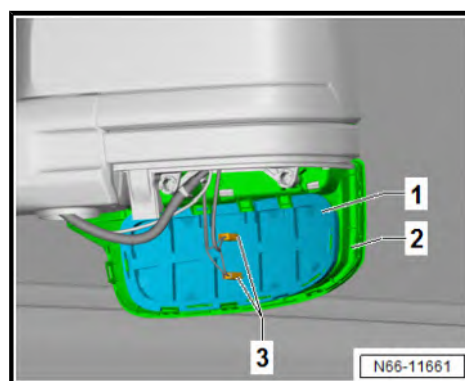
- Disconnect electrical connectors -3- from auxiliary panoramic mirror -1-.
- Remove mirror trim -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ ["6.1 Assembly overview - exterior mirror", page 293](#)





7 Wheel housing liner

⇒ [“7.1 Assembly overview - front wheel housing liner”, page 299](#)

⇒ [“7.2 Removing and installing front wheel housing liner”, page 299](#)

7.1 Assembly overview - front wheel housing liner

1 - Wheel housing liner

- ☐ Front
- ☐ Removing and installing
⇒ [page 299](#)

2 - Bolts

- ☐ 15 or 16 depending on equipment
- ☐ 3 Nm

3 - Maintenance cover

4 - Bolts

- ☐ Qty. 3
- ☐ Connecting piece for wheel housing liner, front
- ☐ 3 Nm

5 - Bolts

- ☐ Qty. 2
- ☐ Connecting piece, noise insulation
- ☐ 3 Nm

6 - Stone deflector

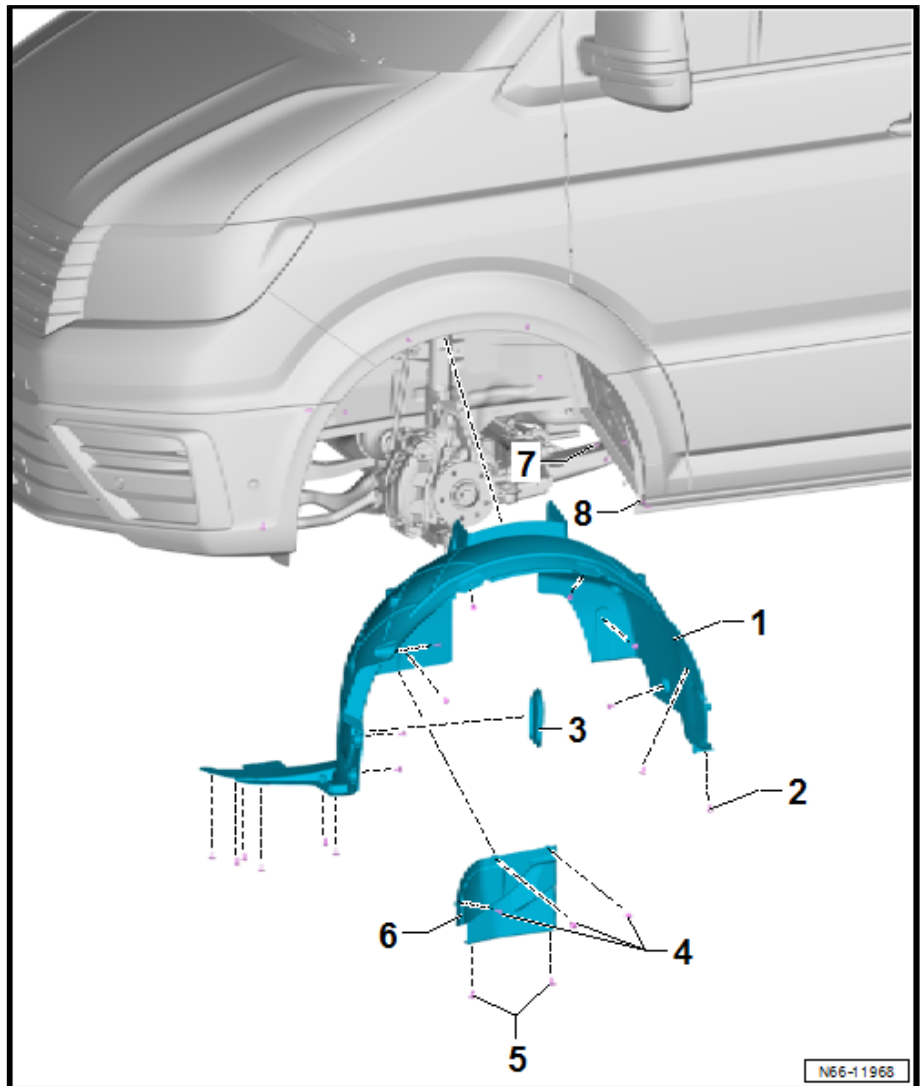
- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 299](#)

7 - Expanding nuts

- ☐ Qty. 4

8 - Expansion nut

- ☐ Side member connecting section



Volkswagen Technical Site: <https://vwts.ru>

7.2 Removing and installing front wheel housing liner

⇒ [“7.2.1 Removing and installing front wheel housing liner”, page 299](#)

⇒ [“7.2.2 Removing and installing stone deflector”, page 300](#)

7.2.1 Removing and installing front wheel housing liner

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

- If fitted, remove noise insulation ⇒ [page 271](#) .
- Unscrew bolts -1-.

Vehicles with stone deflector

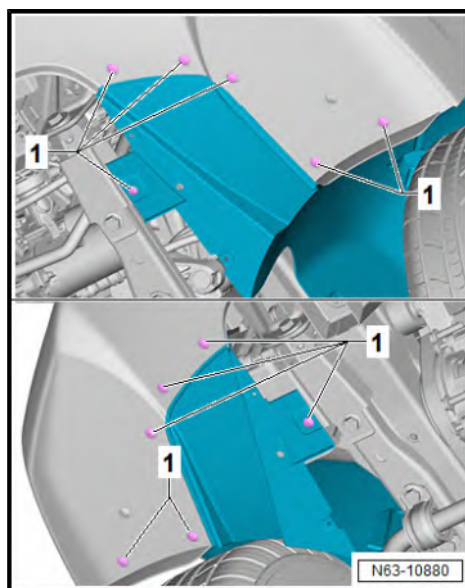
- Remove stone deflector ⇒ [page 300](#) .

Continued for all vehicles



Note

To improve clarity, the front wheel is not shown in the following illustrations.



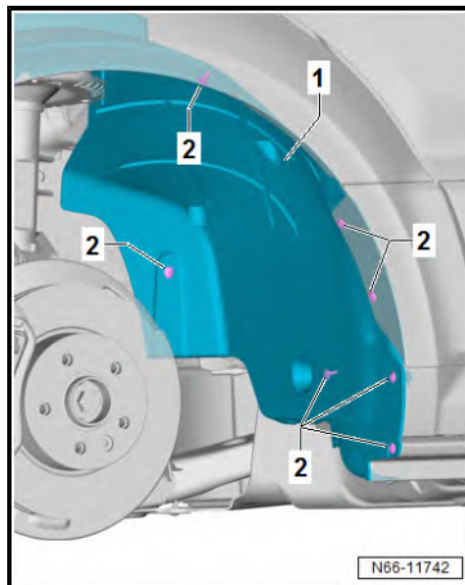
- Unscrew bolts -2-.
- Remove wheel housing liner -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“7.1 Assembly overview - front wheel housing liner”, page 299](#)

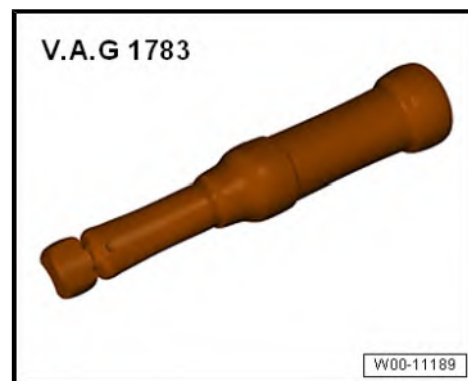


7.2.2 Removing and installing stone deflector

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

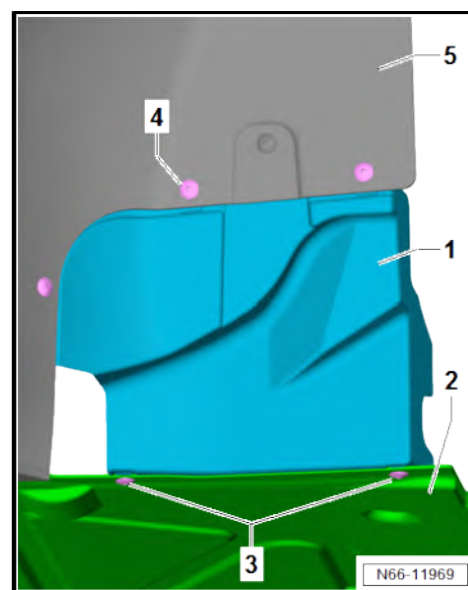
- Unscrew bolts -4- from wheel housing liner -5-.
- Unscrew bolts -3- from noise insulation -2-.
- Remove stone deflector -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["7.1 Assembly overview - front wheel housing liner", page 299](#)





8 Lettering and emblems

- ⇒ ["8.1 Assembly overview - lettering and emblems", page 302](#)
- ⇒ ["8.2 Dimensions - lettering and emblems on rear", page 308](#)
- ⇒ ["8.3 Dimensions - lettering and emblems on sides", page 311](#)
- ⇒ ["8.4 Renewing lettering and badges at rear", page 311](#)
- ⇒ ["8.5 Renewing lettering and badges", page 314](#)
- ⇒ ["8.6 Removing and installing lettering and emblems for radiator grille", page 315](#)

8.1 Assembly overview - lettering and emblems

- ⇒ ["8.1.1 Assembly overview - lettering and badges, Crafter", page 302](#)
- ⇒ ["8.1.2 Assembly overview - lettering and badges, e-Crafter", page 304](#)
- ⇒ ["8.1.3 Assembly overview - lettering and badges, Crafter double cab", page 305](#)
- ⇒ ["8.1.4 Assembly overview - lettering and badges, MAN TGE", page 306](#)
- ⇒ ["8.1.5 Assembly overview - lettering and badges, MAN TGE double cab", page 307](#)

8.1.1 Assembly overview - lettering and badges, Crafter



1 - Lettering

- ☐ Removing and installing
⇒ [page 315](#)

2 - Badge

- ☐ Removing and installing
⇒ [page 315](#)

3 - Bolts

- ☐ Qty. 2
- ☐ 4 Nm

4 - Cover

- ☐ Right rear window
- ☐ Removing and installing
⇒ [page 311](#)

5 - Floor panel

- ☐ For badge
- ☐ Removing and installing
⇒ [page 311](#)

6 - Bolt

- ☐ 10 Nm

7 - Badge

- ☐ Removing and installing
⇒ [page 311](#)

8 - Cover

- ☐ Left rear window
- ☐ Removing and installing
⇒ [page 311](#)

9 - Bolt

- ☐ 2 Nm

10 - Lettering

- ☐ Depending on the engine designation
- ☐ Renewing ⇒ [page 314](#)
- ☐ Dimensions ⇒ [page 308](#)

11 - Lettering

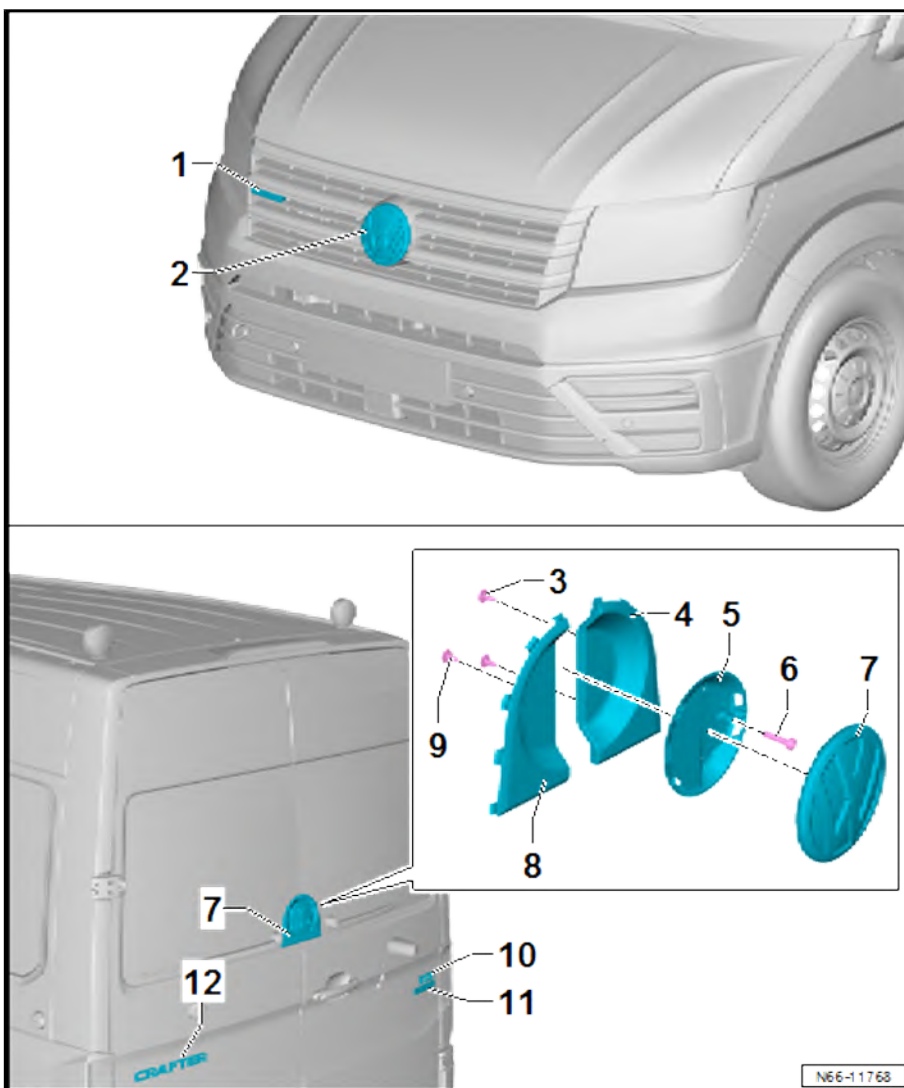
- ☐ Additional model designation

The lettering differs depending on the model designation, for allocation refer to ⇒ Electronic parts catalogue (ETKA) .

- ☐ Renewing ⇒ [page 314](#)
- ☐ Dimensions ⇒ [page 308](#)

12 - Lettering

- ☐ Model designation.
- ☐ Renewing ⇒ [page 314](#)
- ☐ Dimensions ⇒ [page 308](#)





8.1.2 Assembly overview - lettering and badges, e-Crafter

1 - Lettering

- ❑ Removing and installing
⇒ [page 315](#)

2 - Badge

- ❑ Removing and installing
⇒ [page 315](#)

3 - Bolts

- ❑ Qty. 2
- ❑ 4 Nm

4 - Cover

- ❑ Right rear window
- ❑ Removing and installing
⇒ [page 311](#)

5 - Floor panel

- ❑ For badge
- ❑ Removing and installing
⇒ [page 311](#)

6 - Bolt

- ❑ 10 Nm

7 - Badge

- ❑ Removing and installing
⇒ [page 311](#)

8 - Cover

- ❑ Left rear window
- ❑ Removing and installing
⇒ [page 311](#)

9 - Bolt

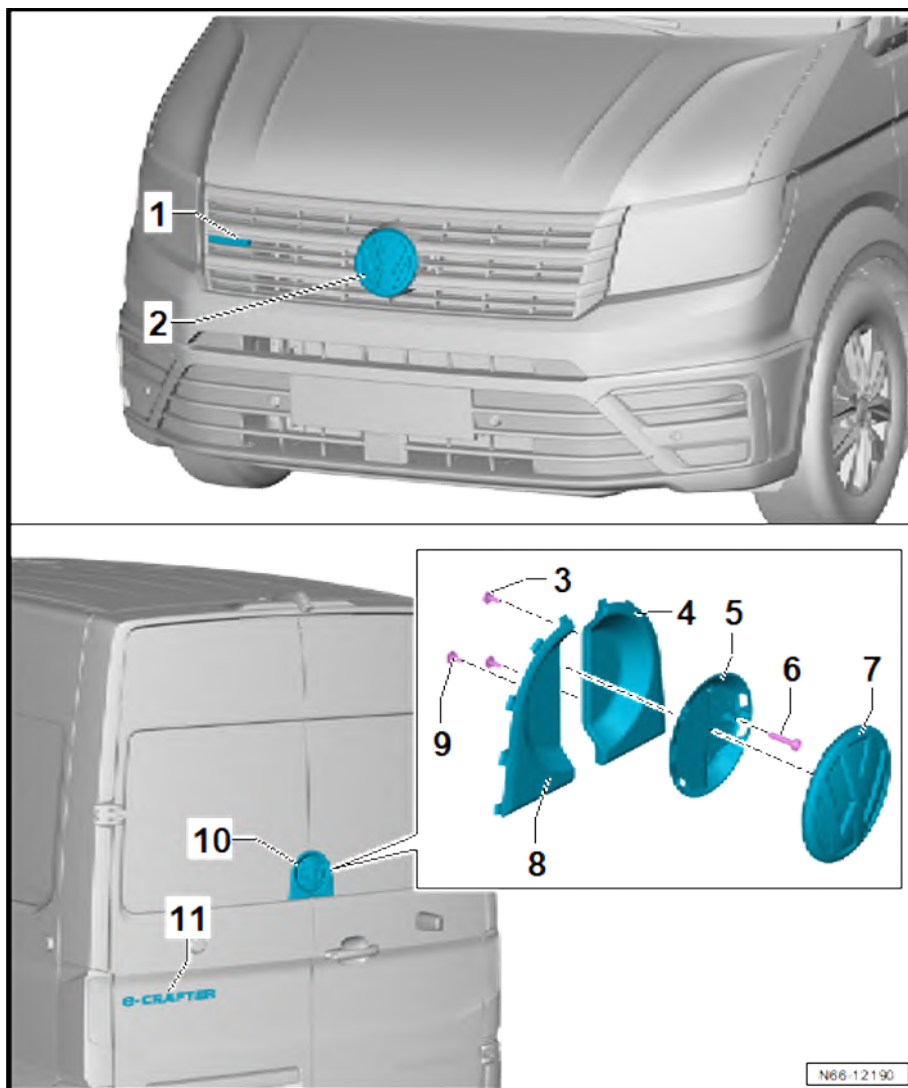
- ❑ 2 Nm

10 - Badge

- ❑ Removing and installing
⇒ [page 311](#)

11 - Lettering

- ❑ Model designation.
- ❑ Renewing ⇒ [page 314](#)
- ❑ Dimensions ⇒ [page 308](#)





8.1.3 Assembly overview - lettering and badges, Crafter double cab

1 - Lettering

- ☐ Removing and installing
⇒ [page 315](#)

2 - Badge

- ☐ Removing and installing
⇒ [page 315](#)

3 - Lettering

- ☐ Model designation.
- ☐ Renewing ⇒ [page 314](#)
- ☐ Dimensions
⇒ [page 308](#)

4 - Badge

- ☐ Tailgate
- ☐ Removing and installing
⇒ [page 314](#)

5 - Lettering

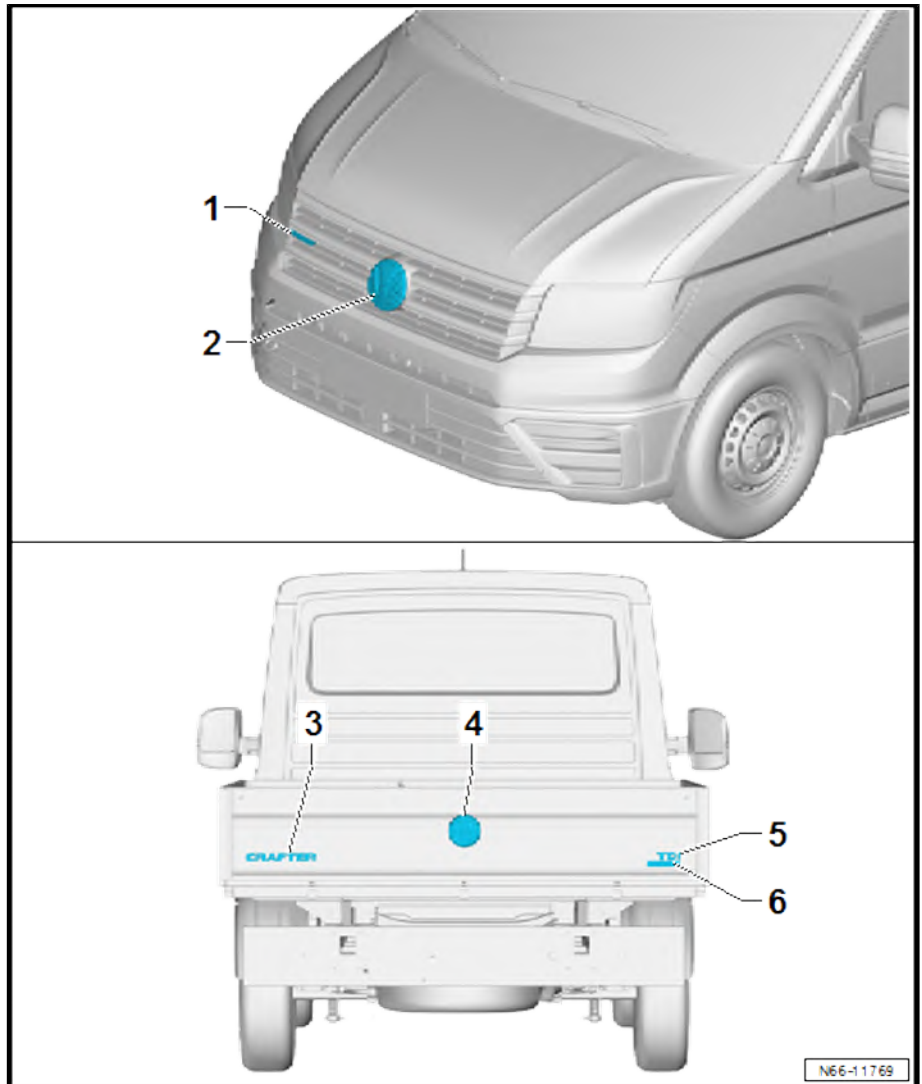
- ☐ Depending on the engine designation
- ☐ Renewing ⇒ [page 314](#)
- ☐ Dimensions
⇒ [page 308](#)

6 - Lettering

- ☐ Additional model designation

The lettering differs depending on the model designation, for allocation refer to ⇒ Electronic parts catalogue (ETKA) .

- ☐ Renewing ⇒ [page 314](#)
- ☐ Dimensions
⇒ [page 308](#)





8.1.4 Assembly overview - lettering and badges, MAN TGE

1 - Lettering

- ❑ Removing and installing
⇒ [page 315](#)

2 - Badge

- ❑ Removing and installing
⇒ [page 315](#)

3 - Lettering

- ❑ Wing and door
- ❑ Additional model designation
- ❑ Different versions; for allocation for MAN TGE, refer to ⇒ webMANTIS
- ❑ Renewing ⇒ [page 314](#)
- ❑ Dimensions
⇒ [page 311](#)

4 - Badge

- ❑ Rear
- ❑ Removing and installing
⇒ [page 311](#)

5 - Lettering

- ❑ Model designation.
- ❑ Renewing ⇒ [page 314](#)
- ❑ Dimensions

6 - Cover

- ❑ Left rear window
- ❑ Removing and installing
⇒ [page 311](#)

7 - Bolt

- ❑ 2 Nm

8 - Bolts

- ❑ Qty. 2
- ❑ 4 Nm

9 - Cover

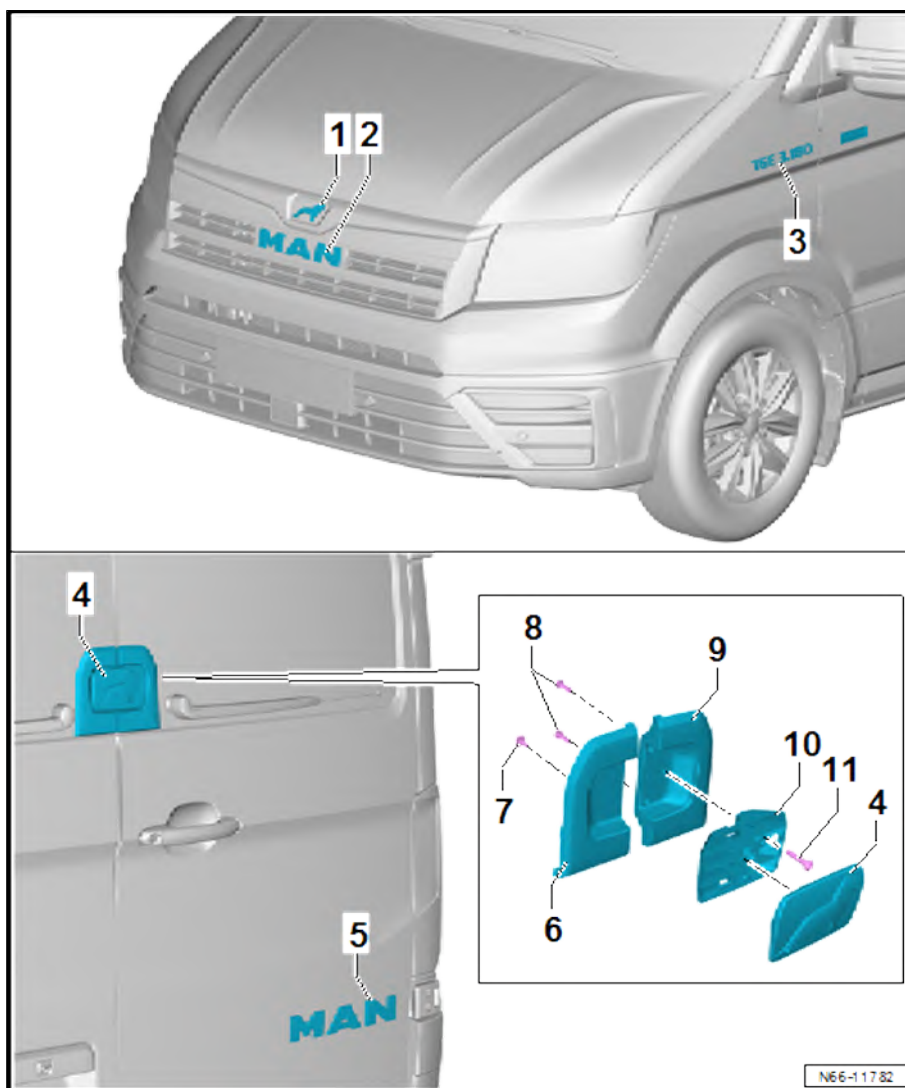
- ❑ Right rear window
- ❑ Removing and installing ⇒ [page 311](#)

10 - Floor panel

- ❑ For badge
- ❑ Removing and installing ⇒ [page 311](#)

11 - Bolt

- ❑ 10 Nm





8.1.5 Assembly overview - lettering and badges, MAN TGE double cab

1 - Lettering

- ❑ Removing and installing
⇒ [page 315](#)

2 - Badge

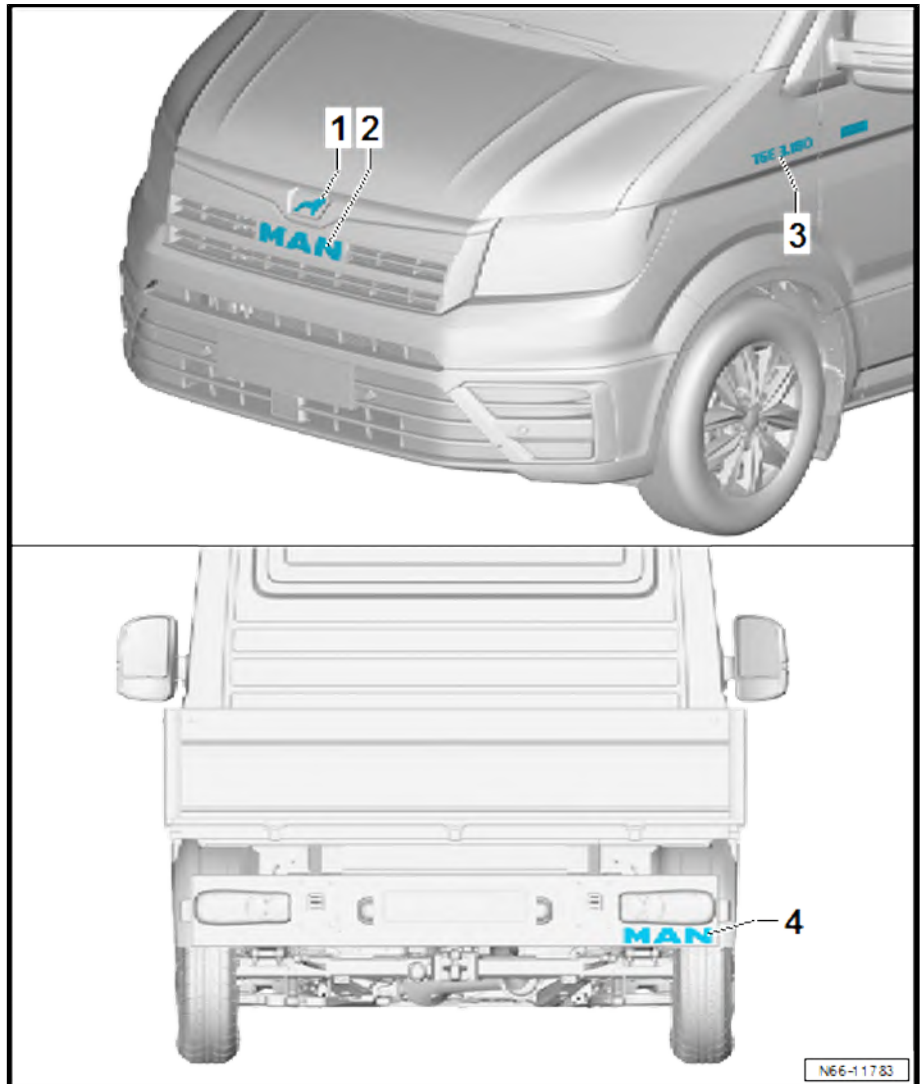
- ❑ Removing and installing
⇒ [page 315](#)

3 - Lettering

- ❑ Wing and door
- ❑ Additional model designation
- ❑ Different versions; for allocation for MAN TGE, refer to ⇒ webMANTIS
- ❑ Renewing ⇒ [page 314](#)
- ❑ Dimensions
⇒ [page 311](#)

4 - Lettering

- ❑ Model designation.
- ❑ Renewing ⇒ [page 314](#)
- ❑ Dimensions
⇒ [page 308](#)





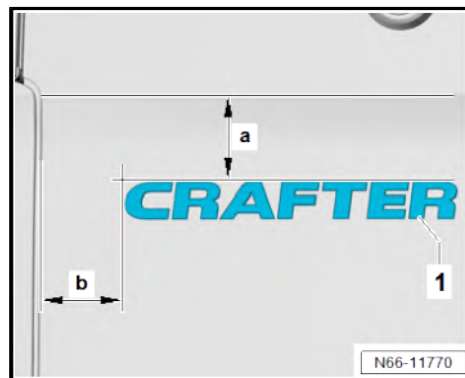
8.2 Dimensions - lettering and emblems on rear

- ⇒ ["8.2.1 Dimensions - lettering on left, Crafter", page 308](#)
- ⇒ ["8.2.2 Dimensions - lettering on left, e-Crafter", page 308](#)
- ⇒ ["8.2.3 Dimensions - lettering on right", page 309](#)
- ⇒ ["8.2.4 Dimensions - lettering on right, TGE", page 309](#)
- ⇒ ["8.2.5 Dimensions - lettering on left, double cab, Crafter", page 309](#)
- ⇒ ["8.2.6 Dimensions - lettering on right, double cab", page 310](#)
- ⇒ ["8.2.7 Dimensions - lettering on right, double cab, TGE", page 310](#)

8.2.1 Dimensions - lettering on left, Crafter

Lettering -1- indicating model

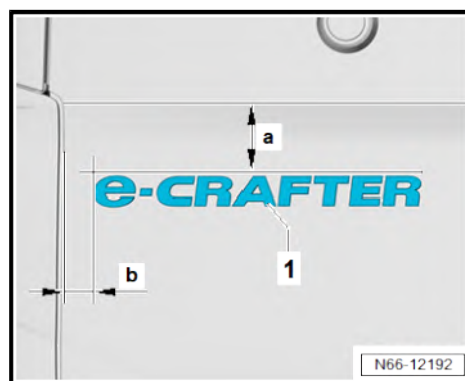
- ◆ Dimension -a- 47 mm
- ◆ Dimension -b- 80 mm



8.2.2 Dimensions - lettering on left, e-Crafter

Lettering -1- indicating model

- ◆ Dimension -a- 47 mm
- ◆ Dimension -b- 80 mm



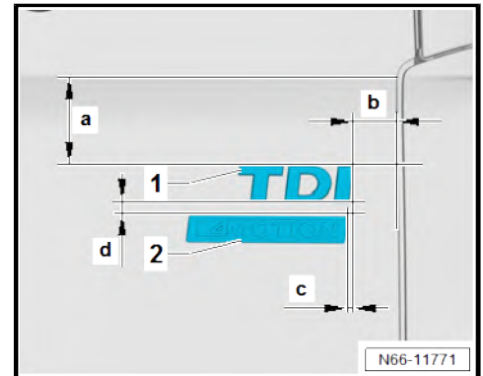


8.2.3 Dimensions - lettering on right

Lettering -1- indicating engine

Lettering -2- indicating additional model designation

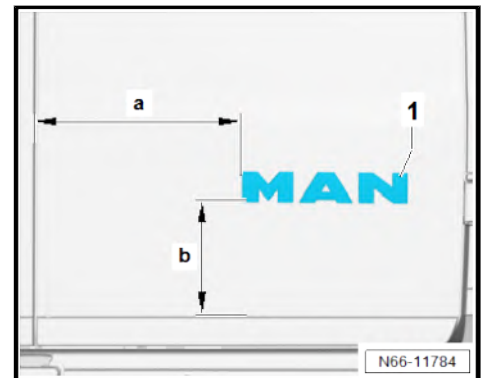
- ◆ Dimension -a- 56 mm
- ◆ Dimension -b- 36 mm
- ◆ Dimension -c- 5 mm
- ◆ Dimension -d- 10 mm



8.2.4 Dimensions - lettering on right, TGE

Lettering -1- indicating additional model designation

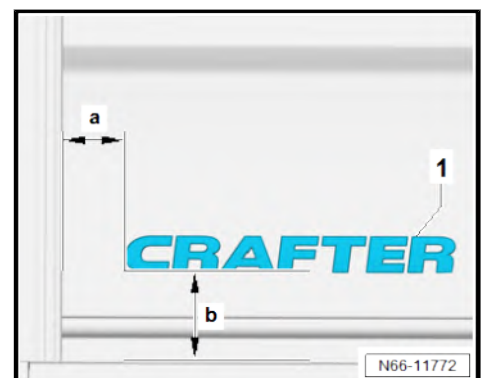
- ◆ Dimension -a- 388 mm
- ◆ Dimension -b- 212 mm



8.2.5 Dimensions - lettering on left, double cab, Crafter

Lettering -1- indicating model

- ◆ Dimension -a- 47 mm
- ◆ Dimension -b- 80 mm



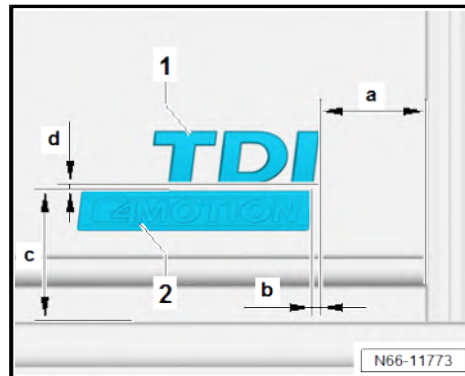


8.2.6 Dimensions - lettering on right, double cab

Lettering -1- indicating engine

Lettering -2- indicating additional model designation

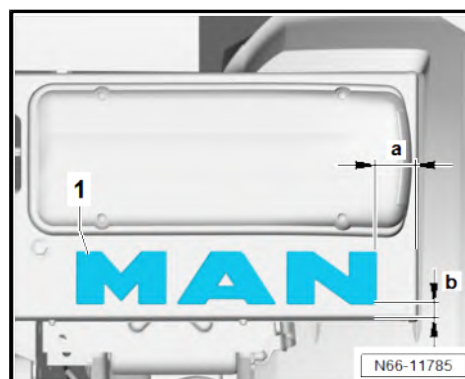
- ◆ Dimension -a- 48 mm
- ◆ Dimension -b- 5.5 mm
- ◆ Dimension -c- 80 mm
- ◆ Dimension -d- 10 mm



8.2.7 Dimensions - lettering on right, double cab, TGE

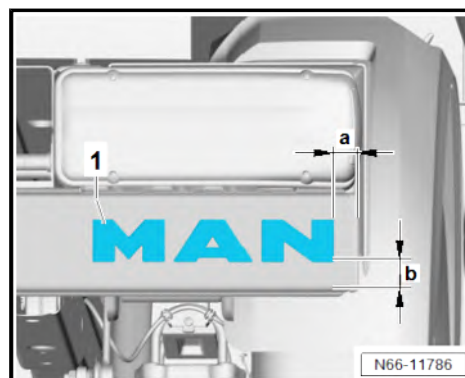
Lettering -1- indicating model for wide carrier

- ◆ Dimension -a- 43 mm
- ◆ Dimension -b- 12 mm



Lettering -1- indicating model for narrow carrier

- ◆ Dimension -a- 46 mm
- ◆ Dimension -b- 44 mm





8.3 Dimensions - lettering and emblems on sides

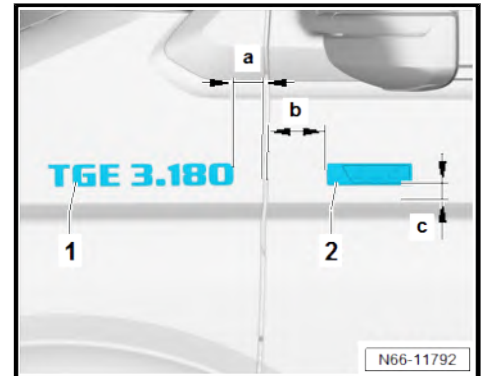
⇒ ["8.3.1 Dimensions - lettering and badges on sides, TGE", page 311](#)

8.3.1 Dimensions - lettering and badges on sides, TGE

Lettering -1- on wing

Lettering -2- on door

- ◆ Dimension -a- 42 mm
- ◆ Dimension -b- 86 mm
- ◆ Dimension -c- 21 mm



8.4 Renewing lettering and badges at rear

⇒ ["8.4.1 Removing and installing rear window cover, Crafter", page 311](#)

⇒ ["8.4.2 Removing and installing rear window cover, TGE", page 312](#)

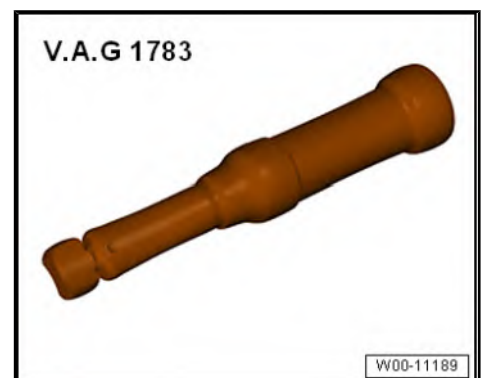
⇒ ["8.4.3 Removing and installing badges on rear, Crafter", page 312](#)

⇒ ["8.4.4 Removing and installing badges on rear, TGE", page 313](#)

8.4.1 Removing and installing rear window cover, Crafter

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-





Removing

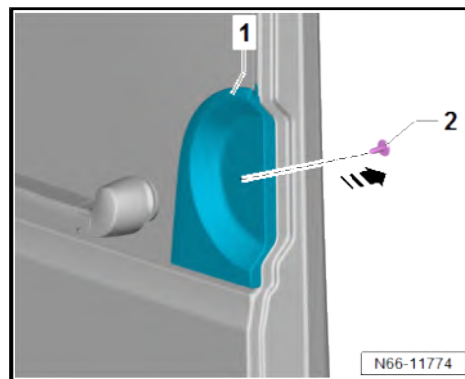
- Unscrew bolt -2-.
- Remove rear window cover -1- in direction of -arrow-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ ["8.1 Assembly overview - lettering and emblems", page 302](#)



8.4.2 Removing and installing rear window cover, TGE

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-



Removing

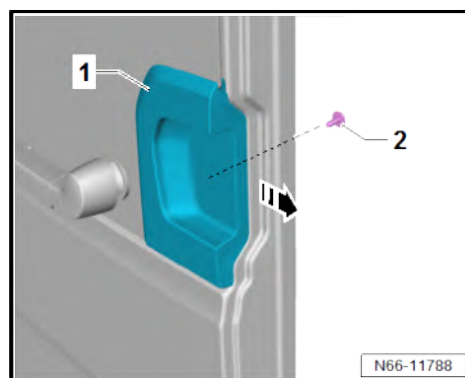
- Unscrew bolt -2-.
- Remove rear window cover -1- in direction of -arrow-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ ["8.1 Assembly overview - lettering and emblems", page 302](#)

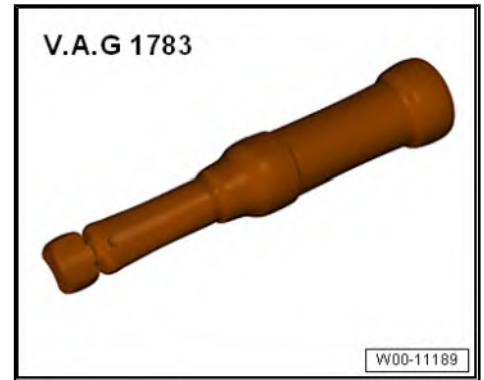


8.4.3 Removing and installing badges on rear, Crafter

Special tools and workshop equipment required

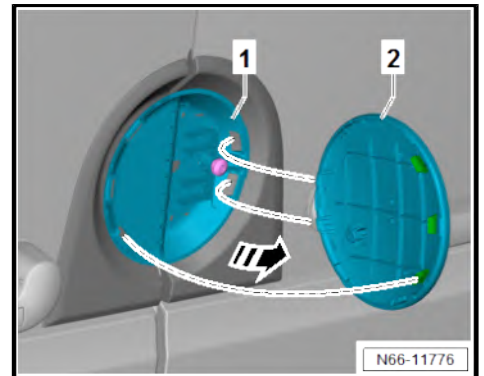


- ◆ Torque wrench - V.A.G 1783-



Removing

- Unclip badges -2- in direction of -arrow- from base plate -1-.



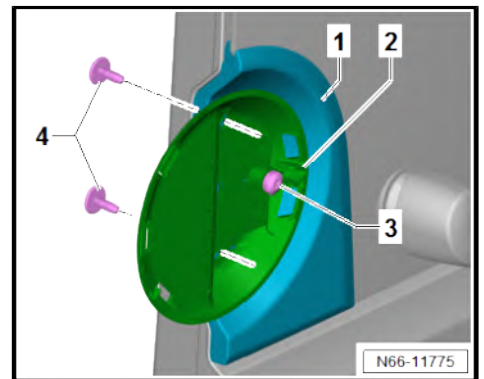
- Unscrew bolt -3-.
- Remove base plate -2-.
- Unscrew bolts -4-.
- Remove rear window cover -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

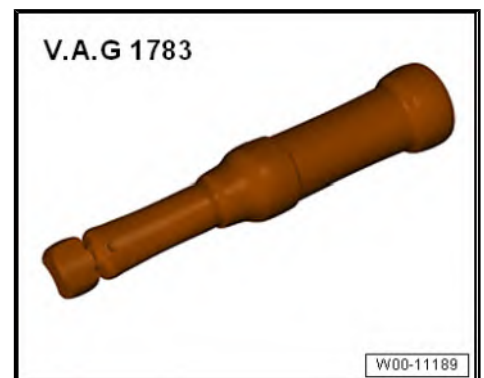
- ◆ ➤ ["8.1 Assembly overview - lettering and emblems", page 302](#)



8.4.4 Removing and installing badges on rear, TGE

Special tools and workshop equipment required

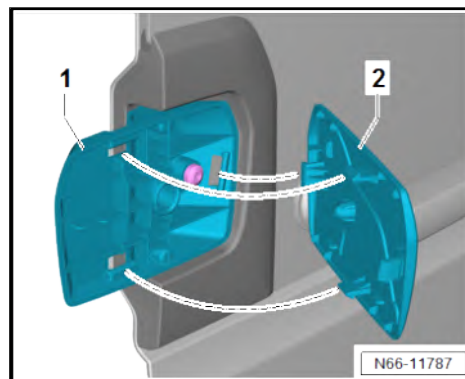
- ◆ Torque wrench - V.A.G 1783-





Removing

- Unclip badges -2- in direction of -arrow- from base plate -1-.



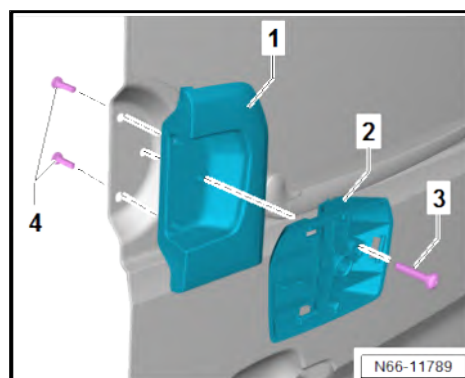
- Unscrew bolt -3-.
- Remove base plate -2-.
- Unscrew bolts -4-.
- Remove rear window cover -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

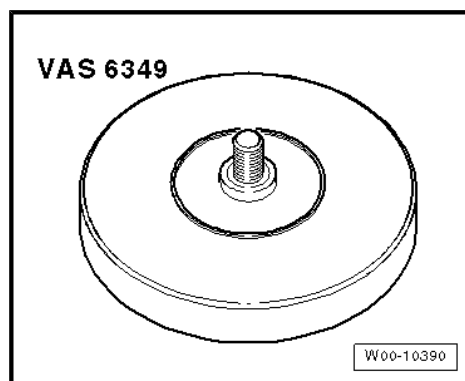
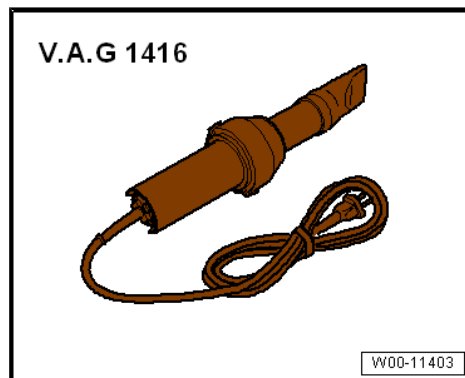
- ♦ ⇒ ["8.1 Assembly overview - lettering and emblems", page 302](#)



8.5 Renewing lettering and badges

Special tools and workshop equipment required

- ♦ Hot-air blower - V.A.G 1416-
- ♦ Adhesive strip remover - VAS 6349-



Procedure

When removing cover, adhesive tape may become damaged.



If using new adhesive tape as prescribed in ⇒ Electronic parts catalogue (ETKA) and ⇒ webMANTIS for MAN TGE, the trim strip can be reused.

- Check clips for damage and renew if necessary.

If a cover is installed for the first time on a new vehicle or on newly painted surfaces, the procedure in ⇒ Workshop manual "Paint"; ⇒ Paintwork repairs; ⇒ Paint finish; ⇒ Mouldings & film should be observed.

If a cover is removed and reattached, use only adhesive remover to remove the adhesive residue.

- Remove existing adhesive residue of adhesive tape using adhesive strip remover - VAS 6349- .
- Bonding surfaces must be free of dust and grease.
- Install cover immediately after cleaning.
- Do not remove protective foil until immediately before installation.
- The working temperature is approx. 21°C.

8.6 Removing and installing lettering and emblems for radiator grille

⇒ ["8.6.1 Removing and installing lettering for radiator grille, Crafter", page 315](#)

⇒ ["8.6.2 Removing and installing lettering for radiator grille, e-Crafter", page 316](#)

⇒ ["8.6.3 Removing and installing badge for radiator grille, Crafter", page 316](#)

⇒ ["8.6.4 Removing and installing lettering for radiator grille, TGE", page 316](#)

⇒ ["8.6.5 Removing and installing badge for radiator grille, TGE", page 317](#)

8.6.1 Removing and installing lettering for radiator grille, Crafter

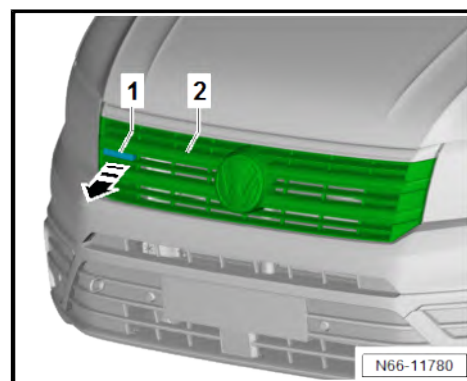
Removing

- Remove bumper cover ⇒ [page 179](#) .
- Unclip lettering -1- in direction of -arrow- from fasteners in radiator grille -2-.

Installing

Install in reverse order of removal, observing the following:

- Press lettering -1- onto radiator grille -2- until retaining hooks engage audibly.





8.6.2 Removing and installing lettering for radiator grille, e-Crafter

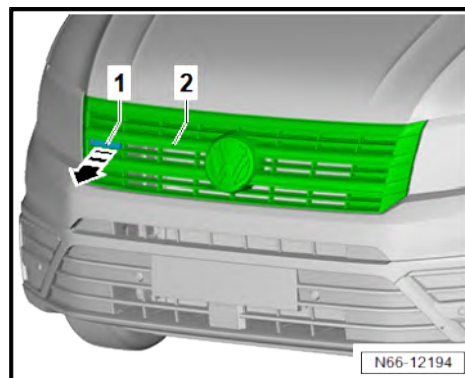
Removing

- Remove bumper cover ➔ [page 179](#) .
- Unclip lettering -1- in direction of -arrow- from fasteners in radiator grille -2-.

Installing

Install in reverse order of removal, observing the following:

- Press lettering -1- onto radiator grille -2- until retaining hooks engage audibly.



8.6.3 Removing and installing badge for radiator grille, Crafter

Removing



Note

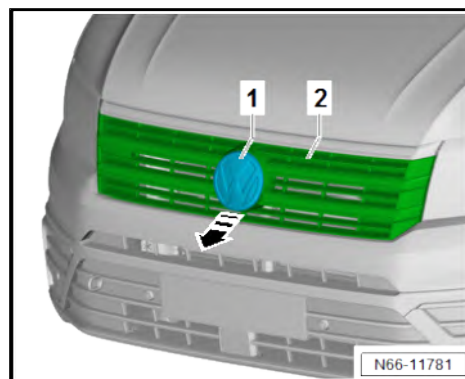
Badge -1- cannot be removed without being damaged.

- Using a screwdriver, unclip badge -1- from radiator grille -2-.

Installing

Install in reverse order of removal, observing the following:

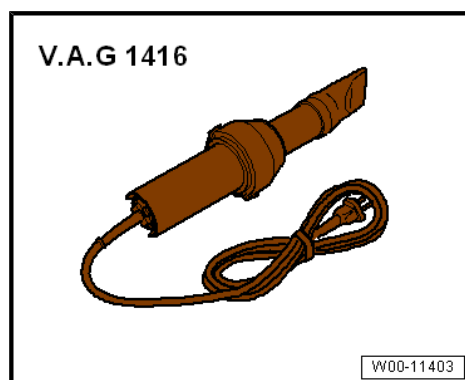
- Press front badge -1- on radiator grille -2- until it engages audibly.



8.6.4 Removing and installing lettering for radiator grille, TGE

Special tools and workshop equipment required

- ♦ Hot-air blower - V.A.G 1416-





Removing



Note

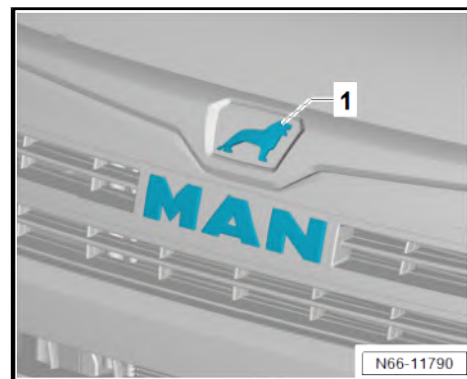
Bonded lettering or emblems cannot be removed without being damaged.

- Before removing lettering -1-, heat it with a hot air blower - V.A.G 1416- .
- Lift off lettering -1- from radiator grille.

Installing

Install in reverse order of removal, observing the following:

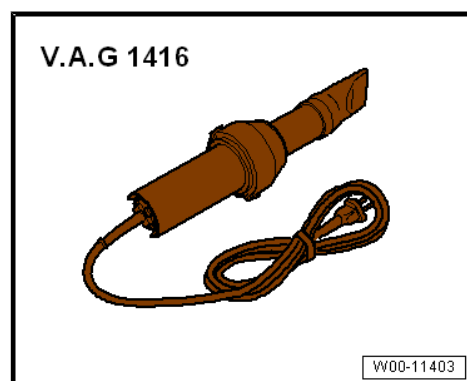
- Follow installation instructions ⇒ [page 314](#) .
- Position lettering -1- in depression in radiator grille and press on firmly.



8.6.5 Removing and installing badge for radiator grille, TGE

Special tools and workshop equipment required

- ◆ Hot-air blower - V.A.G 1416-



Removing



Note

Bonded lettering or emblems cannot be removed without being damaged.

- Before removing badge -1-, heat it with hot air blower - V.A.G 1416- .
- Lift off badge -1- from radiator grille.

Installing

Install in reverse order of removal, observing the following:

- Follow installation instructions ⇒ [page 314](#) .
- Position badge -1- on radiator grille and press on firmly.



9 Towing bracket

⇒ ["9.1 Assembly overview - towing bracket", page 318](#)

⇒ ["9.2 Removing and installing towing bracket", page 319](#)

9.1 Assembly overview - towing bracket

⇒ ["9.1.1 Assembly overview - towing bracket", page 318](#)

⇒ ["9.1.2 Assembly overview - towing bracket, double cab", page 319](#)

9.1.1 Assembly overview - towing bracket

1 - Towing bracket

- ☐ Removing and installing

2 - Bolts

- ☐ Renew after removal
- ☐ Qty. 8
- ☐ 70 Nm +270°

3 - Ball head

- ☐ Removing and installing

4 - Bolts

- ☐ Renew after removal
- ☐ Qty. 4
- ☐ 50 Nm +90°

5 - Cover

6 - Double cab towing bracket

- ☐ Removing and installing

7 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 20 Nm

8 - Step

- ☐ For towing bracket.
- ☐ Removing and installing

9 - Ball head

- ☐ Removable

10 - Bolts

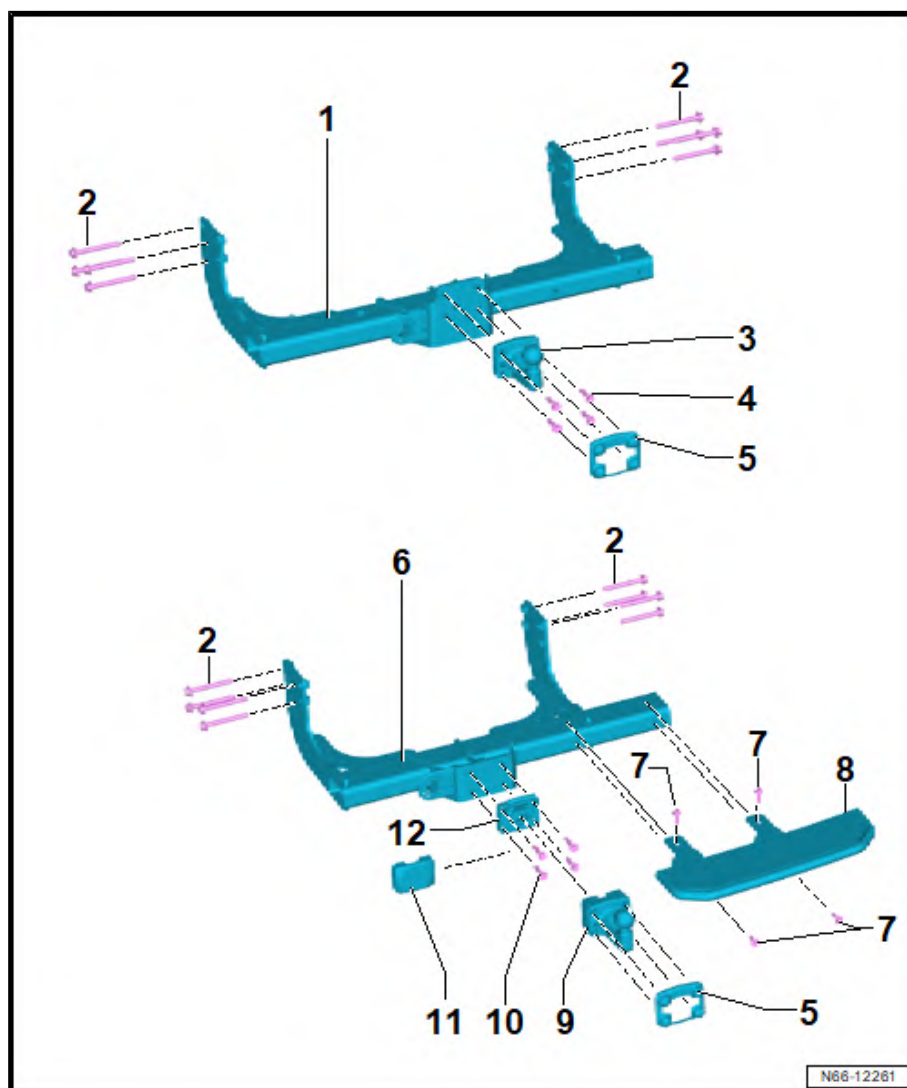
- ☐ Renew after removal
- ☐ Qty. 4
- ☐ 50 Nm +90°

11 - Cover

- ☐ For base plate

12 - Base plate

- ☐ Removing and installing





9.1.2 Assembly overview - towing bracket, double cab

1 - Towing bracket

- ☐ Removing and installing

2 - Bolts

- ☐ Renew after removal
- ☐ Qty. 8
- ☐ 70 Nm +270°

3 - Bolts

- ☐ Qty. 4
- ☐ 40 Nm

4 - Securing bracket

- ☐ Left and right

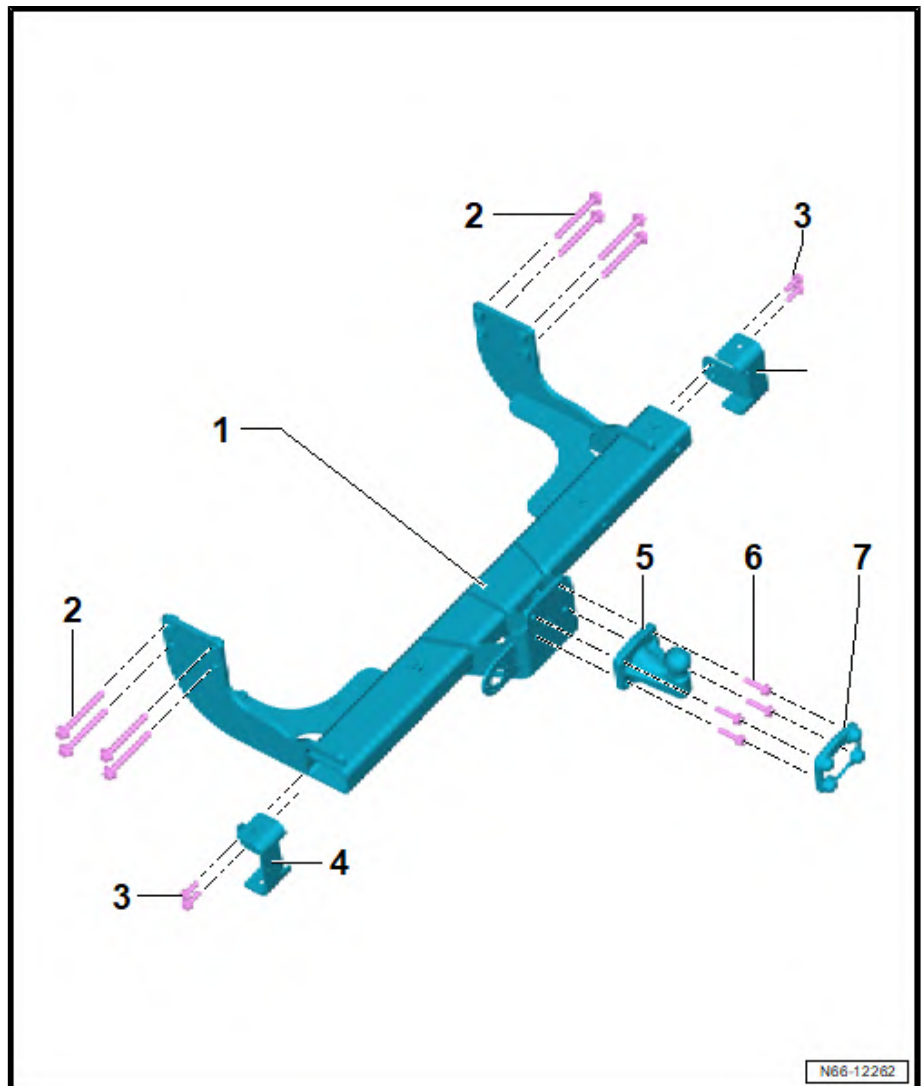
5 - Ball head

- ☐ Removing and installing

6 - Bolts

- ☐ Qty. 4
- ☐ 50 Nm +90°

7 - Cover



9.2 Removing and installing towing bracket

⇒ [“9.2.1 Removing and installing towing bracket”, page 319](#)

⇒ [“9.2.2 Removing and installing towing bracket, double cab”, page 320](#)

⇒ [“9.2.3 Removing and installing base plate”, page 321](#)

⇒ [“9.2.4 Removing and installing ball head”, page 322](#)

9.2.1 Removing and installing towing bracket

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1332-



- ◆ Engine and gearbox jack - VAS 6931-



Removing

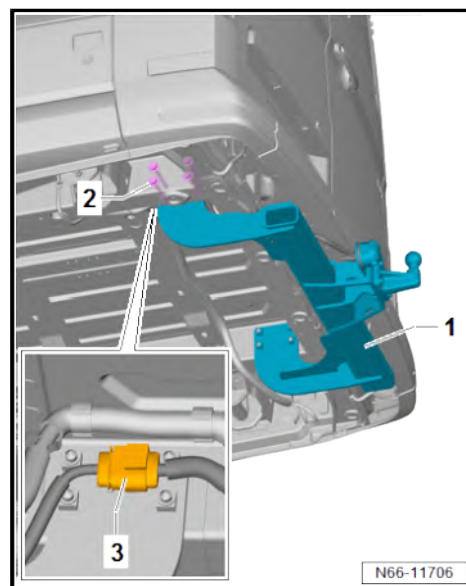
- Disconnect electrical connector -3-.
- The following work steps must be performed by a second mechanic.
- Place engine and gearbox jack - VAS 6931- under towing bracket -1- until it makes contact.
- Have a second mechanic secure towing bracket -1- on engine and gearbox jack - VAS 6931- .
- Remove bolts -2- on left and right.
- Remove towing bracket -1- together with a second mechanic from engine and gearbox jack - VAS 6931- .

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“9.1 Assembly overview - towing bracket”, page 318](#)



9.2.2 Removing and installing towing bracket, double cab

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1332-



- ◆ Engine and gearbox jack - VAS 6931-



Removing

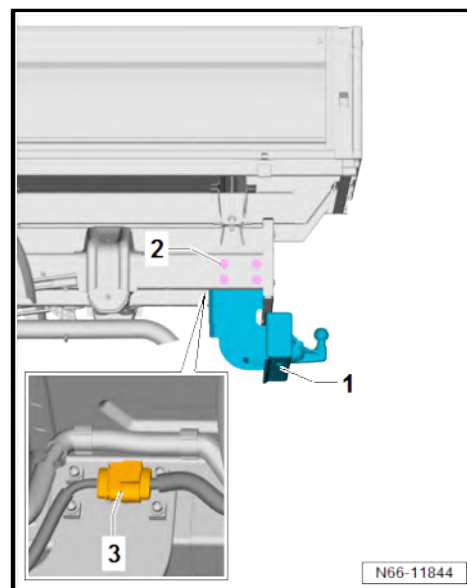
- Disconnect electrical connector -3-.
- The following work steps must be performed by a second mechanic.
- Place engine and gearbox jack - VAS 6931- under towing bracket -1- until it makes contact.
- Have a second mechanic secure towing bracket -1- on engine and gearbox jack - VAS 6931- .
- Remove bolts -2- on left and right.
- Remove towing bracket -1- together with a second mechanic from engine and gearbox jack - VAS 6931- .

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["9.1 Assembly overview - towing bracket", page 318](#)



9.2.3 Removing and installing base plate

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1332-



Removing

Vehicles with cap

- Remove cap.

Vehicles with removable ball head

- Remove ball head ➔ [page 322](#) .

Continued for all vehicles

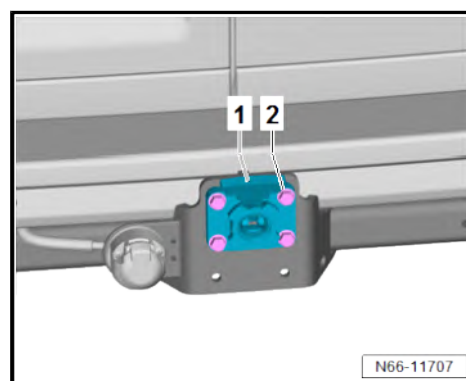
- Unscrew bolts -2-. When doing this, remove base plate -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [“9.1 Assembly overview - towing bracket”, page 318](#)



9.2.4 Removing and installing ball head

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-



Removing

- Detach cap -3-.



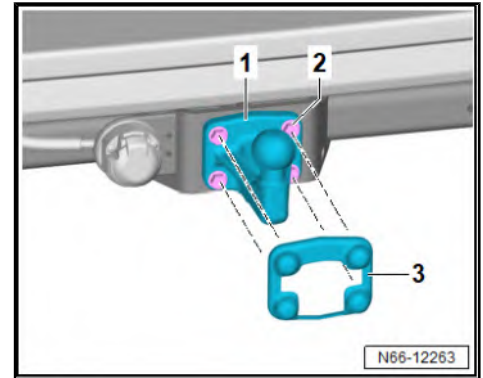
- Unscrew bolts -2-. When doing this, remove ball head -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["9.1 Assembly overview - towing bracket", page 318](#)



9.2.5 Removing and installing step

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

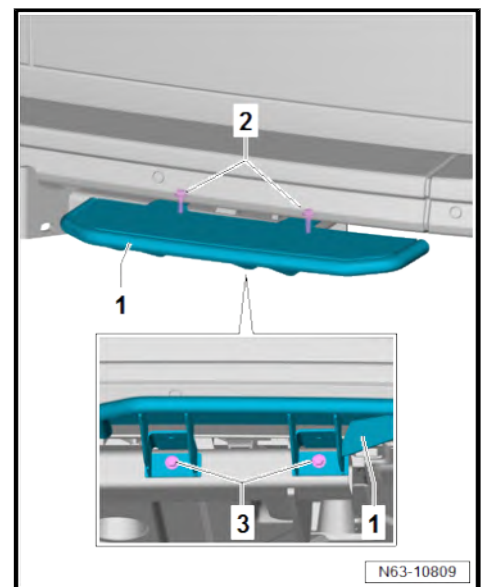
- Remove centre bumper cover ⇒ [page 192](#) .
- Unscrew bolts -2- and -3-.
- Unscrew bolts -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["9.1 Assembly overview - towing bracket", page 318](#)



10 Spare wheel bracket

⇒ ["10.1 Assembly overview - spare wheel bracket", page 324](#)

⇒ ["10.2 Removing and installing spare wheel bracket", page 325](#)

⇒ ["10.3 Removing and installing tether", page 327](#)

⇒ ["10.4 Removing and installing spare wheel winch bracket", page 328](#)

10.1 Assembly overview - spare wheel bracket

⇒ ["10.1.1 Assembly overview - spare wheel bracket, panel van or Kombi", page 324](#)

⇒ ["10.1.2 Assembly overview - spare wheel bracket, single or double cab", page 325](#)

10.1.1 Assembly overview - spare wheel bracket, panel van or Kombi

1 - Bolts

- ☐ Qty. 3
- ☐ 30 Nm

2 - Spare wheel winch

- ☐ Together with spare wheel bracket
- ☐ Removing and installing
⇒ [page 325](#)

3 - Hook

- ☐ Together with tether

4 - Threaded stud

- ☐ 55 Nm

5 - Spare wheel bracket

- ☐ Together with spare wheel winch
- ☐ Removing and installing
⇒ [page 325](#)

6 - Bolt

- ☐ 55 Nm

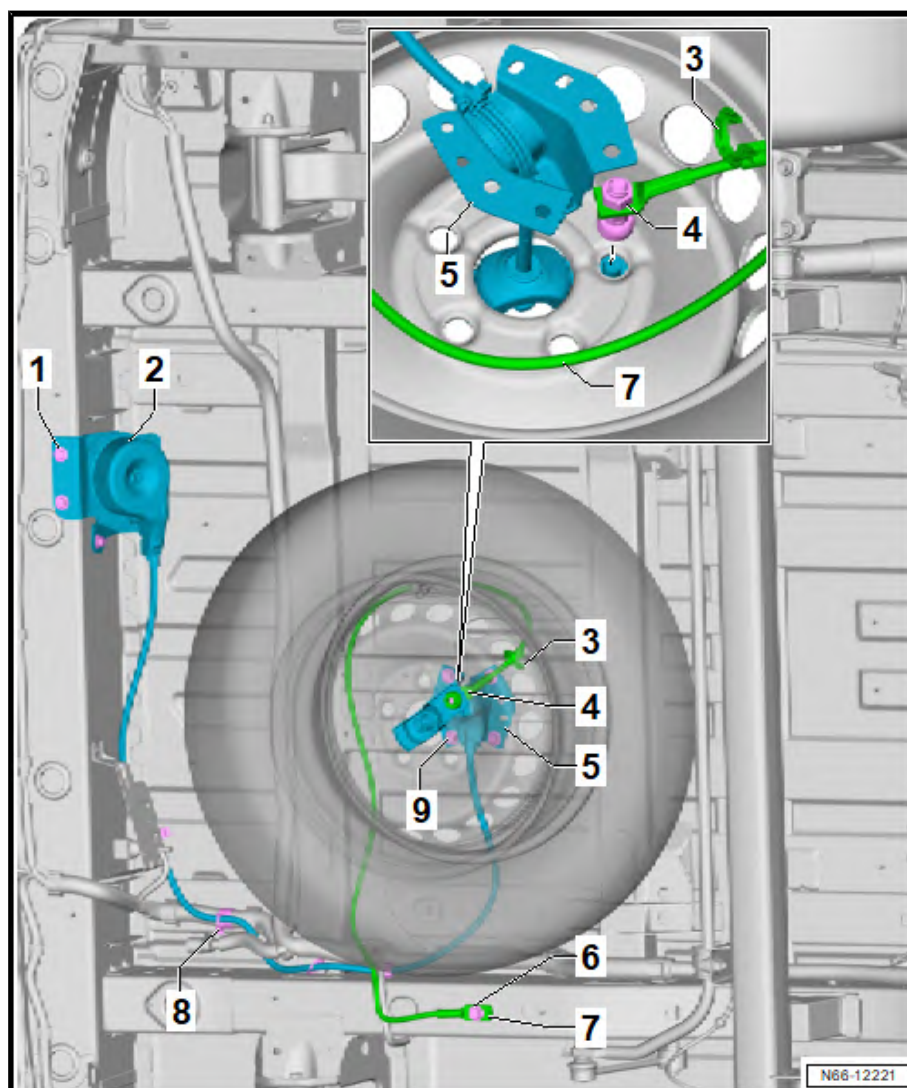
7 - Tether

- ☐ Removing and installing
⇒ [page 327](#)

8 - Cable holder

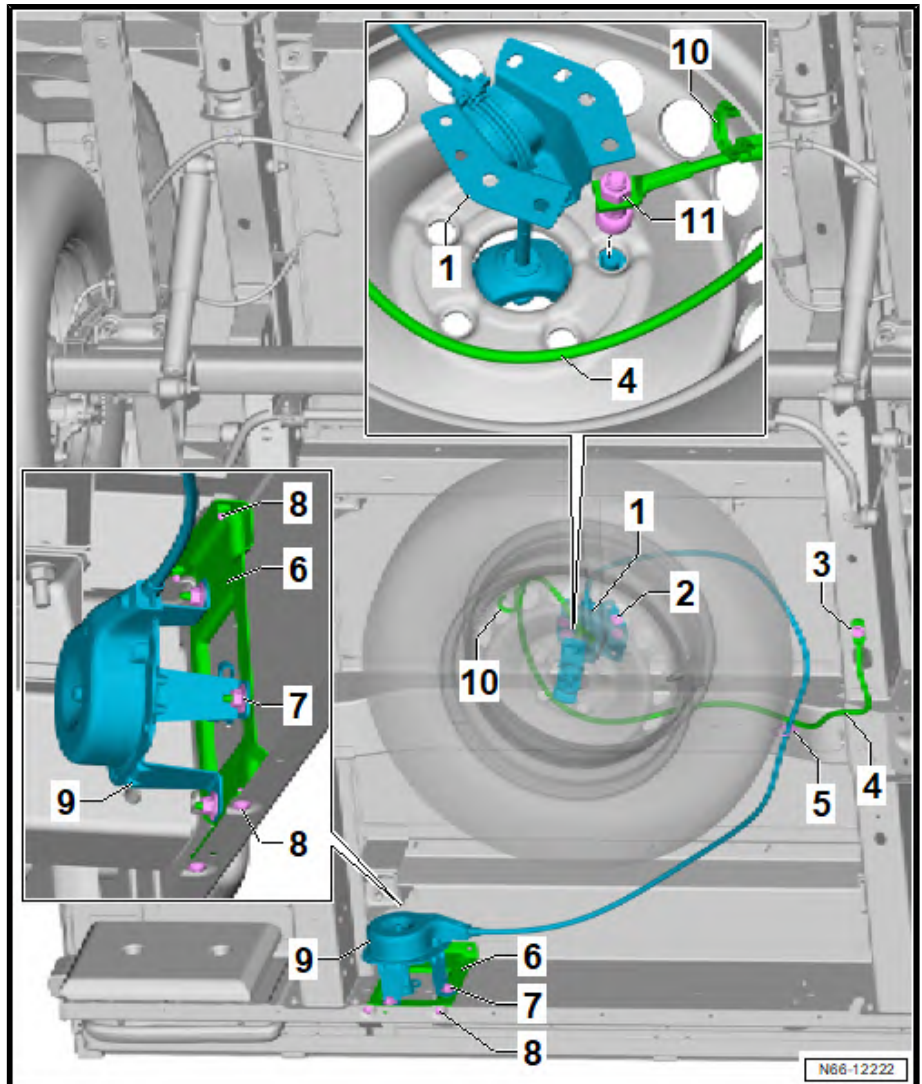
9 - Bolts

- ☐ Qty. 4
- ☐ 30 Nm



10.1.2 Assembly overview - spare wheel bracket, single or double cab

- 1 - Spare wheel bracket**
 - ☐ Together with spare wheel winch
 - ☐ Removing and installing
⇒ [page 325](#)
- 2 - Bolts**
 - ☐ Qty. 4
 - ☐ 30 Nm
- 3 - Bolt**
 - ☐ 55 Nm
- 4 - Tether**
 - ☐ Removing and installing
⇒ [page 327](#)
- 5 - Cable holder**
- 6 - Bracket**
 - ☐ Spare wheel winch
 - ☐ Removing and installing
⇒ [page 328](#)
- 7 - Nuts**
 - ☐ Qty. 3
 - ☐ 20 Nm
- 8 - Bolts**
 - ☐ Qty. 4
 - ☐ 8 Nm
- 9 - Spare wheel winch**
 - ☐ Together with spare wheel bracket
 - ☐ Removing and installing
⇒ [page 325](#)
- 10 - Hook**
 - ☐ Together with tether
- 11 - Threaded stud**
 - ☐ 55 Nm



10.2 Removing and installing spare wheel bracket

⇒ [“10.2.1 Removing and installing spare wheel bracket, panel van or Kombi”, page 325](#)

⇒ [“10.2.2 Removing and installing spare wheel bracket, single or double cab”, page 326](#)

10.2.1 Removing and installing spare wheel bracket, panel van or Kombi

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331-



Removing

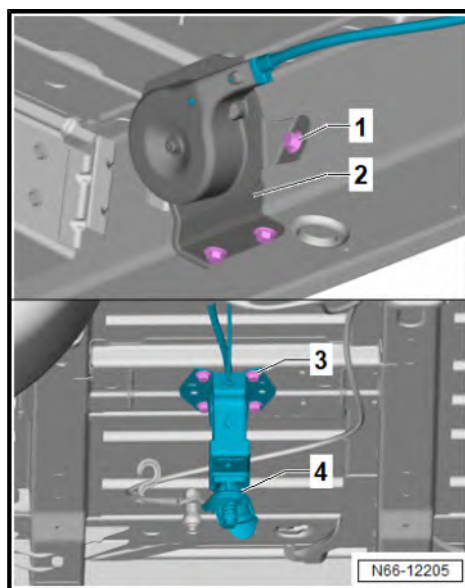
- Remove spare wheel ⇒ Maintenance ; Booklet 10.3 .



Note

Depending on equipment (different body derivatives), there may be slight deviations from the illustrations below.

- If fitted, detach cable retainers or Bowden cable clips of spare wheel winch.
- Unscrew bolts -1-. When doing this, remove spare wheel winch -2-.
- Unscrew bolts -3-. When doing this, remove spare wheel bracket -4- together with spare wheel winch -2-.



Installing

Install in reverse order of removal, observing the following:

- Check spare wheel bracket for contamination, and clean it if necessary ⇒ Maintenance ; Booklet 10.3 .

Specified torques

- ◆ ⇒ [“10.1 Assembly overview - spare wheel bracket”, page 324](#)

10.2.2 Removing and installing spare wheel bracket, single or double cab

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





Removing

- Remove spare wheel ⇒ Maintenance ; Booklet 10.3 .



Note

Depending on equipment (different body derivatives), there may be slight deviations from the illustrations below.

- If fitted, detach cable retainers or Bowden cable clips of spare wheel winch.
- Unscrew nuts -1-.
- Remove spare wheel winch -2-.
- Unscrew bolts -3-. When doing this, remove spare wheel bracket -4- together with spare wheel winch -2-.

Installing

Install in reverse order of removal, observing the following:

- Check spare wheel bracket for contamination, and clean it if necessary ⇒ Maintenance ; Booklet 10.3 .

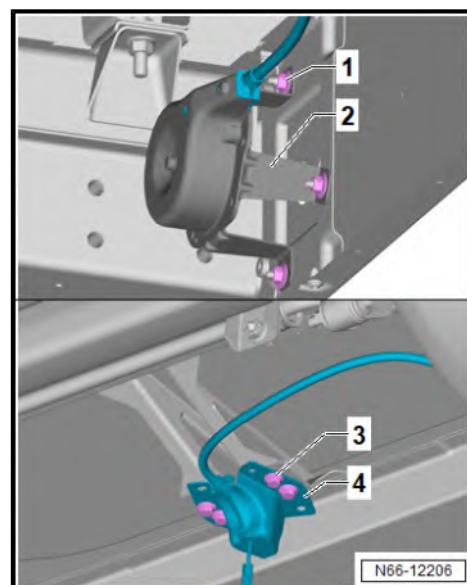
Specified torques

- ◆ ⇒ ["10.1 Assembly overview - spare wheel bracket", page 324](#)

10.3 Removing and installing tether

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1332-





Removing

- Remove spare wheel ⇒ Maintenance ; Booklet 10.3 .



Note

Depending on equipment (different body derivatives), there may be slight deviations from the illustrations below.

- Unscrew bolt -1- When doing this, remove tether -2-.

Installing

Install in reverse order of removal, observing the following:

- Check spare wheel bracket for contamination, and clean it if necessary ⇒ Maintenance ; Booklet 10.3 .

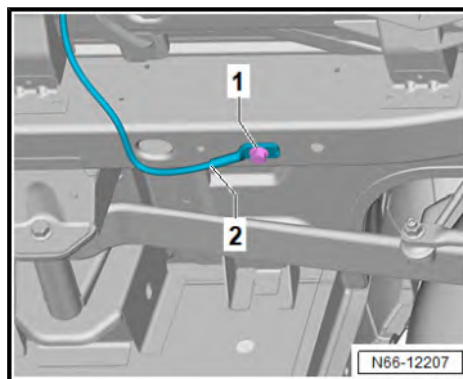
Specified torques

- ♦ ⇒ [“10.1 Assembly overview - spare wheel bracket”, page 324](#)

10.4 Removing and installing spare wheel winch bracket

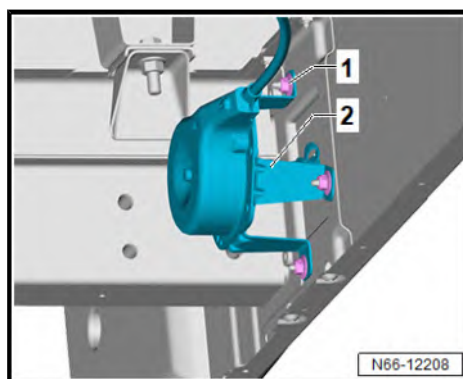
Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-



Removing

- Unscrew nuts -1-.
- Remove spare wheel winch -2-.





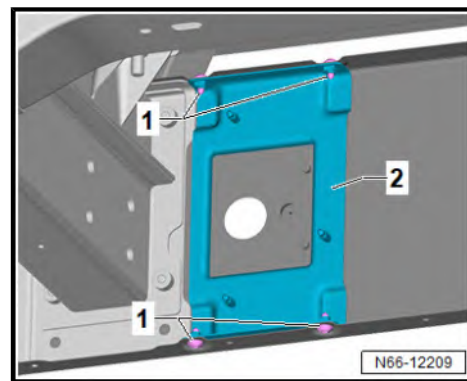
- Unscrew bolts -1-. When doing this, remove spare wheel winch bracket -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“10.1 Assembly overview - spare wheel bracket”, page 324](#)





11 Mudflap

⇒ [“11.1 Assembly overview - mudflap”, page 330](#)

⇒ [“11.2 Removing and installing mudflaps”, page 330](#)

11.1 Assembly overview - mudflap

1 - Mudflap

- ☐ Front
- ☐ Removing and installing
⇒ [page 330](#)

2 - Bolts

- ☐ Qty. 3
- ☐ 1.5 Nm

3 - Bolts

- ☐ Depending on equipment
- ☐ Qty. 2
- ☐ 1.5 Nm

4 - Mudflap

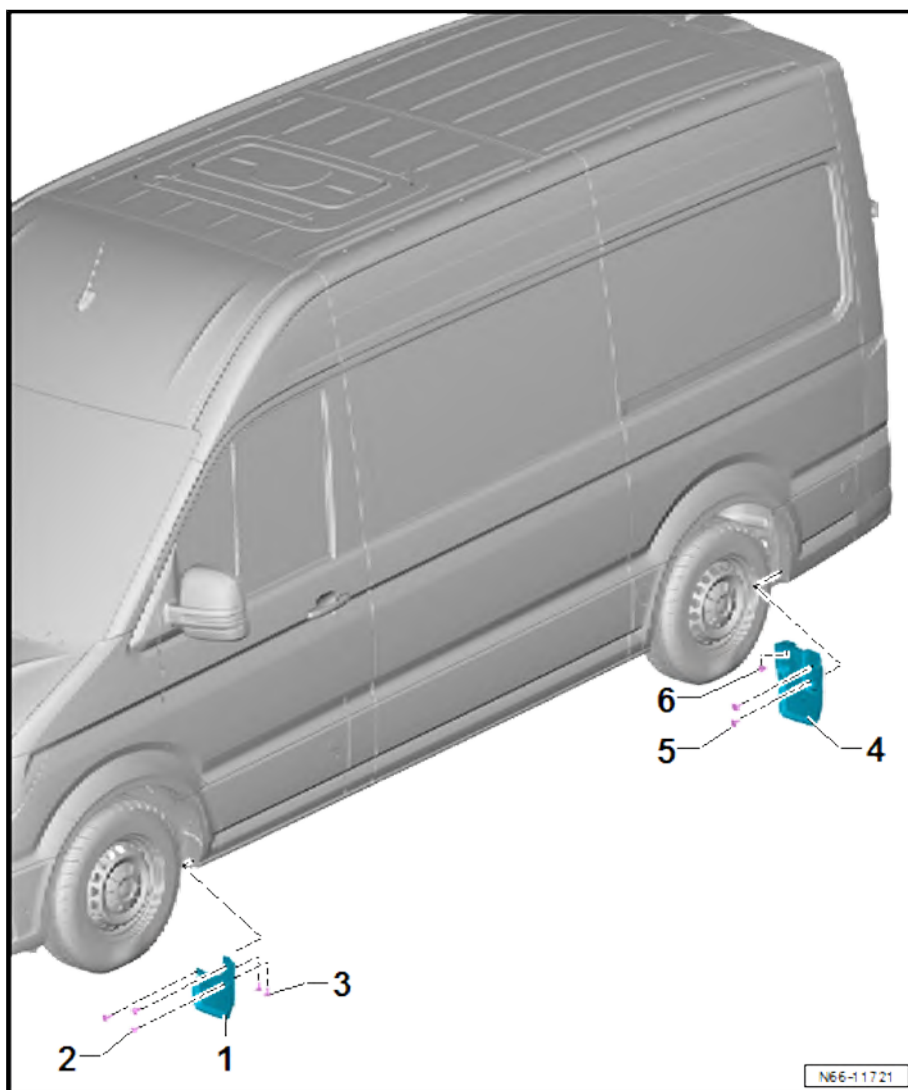
- ☐ Rear
- ☐ Removing and installing
⇒ [page 330](#)

5 - Bolts

- ☐ Qty. 2
- ☐ 1.5 Nm

6 - Nut

- ☐ 1.5 Nm



11.2 Removing and installing mudflaps

⇒ [“11.2.1 Removing and installing mudflaps, front”, page 330](#)

⇒ [“11.2.2 Removing and installing mudflaps, rear”, page 331](#)

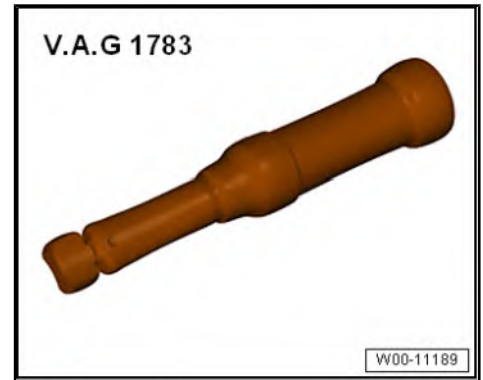
⇒ [“11.2.3 Removing and installing mudflap, double cab”, page 332](#)

11.2.1 Removing and installing mudflaps, front

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

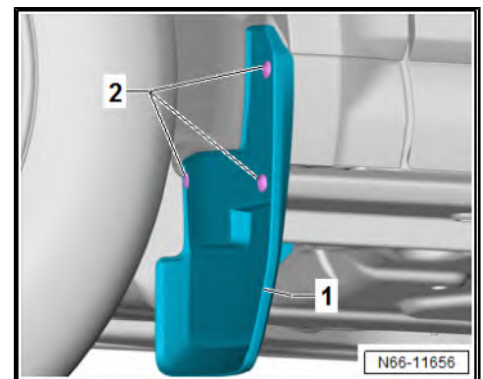
- Unscrew bolts -2-. When doing this, remove mudflap -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

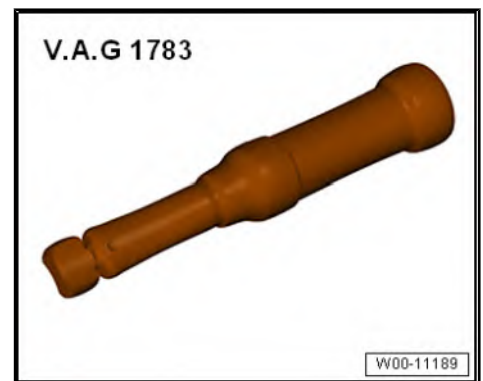
- ◆ ⇒ ["11.1 Assembly overview - mudflap", page 330](#)



11.2.2 Removing and installing mudflaps, rear

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing



Note

For reasons of clarity, the rear wheel is not shown.

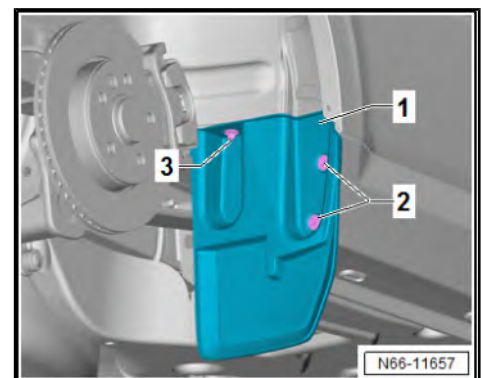
- Unscrew bolts -2-.
- Unscrew nut -3-. When doing this, remove mudflap -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["11.1 Assembly overview - mudflap", page 330](#)





11.2.3 Removing and installing mudflap, double cab

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1783-



Removing

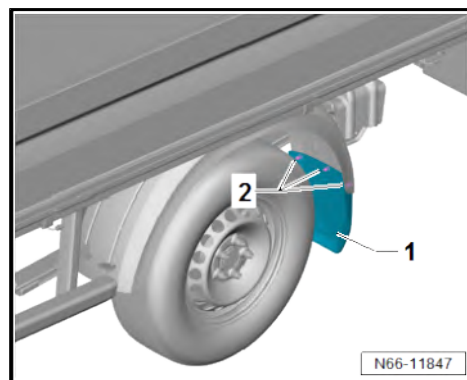
- Unscrew nuts -2-. When doing this, remove mudflap -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["12.7 Assembly overview - wheel housing", page 341](#)





12 Load floor body

- ⇒ ["12.1 Contact corrosion", page 333](#)
- ⇒ ["12.2 Assembly overview - dropside bed", page 333](#)
- ⇒ ["12.3 Assembly overview - dropside body", page 337](#)
- ⇒ ["12.4 Assembly overview - rear cross member", page 338](#)
- ⇒ ["12.5 Assembly overview - carrier", page 339](#)
- ⇒ ["12.6 Assembly overview - skid plate", page 340](#)
- ⇒ ["12.7 Assembly overview - wheel housing", page 341](#)
- ⇒ ["12.8 Removing and installing dropside bed", page 341](#)
- ⇒ ["12.9 Removing and installing tailboard, front", page 342](#)
- ⇒ ["12.11 Removing and installing console", page 343](#)
- ⇒ ["12.12 Removing and installing carrier", page 344](#)
- ⇒ ["12.13 Removing and installing skid plate", page 344](#)
- ⇒ ["12.14 Removing and installing strut", page 345](#)
- ⇒ ["12.15 Removing and installing wheel housing", page 346](#)

12.1 Contact corrosion

The add-on parts, side boards and tail boards of the dropside model are made from aluminium.

When these parts are installed, special care must be taken regarding contact corrosion.

Contact corrosion can occur if unsuitable fastening elements (bolts, nuts, washers and pop rivets) are used.

For this reason, only connecting elements with a special surface coating (Dacromet) have been fitted. These elements can be recognised by their green colour.

If unsure about the reusability of a specific part, always fit a new part.

Use only genuine parts.

Damage resulting from contact corrosion is not covered by the warranty.

12.2 Assembly overview - dropside bed

- ⇒ ["12.2.1 Assembly overview - dropside bed, tailboards", page 333](#)
- ⇒ ["12.2.2 Assembly overview - dropside bed, up to calendar week 35/2017", page 335](#)
- ⇒ ["12.2.3 Assembly overview - dropside bed, as of calendar week 35/2017", page 336](#)

12.2.1 Assembly overview - dropside bed, tailboards



1 - Tailgate

- ☐ Rear
- ☐ Removing and installing

2 - Pop rivets

- ☐ Qty. 4

3 - Cover

- ☐ Top left and right

4 - Locking ring

- ☐ For left and right on tail-board

5 - Cover

- ☐ Bottom left and right

6 - Pop rivets

- ☐ Qty. 2

7 - Arrester cable

- ☐ Qty. 2

8 - Bolts

- ☐ Qty. 2
- ☐ 10 Nm

9 - Hinges

- ☐ Qty. 3

10 - Bolts

- ☐ Qty. 2
- ☐ 10 Nm

11 - Pop rivets

- ☐ Qty. 4

12 - Step

- ☐ Qty. 1
- ☐ For tailgate

13 - Tailgate

- ☐ Left and right
- ☐ Removing and installing

14 - Locking ring

- ☐ For left and right on tailboard

15 - Cover

- ☐ Bottom left and right

16 - Pop rivets

- ☐ Qty. 2

17 - Cover

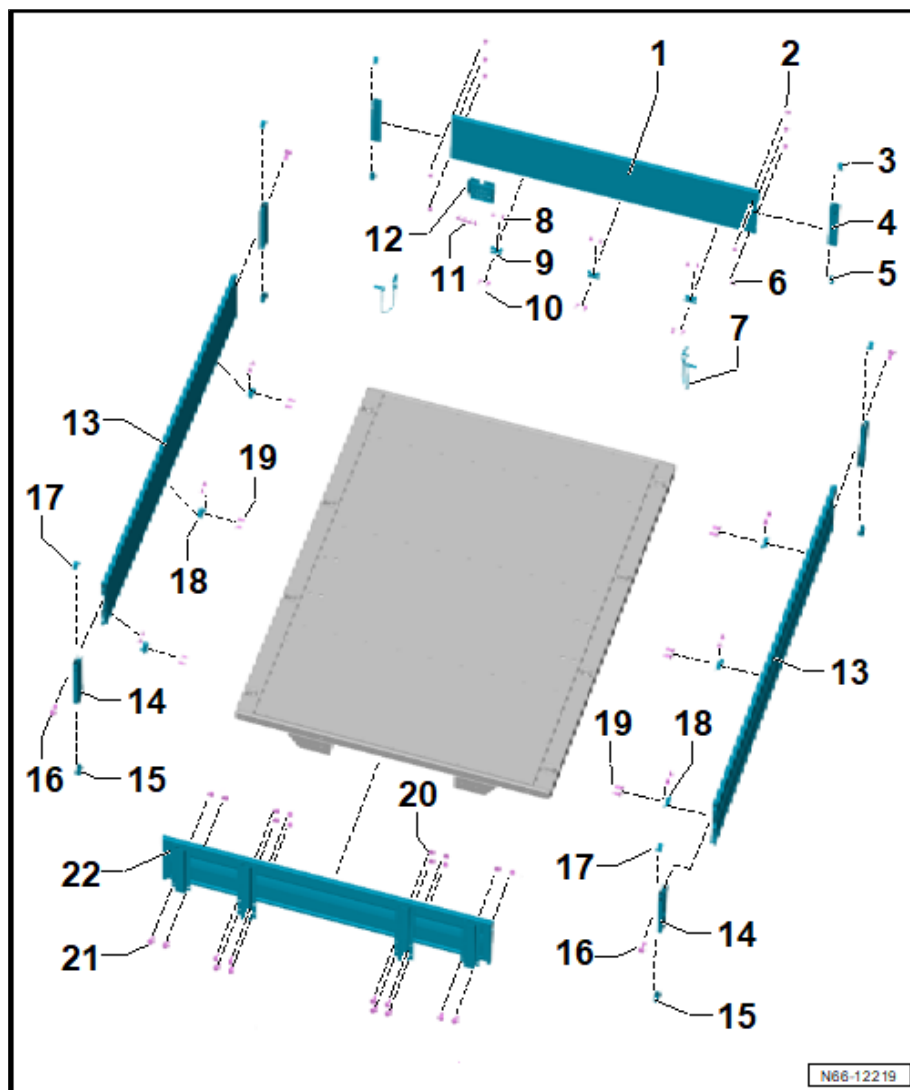
- ☐ Top left and right

18 - Hinge

- ☐ Qty. 4

19 - Bolts

- ☐ 4 per hinge
- ☐ 10 Nm





20 - Nuts

- ☐ Qty. 12
- ☐ 85 Nm

21 - Bolts

- ☐ Qty. 12
- ☐ 85 Nm

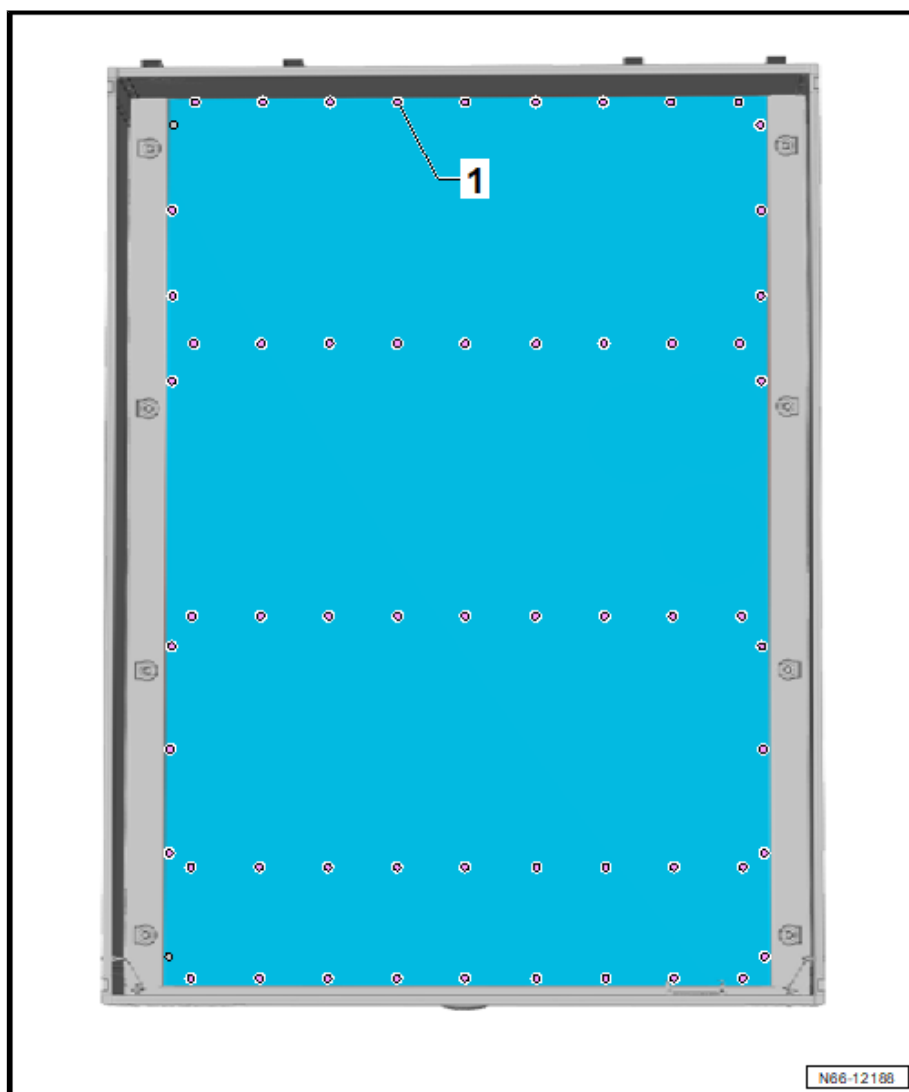
22 - Front board

- ☐ Removing and installing

12.2.2 Assembly overview - dropside bed, up to calendar week 35/2017

1 - Bolts

- ☐ Qty. 44
- ☐ 12 Nm





12.2.3 Assembly overview - dropside bed, as of calendar week 35/2017

1 - Bolts

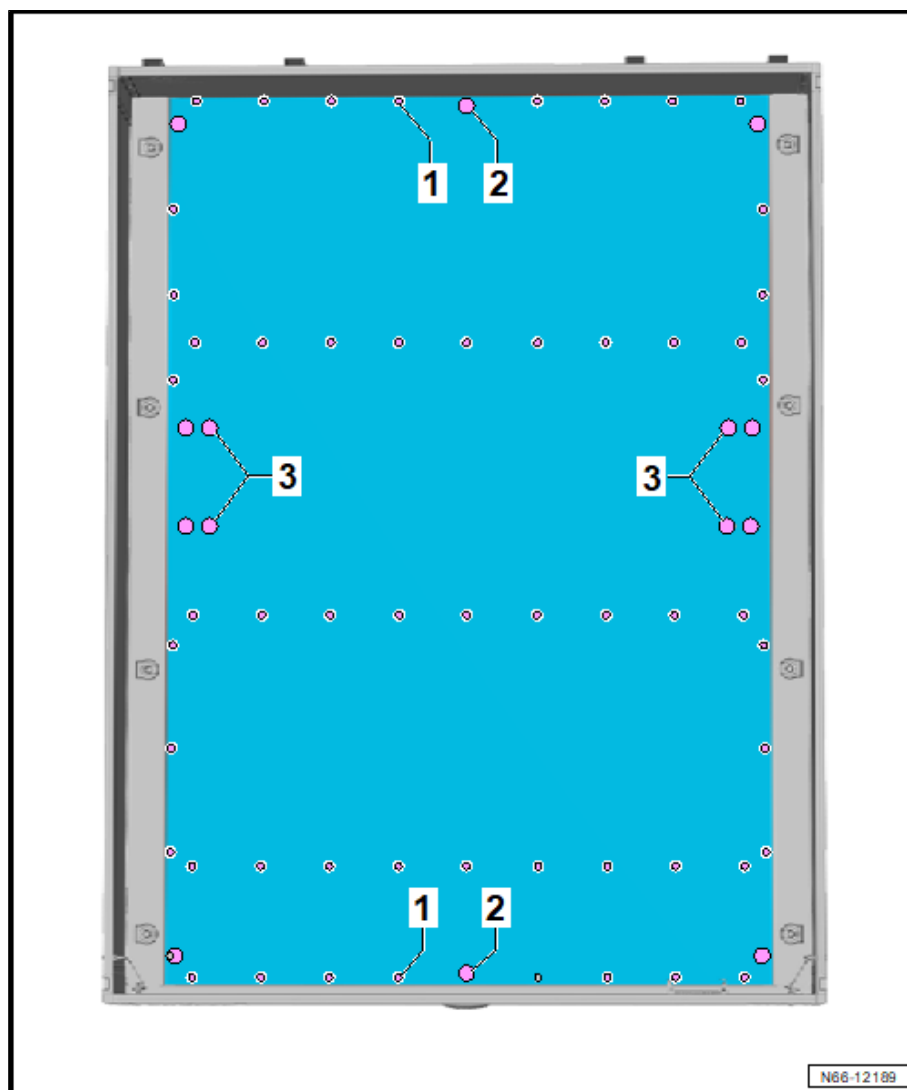
- Qty. 34
- 12 Nm

2 - Bolts

- Qty. 6
- 12 Nm

3 - Bolts

- Qty. 8
- 12 Nm





12.3 Assembly overview - dropside body

1 - Load floor body

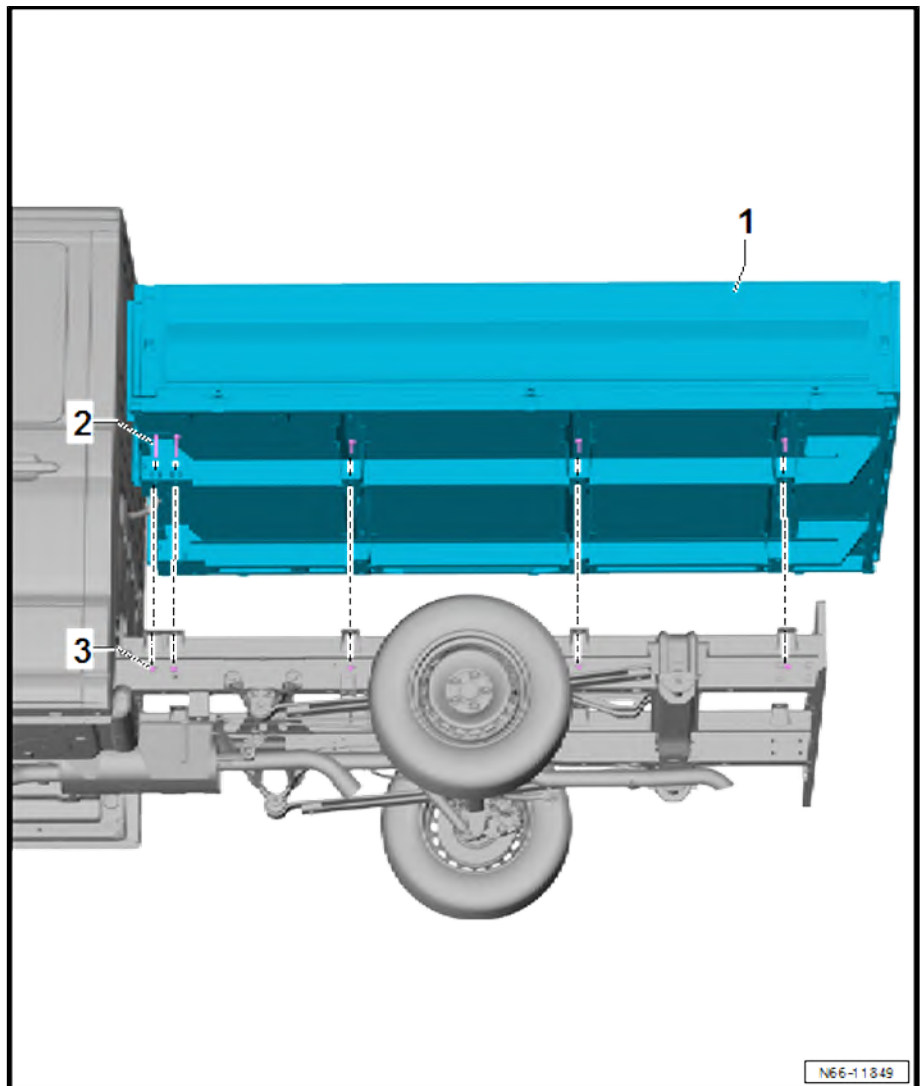
- ☐ Removing and installing
⇒ [page 341](#)

2 - Bolts

- ☐ Renew after removal
- ☐ 9, 11, 13 or 15 pieces per side
- ☐ Number of bolts depends on dropside bed length

3 - Nuts

- ☐ Renew after removal
- ☐ 9, 11, 13 or 15 pieces per side
- ☐ Number of nuts depends on dropside bed length
- ☐ 70 Nm +180°





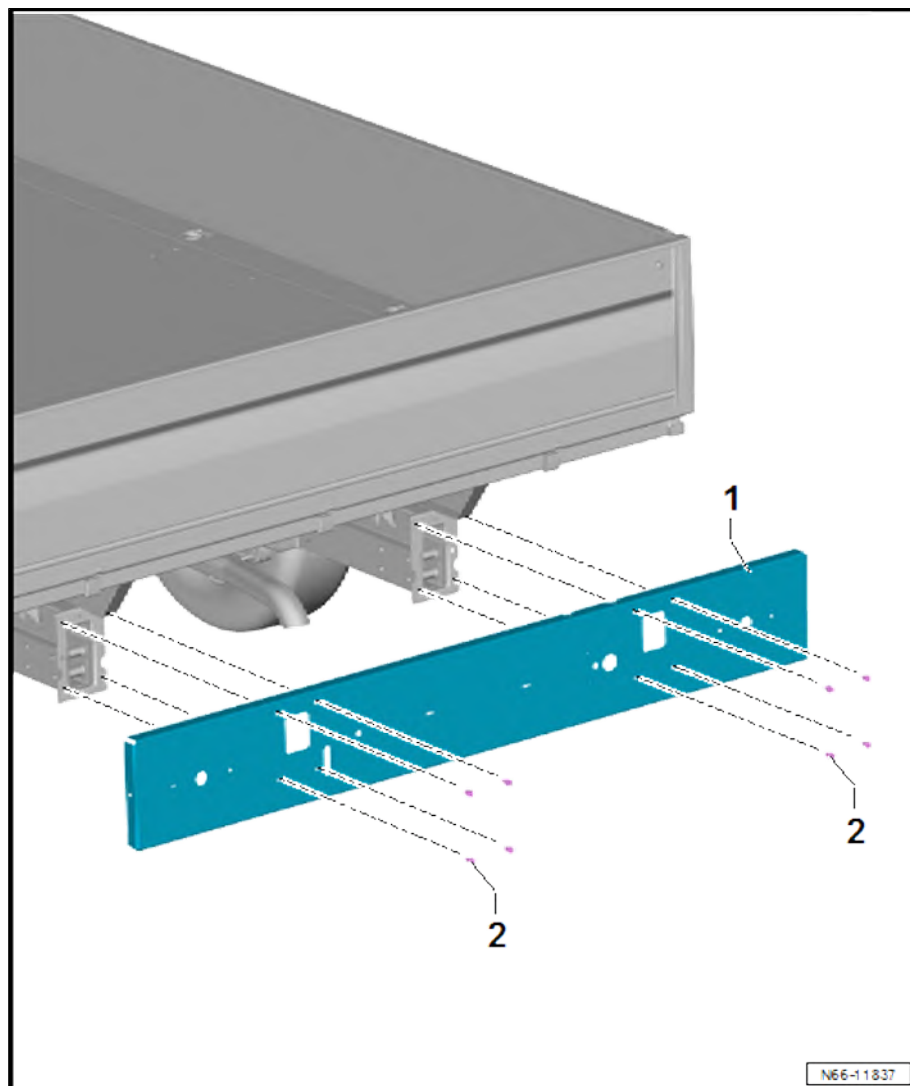
12.4 Assembly overview - rear cross member

1 - Rear cross member

- ☐ Different versions; for allocation, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ web-MANTIS for MAN TGE.
- ☐ Removing and installing

2 - Bolts

- ☐ Qty. 4 on each side
- ☐ 20 Nm





12.5 Assembly overview - carrier

1 - Mounting plate

- ❑ Removing and installing
⇒ [page 344](#)

2 - Bolts

- ❑ Qty. 2 on each side
- ❑ 40 Nm

3 - Nuts

- ❑ Qty. 2 on each side
- ❑ Specified torque⇒ Electrical system; Rep. gr. 94 ; Tail lights; Assembly overview - tail lights in body

4 - Stone deflector

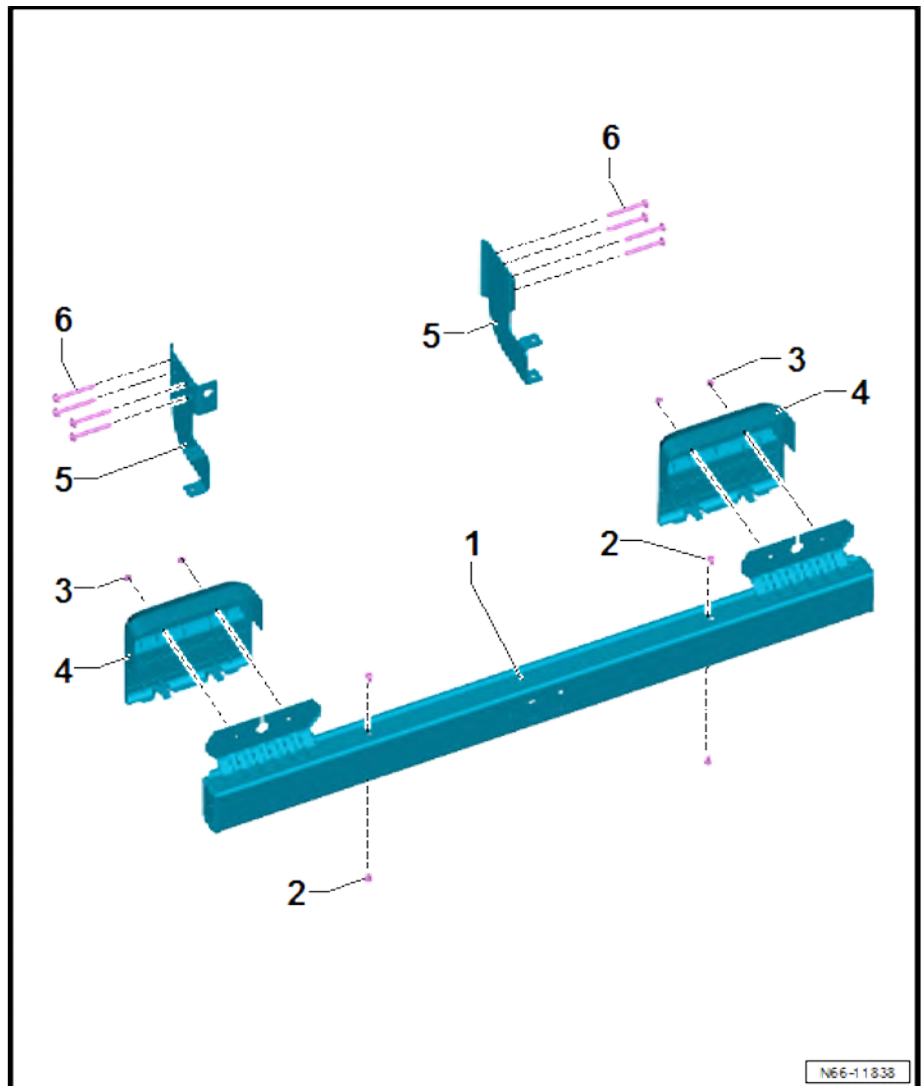
- ❑ Different versions; for allocation, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ web-MANTIS for MAN TGE.

5 - Console

- ❑ Left and right

6 - Bolts

- ❑ Renew after removal
- ❑ Qty. 4 on each side
- ❑ 70 Nm +270°





12.6 Assembly overview - skid plate

1 - Struts

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 345](#)

2 - Bolts

- ☐ 3 per strut
- ☐ 20 Nm

3 - Nuts

- ☐ Qty. 2
- ☐ 20 Nm

4 - Bolts

- ☐ Qty. 2

5 - End piece

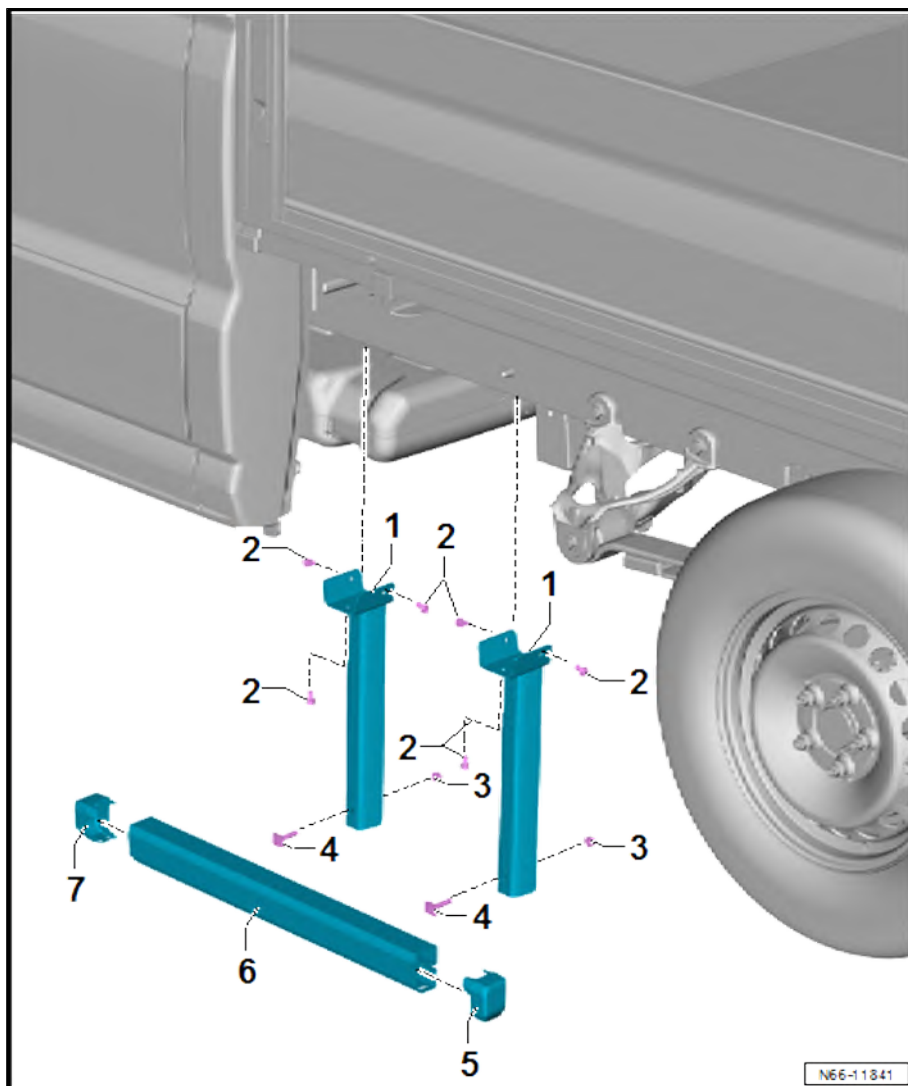
- ☐ Right

6 - Skid plate

- ☐ Removing and installing
⇒ [page 344](#)

7 - End piece

- ☐ Left





12.7 Assembly overview - wheel housing

1 - Wheel housing

- ☐ Removing and installing
⇒ [page 346](#)

2 - Bolts

- ☐ Qty. 3

3 - Mudflap

- ☐ Removing and installing
⇒ [page 330](#)

4 - Nuts

- ☐ Qty. 3
- ☐ 2.5 Nm

5 - Bolts

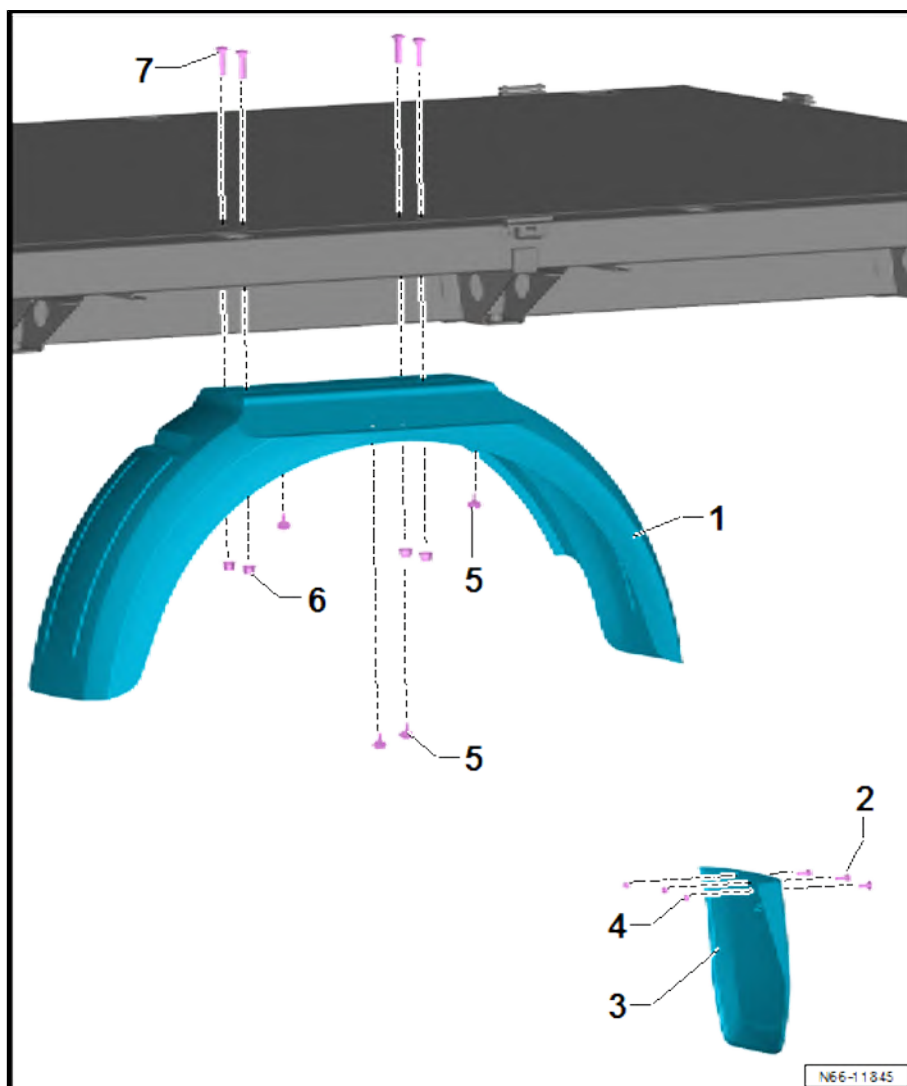
- ☐ Qty. 4
- ☐ 3 Nm

6 - Nuts

- ☐ Qty. 4
- ☐ 12 Nm

7 - Bolts

- ☐ Qty. 4



12.8 Removing and installing dropside bed

Special tools and workshop equipment required

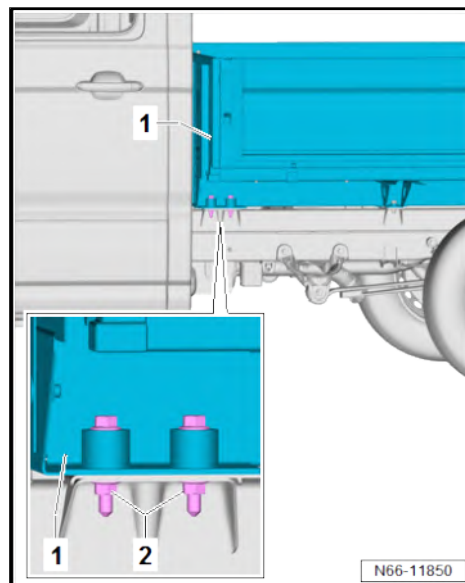
- ◆ Torque wrench - V.A.G 1332-





Removing

- Unscrew nuts -2- at front.
- Remove bolts from dropside bed -1-.



- Unscrew nut -2-.



Note

Number of bolted connections in rear dropside bed depends on dropside bed length.

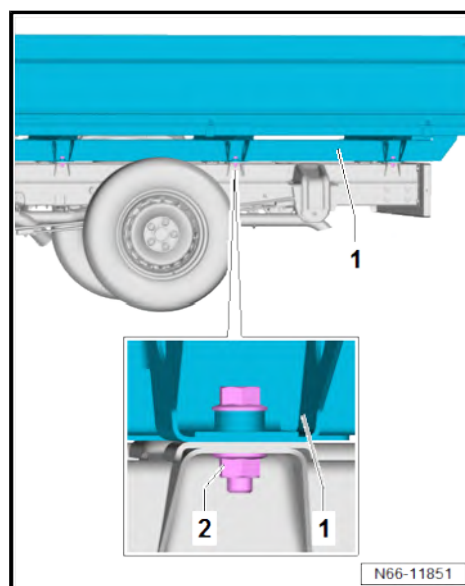
- Remove bolt upwards from dropside bed -1-.
- Lift off dropside bed -1- from vehicle. Several fitters are needed.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [“12.3 Assembly overview - dropside body”, page 337](#)



12.9 Removing and installing tailboard, front

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1332-





Removing

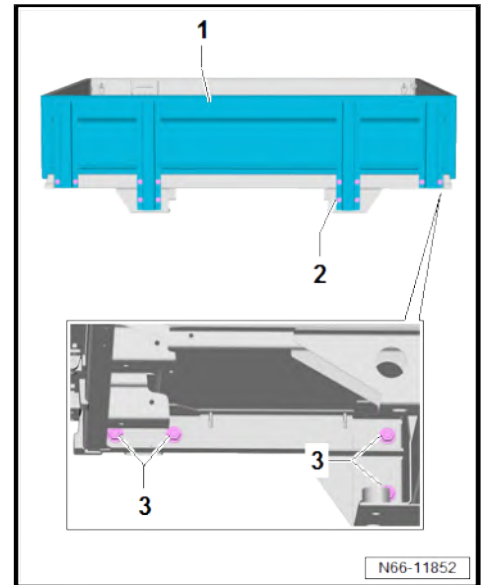
- Remove dropside bed ➤ [page 341](#) .
- Remove bolts -2- on left and right.
- Unscrew nuts -3- on left and right.
- Remove front tailboard -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [“12.2 Assembly overview - dropside bed”, page 333](#)



12.10 Removing and installing tailgate

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

- The following work steps must be performed by a second mechanic.



Note

Number of hinges depends on length of tailboard.

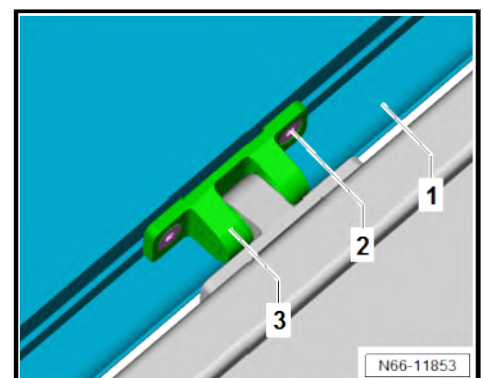
- Unscrew bolts -2-.
- Remove tailboard -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [“12.2 Assembly overview - dropside bed”, page 333](#)



12.11 Removing and installing console

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1332-



Removing

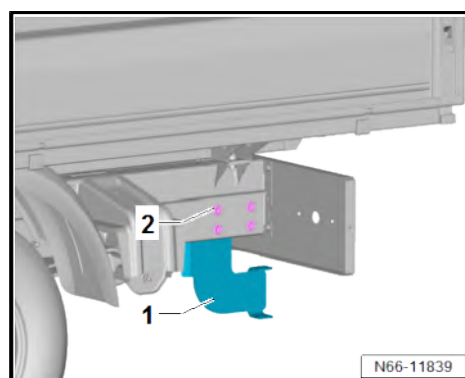
- Remove carrier ➔ [page 344](#) .
- Unscrew bolts -2-.
- Remove console -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ ["12.5 Assembly overview - carrier", page 339](#)



12.12 Removing and installing carrier

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-



Removing

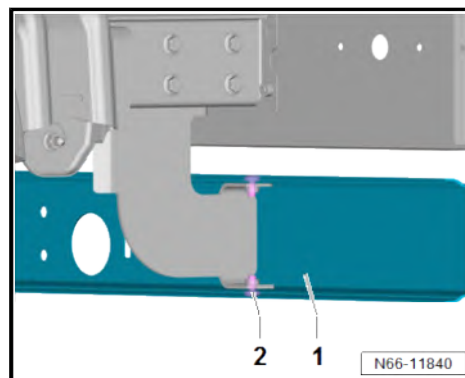
- Remove tail lights ➔ Electrical system; Rep. gr. 94 ; Tail lights; Removing and installing tail light .
- Remove bolts -2- on left and right.
- Remove carrier -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ ["12.5 Assembly overview - carrier", page 339](#)

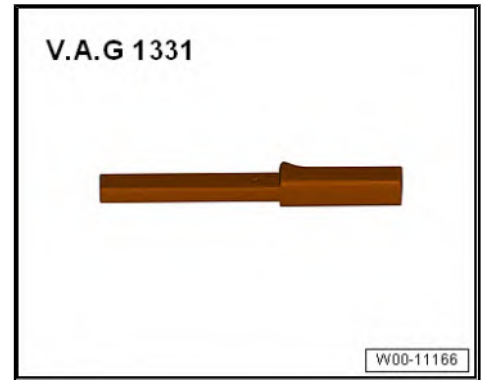


12.13 Removing and installing skid plate

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331-



Removing

- Unscrew nuts -2- on left and right.
- Remove skid plate -1-.

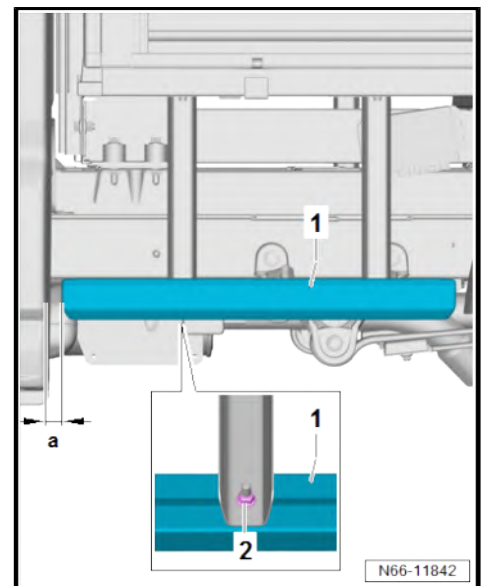
Installing

Install in reverse order of removal, observing the following:

- Adjust skid plate -1- to -dimension a- with gap of 28 mm to cab back wall.

Specified torques

- ◆ ⇒ ["12.6 Assembly overview - skid plate", page 340](#)



12.14 Removing and installing strut

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





Removing

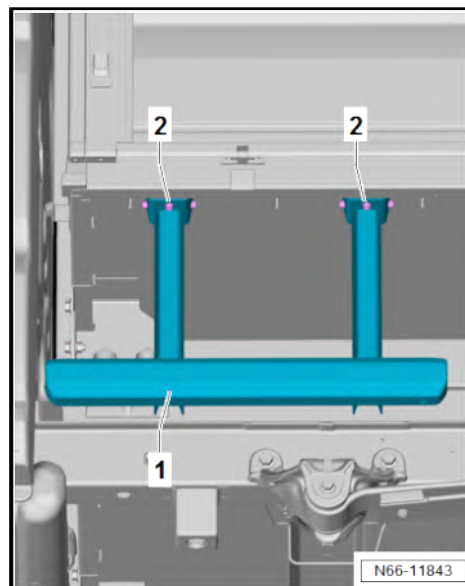
- Remove bolts -2- on left and right.
- Remove struts with skid plate -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ ["12.6 Assembly overview - skid plate", page 340](#)



12.15 Removing and installing wheel housing

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1783-



- ♦ Torque wrench - V.A.G 1331-





Removing

- Unscrew bolts -3-.
- Unscrew nuts -2-.
- Remove wheel housing -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ ["12.7 Assembly overview - wheel housing", page 341](#)

